Software Requirements Specification for Squigglr: A Browser-Based Visual-Motor Practice and Progress-Tracking Platform

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Contents

Introduction 1

1 Purpose of the Project 3
- 1.1 User Business . . . 3
- 1.2 Goals of the Project . . . 3

2 Stakeholders 8
- 2.1 Client . . . 9
- 2.2 Customer . . . 9
- 2.3 Other Stakeholders . . . 9
- 2.4 Hands-On Users of the Project . . . 10
- 2.5 Personas . . . 14
- 2.6 Priorities Assigned to Users . . . 16
- 2.7 User Participation . . . 16
- 2.8 Maintenance Users and Service Technicians . . . 17

3 Mandated Constraints 17
- 3.1 Solution Constraints . . . 18
- 3.2 Implementation Environment of the Current System . . . 20
- 3.3 Partner or Collaborative Applications . . . 20
- 3.4 Off-the-Shelf Software . . . 20
- 3.5 Anticipated Workplace Environment . . . 20
- 3.6 Schedule Constraints . . . 20
- 3.7 Budget Constraints . . . 21
- 3.8 Enterprise Constraints . . . 21

4 Naming Conventions and Terminology 22
- 4.1 Glossary and Acronyms . . . 22
- 4.2 Symbolic Constants . . . 28

5 Relevant Facts And Assumptions 33
- 5.1 Relevant Facts . . . 33
- 5.2 Business Rules . . . 36
- 5.3 Assumptions . . . 39

6 The Scope of the Work **44**
6.1 The Current Situation . . . 44
6.2 The Context of the Work . . . 45
6.3 Work Partitioning . . . 46
6.4 Specifying a Business Use Case (BUC) . . . 47

7 Business Data Model and Data Dictionary **48**
7.1 Business Data Model . . . 49
7.2 Data Dictionary . . . 51

8 The Scope of the Product **65**
8.1 Product Boundary . . . 65
8.2 Product Use Case Table . . . 68
8.3 Individual Product Use Cases (PUC's) . . . 70
8.3.1 UC-001: Create a child's profile . . . 70
8.3.2 UC-002: Choose a lesson . . . 71
8.3.3 UC-003: Practise a guided lesson . . . 72
8.3.4 UC-004: Start the child at a later stage . . . 73
8.3.5 UC-005: Respond to a suggestion to go back a stage . . . 74
8.3.6 UC-006: Create and edit artwork . . . 75
8.3.7 UC-007: Refine artwork and restore the original . . . 75
8.3.8 UC-008: Record an attempt . . . 76
8.3.9 UC-009: Review progress . . . 77
8.3.10 UC-010: Review history and change settings . . . 77

9 Functional Requirements **78**
9.1 Learning Activity Requirements . . . 78
9.2 Formal Specification . . . 90
9.3 Creative Drawing Requirements . . . 96
9.4 Attempt Record Requirements . . . 110
9.5 Progress Summary Requirements . . . 116

10 Look and Feel Requirements **122**
10.1 Appearance Requirements . . . 123
10.2 Style Requirements . . . 124

11 Usability and Humanity Requirements **126**
11.1 Ease of Use Requirements . . . 126
11.2 Personalization and Internationalization Requirements . . . 127
11.3 Learning Requirements . . . 128
11.4 Understandability and Politeness Requirements . . . 129
11.5 Accessibility Requirements . . . 131

12 Performance Requirements **134**
12.1 Speed and Latency Requirements . . . 134
12.2 Safety-Critical Requirements . . . 135
12.3 Precision or Accuracy Requirements . . . 141
12.4 Robustness or Fault-Tolerance Requirements . . . 142
12.5 Capacity Requirements . . . 143
12.6 Scalability or Extensibility Requirements . . . 143
12.7 Longevity Requirements . . . 143

13 Operational and Environmental Requirements **144**
13.1 Expected Physical Environment . . . 144
13.2 Wider Environment Requirements . . . 144
13.3 Requirements for Interfacing with Adjacent Systems . . . 145
13.4 Productization Requirements . . . 145
13.5 Release Requirements . . . 146

14 Maintainability and Support Requirements **146**
14.1 Maintenance Requirements . . . 146
14.2 Supportability Requirements . . . 147
14.3 Adaptability Requirements . . . 148

15 Security Requirements **148**
15.1 Access Requirements . . . 148
15.2 Integrity Requirements . . . 149
15.3 Privacy Requirements . . . 150
15.4 Audit Requirements . . . 150
15.5 Immunity Requirements . . . 151

16 Cultural Requirements **151**

17 Compliance Requirements **152**
17.1 Legal Requirements 152
17.2 Standards Compliance Requirements 153

18 Open Issues **153**

19 Off-the-Shelf Solutions **157**
19.1 Ready-Made Products 159
19.2 Reusable Components 163
19.3 Products That Can Be Copied 168

20 New Problems **169**
20.1 Effects on the Current Environment 169
20.2 Effects on the Installed Systems 171
20.3 Potential User Problems 173
20.4 Limitations in the Anticipated Implementation Environment That May Inhibit the New Product 175
20.5 Follow-Up Problems 177

21 Tasks **180**
21.1 Project Planning 180
21.2 Planning of the Development Phases 180

22 Migration to the New Product **188**
22.1 Requirements for Migration to the New Product 188
22.2 Data That Has to be Modified or Translated for the New System 189

23 Costs **189**

24 User Documentation and Training **192**
24.1 User Documentation Requirements 192
24.2 Training Requirements 195

25 Waiting Room **195**

26 Ideas for Solution **197**

27 Likely and Unlikely Changes **200**
27.1 Likely Changes 200
27.2 Unlikely Changes 202

28 Traceability **204**
28.1 Goal/source to use case to requirement 204
28.2 Functional to nonfunctional requirements 205
28.3 Hazards to mitigating requirements 206

29 Appendix — Requirement Sources **216**
29.1 Key Interview Statements 219
29.2 Key Questionnaire Statements 221

30 Appendix — Specialist Interview Transcript **223**

31 Appendix — Specialist Questionnaire Responses **245**

32 Appendix — Generative AI Use Disclosure **249**
32.1 AI Tools Used 249
32.2 How and Where Used 249
32.3 Verification of AI-Generated Content 251

33 Appendix — Reflection **255**

Revision History

Date	Developer(s)	Change
13 October 2026	Rayan Nasrallah	Added the de-identified specialist interview transcript as an appendix (#84).
14 October 2026	Rayan Nasrallah	Wrote Section 1, Purpose of the Project, with the six goals and their constants (#85).
14 October 2026	Rayan Nasrallah	Wrote Section 2, Stakeholders, including personas (#86).
14 October 2026	Rayan Nasrallah	Wrote Section 3, Mandated Constraints (#87).
14 October 2026	Rayan Nasrallah	Wrote Appendix A, Requirement Sources (#89).
14 October 2026	Rayan Nasrallah	Linked interview citations to transcript turns (#95).
14 October 2026	Rayan Nasrallah	Wrote Section 5, Relevant Facts and Assumptions, with business rules (#88).
14 October 2026	Rayan Nasrallah	Made child-dependent thresholds adult-configurable, added the improvement rule and SRC-008 (#102).
14 October 2026	Rayan Nasrallah	Wrote Section 18, Open Issues, and Section 27, Likely and Unlikely Changes (#90).
14 October 2026	Rayan Nasrallah	Added safety-critical requirements SAF-RAY-001 to SAF-RAY-008 to Section 12.2 (#91).
14 October 2026	Moustafa Moustafa	Wrote Section 8.2, Product Use Case Table (#108).
15 October 2026	Rayan Nasrallah	Added the specialist questionnaire responses as SRC-009, with an appendix.

Date	Developer(s)	Change
15 October 2026	Ahmed Elzaria	Completed SRS front matter and conventions, introduced the glossary and constants register, and prepared the shared requirement structure (#131).
15 October 2026	Rayan Nasrallah	Added my reflection responses to the appendix and my generative AI use disclosure (#139).
15 October 2026	Ahmed Elzaria	Specified guided learning activities and formal lesson behaviour; clarified pass-based advancement within adult-selected lessons (#132).
15 October 2026	Ahmed Elzaria	Added safety-critical requirements SAF-AHM-001 and SAF-AHM-002 for drawing-input integrity (#135).
15 October 2026	Moustafa Moustafa	Let the revision history table continue across pages (#108).
15 October 2026	Moustafa Moustafa	Wrote Section 6, The Scope of the Work, with the context diagram (#109).
15 October 2026	Moustafa Moustafa	Wrote Section 8.1, Product Boundary (#110).
16 October 2026	Ahmed Elzaria	Added the requirement-to-phase allocation and deferred-feature register in Sections 21 and 25 (#133).
16 October 2026	Moustafa Moustafa	Wrote Section 8.3, Individual Product Use Cases (#111).
16 October 2026	Moustafa Moustafa	Showed the specialists' disagreement on input in SRC-005.04 and added SRC-005.13 (#112).
16 October 2026	Shamil Canbolat	Added safety-critical requirements SAF-SHA-001 to SAF-SHA-004 for security, data integrity and AI-assisted refinement (#82)

Date	Developer(s)	Change
16 October 2026	Moustafa Moustafa	Wrote Section 10, Look and Feel Requirements, with NFR-UX-001 to NFR-UX-007, three constants and the WCAG 2.2 reference; added the supervisor's WCAG advice to SRC-008 (#113).
16 October 2026	Rayan Nasrallah	Let refinement be turned off in G.4, BR-005, ULC-005, the stakeholder needs and the glossary, following the questionnaire (SRC-009.01).
16 October 2026	Moustafa Moustafa	Wrote Section 11, Usability and Humanity Requirements, with NFR-UX-008 to NFR-UX-024 and five constants, and updated OI-08 (#114).
16 October 2026	Moustafa Moustafa	Added safety-critical requirements SAF-MOU-001 to SAF-MOU-003 and linked NFR-UX requirements to hazards H-MOU-001 to H-MOU-006 (#115).
16 October 2026	Muhammad Huzaifah	Wrote Section 9.3, Creative Drawing Requirements, with FR-CRT-001 to FR-CRT-030 (#122).
16 October 2026	Muhammad Huzaifah	Wrote the Attempt Record (FR-REC) and Progress Summary (FR-PRG) Requirements in Sections 9.4 and 9.5 (#123).
16 October 2026	Muhammad Huzaifah	Wrote Section 19, Off-the-Shelf Solutions, with the product comparison, candidate components and the bibliography (#124).
16 October 2026	Muhammad Huzaifah	Wrote Sections 20, 22, 23, 24 and 26: New Problems, Migration to the New Product, Costs, User Documentation and Training, and Ideas for Solution (#125).

Date	Developer(s)	Change
17 October 2026	Moustafa Moustafa	Added my reflection responses to the appendix and my generative AI use disclosure (#117).
17 October 2026	Muhammad Huzaifah	Wrote Section 7, Business Data Model and Data Dictionary, with the data model figure, the data constraints and the data dictionary (#121).
17 October 2026	Shamil Canbolat	Added sections 13-17, the remainder of section 12, and my AI use disclosure (#82).
18 October 2026	Shamil Canbolat	Added section 28 and my reflection questions and performed final QA review (#82).
18 October 2026	Muhammad Huzaifah	Added my reflection responses to the appendix and my generative AI use disclosure.
19 October 2026	Moustafa Moustafa	Final QA: renumbered the session-timeout requirement SAF-SHA-006 and restored NFR-CULT-002, so no identifier is reused; removed vendor names from requirements; fixed SAF-SHA-003, the sources table layout, UC-005, child navigation and reflection answers 5 and 6 (#82).
29 October 2026	Rayan Nasrallah	Pointed each TBD performance constant to the VnV Plan; added a first-stroke stylus check to SAF-RAY-002; added SAF-RAY-009 for idle lessons; replaced children's first names and usernames with an identifier and an avatar; changed "restarted" to "started again"; changed FIPPA to MFIPPA (#224).

Date	Developer(s)	Change
30 October 2026	Rayan Nasrallah	Addressed peer review: clarified that keeping work through a lost connection is in scope (ASM-005); said which parts of FACT-007 the team accepted; explained the `ATTEMPT_LIMIT` check and what `ATTEMPT_LIMIT_MAX` counts (#184, #185, #190).
30 October 2026	Ahmed Elzaria	Addressed peer review: gave `REFINE_TOLERANCE` a unit and said how and when it is set, as planned in the VnV Plan (#193).
30 October 2026	Ahmed Elzaria	Set `ADULT_REVIEWERS` and `REVIEW_PASS_RATE`, and said how and when `RECUR_MIN` is set, as planned in the VnV Plan (#247).
1 November 2026	Ahmed Elzaria	Addressed peer review: an unanswered go-back suggestion now stays pending in every section (FR-LRN-019, UC-005, SAF-RAY-001, the product input table, the data dictionary and Section 9.2), and "ignored" is the status of a suggestion cleared before any answer (#176).
1 November 2026	Ahmed Elzaria	Addressed peer review: moved SAF-RAY-001 to PH-1, with the lesson and starting stage choice it protects, and added the adult check to the fit criterion of FR-LRN-016 (#177).

Date	Developer(s)	Change
1 November 2026	Ahmed Elzaria	Addressed peer review: defined the stage result and a complete attempt in black-box terms in Section 9.2, with `TRACE_TOL` and `PART_COVER`; listed the initial lesson catalogue and each lesson's required parts; added NFR-PERF-010 so the stage result must match its definition (#175).
1 November 2026	Moustafa Moustafa	Changed the fit criterion of NFR-UX-006 to the share of reviewers answering 4 or 5, as T-UX-006 in the VnV Plan does, and removed `STYLE_RATING_MIN` (#275).

Introduction

Purpose and intended readers. This Software Requirements Specification (SRS) defines the externally observable behaviour, constraints and quality requirements of Squigglr. It establishes a shared basis for the development team, the supervisor, reviewers and future maintainers to understand, design, implement and verify the product. It specifies *what* Squigglr must provide rather than prescribing an implementation, except where a stakeholder has imposed a constraint. The project serves children practising drawing on a stylus-enabled touchscreen and the supporting adults who configure activities and review practice records (SRC-001, SRC-004). The scope, stakeholders, goals and constraints are established in Sections 1–3; requirements are refined in Section 9 and the subsequent quality sections.

Template and adaptations. This document follows the Volere Requirements Specification Template by Suzanne Robertson and James Robertson [33]; its LaTeX skeleton was adapted from the course's Volere template. The team retains the original headings except where changes clarify the capstone scope. In particular, Section 4 includes a new symbolic-constants register; Section 9 separates learning activities, formal specification, creative drawing, attempt records and progress summaries; and Section 27 records likely and unlikely changes. Section 28 provides cross-document traceability, while Appendix 29 records requirement evidence. A separate Hazard Analysis covers risks, failures and mitigation measures rather than repeating Volere's risks material. The source register includes anonymized interview paraphrases with speaker codes and timestamps, and questionnaire statements identified by respondent and question code. The de-identified interview transcript is reproduced in Appendix 30, providing direct links to the supporting interview turns (SRC-005).

Conventions and requirement format. Project goals use `G.1`–`G.6`; constraints use `CON-###`; facts `FACT-###`; business rules `BR-###`; assumptions `ASM-###`; and fictional personas `PER-#`. Requirement sources use `SRC-###`; numbered statements within a source use `SRC-###.##`; and interview participants use anonymized codes `P1`, `P2` and `P3`. Product use cases use `UC-###`; open questions use `OI-##`; and likely and unlikely changes use `LC-###` and `ULC-###`, respectively. Functional requirements have the prefixes `FR-LRN`,

`FR-CRT`, `FR-REC` and `FR-PRG`, each followed by a three-digit suffix. Non-functional requirements use their section's assigned prefix. Hazard Analysis uses owner-prefixed identifiers such as `H-AHM`, likewise followed by a three-digit suffix. Identifiers, once assigned, are not renumbered. Each requirement contains a short title, one testable "shall" statement, rationale where needed, a fit criterion, a priority, a phase, evidence sources and links to related use cases, requirements or hazards. *Must*, *Should* and *Could* identify delivery priority. Delivery phases are *PH-1* (PoC demo: Nov. 23, 2026), *PH-2* (Rev 0: Feb. 1, 2027), *PH-3* (final demo: Mar. 15, 2027), and *Future*. These phase identifiers are distinct from the interview participant codes; the requirement-by-phase plan belongs in Section 21. Symbolic constants are written in monospaced capitals (for example, `STAGE_PASS`) and defined once in Section 4.2. A value marked *TBD* is not an approved threshold; unresolved values appear in the constants register, and are also linked to an open issue in Section 18 where such an issue exists. Performance and maintainability values marked TBD are not unknowns: the VnV Plan states how each will be set by repeated measurement once the system can be tested. The same holds for `REFINE_TOLERANCE`, which the VnV Plan sets from adult-made drawings. Diagrams, if used, must include a legend identifying their symbols and arrow meanings. Section 9.2 defines the state-machine notation used to specify guided lesson behaviour.

Document roadmap. Sections 1–5 establish the problem, stakeholders, constraints, terminology and assumptions. Sections 6–9 describe the work and product boundaries, data and use cases, and functional behaviour. Sections 10–17 specify quality and compliance requirements. Sections 18–26 document open questions, reusable solutions, project implications, implementation phases and deferred features. Sections 27–28 cover change expectations and traceability. The appendices contain the requirement sources register, de-identified specialist interview transcript, follow-up specialist questionnaire (SRC-009), generative AI-use disclosure and team reflection. Each section range links to its first section; the table of contents provides links to individual sections.

1 Purpose of the Project

This section states the problem Squigglr addresses, who has it, and the goals the project commits to. The problem and goals come from the Problem Statement and Goals (PSG) document (SRC-001).

1.1 User Business

Visual-motor integration is the ability to coordinate what a person sees with how they move. Children rely on it when drawing, copying shapes and learning to write (SRC-001). Kindergarten and primary school learners need chances to practise these skills at a suitable level and to complete activities with less help over time. Older children who need continued support, including some children with autism, attention-deficit/hyperactivity disorder (ADHD) or other developmental or learning differences, share this need. A diagnosis alone does not determine the support a child requires (SRC-001).

The adults around these children have a related problem. Parents, guardians, caregivers and educators need to understand which difficulties a child met and how much guidance the child needed. A finished drawing alone does not show whether the child completed it independently, relied on tracing guides, or corrected mistakes during the attempt (SRC-001). Squigglr is a website, used on a touchscreen device with a stylus, that offers guided drawing practice, creative drawing, and records that help adults understand that practice. It makes no diagnosis or treatment claims (SRC-001).

The gap left by current alternatives, as the PSG frames it, is that a drawing's final result is kept without the process behind it: the guidance used, the difficulties met and the corrections made. Squigglr's value is in keeping that process visible to supporting adults while children practise with gradually less support. The comparison with existing products appears in the Off-the-Shelf Solutions section of this document.

1.2 Goals of the Project

Table 2 lists the six project goals, G.1 to G.6, using the PSG's goal titles (SRC-001). Each goal is stated as Purpose, Advantage and Measurement: what the goal is, the benefit it brings and to whom, and how the team will check that it is met. Following the teaching assistant's (TA's) request for

measurable goals (SRC-003) and the supervisor's direction that no requirements are gathered from children (SRC-004), every check is done by the team or by adult reviewers using prepared or adult-made attempts, never with child participants. The acceptance parameters are named in Section 4.2. On the supervisor's advice (SRC-008), parameters that depend on the child are selected by a supporting adult within specified ranges; tests check that the selected settings and defaults are respected. Each lesson (for example, drawing a square) moves through fixed stages that give gradually less help: a demonstration, tracing a full outline, drawing between endpoints only, and drawing with no guide. This sequence follows the activity the team presented to the domain specialists (SRC-005, Moustafa, 6:03).

G.3 was revised after the PSG was submitted. The submitted PSG said Squigglr adjusts guidance and difficulty automatically. Following the domain specialist interview (SRC-005, P1, 28:37; P2, 29:18) and the Oct. 7 TA discussion (SRC-006), the team assigned *lesson selection, starting stage selection and discretionary stage changes* to a supporting adult (G.3). A passing drawing attempt still advances automatically to the next stage *within the lesson the adult selected* (G.2, BR-002); completing a lesson never automatically selects another lesson.

Table 2: Project goals

Goal	**Purpose, advantage and measurement**
G.1 Support structured drawing practice	*Purpose:* every lesson in the agreed catalogue covers lines, shapes or simple drawings, starts with a demonstration, and gives feedback and a retry at each drawing stage (SRC-001). *Advantage:* structured practice at a suitable level supports visual-motor development (SRC-001), and once an adult starts a lesson the child practises without constant adult help (SRC-005, P1, 34:13). Within a lesson the child does not navigate; Squigglr moves between stages. Serves children. *Measurement:* met when the team completes every stage of every catalogue lesson with prepared correct attempts, each attempt passes, and each drawing stage accepts a drawing, gives feedback and allows a retry.

Goal	Purpose, advantage and measurement
G.2 Support progression towards independent drawing	*Purpose:* within an adult-selected lesson, Squigglr automatically advances to the next drawing stage when an attempt meets that stage's pass criteria (`STAGE_PASS`, chosen by the adult within a team-set range). Passing the final independent stage completes the current lesson; it never starts a different lesson. An attempt that does not meet the criteria is not passed, and the child can retry. This checks the drawing against fixed criteria and does not assess the child (SRC-001). *Advantage:* children need to perform activities with gradually less help (SRC-001); a child practises one level until proficient before moving on (SRC-005, P1, 28:37), and children reach these skills at different points (SRC-005, P1, 20:27). Serves children and supporting adults. *Measurement:* met when (a) prepared correct attempts advance from outline tracing to endpoint guidance and from endpoint guidance to independent drawing, and a correct independent attempt completes only the current lesson; (b) every prepared incorrect attempt, such as one with a missing side or a wrong endpoint, is not passed and its predefined difficulty is reported; and (c) every saved attempt records the stage it was made at, with no mislabelled record; and (d) for prepared attempts scored just above and just below the chosen `STAGE_PASS`, at its minimum, default and maximum, each attempt passes or fails as expected.

Goal	Purpose, advantage and measurement
G.3 Let supporting adults control support and difficulty	*Purpose:* adults choose which lesson a child practises and may let the child start at a later stage. Changing to another lesson, skipping a stage or moving backward requires an adult's authorization; routine forward advancement on a passing attempt within the chosen lesson does not (G.2). Squigglr may suggest returning to an earlier stage after `DIFFICULTY_STREAK` failed attempts in a row; the change applies only when an adult accepts it, and a declined or ignored suggestion changes nothing. To avoid endless repetition, the lesson ends after `ATTEMPT_LIMIT` (larger than `DIFFICULTY_STREAK`) unsuccessful attempts at one stage, and the furthest stage reached is saved. If the child's attempts at a stage are improving, Squigglr holds back the suggestion and allows `ATTEMPT_EXTENSION` more attempts, up to `ATTEMPT_LIMIT_MAX` in total, so that the limit does not cut off a child who is getting closer to passing. Improvement compares the child's own recent attempts with each other against the stage's fixed criteria; it is not shown as a score and does not assess the child. The adult chooses these values for each child within a team-set range (SRC-008). *Advantage:* automatically choosing a new lesson or changing the learner's starting level without adult direction would require an out-of-scope readiness assessment; the specialists advised adult control of such decisions (SRC-005, P1, 28:37; P2, 29:18; SRC-006). Serves children and supporting adults. *Measurement:* met when, for prepared sequences of successful, unsuccessful and mixed attempts, every suggestion shown matches the agreed criteria, each lesson without improvement ends exactly at `ATTEMPT_LIMIT` and each improving one is extended by `ATTEMPT_EXTENSION` and never passes `ATTEMPT_LIMIT_MAX`, with its progress saved, no lesson or starting stage changes without an adult's action, and these hold at the minimum, default and maximum of each value.

Goal	Purpose, advantage and measurement
G.4 Support editable creative drawing and refinement	*Purpose:* in the creative space, Squigglr refines a child's drawing of a shape or object in the agreed supported set by adjusting the child's own strokes; the result stays editable, the original can be restored, and refinement can be turned off (SRC-001; Moustafa, 6:03; SRC-009.01). *Advantage:* creative drawing gives room for self expression, and editability keeps the child in control of the work (SRC-001). Serves children. *Measurement:* using prepared drawings of each item in the supported set, met when for every drawing (a) every point of the refined drawing lies within `REFINE_TOLERANCE` (TBD) of the original strokes; (b) the refined parts can be moved, resized, recoloured, erased and undone; (c) restoring returns the original strokes exactly; (d) a failed refinement request leaves the drawing unchanged; and (e) with refinement turned off, the child's strokes are kept exactly as drawn.
G.5 Preserve meaningful activity records	*Purpose:* each attempt's record keeps the stage, every predefined difficulty observed, later corrections and the final outcome; each lesson's record keeps the furthest stage reached and the successful attempts at each stage (SRC-001). *Advantage:* a finished drawing alone hides how much guidance was used and the corrections made (SRC-001). Serves supporting adults. *Measurement:* met when, for prepared attempts and lessons with known actions, stages and outcomes, every saved record matches its known contents, with no difficulty lost after a correction.

Goal	Purpose, advantage and measurement
G.6 Make progress understandable to supporting adults	*Purpose:* summaries show a child's progress over time, separate attempts made without a guide from attempts made with one, and link to the attempts behind them (SRC-001). *Advantage:* adults need to see changes in independent drawing to decide on further practice (SRC-001); the specialists suggested giving the adult a view of the child's progress alongside control of activities (SRC-005, P2, 29:18). Serves parents, guardians, caregivers and educators. *Measurement:* met when, for prepared histories showing improvement, recurring difficulties and changing stages, every summary separates the two kinds of attempt correctly and links to the right attempts; and at least `ADULT_REVIEWERS` adult reviewers interpret example summaries and locate the supporting attempts, with at least `REVIEW_PASS_RATE` of their tasks succeeding.

The symbolic constants used to evaluate these goals, including values configurable by supporting adults, are recorded in the central register (Table 10, Section 4.2). Unknown adult-configurable ranges, midpoint defaults and the improvement margin are defined in Section 4.2 (SRC-008).

2 Stakeholders

This section identifies who has an interest in Squigglr, who uses it, and how each group informed the requirements. Sources are cited by their Requirement Source (SRC) identifiers from Appendix A. The development team is not listed as a stakeholder.

Each stakeholder is classed by how directly they use Squigglr. *Primary* stakeholders use it directly: children, parents, guardians and caregivers, and educators and learning support staff. *Secondary* stakeholders shape it or are affected by it without using it: the client, school administrators, and the domain specialists. *Tertiary* stakeholders have an indirect interest: occupational therapists and educational specialists, and the TA and course instructors.

2.1 Client

The client, a secondary stakeholder, is Dr. Christopher Anand, the project supervisor, who acts in a client-like role (SRC-003). He provides guidance on scope, feasibility, and technical direction (SRC-001). His binding decisions are recorded in the Mandated Constraints section (SRC-004).

2.2 Customer

Squigglr has no paying customer; it is a capstone project and no sale or purchase is planned. The adults who would choose to adopt it act as customers in practice: parents, guardians, and caregivers for home use, and educators and school administrators for school use (SRC-001).

2.3 Other Stakeholders

Table 3 lists the stakeholders other than the client and the hands-on users, with the knowledge each provides and how much they are involved and can influence the project. Conflicts between stakeholders are handled in the Priorities Assigned to Users section.

Table 3: Other stakeholders: knowledge needed, involvement and influence

Stakeholder	**Interest and knowledge needed**	**Involvement and influence**
School administrators (secondary)	Suitability for the school setting, device availability, and the resources needed (SRC-001).	Not yet involved. May influence adoption if Squigglr is considered for school use.

Stakeholder	Interest and knowledge needed	Involvement and influence
Domain specialists P1, P2 and P3 (secondary)	Three specialists with doctorates in human-computer interaction and related engineering fields who build software for children with autism, attention-deficit/hyperactivity disorder (ADHD) and other learning needs. They provide expert knowledge of the intended users (SRC-005). They preferred not to be named and are de-identified (SRC-006).	Interviewed on October 1, 2026 (SRC-005). They offered to help guide a future pilot (SRC-005, P1, 24:03), which supports involving them in verification and validation (SRC-006). They advise; they do not decide.
Occupational therapists and educational specialists (tertiary)	Progress milestones, benchmarks and feedback wording, as the specialists recommended (SRC-005, P2, 31:42).	Not yet consulted.
TA and course instructors (tertiary)	Assess the project deliverables.	Review each deliverable; they influence documentation, not product behaviour.

2.4 Hands-On Users of the Project

Squigglr has two groups of hands-on users, both primary stakeholders: children, who complete drawing activities, and supporting adults, who choose activities and review records.

Children. Kindergarten and primary school learners, and older children who need continued support, including children with autism, ADHD, or other developmental or learning differences (SRC-001). The intended age range is not yet defined; it is recorded as an open item, `AGE_RANGE` (TBD). Their characteristics vary widely, and a diagnosis alone does not determine the support a child needs (SRC-001; Rayan, 2:04). The characteristics below matter most for the requirements, because they shape how children read,

draw and move through the application without help:

- **Reading ability:** some children cannot read yet, or may not understand written instructions. Squigglr has to communicate in ways that do not depend on reading, such as images and icons (SRC-005, P1, 34:13) and spoken audio that can be muted (SRC-005, P2, 37:39).
- **Motor skills:** fine-motor ability varies; some children find holding a stylus difficult, and grip can affect how they draw (SRC-005, P3, 25:36). Fine-motor control is part of visual-motor practice, so children need to practise holding a tool, moving from larger tools to smaller ones closer to what schools use (SRC-005, P1, 26:46). Squigglr therefore uses a stylus as its main input. Drawing with a finger is the easiest starting point (SRC-005, P1, 26:46), and finger input is a stretch goal. Some children may also have gross-motor or vision challenges (SRC-005, P2, 19:05).
- **Support needs:** once an adult starts a lesson, he should be able to practise without constant adult help (SRC-005, P1, 34:13). He does not need to navigate, exit or restart a lesson. Adults decide when a child moves to a new lesson or skips a stage (SRC-005, P1, 28:37).
- **Device experience:** many young children already know how to navigate touchscreen applications on tablets (SRC-005, P1, 34:13).

Supporting adults. Parents, guardians, caregivers, educators, and learning-support staff. They set up accounts, choose lessons and starting stages, and review attempt records and progress summaries on a tablet or computer (SRC-001; P1, 34:13). Their technical experience is assumed to be that of an everyday web user (assumption). They need understandable information without a single overall score, diagnostic labels, or comparisons with other children (SRC-005, Moustafa, 6:03).

Experience levels. Table 4 rates each user group's subject matter and technological experience on a scale of novice, journeyman and master.

Table 4: Experience of each hands-on user group

User group	**Subject matter (drawing and visual-motor practice)**	**Technology**
Children	Novice: they are still learning to draw lines and shapes (SRC-001).	Novice to journeyman: many can already navigate tablet applications (SRC-005, P1, 34:13), but some need adult help.
Parents, guardians, caregivers	Novice: not assumed to have training in visual-motor development (assumption).	Journeyman: everyday web users (assumption).
Educators, learning-support staff	Journeyman: they support learners' skill development as part of their work (assumption).	Journeyman: they use school-provided web tools (assumption).

Needs, concerns and their effect on requirements. Table 5 records what each user group needs or worries about, and how this section expects the requirements to respond. The concerns come from the sources cited; none were gathered from children (SRC-004).

Table 5: User needs and concerns, with the requirement response

User	**Need or concern (source)**	**Expected requirement response**
Children	May not read; need to practise without constant adult help (SRC-005, P1, 34:13).	An adult starts the lesson; within it the child does not navigate, and Squigglr moves between stages.
Children	Grip and fine-motor ability vary (SRC-005, P3, 25:36).	Stylus is the main input; finger input is a stretch goal.

User	Need or concern (source)	Expected requirement response
Children (guided lessons)	Attempts must show what the child actually drew (SRC-005, Moustafa, 6:03).	Records keep the child's actual strokes; no corrected shape is counted as their work.
Children (creative space)	Get a cleaner version of their own drawing that is still theirs (SRC-005, Moustafa, 6:03).	Refinement adjusts the child's strokes rather than replacing them; the result stays editable, the original can be restored, and refinement can be turned off.
Children	Two children with the same diagnosis may need different support (SRC-005, Rayan, 2:04).	Adults choose the lesson and starting stage for each child; nothing is set by diagnosis.
Caregivers	Become annoyed if they must intervene constantly (SRC-005, P1, 34:13).	Children can practise without constant adult help.
Caregivers, educators	Want to decide when a child moves to a new lesson or needs a different starting level (SRC-005, P1, 28:37; P2, 29:18).	Adults choose the lesson and starting stage and approve skipping or returning to an earlier stage. Passing an attempt advances only within that lesson.
Caregivers, educators	Need to see how far the child got in each lesson, how many attempts succeeded at each stage, and which difficulties recur (SRC-001).	Records save the furthest stage reached and the successful attempts at each stage. To avoid endless repetition, a lesson ends after `ATTEMPT_LIMIT` unsuccessful attempts at one stage, and the progress is saved. No single overall score or labels.
Administrators	Installing applications on school devices is difficult (SRC-003, SRC-004).	Browser-based; no installation.

2.5 Personas

The three personas below are **fictional**. They do not describe any real child or adult. Personas are used because requirements cannot be gathered from children directly, and the supervisor recommended them for this case (SRC-004). There are at most three, and each is a short life story of a person. Details that come from a source cite it; all other details were invented by the team to make the persona concrete. No information was collected from children (SRC-004). Each persona describes one individual; none represents all children or adults in its group.

The personas describe people, not the product. They leave out design choices such as screen layout or feedback wording, so they stay valid if those choices change. Each persona has a fixed identifier (PER-1 to PER-3) so that scenarios, use cases, usability tests and later documents can refer to it. The three personas are linked: Adam is Sarah's son and one of Janice's learners, so one scenario can follow him between home and school. Invented details should be checked with caregivers and educators when the opportunity arises; if one changes, the scenarios that depend on it must be reviewed.

PER-1 (learner): Adam. Adam is 15 years old; older children who need continued support are in scope (SRC-001). He lives with his parents and no siblings. Adam is autistic and has ADHD. He loves Spider-Man, colours Spider-Man pictures at home, and tends to fixate on things he enjoys.

- **Communication:** he is verbal but speaks in short, simple sentences, and most of his responses are memorized. He can say how he feels or what he wants, such as being hungry or wanting to play, but cannot hold the conversations typical for his age. He understands spoken words and instructions well.
- **Reading and writing:** he reads and writes only with a parent's support. He can read a few short words he recognizes, such as "yes" and "no", but not longer words or sentences.
- **Learning:** he learns best by watching someone do a task first and then copying it, rather than being told what to do without an example.
- **Attention:** he loses focus easily. For skill-building tasks such as reading or drawing, a caregiver needs to watch over him throughout. When

he colours, which he enjoys, he can work alone for a while, but needs encouragement every now and then to stay focused and finish.

- **Frustrations:** he becomes agitated when a task is repeated over and over.
- **Attitude to technology and tools:** he knows how to use an iPad. He has no experience with a stylus, but uses crayons, pens, pencils, coloured pencils and markers.
- **Goals:** finish a picture of something he likes and show it to his family.
- **Context of use:** at home with his mother nearby, and at school in sessions with Janice.

PER-2 (caregiver): Sarah. Sarah is Adam's mother and works full time. She wants Adam to keep building the drawing and fine-motor skills that support writing. She is not a specialist and learns about Adam's needs from his teachers and therapists.

- **Attitude to technology:** an everyday web user who wants things to work without setup.
- **Goals:** let Adam practise without being called over constantly (SRC-005, P1, 34:13); choose what Adam practises and decide when he moves on (SRC-005, P1, 28:37).
- **Review habits:** looks through Adam's practice history on a laptop once a week to see which stages he reached and what keeps going wrong (SRC-001). Shares what she sees with Janice at parent meetings.
- **Concerns:** does not want Adam reduced to a single score, labelled or compared with other children (SRC-005, Moustafa, 6:03); wants Adam's data kept private.

PER-3 (educator): Janice. Janice is a teacher at a school for students with special needs who works with several learners each day, including Adam. She has a short slot with each learner and little time to set things up between them.

- **Environment:** the school's devices are managed centrally, so she cannot install applications (SRC-003, SRC-004).
- **Attitude to technology:** comfortable with the school's web tools, but has no administrator rights.
- **Goals:** set a different activity for each learner (SRC-005, P2, 29:18); compare a learner's records with what she observes in class (SRC-001).
- **Concerns:** switching between learners quickly, and making sure one learner's records are never shown to another.

2.6 Priorities Assigned to Users

Table 6 assigns priorities. Teachers and students are the main priority (SRC-003). Rationale: children are the users the activities serve; educators and caregivers choose activities and act on the records, so their needs come next. Administrators affect adoption but not daily use, and the specialists advise rather than use Squigglr (SRC-006). When needs conflict, a child's ability to practise independently is balanced with adult control of lesson selection and discretionary stage changes; passing a stage advances within the selected lesson (G.2; SRC-005, P1, 28:37).

Table 6: User priorities

Priority	**Users**
Key	Children (learners); educators and learning-support staff (SRC-003).
Key	Parents, guardians, and caregivers (SRC-001); ranked key because they choose activities and review records at home.
Secondary	School administrators; domain specialists (SRC-006).

2.7 User Participation

Children do not take part in requirements gathering; their needs are inferred from adults and from personas (SRC-004). The requirements were informed by:

- the supervisor meeting (SRC-004) and TA meetings (SRC-003, SRC-006);
- a recorded interview with domain specialists P1–P3 (SRC-005), held because children could not be asked directly (SRC-004) and the specialists build software for the intended users (SRC-006);
- peer review of the Problem Statement and Development Plan (SRC-007);
- written answers from two of the three specialists to a follow-up questionnaire (SRC-009).

Parents, caregivers, and educators have not yet been consulted (SRC-005, Rayan, 12:31); this is an open item. Planned later participation includes adult reviewers in usability sessions (SRC-001), possible feedback from adults with autism or ADHD (SRC-005, P2, 37:39), and the specialists' offered support for a pilot (SRC-005, P1, 24:03). Any study involving children requires ethics approval first (SRC-004).

2.8 Maintenance Users and Service Technicians

During the project, the development team maintains Squigglr, including activity content and the hosted website. No maintenance users or service technicians outside the team have been identified. Who maintains Squigglr after the project ends is an open item (TBD).

3 Mandated Constraints

This section lists constraints imposed on Squigglr from outside the team: by the supervisor, the teaching assistant (TA), the domain specialists interviewed (SRC-005), the course and the university. They are not design choices. Each constraint has an identifier of the form CON-###, a description, a rationale, the party that imposed it, a fit criterion that shows whether it is met, and its sources from the Requirement Sources register (SRC-###). The constraints limit what the team may build; they do not choose a technology, library or design. For example, CON-001 requires a website but does not name a framework. Table 7 summarizes them.

Table 7: Summary of mandated constraints

ID	Constraint	Source
CON-001	Squigglr shall be a website used in a web browser.	SRC-003, SRC-004
CON-002	Squigglr shall require no installation on the user's device.	SRC-003, SRC-004
CON-003	Squigglr shall work on touchscreen devices with stylus input.	SRC-001, SRC-003–SRC-005
CON-004	Deliverables shall meet the course deadlines in Table 8.	SRC-002
CON-005	The project shall gather no information from or about children for requirements or evaluation.	SRC-004
CON-006	The project shall run no study involving children.	SRC-004, SRC-005
CON-007	Squigglr shall not change lessons, skip guidance stages or move backward without adult authorization; pass-based forward movement within the selected lesson is allowed.	SRC-001, SRC-005, SRC-006

3.1 Solution Constraints

The supervisor, the TA and the domain specialists set the form of the solution.

CON-001 Squigglr shall be a website used in a web browser.
Rationale: Installing apps on school devices is difficult, and for Canadian schools a website is much easier to adopt than an app-store app.
Imposed by: supervisor and TA.
Fit criterion: Squigglr opens and runs from a web address in a browser.
Source: SRC-003, SRC-004.

CON-002 Squigglr shall require no installation on the user's device.
Rationale: Shared school devices are often managed by the school, so adults may not be able to install software. Opening a web address is enough to start.
Imposed by: supervisor and TA.
Fit criterion: a first-time user starts an activity without installing any

app, plugin or browser extension.
Source: SRC-003, SRC-004.

CON-003 Squigglr shall work on touchscreen devices with stylus input.
Rationale: Squigglr targets visual-motor skills, which include fine motor control. Children build that control by practising with a held tool, moving from large tools to smaller ones like pencils (SRC-005, P1, 26:46). Grip varies between children and affects how they draw (SRC-005, P3, 25:36), so finger input is kept as a stretch goal (SRC-001).
Imposed by: domain specialists, within the scope agreed with the supervisor and TA.
Fit criterion: each activity can be completed with a stylus on a touchscreen device; the device and browser combinations are TBD.
Source: SRC-001, SRC-003, SRC-004, SRC-005.

CON-007 Squigglr shall not select another lesson, skip a guidance stage or move backward to an earlier stage unless a supporting adult authorizes that change. Normal forward movement to the next stage following a passing attempt *within the selected lesson* is not an unauthorized change (G.2, BR-002).
Rationale: Choosing what a child is ready to practise requires judgement by the adult who knows the child (SRC-005, P1, 28:37; P2, 29:18). This restriction applies to lesson selection and discretionary stage changes, not to the fixed pass-based steps already defined for an adult-selected lesson. Squigglr may suggest going back a stage, but an adult must accept (G.3).
Imposed by: domain specialists' advice on adult choice, clarified by the team's progression decision.
Fit criterion: test that passing a tracing or endpoint stage advances within the current lesson without an additional adult action; passing the final stage completes that lesson without starting another. Verify that switching lessons, skipping and returning to an earlier stage cannot occur without an adult choice or acceptance.
Source: SRC-001, SRC-005 (P1, 28:37; P2, 29:18), SRC-006; team decision documented in #132.

3.2 Implementation Environment of the Current System

Not applicable. There is no current system that Squigglr replaces or must run alongside; it is a new product.

3.3 Partner or Collaborative Applications

Not applicable. No stakeholder requires Squigglr to work with a specific partner application. Candidate integrations, if any, belong in the Off-the-Shelf Solutions section.

3.4 Off-the-Shelf Software

Not applicable. No stakeholder mandates any specific off-the-shelf software. Candidate libraries and services are discussed in the Off-the-Shelf Solutions section, not here.

3.5 Anticipated Workplace Environment

Squigglr will be used at home, in classrooms and in other places where children practise drawing, such as rehabilitation centres (SRC-005, P2, 21:25), usually with an adult nearby. School devices may be shared by several children (SRC-004). These places give context for later requirements; they are not a constraint, since the team will not test Squigglr in each of them.

3.6 Schedule Constraints

The course sets the project's deadlines. Table 8 lists the course deadlines relevant to this document, taken from the Development Plan (SRC-002) course timeline. Dates use the America/Toronto time zone. The Development Plan holds the full timeline.

CON-004 Deliverables shall meet the course deadlines in Table 8.

Rationale: The capstone course fixes these dates; the team cannot change them.

Imposed by: the capstone course.

Fit criterion: each deliverable is submitted by its date.

Source: SRC-002.

Table 8: Course deadlines (from the Development Plan)

Date	**Deliverable**
Oct. 19, 2026	SRS and Hazard Analysis
Oct. 22, 2026	SRS peer-review issues
Nov. 2, 2026	Verification and Validation (V&V) Plan
Nov. 16, 2026	Design Revision −1 and fall custom document
Nov. 23–Dec. 4, 2026	Proof of Concept (POC) demonstration window
Jan. 18, 2027	Design documentation
Feb. 1–11, 2027	Rev 0 presentation sessions
Mar. 8, 2027	V&V Report and winter custom document
Mar. 15–21, 2027	Final demonstration sessions
Mar. 31, 2027	Final documentation

3.7 Budget Constraints

TBD. The Development Plan states no budget for the project, and no stakeholder has set one. It notes only that service limits and possible costs of external services must be monitored. The budget remains an open issue to confirm with the TA.

3.8 Enterprise Constraints

The supervisor and the university set rules that protect children.

CON-005 The project shall gather no information from or about children for requirements or evaluation.
Rationale: The users are children, and the supervisor directed that no information about them be collected. Requirements come from adults and personas, and every check uses attempts made by adults. This constraint covers the project's work; how Squigglr itself protects the practice records it keeps is a privacy requirement, not this constraint.
Imposed by: supervisor.
Fit criterion: no requirement source, test or evaluation record contains data from or about a child.
Source: SRC-004.

CON-006 The project shall run no study involving children.
Rationale: Any study run through the university that involves children

requires ethics board approval, which the project does not seek (SRC-005, P2, 22:56; P1, 33:00). Evaluation is done with adults only (CON-005).
Imposed by: supervisor and the university, as the domain specialists also advised.
Fit criterion: no study with child participants takes place during the project.
Source: SRC-004, SRC-005.

4 Naming Conventions and Terminology

This section defines the vocabulary, identifiers and constants used throughout the specification.

4.1 Glossary and Acronyms

Table 9 defines the terms and abbreviations used in the goals, business rules, stakeholder evidence and requirements. Source identifiers link to Appendix 29.

Table 9: Project terminology and acronyms

Term or acronym	Definition	Context
Visual-motor integration	Coordination of visual information with movement, relevant to drawing and writing readiness activities.	Section 1
Guided lesson	A structured activity in the agreed lesson catalogue in which drawing guidance is provided in successive stages.	G.1–G.3
Stage (guidance level)	A step within a lesson that provides a specified amount of help, from demonstration through independent drawing. A passing drawing attempt advances automatically to the next stage within the chosen lesson; adult authorization is needed to skip or return.	G.2–G.3

Term or acronym	Definition	Context
Attempt	One try at the drawing task of a lesson stage; its strokes, observed difficulties and outcome are represented in the practice record.	BR-002, BR-004
Attempt record	The record of the actual strokes, stage, predefined observations and outcome of an attempt; guided records are not replaced by refined shapes.	BR-004
Lesson record	The record kept each time a learner starts a lesson: the lesson, the starting stage, the furthest stage reached, the successful attempts at each stage and how the lesson ended.	Section 9.4
Interrupted attempt	An attempt that an adult stops, by ending the session or switching learner, after at least one stroke and before its outcome is decided; its attempt record has no outcome.	Section 9.4
Unfinished lesson	A lesson that stops before its last stage is passed and before it reaches an attempt limit; its lesson record is marked as unfinished.	Section 9.4
Progress summary	The view in which a supporting adult reviews one learner's practice over time and opens the attempt records behind it.	Section 9.5
Recurring difficulty	A predefined difficulty observed in at least `RECUR_MIN` of a learner's attempts at the same stage of the same lesson.	Section 9.5
Attempt made with or without a guide	An attempt made at the tracing or endpoint stage is made with a guide; an attempt at the stage with no guide is made without one.	Section 9.5
Creative refinement	Automatic, minimal adjustment of the child's strokes in the creative space only. The drawing remains editable, and the original can be restored, or refinement turned off, by the child or, if needed, by a supporting adult.	BR-005
Creative space	The part of Squigglr where a child draws freely, beside the guided lessons. Refinement happens only here.	Section 9.3

Term or acronym	Definition	Context
Creative drawing	One drawing in the creative space, made of strokes; each stroke has a path, a colour and a width.	Section 9.3
Refined drawing	The result of one refinement, which adjusts the paths of the child's own strokes.	Section 9.3
Original (of a refinement)	The creative drawing exactly as it was immediately before that refinement was applied: the same strokes in the same order, with the same points, colours and widths.	Section 9.3
Supported set	The agreed set of shapes and simple objects that Squigglr refines; its contents are open issue OI-03.	OI-03
Supporting adult	A parent, guardian, caregiver, educator or learning-support staff member who configures activities and reviews records, as permitted by the account rules.	Section 2
Lesson choice	The lesson and starting stage that a supporting adult has chosen for a learner.	Section 7.2
Learner settings	The values that tailor practice to one learner: the adult-set values, the audio setting, the accessibility settings and the refinement setting.	Section 7.2
Team-set values	The ranges and fixed values that the team sets for every learner (BR-009).	Section 4.2
Fit criterion	An observable or measurable acceptance check used to verify whether a requirement or constraint is met.	Section 3
Open-weight model	A trained model whose weights are published, so that a team can run it on its own hardware.	Section 19.2
Vision-language model	A model that takes both images and text as input.	Section 19.2
Zero data retention	An arrangement, which the provider must approve, under which the provider does not keep the data sent to it.	Section 19.2
ADHD	Attention-deficit/hyperactivity disorder.	Section 1
AI	Artificial intelligence.	BR-007

Term or acronym	Definition	Context
API	Application programming interface; a defined way for one program to request a service from another.	Section 19.2
ASM	Assumption identifier; `ASM-###`.	Section 5.3
BR	Business rule identifier; `BR-###`.	Section 5.2
BSD	Berkeley Software Distribution; a software licence that allows reuse if the copyright notice is kept.	Section 19.2
BUC	Business use case.	Document roadmap
CC BY	Creative Commons Attribution; a licence that allows copying and reuse with credit to the creator.	Section 19.2
CI	Continuous integration; automated checks run as changes are integrated into the project.	Section 14
CLI	Command-line interface.	AI-use disclosure
CON	Mandated constraint identifier; `CON-###`.	Section 3
COPPA	Children's Online Privacy Protection Act (United States).	Section 17
CSS	Cascading Style Sheets, a language for describing the presentation of web pages.	Section 10
EARS	Easy Approach to Requirements Syntax.	Team reflection
FACT	Relevant fact identifier; `FACT-###`.	Section 5.1
FR	Functional requirement.	Section 9
GB	Gigabyte.	Section 23
GDPR	General Data Protection Regulation (European Union).	Section 17
GNU	GNU's Not Unix, the name of a free software project.	Section 19.1
GPL	GNU General Public License; its conditions also apply to code copied from a program released under it.	Section 19.1
GPU	Graphics processing unit.	Section 19.2
HA	Hazard Analysis, a separate document identifying failures, losses, mitigation measures and related requirements.	Introduction

Term or acronym	Definition	Context
HCI	Human-computer interaction.	Team reflection
HTTPS	Hypertext Transfer Protocol Secure, the encrypted form of HTTP used for network communication.	Section 13
IDE	Integrated development environment.	AI-use disclosure
JSON	JavaScript Object Notation; a text format for structured data.	Section 19.2
LC	Likely change identifier; `LC-###`.	Section 27.1
LLM	Large language model.	AI-use disclosure
MB	Megabyte.	Section 20
MFIPPA	Municipal Freedom of Information and Protection of Privacy Act (Ontario).	Section 17
MG	Module Guide, a high-level description of a system's modules and their relationships.	Module Guide document
MIS	Module Interface Specification, a detailed description of module interfaces and contracts.	Module Interface Specification document
MIT	Massachusetts Institute of Technology licence; a software licence that allows reuse if the copyright notice is kept.	Section 19.2
NFR	Nonfunctional requirement; a required quality or constraint on system performance or operation.	Section 12
OI	Open issue identifier; `OI-##`.	Section 18
PDF	Portable Document Format.	Team reflection
PER	Fictional persona identifier; `PER-#`.	Section 2.5
P1–P3	Anonymized identifiers for the three domain specialists interviewed as part of requirements elicitation; not delivery phases.	Appendix A: Interview statements
PII	Personally identifiable information.	Section 15
PIPEDA	Personal Information Protection and Electronic Documents Act (Canada).	Section 17

Term or acronym	Definition	Context
POC / PoC	Proof of concept; the planned capstone feasibility demonstration, not a production release.	Schedule
PSG	Problem Statement and Goals project document.	Section 1
PUC	Product use case.	Section 8
QA	Quality assurance.	Team reflection
SFWRENG	Software Engineering, the course subject code used by McMaster University's Faculty of Engineering.	Team reflection
SQL	Structured Query Language, used to query and manage relational databases.	Section 15
SRS	Software Requirements Specification, this document.	Introduction
SRC	Requirement source identifier; `SRC-###`. Numbered source statements use `SRC-###.##`.	Appendix A
TA	Teaching assistant.	Section 1
TBD	To be determined; a value or decision not yet approved.	Section 4.2
UC	Product use case identifier; `UC-###`.	Section 8
UI/UX	User interface and user experience.	Section 24
ULC	Unlikely change identifier; `ULC-###`.	Section 27.2
URL	Uniform Resource Locator, the address of a resource on a network.	Section 13
V&V	Verification and validation.	Schedule
VS	Visual Studio Code (VS Code), a source-code editor.	AI-use disclosure
W3C	World Wide Web Consortium, which publishes web standards such as Pointer Events.	Section 19.2
WCAG	Web Content Accessibility Guidelines; selected criteria may inform project accessibility requirements, without a claim of complete conformance (SRC-008).	OI-08
XSS	Cross-site scripting, an attack that injects malicious scripts into web content.	Section 15

4.2 Symbolic Constants

Table 10 specifies the parameters used in guided lessons, creative refinement and progress summary verification. The eight adult-configurable parameters have an allowed range and a default value for each learner profile (BR-008, SRC-008). Their defaults are the midpoints of their respective ranges. Each configuration shall satisfy `ATTEMPT_LIMIT` > `DIFFICULTY_STREAK` and `ATTEMPT_LIMIT_MAX` ≥ `ATTEMPT_LIMIT`, and `LESSON_IDLE_PROMPT` < `LESSON_IDLE_END` < `IDLE_TIMEOUT`. Drawing stage results are expressed on a normalized scale from 0 to 100; higher results indicate closer agreement with the stage's fixed criteria. The result is not a measure of the child's ability (BR-006). Attempt count settings shall be integers, and values violating the allowed ranges or the two cross-setting inequalities shall be rejected (BR-008).

Basis and verification of numerical values. The project team selected the initial ranges, defaults, five-point `IMPROVE_MARGIN` and 0–100 result scale in a team discussion on 15 October 2026 concerning issue #132. These are team-selected engineering starting values, not thresholds established by child testing or supplied numerically by the supervisor. The supervisor's 14 October guidance (SRC-008) requires adult-configurable ranges, midpoint defaults and example drawings illustrating both ends of the `STAGE_PASS` range. The chosen values and the drawing-to-score interpretation are checked using examples made by adults at the lower and upper boundaries of each configurable range (OI-05; CON-005–006).

Table 10: Central symbolic constants register

Constant	Meaning, unit, value and traceability
`STAGE_PASS`	Minimum stage result threshold for a complete attempt to pass a drawing stage. The stage result (Section 9.2) is the smaller of the share of the lesson's figure that was drawn and the share of the drawing that lies on the figure, so a result of 80 means at least 80% of each. *Unit:* points out of 100. *Range:* 70–90. *Default:* 80. *Trace:* G.2, BR-002, BR-008, OI-05.

Constant	Meaning, unit, value and traceability
TRACE_TOL	Largest distance at which a drawn point counts as on the lesson's figure, and a part of the figure counts as drawn. *Unit:* percent of the figure's size (the diagonal of its bounding box), so it does not depend on the screen. *Value:* proposed 5; confirmed by the team in PH-1, before the first evaluation test, on adult-made drawings (BR-009), as the VnV Plan calibrates its prepared attempts. *Trace:* G.2, FR-LRN-011, FR-LRN-013.
PART_COVER	Smallest share of the length of a required part of the figure that must be drawn for the part to count as present. *Unit:* percent. *Value:* proposed 90; confirmed with TRACE_TOL. *Trace:* G.2, FR-LRN-011.
DIFFICULTY_STREAK	Consecutive unsuccessful attempts before Squigglr may suggest returning to an earlier stage. *Unit:* failed attempts. *Range:* 3–5. *Default:* 4. *Trace:* G.3, BR-003, OI-05.
ATTEMPT_LIMIT	Number of unsuccessful attempts that triggers the lesson ending or extension decision. *Unit:* failed attempts. *Range:* 6–10. *Default:* 8; must exceed DIFFICULTY_STREAK. With the current ranges this always holds; the check stays in case the ranges change. *Trace:* G.3, BR-002, OI-05.
IMPROVE_WINDOW	Number of evaluable attempts in each comparison window used by the improvement rule. *Unit:* attempts. *Range:* 1–3. *Default:* 2. *Trace:* G.3, BR-002, OI-05.
IMPROVE_MARGIN	Minimum increase in average stage result points across consecutive comparison windows that qualifies as improvement. *Unit:* points out of 100. *Value:* 5 points (fixed by the team). *Trace:* G.3, BR-002, OI-05.
ATTEMPT_EXTENSION	Additional evaluable attempts permitted when recent results satisfy the improvement criterion. *Unit:* attempts. *Range:* 1–3. *Default:* 2. *Trace:* G.3, BR-002, OI-05.

Constant	Meaning, unit, value and traceability
ATTEMPT_LIMIT_MAX	Maximum number of evaluable attempts at a stage, including extensions. Because a pass moves the child to the next stage, at most one of these is a pass and it is always the last; a maximum of 10 allows 9 failed attempts and a pass, or 10 failed attempts. *Unit:* attempts. *Range:* 8–12. *Default:* 10; must be at least ATTEMPT_LIMIT. *Trace:* G.3, BR-002, OI-05.
LESSON_IDLE_PROMPT	Time with no input during a guided lesson after which Squigglr shows the child a check-in. *Unit:* seconds. *Range:* team-defined; TBD. *Default:* middle of the range; must be less than LESSON_IDLE_END. *Trace:* BR-008, SAF-RAY-009.
LESSON_IDLE_END	Time with no input during a guided lesson after which Squigglr ends the lesson. *Unit:* seconds. *Range:* team-defined; TBD. *Default:* middle of the range; must be less than IDLE_TIMEOUT. *Trace:* BR-008, SAF-RAY-009.
REFINE_TOLERANCE	Maximum permitted distance between refined strokes and the child's original strokes in the creative space. *Unit:* CSS pixels on the drawing area used for testing. *Value:* TBD; set by the team in PH-1, before the first refinement test, by judging adult-made drawings of the supported set (BR-009), then confirmed by supporting adults in the usability sessions (H-MUH-006), as planned in the VnV Plan (Symbolic Parameters appendix). *Trace:* G.4, BR-005.
ADULT_REVIEWERS	Minimum number of adult reviewers for progress summary understandability checks. *Unit:* people (count). *Value:* 5, set in the VnV Plan (Symbolic Parameters appendix). *Trace:* G.6.
REVIEW_PASS_RATE	Minimum proportion of progress summary review tasks completed successfully by adults. *Unit:* proportion. *Value:* 0.8, set in the VnV Plan before any review session (Symbolic Parameters appendix). *Trace:* G.6.

Constant	Meaning, unit, value and traceability
RECUR_MIN	Minimum number of a learner's attempts, at the same stage of the same lesson within a progress summary's period, in which one predefined difficulty must be observed for the summary to list it as recurring. *Unit:* attempts (integer ≥ 2). *Value:* TBD; proposed 3, and confirmed with the specialists in PH-2, as planned in the VnV Plan (Symbolic Parameters appendix). *Trace:* G.6, FR-PRG-004.
AGE_RANGE	Intended lower and upper ages of learners served by Squigglr. *Unit:* years. *Value:* lower and upper bounds TBD. *Trace:* Section 2, G.1.
CHILD_CONTROLS_MAX	Maximum number of controls a child can use on one child-facing screen. *Unit:* controls (count). *Value:* team-defined; TBD. *Trace:* NFR-UX-001.
UX_REVIEWERS	Minimum number of adult reviewers in the look-and-feel and usability reviews. *Unit:* people (count). *Value:* team-defined; TBD. *Trace:* NFR-UX-003, NFR-UX-006, NFR-UX-007, NFR-UX-009, NFR-UX-013, NFR-UX-014.
UX_PASS_RATE	Minimum proportion of reviewers who must complete a usability or learning task, or answer 4 or 5 on a five-point rating. *Unit:* proportion. *Value:* team-defined; TBD. *Trace:* NFR-UX-006, NFR-UX-009, NFR-UX-013, NFR-UX-014.
SETUP_TIME_MAX	Maximum time for a first-time adult to create a child's profile and start a lesson. *Unit:* minutes. *Value:* team-defined; TBD. *Trace:* NFR-UX-009.
FEEDBACK_WORDS_MAX	Maximum length of a message to the child when spoken. *Unit:* words. *Value:* team-defined; TBD. *Trace:* NFR-UX-015.
TARGET_SIZE_MIN	Minimum touch target size of a control on a child-facing screen. *Unit:* CSS pixels. *Value:* 44 by 44, from WCAG 2.2 success criterion 2.5.5. *Trace:* NFR-UX-018.

Constant	Meaning, unit, value and traceability
INTERFACE_LANGUAGE	Language of all text in adult views. *Value:* TBD. *Trace:* ASM-007, NFR-UX-012.
IDLE_TIMEOUT	Duration of user inactivity after which Squigglr ends the session and logs out the user. *Unit:* minutes. *Value:* team-defined; TBD. *Trace:* SAF-SHA-006.
ADULT_CHECK	Whether a supporting adult must confirm an adult-only action before it can be carried out. *Unit:* Boolean. *Value:* required for adult-only actions. *Trace:* SAF-RAY-001, SAF-RAY-008.
SUPPORTED_DEVICES	Supported browser and device combinations on which Squigglr is expected to function properly. *Unit:* device/browser set. *Value:* team-defined list; TBD. *Trace:* SAF-RAY-002, H-RAY-002.
FAILED_SAVES	Number of consecutive failed save attempts used in the fit criterion sample for save failure notification tests. *Unit:* save attempts (count). *Value:* team-defined; TBD. *Trace:* SAF-SHA-003.
FEEDBACK_LATENCY_MAX	Maximum allowed time between the completion of a stylus stroke and visual acknowledgement of that stroke during a lesson. *Unit:* milliseconds. *Value:* TBD; set by repeated measurement, as planned in the VnV Plan (Symbolic Parameters appendix). *Trace:* NFR-PERF-001.
REFINE_LATENCY_MAX	Maximum allowed time between system's AI refinement request and visual acknowledgement of that request strokes (or falling back to unrefined strokes). *Unit:* seconds. *Value:* TBD; set by repeated measurement, as planned in the VnV Plan (Symbolic Parameters appendix). *Trace:* NFR-PERF-002.
SUMMARY_LOAD_MAX	Maximum allowed time for an adult's progress summary page to become fully visible after request. *Unit:* seconds. *Value:* TBD; set by repeated measurement, as planned in the VnV Plan (Symbolic Parameters appendix). *Trace:* NFR-PERF-003.

Constant	Meaning, unit, value and traceability
LOCATE_TIME_MAX	Maximum time a team member unfamiliar with the codebase should need to locate the module responsible for a given stage behaviour. *Unit:* minutes. *Value:* TBD; set by repeated measurement, as planned in the VnV Plan (Symbolic Parameters appendix). *Trace:* NFR-MAINT-002.
CONCURRENT_ CAPACITY	Maximum number of simultaneous active users the system shall support without degrading response times below the thresholds in Section 12.1. *Unit:* concurrent sessions (count). *Value:* TBD; set by repeated measurement, as planned in the VnV Plan (Symbolic Parameters appendix). *Trace:* NFR-PERF-008.

5 Relevant Facts And Assumptions

This section records what is known to be true now, the rules that govern how Squigglr may be used, and what the team believes but has not verified. Facts and assumptions are kept apart so that a reader can see which requirements rest on unconfirmed beliefs. The Hazard Analysis uses the assumptions as possible causes of hazards: if an assumption turns out to be false, the requirements that depend on it must be reviewed. Each entry identifies the goals, constraints or product behaviours it supports; Section 9 specifies the corresponding functional requirements.

5.1 Relevant Facts

Each fact below is true at the time of writing and is supported by a source in Appendix 29. Facts from the interview are expert statements, not observations of children.

FACT-001

Visual-motor integration is linked to handwriting legibility in kindergarten children, but the link is a correlation; it does not show that practising one improves the other (SRC-001; Rayan, 2:04).
Relevance: Squigglr makes no claim that its practice improves handwriting or visual-motor skills (BR-006).

FACT-002

The study of structured, graded practice cited in the Problem Statement had no control group. It supports the idea of graded practice but does not show that Squigglr will produce the same results (SRC-001; Rayan, 2:04).

Relevance: effectiveness is not a goal or a fit criterion of this project.

FACT-003

Motor and handwriting difficulties occur in some autistic children and some children with ADHD, with severity ranging from mild to severe; not every child in these groups has them (SRC-001; Rayan, 2:04).

Relevance: guidance and difficulty are set per child by an adult, not per diagnosis (G.3).

FACT-004

The specialists stated that deciding automatically when a child is ready for the next level requires assessment, which is too complex for the project's time frame, and that an adult who knows the child should decide (SRC-005, P1, 28:37; P2, 29:18).

Relevance: basis for adult controlled lesson selection and nonroutine stage changes (CON-007, G.3, BR-003); the team retains fixed pass-based forward progression within the selected lesson (G.2).

FACT-005

The specialists stated that how a child holds a tool affects how they draw, that finger input is the easiest start, and that fine-motor practice needs a held tool, moving from large to small (SRC-005, P3, 25:36; P1, 26:46).

Relevance: basis for stylus input (CON-003) with finger input as a stretch goal.

FACT-006

The specialists stated that tracing in the air in front of a camera practises gross rather than fine motor skills, and that it should come after the touchscreen activities when time is limited (SRC-005, P1, 13:42; P1, 16:13; P2, 16:45).

Relevance: camera finger tracing stays a stretch goal.

FACT-007

The specialists stated that young children may not read yet but navi-

gate games through images and icons, and that a child should be able to enter a lesson, practise, restart and move back and forth without constant adult help, while adults handle setup (SRC-005, P1, 34:13).
Relevance: G.1 and the usability requirements. The team accepted navigation through images and icons and practice without constant adult help. It did not accept the child restarting a lesson or moving between stages: within a lesson Squigglr moves between stages, and the adult chooses and starts lessons (CON-007, BR-003).

FACT-008

A study that involves children requires ethics approval; meeting adult experts for feedback, without involving children, does not (SRC-004; SRC-005, P2, 22:56; P1, 33:00).
Relevance: basis for CON-005 and CON-006.

FACT-009

Installing applications on school-managed devices is difficult, and a website is easier for Canadian schools to adopt (SRC-003, SRC-004).
Relevance: basis for CON-001 and CON-002.

FACT-010

The specialists recommended starting from the Web Content Accessibility Guidelines (WCAG), which one participant stated are adopted in law in many countries, and then applying design heuristics for autism from the literature (SRC-005, P2, 39:26).
Relevance: accessibility and standards requirements (NFR-UX-018–NFR-UX-023, NFR-COMP-005).

FACT-011

The course fixes the project's deadlines (CON-004, SRC-002).
Relevance: scope decisions, such as keeping assessment and camera tracing out of the core product, follow from the time available.

FACT-012

External requirements evidence includes a 43-minute interview with three domain specialists and a follow-up questionnaire answered by two of them. No parent, educator, school administrator or occupational therapist has been consulted, and no child has taken part in any source (SRC-005, SRC-009; SRC-004).

Relevance: requirements based on what parents and educators need rest on assumptions (ASM-006) until those groups are consulted.

FACT-013

The interview did not cover several topics that affect the requirements: how Squigglr compares with existing drawing apps (one participant noted only that many exist (SRC-005, P2, 35:36)), differences between home and school use, the relative priority of the features, and how practice records should be kept, shared or deleted (SRC-005).
The follow-up questionnaire later covered refinement in the creative space and letter activities (SRC-009).
Relevance: the requirements on these topics rest on assumptions (ASM-010 to ASM-012) and are open to change.

5.2 Business Rules

The rules below govern who may do what with Squigglr and what Squigglr must never do. They hold no matter how Squigglr is designed. Each rule has an identifier of the form BR-###, a rationale and its sources.

BR-001

Only the adult who holds an account, a parent or an educator, may create, view and manage the learner profiles under it and their records. One account can hold several learner profiles. A learner profile holds only a unique identifier that Squigglr generates and an avatar, chosen by the adult or, with the adult, by the child; no name or other information about the child is collected.
Rationale: practice records describe a child's difficulties and are personal information. Keeping each profile to an identifier and an avatar lets an adult, and a child who cannot read, tell children apart for example on a shared device, while collecting as little about them as possible. How records are protected is a privacy requirement.
Source: SRC-001, SRC-004; G.5, G.6.

BR-002

Within an adult-selected lesson, Squigglr automatically moves the child to the next guidance stage when an attempt meets the current drawing stage's pass criteria (`STAGE_PASS`). Passing the independent drawing stage completes the lesson; it does not select or start another lesson.

After ATTEMPT_LIMIT (larger than DIFFICULTY_STREAK) unsuccessful attempts at one stage, the lesson ends, and the furthest stage reached and the number of successful attempts at each stage are saved. If, when ATTEMPT_LIMIT is reached, the average result of the last IMPROVE_WINDOW attempts is better than that of the IMPROVE_WINDOW attempts before them by at least IMPROVE_MARGIN, the child is improving and Squigglr grants a single extension of up to ATTEMPT_EXTENSION additional evaluable attempts. The extension cannot be granted a second time at the same stage, regardless of later result changes. Total evaluable attempts at a stage never exceed ATTEMPT_LIMIT_MAX.
Rationale: moving forward on a pass checks the drawing against fixed criteria and does not assess the child (G.2). Ending the lesson stops a child from repeating a stage they cannot pass without limit, and saving the progress lets the adult decide what comes next. Extending the limit while the child improves avoids cutting off practice that is working, and ATTEMPT_LIMIT_MAX keeps the stop. The comparison uses only the child's own attempts against the stage's fixed criteria, so it does not assess the child.
Source: G.2, G.3, G.5.

BR-003

A supporting adult chooses which lesson the child practises and its starting stage. Squigglr never changes lessons, skips a stage or moves back a stage on its own. Passing an attempt may advance the child to the next stage *within that same lesson* as defined by BR-002. After DIFFICULTY_STREAK failed attempts in a row, Squigglr may suggest going back a stage, unless the child is improving (BR-002); the change happens only if an adult accepts.
Rationale: deciding readiness needs assessment that is out of scope (FACT-004), and the adult knows the child's level.
Source: CON-007; G.3; SRC-005, P1, 28:37; P2, 29:18.

BR-004

A guided lesson record keeps the child's actual strokes. Squigglr never replaces them with a corrected or ideal shape.
Rationale: adults judge progress from the records; a record that shows a corrected shape would mislead them.
Source: G.5; SRC-005, Moustafa, 6:03.

BR-005

Refinement of strokes happens only in the creative space. It is automatic and minimal, the refined drawing stays editable, and the child's original drawing can be restored. Refinement can also be turned off for the creative space. Restoring and turning off are done by the child or, if the child cannot, by the adult.
Rationale: the work must remain the child's own, and refinement must never reach the records adults rely on (BR-004). Refinement needs no action from the child, and an adult can restore the original or turn refinement off for a child who cannot use the controls. Both questionnaire respondents suggested that refinement be optional or adjustable (SRC-009.01).
Source: G.4; SRC-001; SRC-009.01.

BR-006

Squigglr gives no diagnosis, no judgement of a child's abilities, no comparison with other children, and no claim that its practice improves any skill.
Rationale: the team are not occupational therapists and cannot judge a child's abilities (SRC-008), and the evidence does not support effectiveness claims (FACT-001, FACT-002).
Source: SRC-001, SRC-008; SRC-005, Moustafa, 6:03.

BR-007

Feedback during a guided activity comes from observations the team defines in advance, such as a missed part of an outline. It is not personalized advice generated by an AI model.
Rationale: predefined observations can be reviewed and checked before release; generated advice to a child cannot.
Source: SRC-001; SRC-005, Moustafa, 6:03.

BR-008

The supporting adult sets `STAGE_PASS`, `DIFFICULTY_STREAK`, `ATTEMPT_LIMIT`, `IMPROVE_WINDOW`, `ATTEMPT_EXTENSION`, `ATTEMPT_LIMIT_MAX`, `LESSON_IDLE_PROMPT` and `LESSON_IDLE_END` for each child. When an adult creates a child profile, Squigglr asks them to choose each value, and for `STAGE_PASS` shows example drawings of what its minimum and maximum look like. The adult can change the values later. A value the adult does not choose is set to the middle of its range. Squigglr

accepts only values within the range, only an `ATTEMPT_LIMIT` larger than `DIFFICULTY_STREAK`, and only an `ATTEMPT_LIMIT_MAX` at least `ATTEMPT_LIMIT`.
Rationale: the adult knows the child, and a value that suits one child may not suit another (FACT-003). The team sets only the minimum and maximum, so no single value has to be shown right for every child.
Source: SRC-008; CON-007; Table 10.

BR-009

The team sets the minimum and maximum of each adult-set value, and the fixed values in Table 10, using adult-made drawings, never from testing with children.
Rationale: no study with children takes place (CON-005, CON-006).
Source: SRC-004, SRC-008; Table 10.

5.3 Assumptions

The assumptions below are beliefs the team holds but has not verified. Each has an identifier of the form ASM-###, a rationale, what depends on it, and the effect if it is false. Assumptions marked as relevant to safety are inputs to the Hazard Analysis.

ASM-001

An adult is present when a child starts a guided lesson, to choose the lesson and the starting stage.
Rationale: the specialists described adults handling setup and deciding progression (SRC-005, P1, 28:37; P1, 34:13). How often this happens at home or in class is unknown.
Depends on it: G.3, BR-003, CON-007.
If false: a child may start at a level an adult did not choose. Relevant to safety.

ASM-002

A child may continue practising without an adult watching, but an adult nearby remains responsible for the child's physical safety, wellbeing and time on the device. Squigglr cannot observe or prevent physical harm, distress or overuse.
Rationale: the specialists advised that children should practise without constant adult help (SRC-005, P1, 34:13), and the Problem Statement

expects adult support as each learner needs (SRC-001).
Depends on it: G.1, the safety requirements (requirement IDs to be linked when written).
If false: the safety requirements must cover unsupervised use, for example limits on session length or sound. Relevant to safety.

ASM-003

Homes and schools that use Squigglr have a touchscreen device with a stylus that works in a supported browser.
Rationale: the stylus is the main input (CON-003), and the interview example used a tablet with a stylus (SRC-005, Rayan, 24:56). Which devices families and schools own is not known. The supported device and browser combinations are TBD.
Depends on it: CON-003, G.1, G.2.
If false: children without a stylus can still use Squigglr with a finger, but not as intended: the stage checks and value ranges are set for stylus input, and tuning them for finger input is a stretch goal.

ASM-004

Supported browsers report stylus strokes in enough detail (position and order of points) for Squigglr to check an attempt against `STAGE_PASS` and to store the actual strokes.
Rationale: the stage checks and the records both depend on the strokes; this has not been tested on any device yet.
Depends on it: G.2, G.5, BR-004.
If false: an attempt may be judged wrongly or stored inaccurately, which would mislead the adult. The team may then need to deliver Squigglr as an installed app, if input there is more reliable, which would change CON-001. Relevant to safety.

ASM-005

A network connection is available during normal use, for loading activities and for saving and viewing records. Keeping work through a temporary loss of connection is in scope (SAF-RAY-003, NFR-PERF-005, FR-CRT-025); starting or running lessons with no connection is out of the initial scope.
Rationale: stated in the Problem Statement (SRC-001); school and home networks have not been checked.
Depends on it: G.5, G.6, CON-001.

If false: attempts may be lost when the connection drops, and an adult may decide from incomplete records. Relevant to safety.

ASM-006

The supporting adult can judge from the records when a child should move on, stay or go back.

Rationale: the specialists placed this decision with a parent or educator who knows the child (SRC-005, P1, 28:37). Squigglr shows the adult every value it records, such as each attempt's result and the `STAGE_PASS` and `ATTEMPT_LIMIT` in use for that attempt, and the adult chooses which of them their view shows, so the team does not have to pick which measures matter.

Depends on it: G.3, G.6, CON-007, BR-003.

If false: a child may be kept at a level that is too hard or too easy; the adult views would need more guidance. Relevant to safety.

ASM-007

The adult who uses the adult views can read the language of the interface. The interface language is TBD. Children do not need to read to use the child views.

Rationale: adult tasks such as reviewing records need text, while child navigation uses images and icons (FACT-007).

Depends on it: G.6 and NFR-UX-012.

If false: adults may misread records or settings.

ASM-008

On a shared device, an adult makes sure the right child is selected before a session and that another child cannot open the previous child's records.

Rationale: school devices may be shared by several children (SRC-004). One adult account can hold several learner profiles (BR-001).

Depends on it: BR-001, G.5.

If false: attempts may be saved to the wrong child, or a child's records may be seen by others. Relevant to safety and privacy.

ASM-009

The adults who use Squigglr for a child have the right to see that child's records, and any consent this needs is obtained outside Squigglr.

Rationale: Squigglr cannot verify a parent's or school's authority over

a child; it only knows the adult who holds the account (BR-001).
Depends on it: BR-001, the privacy requirements (requirement IDs to be linked when written).
If false: an unauthorised adult could see a child's records. Relevant to privacy.

ASM-010

Lines, basic shapes and simple drawings are suitable practice content for the intended children, without letters.
Rationale: the specialists did not object to this content, and nurseries focus on these skills (SRC-005, P1, 20:27). Two specialists saw letters as a later stage after shapes (SRC-009.04), so they remain a stretch goal (SRC-001).
Depends on it: G.1, G.2.
If false: the activity set changes.

ASM-011

Children and adults value automatic refinement in the creative space, and children understand that the refined drawing is still theirs.
Rationale: questionnaire respondents supported subtle, optional refinement that preserves the child's drawing (SRC-009.01), but their responses do not establish children's or caregivers' preferences.
Depends on it: G.4, BR-005.
If false: refinement may discourage a child, and G.4 would need to change.

ASM-012

Squigglr serves the same needs at home and in school, so one set of requirements covers both.
Rationale: the specialists named homes, kindergartens, schools and rehabilitation centres as possible settings (SRC-005, P2, 21:25) but did not discuss differences between them (FACT-013).
Depends on it: G.3, G.6, BR-001.
If false: separate requirements may be needed, for example for an educator managing several children.

ASM-013

Schools allow access to the Squigglr website on their managed devices and networks.

Rationale: a website is easier to adopt than an installed app (FACT-009), but no school has confirmed that it would not block the site.
Depends on it: CON-001, CON-002.
If false: school use needs approval from administrators.

ASM-014

Strokes drawn by adults, or prepared examples, are close enough to children's drawing to verify stage checks and refinement.
Rationale: evaluation uses adults only (CON-005, CON-006).
Depends on it: G.2, G.4, BR-009, and the verification plan.
If false: the range of `STAGE_PASS` and the value of `REFINE_TOLERANCE` may be wrong for children, so attempts may be judged wrongly. Relevant to safety.

ASM-015

An adult can turn off audio feedback, or stop a session, if sound or on-screen feedback upsets a child.
Rationale: the specialists recommended that users can turn audio off (SRC-005, P2, 37:39). Children react to sensory feedback in different ways, and the team has no information on which feedback upsets which children.
Depends on it: NFR-UX-010, SAF-RAY-005, FR-REC-005 and FR-REC-007.
If false: feedback must be safe by default rather than relying on an adult. Relevant to safety.

ASM-016

A supporting adult can choose adult-set values (Table 10) that suit their child, helped by the example drawings for `STAGE_PASS`, or leave the defaults.
Rationale: the supervisor advised leaving these values to the adult (SRC-008); no adult has tried the choice yet.
Depends on it: G.2, G.3, BR-008.
If false: a child may be held at a stage too long or moved on too soon, which can frustrate them. Relevant to safety.

ASM-017

A rise in the average result of a child's recent attempts by at least

`IMPROVE_MARGIN` reflects real improvement, not chance variation between drawings.
Rationale: drawing results vary from one attempt to the next; the team will check the margin with prepared adult-made sequences, not with children (CON-005, CON-006).
Depends on it: G.3, BR-002.
If false: a lesson may be extended for a child who is not improving, or not extended for one who is. `ATTEMPT_LIMIT_MAX` still ends the lesson. Relevant to safety.

6 The Scope of the Work

This section describes the work that Squigglr will become part of: children practising drawing, and the adults around them choosing that practice and following the child's progress. It covers how the work is done now, the boundary of the work and what crosses it, the business events the work responds to, and the business use case for each event. Squigglr's own part of the work is in Section 8.

6.1 The Current Situation

Children practise drawing lines, shapes and simple drawings at home, in nurseries and kindergartens, in schools and in rehabilitation centres (SRC-005, P1, 20:27; P2, 21:25). They practise with tools they hold, starting with large ones that are easier for their hands and moving to smaller ones, such as pencils, that are closer to what they will use in school (SRC-005, P1, 26:46). Some children also take part in therapy programmes run by occupational therapists (SRC-005, P1, 32:42). Structured programmes exist that grade the same activity into easier and harder levels (SRC-001), and many drawing applications are available for children (SRC-005, P2, 35:36). The comparison with existing products is in the Off-the-Shelf Solutions section.

The adults who support a child follow progress mainly through what the child produces. A finished drawing does not show whether the child made it alone, relied on tracing guides, or corrected mistakes along the way (SRC-001). The adult who knows the child decides when the child is ready to move on (FACT-004). The work as it is done now therefore has three gaps that the project goals address:

- practice is not always graded so that help fades as the child improves (G.1, G.2);
- the adult's decision on the next step rests on finished drawings, not on a record of how they were made (G.3, G.5);
- nothing summarizes the child's practice over time for the adult (G.6).

6.2 The Context of the Work

Figure 1 shows the boundary of the work and the adjacent actors that exchange information with it, and Table 11 lists every data flow that crosses the boundary. The work includes Squigglr and the decisions supporting adults make with it, such as which lesson a child practises. The child and the supporting adult are drawn outside the boundary because they start the work and receive its results. The Squigglr team is an adjacent actor because it maintains the lesson catalogue and sets the ranges of the adult-set values (BR-009). Children and adults reach the work through a touchscreen device with a stylus in a web browser (CON-001, CON-003). The device carries the flows; it is not an adjacent actor because it neither starts the work nor uses its results.

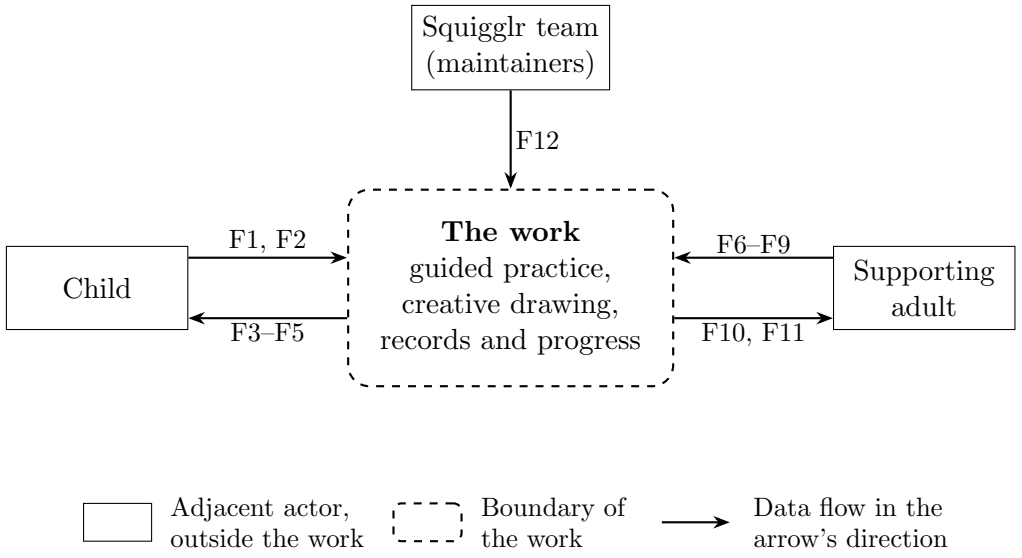


Figure 1: Context of the work. Each arrow carries the data flows named on it, which Table 11 defines.

Table 11: Data flows across the boundary of the work

Flow	From	To	Data
F1	Child	The work	Strokes drawn for an attempt or in the creative space.
F2	Child	The work	In the creative space, a request to turn refinement off or to restore the original drawing.
F3	The work	Child	Demonstrations and tracing guides for each stage.
F4	The work	Child	Feedback on an attempt, shown on screen and, unless muted, as audio.
F5	The work	Child	The refined drawing in the creative space.
F6	Supporting adult	The work	A child's profile details (an avatar) and adult-set values (BR-001, BR-008).
F7	Supporting adult	The work	The lesson and starting stage chosen for a child, and the start of that lesson (BR-003).
F8	Supporting adult	The work	A decision on a suggestion to go back a stage: accept or decline.
F9	Supporting adult	The work	Changes to a child's settings (accessibility, audio, whether refinement is on, adult-set values), or a request to restore an original drawing.
F10	The work	Supporting adult	A suggestion to go back a stage after repeated difficulty (BR-003).
F11	The work	Supporting adult	Progress summaries and the attempt records behind them.
F12	Squigglr team	The work	The lesson catalogue, the ranges of the adult-set values, and the fixed values (BR-009).

6.3 Work Partitioning

Table 12 partitions the work by business event: a happening outside the work, or a condition reached within it, that the work must respond to. Each event's inputs and outputs are data flows from Table 11, and each event has one business use case (BUC) in Section 6.4, with the same number.

Table 12: Business events

ID	Business event	Input	Output	BUC
BE-001	An adult wants a child to start practising with Squigglr.	F6	–	BUC-001
BE-002	An adult decides what the child should practise next.	F7	–	BUC-002
BE-003	An adult starts a guided lesson for the child.	F1, F7	F3, F4	BUC-003
BE-004	A child fails `DIFFICULTY_STREAK` attempts in a row at one stage without improving.	F8	F10	BUC-004
BE-005	A child wants to draw freely.	F1, F2, F9	F5	BUC-005
BE-006	An adult wants to know how the child's practice is going.	–	F11	BUC-006
BE-007	A child's needs change, so an adult changes the child's settings.	F9	–	BUC-007
BE-008	The team changes the lesson catalogue or the ranges of the adult-set values.	F12	–	BUC-008

6.4 Specifying a Business Use Case (BUC)

Each business use case below is the work's response to the business event with the same number. It describes the whole response, including the parts done by people, and names the product use cases (Table 16) that are Squigglr's part of it.

BUC-001 Add a child (BE-001)

The adult creates a profile for the child and chooses each adult-set value, using the example drawings for `STAGE_PASS`, or leaves the defaults (BR-008). The work records the profile and the values. Product use case: UC-001.

BUC-002 Choose the next practice (BE-002)

The adult judges what the child is ready for, from knowing the child and, once there is practice, from the records (ASM-006). The adult then chooses a lesson and its starting stage. Product use cases: UC-002, UC-004.

BUC-003 Practise a guided lesson (BE-003)
The adult starts the lesson. The child watches the demonstration and draws an attempt at each stage. The work checks each attempt against the stage's criteria, gives feedback, and moves the child on or offers a retry. The lesson ends when the child finishes it or reaches `ATTEMPT_LIMIT`, or `ATTEMPT_LIMIT_MAX` when extended, and every attempt is recorded (BR-002). Product use cases: UC-003, UC-008.

BUC-004 Respond to repeated difficulty (BE-004)
The work suggests going back a stage. The adult decides whether to accept; a declined or ignored suggestion changes nothing (BR-003). Product use case: UC-005.

BUC-005 Draw freely (BE-005)
The child draws and edits in the creative space. The work refines the drawing a little and keeps it editable. The child or the adult can turn refinement off or restore the original (BR-005; SRC-009.01). Product use cases: UC-006, UC-007.

BUC-006 Review progress (BE-006)
The adult reads a summary of the child's practice, opens the attempts behind it, and uses what they see to decide on further practice (BUC-002). The work gives no diagnosis or judgement of the child (BR-006). Product use case: UC-009.

BUC-007 Change a child's settings (BE-007)
The adult opens the child's history or settings and changes accessibility, audio or adult-set values. The work accepts only values within their ranges (BR-008). Product use case: UC-010.

BUC-008 Maintain lessons and ranges (BE-008)
The team adds or changes lessons, or changes the ranges of the adult-set values, using adult-made drawings (BR-009). Squigglr has no product use case for this event; the maintenance requirements cover it.

7 Business Data Model and Data Dictionary

This section defines the business data that Squigglr creates, uses and keeps: what a supporting adult's account, a learner profile, a lesson, a lesson record,

an attempt record and a creative drawing are, how they relate, and what each term in the requirements means. The model is conceptual. Following the Volere template [33], it records the subject matter the business must remember and the rules that hold between its parts; it is not a database design, and it fixes no tables, keys, storage or technology. The names defined here are the names the functional requirements use, in particular those of Sections 9.3, 9.4 and 9.5, and the product use cases UC-001 to UC-010 (Table 16) create, read and change this data. Subsection 7.1 gives the model as a diagram and states the data constraints. Subsection 7.2 defines every class, attribute and data flow.

7.1 Business Data Model

Figure 2 shows the seven classes of business data that Squigglr keeps and how they relate. A supporting adult's account holds learner profiles, and each learner profile has its own learner settings (BR-001, BR-008). The team maintains the lessons, each with its four stages and their predefined difficulties (BR-009; LC-001 in Table 34). Each time an adult starts a lesson for a learner, Squigglr keeps a new lesson record, which holds the attempt records and any go-back suggestions of that lesson (G.3, G.5). The creative space keeps creative drawings, each with its strokes and refinements (G.4, BR-005). A progress summary is not a class: Squigglr derives it from one learner's lesson and attempt records whenever an adult opens it (G.6), so Table 13 defines it as a data flow. That table also gives the attributes of each class and how many of one class relate to another.

The model carries the following data constraints. None is new; each restates as a property of the data a rule or goal stated elsewhere, and each can be checked by inspecting saved data prepared by adults.

- A learner profile holds only a unique identifier and an avatar (BR-001).
- Every adult-set value lies within its team-set range, `ATTEMPT_LIMIT` is larger than `DIFFICULTY_STREAK`, and `ATTEMPT_LIMIT_MAX` is at least `ATTEMPT_LIMIT`; a value the adult does not choose is the middle of its range (BR-008).
- Within one lesson record, the evaluable attempts at one stage never exceed `ATTEMPT_LIMIT_MAX` (BR-002). An interrupted attempt, or one

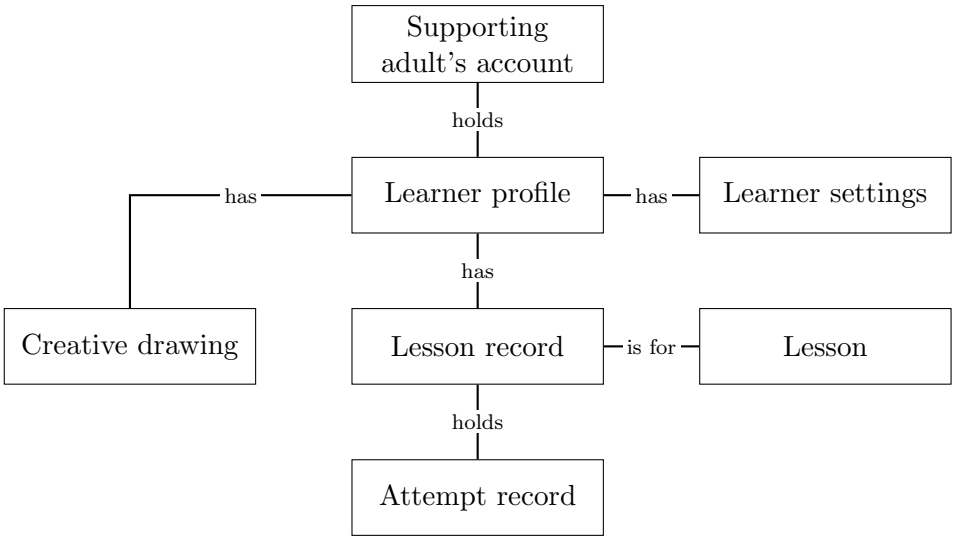


Figure 2: Business data model of Squigglr. Each box is a class of business data, defined with its attributes in Table 13. Each line is a relationship between two classes. Its label reads from the box nearer the top of the figure to the other box, or, for two boxes side by side, from the left box to the right one.

whose input is invalid or too ambiguous to evaluate, is not an evaluable attempt and is not counted (FR-LRN-009, FR-LRN-010).

- The actual strokes of an attempt record are the strokes the child drew; they are never replaced by a corrected, ideal or refined shape (BR-004, BR-005).
- Every observed difficulty of an attempt record is a predefined difficulty of that attempt's stage, and a later correction adds a mark without removing the difficulty (G.5, BR-007).
- A refinement keeps every refined stroke within `REFINE_TOLERANCE` of the original stroke, and it changes neither the number nor the order, colours and widths of the strokes (G.4, BR-005; FR-CRT-011, FR-CRT-012).
- The original strokes of a refinement never change, so a restore returns them exactly (G.4, BR-005).
- A creative drawing is never an attempt record and never counts in a lesson record or a progress summary (BR-004; FR-CRT-028).

7.2 Data Dictionary

Table 13 defines every class of Figure 2, every attribute of a class and every data flow that enters or leaves the work. For a class, the Unit, domain or constraint column also states how it relates to the other classes. Three flows share the name of the data they carry: F5 is the refined drawing, F7 the lesson choice and F10 the go-back suggestion. Entries are sorted by type and then by name. The Meaning column uses the composition notation of the Volere template: "A = B + C" means that A is made of B and C; "{B}" means any number of B; "[B | C]" means exactly one of B or C; and "(B)" means that B may be absent. Units and domains are stated at the business level, and every bound uses a named constant rather than a number; most are in Table 10. The Used by column lists the goals, business rules (Section 5.2), use cases and requirements that use each entry; F1 to F12 are the data flows of Table 11. Older names from earlier documents are given in the Meaning column where they help a reader.

Table 13: Data dictionary

Name	Meaning	Type	Unit, domain or constraint	Used by
Attempt record	The record of one attempt, which is one try at one drawing stage of a lesson. Attempt record = Stage + Actual strokes + {Observed difficulty} + [Outcome | Interrupted mark] + (Stage result) + Attempt end time + Adult-set values in use.	Class	One for each attempt that ends with an outcome, or that an adult interrupts after at least one stroke; belongs to exactly one lesson record. An interrupted attempt does not count towards an attempt limit.	G.5; UC-008; FR-LRN-009; FR-REC-001 to FR-REC-009; FR-PRG-006
Creative drawing	One drawing in the creative space. Creative drawing = {Stroke} + {Refinement} + Refined state.	Class	Belongs to the learner profile that was the active learner when it was drawn; never an attempt record.	G.4; BR-005; UC-006, UC-007; FR-CRT-001 to FR-CRT-030
Learner profile	One child known to an account (called a child's profile in UC-001 and F6). Learner profile = Profile identifier + Avatar.	Class	Holds nothing else about the child; has exactly one learner settings and any number of lesson records and creative drawings; only the adult who holds the account creates, views and manages it.	BR-001; ASM-008; UC-001; F6; FR-REC-006; FR-CRT-023
Learner settings	The values that tailor practice to one learner. Learner settings = Adult-set values + Audio setting + Accessibility settings + Refinement setting.	Class	Exactly one for each learner profile; changed by the adult who holds the account, except that the child can turn refinement off in the creative space (BUC-005).	BR-008; ASM-015; UC-001, UC-007, UC-010; F2, F6, F9; NFR-UX-010, NFR-UX-011

Name	Meaning	Type	Unit, domain or constraint	Used by
Lesson	A catalogue item, such as drawing a square, that teaches one line, shape or simple drawing through stages that give gradually less help (called an activity in earlier documents). Lesson = Lesson name + {Stage + {Predefined difficulty}}.	Class	Exactly four stages, in the order given under Stage; maintained by the team; any number of lesson records are for it.	G.1; ASM-010; BR-009; LC-001; FR-LRN-001; UC-002, UC-003; F12
Lesson record	The record of one start of a lesson by a learner, up to the lesson's end. Lesson record = Lesson + Starting stage + Furthest stage reached + Successful attempts per stage + Extension count + Lesson ending + {Attempt record} + {Go-back suggestion}.	Class	Belongs to exactly one learner profile and is for exactly one lesson; a new lesson record each time the learner starts the lesson; unchanged after the lesson ends, except by deletion.	G.5; BR-002; UC-003; FR-REC-006 to FR-REC-010; FR-PRG-002; SAF-RAY-006
Supporting adult's account	The account of a parent, guardian, caregiver, educator or learning-support staff member who uses Squigglr with one or more children. Supporting adult's account = {Learner profile}.	Class	Held by one adult, who alone sees and manages its learner profiles and their records; how the adult is identified is left to the Access Requirements.	BR-001; ASM-009; ULC-002
Accessibility settings	The accessibility options chosen for a learner.	Attribute	Content to be set by the Accessibility Requirements.	UC-010; F9
Active learner	The learner profile that an adult has selected for the current session, to which attempts, suggestions and creative drawings are saved.	Attribute	One learner profile of the account in use.	ASM-008; FR-CRT-023, FR-CRT-026; SAF-RAY-007

Name	Meaning	Type	Unit, domain or constraint	Used by
Actual strokes	The strokes the child drew in an attempt. Actual strokes = {Stroke}.	Attribute	In the order drawn, as captured; never replaced by a corrected, ideal or refined shape.	BR-004; ULC-005; FR-LRN-007; FR-REC-002; FR-CRT-027
Adult-set values	The eight values a supporting adult chooses for each child (Table 10). Adult-set values = `STAGE_PASS` + `DIFFICULTY_STREAK` + `ATTEMPT_LIMIT` + `IMPROVE_WINDOW` + `ATTEMPT_EXTENSION` + `ATTEMPT_LIMIT_MAX` + `LESSON_IDLE_PROMPT` + `LESSON_IDLE_END`.	Attribute	Each within its team-set range, by default its middle; `ATTEMPT_LIMIT` larger than `DIFFICULTY_STREAK`; `ATTEMPT_LIMIT_MAX` at least `ATTEMPT_LIMIT`. `STAGE_PASS` is a result in points out of 100; `LESSON_IDLE_PROMPT` and `LESSON_IDLE_END` are times in seconds; the others are counts of attempts.	G.2, G.3; BR-008; FR-LRN-017; UC-001, UC-010; F6, F9; SAF-RAY-008
Adult-set values in use	The learner's adult-set values at the time an attempt was made.	Attribute	As for adult-set values; a later change to the learner settings does not alter them.	ASM-006; LC-007; FR-REC-004; FR-PRG-006
Attempt end time	When an attempt ended.	Attribute	Date and time.	FR-REC-001; FR-PRG-002, FR-PRG-011
Audio setting	Whether feedback to the learner includes audio.	Attribute	On or muted.	ASM-015; UC-010; F4, F9; NFR-UX-010

Name	Meaning	Type	Unit, domain or constraint	Used by
Avatar	A picture chosen for a learner, by the adult or by the child, so that learners under one account can be told apart, including by a child who cannot read.	Attribute	One of the pictures Squigglr offers; no photo of the child.	BR-001; ASM-008; F6; SAF-RAY-007
Colour	The colour of a stroke in a creative drawing.	Attribute	One of the colours Squigglr offers; unchanged by refinement.	FR-CRT-005, FR-CRT-011
Correction mark	Whether the child corrected an observed difficulty later in the same attempt (called a change during the attempt in earlier documents).	Attribute	Corrected or not corrected; a correction never removes the observed difficulty.	G.5; FR-REC-003; FR-PRG-004
Extension count	The number of times the attempt limit of a lesson record was extended because the child was improving.	Attribute	Count of extensions, at most one at each stage; evaluable attempts at one stage never exceed `ATTEMPT_LIMIT_MAX`.	BR-002; FR-LRN-020; SAF-RAY-006
Furthest stage reached	The latest stage of the lesson that the learner reached in a lesson record.	Attribute	A stage of the lesson.	G.5; BR-002; FR-REC-006; FR-PRG-002

Name	Meaning	Type	Unit, domain or constraint	Used by
Go-back suggestion	A suggestion that the learner go back to an earlier stage, which an adult accepts or declines (called a difficulty suggestion in FR-LRN-019). Go-back suggestion = Suggested stage + Suggestion status.	Attribute	Part of exactly one lesson record; made only after `DIFFICULTY_STREAK` failed attempts in a row at one stage, and not while the child is improving; changes the stage only if an adult accepts it; a declined suggestion is made again only after another full streak, and an unanswered one stays pending, so no second suggestion is made.	G.3; BR-003; CON-007; UC-005; F8, F10; FR-LRN-019; FR-REC-010; SAF-RAY-001, SAF-RAY-007
Interrupted mark	Marks an attempt that an adult stopped, by ending the session or switching learner, after at least one stroke and before its outcome was decided.	Attribute	Set or not set; an interrupted attempt has no outcome.	ASM-008, ASM-015; FR-REC-005; FR-PRG-009
Latest attempt time	The attempt end time of the most recent attempt that a progress summary includes.	Attribute	Date and time.	FR-PRG-011

Name	Meaning	Type	Unit, domain or constraint	Used by
Lesson ending	How the lesson of a lesson record ended.	Attribute	In progress; every stage passed; ended at `ATTEMPT_LIMIT`; ended when a granted extension ran out or `ATTEMPT_LIMIT_MAX` was reached; or unfinished, when an adult stopped it early.	BR-002; FR-LRN-021; FR-REC-007; FR-PRG-002; SAF-RAY-005, SAF-RAY-006
Lesson name	The name of a lesson in the catalogue, such as drawing a square.	Attribute	Names one lesson of the catalogue.	G.1; LC-001
Observed difficulty	A predefined difficulty that Squigglr observed in an attempt. Observed difficulty = Predefined difficulty + Correction mark.	Attribute	A predefined difficulty of the attempt's stage, listed at most once in each attempt record.	G.5; BR-007; FR-REC-003; FR-PRG-004; F4
Original strokes	The strokes of a creative drawing exactly as they were immediately before a refinement: the same strokes in the same order, with the same points, colours and widths. For the first refinement of a drawing, they are the child's own strokes. Original strokes = {Stroke}.	Attribute	At least one stroke; never changed, so that a restore returns them exactly.	G.4; BR-005; FR-CRT-017 to FR-CRT-019, FR-CRT-023

Name	Meaning	Type	Unit, domain or constraint	Used by
Outcome	Whether an attempt passed: its drawing is complete and its result meets the `STAGE_PASS` in use (passed, also called successful), or it does not (not passed, also called unsuccessful or failed).	Attribute	Passed or not passed; absent for an interrupted attempt.	G.2; BR-002; FR-LRN-013; FR-REC-001; F4
Path	The points of a stroke, in the order drawn. Path = {Point}.	Attribute	At least one point.	ASM-004; FR-LRN-007; SAF-AHM-002; FR-REC-002; FR-CRT-011, FR-CRT-012
Point	One position on the drawing area that a stroke passes through.	Attribute	Position on the drawing area, in CSS pixels, the same unit of distance as `REFINE_TOLERANCE`.	ASM-004; FR-CRT-012
Practice-only statement	The statement on every progress summary that it describes practice only and is not an assessment or diagnosis.	Attribute	Fixed text, shown on every summary.	BR-006; SAF-RAY-004
Predefined difficulty	A problem that the team defines in advance and that Squigglr can observe in an attempt, such as a missed part of an outline (called an observation in earlier documents).	Attribute	Defined by the team for one or more stages; never generated for a particular child.	G.5; BR-007; FR-REC-003; FR-PRG-004
Profile identifier	A unique identifier that Squigglr generates for each learner profile.	Attribute	Generated by Squigglr; holds no information about the child.	BR-001; NFR-SEC-004

Name	Meaning	Type	Unit, domain or constraint	Used by
Recurring difficulty	A predefined difficulty that a progress summary lists because it keeps occurring. Recurring difficulty = Predefined difficulty + Stage + Lesson + count of attempts + count corrected.	Attribute	Observed in at least `RECUR_MIN` attempts at the same stage of the same lesson; `RECUR_MIN` is a team-set count, proposed at 3 and confirmed with the specialists in PH-2 (Section 4.2).	G.6; FR-PRG-004
Refined drawing	A creative drawing while a refinement of it is in effect: the strokes the refinement produced, as the child may later edit them.	Attribute	Same number, order, colours and widths of strokes as its original strokes; every refined stroke within `REFINE_TOLERANCE` of its original stroke; stays editable.	G.4; BR-005; FR-CRT-011 to FR-CRT-013; F5
Refined state	Whether a creative drawing holds a refinement that has not been undone or restored.	Attribute	Refined or not refined.	FR-CRT-016
Refinement	One automatic, minimal adjustment of the strokes of a creative drawing, made when the child finishes the drawing. Refinement = Original strokes + Refinement status.	Attribute	Part of exactly one creative drawing and concerns one supported item; applied whole or not at all; only in the creative space, never to a guided lesson.	G.4; BR-005; UC-007; FR-CRT-010 to FR-CRT-021

Name	Meaning	Type	Unit, domain or constraint	Used by
Refinement setting	Whether automatic refinement is on for a learner in the creative space.	Attribute	On or off.	BR-005; UC-007, UC-010; F2, F9; FR-CRT-030; NFR-UX-011
Refinement status	Where one refinement stands.	Attribute	In effect, undone or restored.	FR-CRT-007, FR-CRT-017
Stage	One of the four steps of a lesson, which give gradually less help (also called a guidance level in Table 9). Attempts at the second and third stages are made with a guide and at the fourth without one (FR-LRN-012 calls them assisted and independent).	Attribute	Demonstration; tracing a full outline; drawing between endpoints only; or drawing with no guide, in that order. Attempts are made only at the three drawing stages, not at the demonstration.	G.1, G.2, G.6; BR-002; FR-LRN-002, FR-LRN-003, FR-LRN-012; FR-PRG-003; `STAGE_PASS`
Stage result	The result of an evaluable attempt on the normalized scale of Table 10, where higher results indicate closer agreement with the stage's fixed criteria.	Attribute	Points out of 100; absent for an interrupted attempt.	BR-002; ASM-006; LC-008; FR-LRN-013, FR-LRN-020; FR-REC-001
Starting stage	The stage at which a learner starts a lesson.	Attribute	A stage of the lesson, chosen by an adult.	BR-003; UC-004; FR-LRN-016; FR-REC-006; FR-PRG-010
Stroke	One continuous line drawn with the stylus. Stroke = Path + (Colour + Width).	Attribute	Belongs to exactly one attempt record, creative drawing or refinement; colour and width are kept for strokes in creative drawings.	CON-003; ASM-004; BR-004; FR-REC-002; FR-CRT-001; F1

Name	Meaning	Type	Unit, domain or constraint	Used by
Successful attempts per stage	For each stage of a lesson record, the number of attempts that passed. Successful attempts per stage = {Stage + count of attempts}.	Attribute	Count of attempts.	G.5; BR-002; FR-REC-006; FR-PRG-002
Suggested stage	The stage a go-back suggestion proposes.	Attribute	An earlier stage of the same lesson than the one at which the failed attempts were made; on acceptance the adult chooses the earlier stage.	BR-003; UC-005; FR-LRN-019
Suggestion status	Where a go-back suggestion stands.	Attribute	Pending, accepted, declined or ignored. Pending until an adult answers; ignored when a pass or the end of the lesson clears it before any answer. Only an adult accepts or declines, and a declined or ignored suggestion changes nothing.	G.3; BR-003; UC-005; F8; FR-LRN-019; FR-REC-010
Supported item	A shape or simple object in the agreed supported set, which Squigglr refines.	Attribute	The set is agreed by the team (OI-03); drawings of other items are not refined.	G.4; FR-CRT-015
Width	The width of a stroke in a creative drawing.	Attribute	Unchanged by refinement.	FR-CRT-011

Name	Meaning	Type	Unit, domain or constraint	Used by
Attempt record view	What the adult sees for one attempt record. Attempt record view = every value of the attempt record, with the actual strokes as recorded.	Data flow	Shown only to the adult who holds the account.	ULC-002; UC-009, UC-010; FR-PRG-006; F11
Catalogue update	What the team adds or changes. Catalogue update = {Lesson} + {Supported item} + (Team-set values).	Data flow	Made from adult-made drawings only.	BR-009; LC-001, LC-002; F12
Child navigation	How a child moves through Squigglr outside a guided lesson. Child navigation = [Open the creative space | Leave the creative space].	Data flow	Within a guided lesson the child does not navigate (G.1); never starts, restarts or changes a lesson, the lesson choice or the starting stage.	G.1; FACT-007; BR-003; UC-006
Editing action	A change the child makes to the open creative drawing. Editing action = [Select | Move | Resize | Recolour | Erase | Undo | Redo] + Part. Part = {Stroke}, the strokes the child selects.	Data flow	Acts only on the open creative drawing.	G.4; FR-CRT-002 to FR-CRT-008
Feedback	What Squigglr tells the child about an attempt. Feedback = {Observed difficulty} + Outcome.	Data flow	Drawn only from predefined difficulties, never generated advice; spoken only when the audio setting is on.	G.1; BR-007; ASM-015; FR-LRN-014; F4
Finish mark	The child's mark that a creative drawing is done.	Data flow	Starts exactly one refinement.	UC-007; FR-CRT-010

Name	Meaning	Type	Unit, domain or constraint	Used by
Learner profile details	What an adult enters to add a child. Learner profile details = Avatar + (Adult-set values).	Data flow	From the adult who holds the account only; an adult-set value not entered takes the middle of its range.	BR-001, BR-008; UC-001; F6
Learner selection	An adult's choice of the active learner on a device. Learner selection = Learner profile.	Data flow	One learner profile of the account in use.	ASM-008; SAF-RAY-007
Lesson choice	The lesson and starting stage that an adult chooses for a learner; starting the lesson begins a new lesson record. Lesson choice = Lesson + Starting stage.	Data flow	Made only by an adult; the starting stage is a stage of the chosen lesson.	BR-003; CON-007; FR-LRN-016; UC-002 to UC-004; F7; SAF-RAY-001
Progress summary	A view of one learner's practice over time. Progress summary = Learner profile + {Lesson record} + counts of attempts made with a guide + counts of attempts made without a guide + {Recurring difficulty} + Latest attempt time + Practice-only statement.	Data flow	Derived only from one learner's records, with no data of its own; no overall score; the two kinds of attempt are never added together.	G.6; BR-006; UC-009; FR-PRG-001 to FR-PRG-012; F11
Refinement off request	A child's request, in the creative space, to turn refinement off.	Data flow	Made in the creative space only; never reaches a guided lesson.	BR-005; UC-007; F2; FR-CRT-030

Name	Meaning	Type	Unit, domain or constraint	Used by
Refinement result	The adjusted strokes returned for one refinement by the service that refines drawings (LC-004). Refinement result = {Path}, one for each stroke of the drawing.	Data flow	Applied whole only if it meets FR-CRT-011 and FR-CRT-012; otherwise the drawing is left unchanged.	LC-004; FR-CRT-020, FR-CRT-021
Restore request	A request to restore the original strokes of a creative drawing.	Data flow	From the child for the open drawing, or from the adult who holds the account for any saved drawing.	BR-005; F2, F9; FR-CRT-017 to FR-CRT-019
Settings change	An adult's change to a learner's settings. Settings change = (Adult-set values) + (Audio setting) + (Accessibility settings) + (Refinement setting).	Data flow	From the adult who holds the account only; values outside their ranges are refused.	BR-008; UC-010; F9; SAF-RAY-008
Stage guide	What Squigglr shows the child at a stage. Stage guide = [Demonstration | Full outline | Endpoints].	Data flow	Set by the stage; none at the stage with no guide.	G.1; F3
Strokes drawn	The strokes a child draws for an attempt or in the creative space. Strokes drawn = {Stroke}.	Data flow	Positions and order of points as the browser reports them.	CON-003; ASM-004; FR-LRN-007; SAF-AHM-001; F1
Suggestion response	An adult's answer to a go-back suggestion. Suggestion response = [Accept | Decline].	Data flow	From an adult only.	BR-003; UC-005; F8; FR-LRN-018; FR-REC-010; SAF-RAY-001

Name	Meaning	Type	Unit, domain or constraint	Used by
Summary view choice	What an adult asks a progress summary to show. Summary view choice = {Lesson} + Date range + {Recorded value}.	Data flow	Changes only what is shown, never the records; until the adult chooses, the summary covers all records.	ASM-006; LC-007; FR-PRG-007

8 The Scope of the Product

This section describes Squigglr's own part of the work in Section 6: the inputs and outputs that cross the product's boundary, the product use cases, and the full flow of each use case.

8.1 Product Boundary

Table 14 lists every input Squigglr accepts and Table 15 every output it produces, covering the inputs and outputs in the Problem Statement and Goals (SRC-001) and the data flows in Table 11. Each row names the use cases (Table 16) and requirements that handle it. For each input, the last column states what Squigglr does when the input is invalid or missing, so that every possible input has a defined response.

Table 14: Product inputs

Input	From	Use cases	Requirements	If invalid or missing
A child's profile details and adult-set values	Supporting adult	UC-001, UC-010	FR-LRN-017, SAF-RAY-008	A value outside its range, an `ATTEMPT_LIMIT` not larger than `DIFFICULTY_STREAK`, or an `ATTEMPT_LIMIT_MAX` smaller than `ATTEMPT_LIMIT` is not saved. A value the adult does not choose is set to the middle of its range (SAF-RAY-008).

Input	From	Use cases	Requirements	If invalid or missing
Lesson choice, starting stage and the start of the lesson	Supporting adult	UC-002, UC-004	FR-LRN-016, FR-LRN-018, SAF-RAY-001	Without an adult's choice no lesson starts; a child cannot choose or start one by tapping icons (SAF-RAY-001). On an unsupported device or browser, the adult sees a notice that results may be unreliable (SAF-RAY-002).
Stylus strokes for an attempt in a guided lesson	Child	UC-003, UC-008	FR-LRN-007, FR-LRN-008	An interrupted or unfinished attempt is not passed and does not count toward the attempt limits (FR-LRN-009). Missing, invalid or ambiguous input is not passed and is not counted (FR-LRN-010).
Strokes and edits in the creative space: draw, move, resize, recolour, erase, undo	Child	UC-006	FR-CRT-001–FR-CRT-008	To be stated with the FR-CRT requirements.
Request to turn refinement off or restore the original drawing	Child or supporting adult	UC-007, UC-010	FR-CRT-017–FR-CRT-019, FR-CRT-030; BR-005	A failed refinement leaves the drawing unchanged (G.4).
Decision on a suggestion to go back a stage	Supporting adult	UC-005	FR-LRN-018, FR-LRN-019, SAF-RAY-001	A declined suggestion changes nothing. An unanswered suggestion stays pending without blocking practice until an adult decides (SAF-RAY-001; BR-003); if a pass or the end of the lesson clears it first, it is recorded as ignored. Only an adult can accept it.

Input	From	Use cases	Requirements	If invalid or missing
Changes to a child's settings: accessibility, audio, whether refinement is on	Supporting adult	UC-010	NFR-UX-010, NFR-UX-011	A setting changes only for the child it was changed for (NFR-UX-010, NFR-UX-011); changes need the adult check (SAF-RAY-008).
Request to view a child's history, attempts or progress summaries	Supporting adult	UC-009, UC-010	BR-001; NFR-SEC-001	A request from anyone other than the adult who holds the account is refused, and nothing is shown (BR-001).
Lesson catalogue, ranges of the adult-set values and fixed values	Squigglr team	–	FR-LRN-001, BR-009	Not entered by users; set and checked by the team (BR-009).
Any of the above while the network connection is lost	Child or supporting adult	All	SAF-RAY-003	Each completed attempt is kept on the device until the server confirms it is saved, and the adult is told which attempts could not be saved (SAF-RAY-003).

Table 15: Product outputs

Output	To	Use cases	Requirements
Demonstration and guidance for each stage: full outline, endpoints only, or no guide	Child	UC-003	FR-LRN-002–FR-LRN-006
The active learner, shown in a form a child who cannot read can recognize	Child	UC-003	SAF-RAY-007
Feedback on an attempt, its result, and moving on to the next stage	Child	UC-003	FR-LRN-011–FR-LRN-015, FR-LRN-023

Output	To	Use cases	Requirements
Suggestion to go back a stage, or an extension while the child improves	Supporting adult	UC-005	FR-LRN-019, FR-LRN-020
End of a lesson at `ATTEMPT_LIMIT` or `ATTEMPT_LIMIT_MAX`, with progress saved	Child; supporting adult	UC-003	FR-LRN-021, SAF-RAY-005, SAF-RAY-006
Refined, editable drawing whose original can be restored	Child	UC-006, UC-007	FR-CRT-010 FR-CRT-013 FR-CRT-015 FR-CRT-022
Attempt records: stage, difficulties, corrections and outcome	Supporting adult	UC-008, UC-009	FR-REC-001–FR-REC-008; SAF-RAY-004, SAF-RAY-007
Progress summaries linked to the attempts behind them, with no diagnosis or judgement of the child	Supporting adult	UC-009	FR-PRG-001–FR-PRG-012; FR-LRN-022, SAF-RAY-004
The adult-set values in use, shown with the child's records	Supporting adult	UC-010	SAF-RAY-008
Notices to the adult: unsupported device or browser, attempts not saved	Supporting adult	All	SAF-RAY-002, SAF-RAY-003

8.2 Product Use Case Table

Table 16 lists the ten product use cases, UC-001 to UC-010. Each row names the actor who starts or carries out the use case, the trigger that starts it, what the actor wants when it ends (the user goal) and the project goals it serves (Table 2). Squigglr is named as the actor where the product starts the use case itself. The identifiers are fixed and never renumbered; requirements trace to them, and Section 8.3 gives each use case's full flows. Names such as `STAGE_PASS` are named constants defined in the constants register (Table 10).

Table 16: Product use cases

ID	Actor: use case	Trigger	User goal	Goals
UC-001	*Supporting adult:* create a child's profile	The adult adds a child to Squigglr.	The child's profile exists, and every adult-set value is within its range, or at the middle of the range if the adult did not choose one.	G.3
UC-002	*Supporting adult:* choose a lesson	The adult chooses a lesson for a child.	The chosen lesson is ready for the adult to start.	G.1, G.3
UC-003	*Child:* practise a guided lesson	An adult starts a lesson for the child.	The child moves to the next stage after an attempt meets `STAGE_PASS`, or retries; the lesson ends when the child finishes it or reaches `ATTEMPT_LIMIT`, or `ATTEMPT_LIMIT_MAX` when extended (BR-002), and the furthest stage reached is saved.	G.1, G.2, G.3
UC-004	*Supporting adult:* start the child at a later stage	The adult judges that the child needs less help in a lesson.	The lesson starts at the stage the adult chose.	G.3
UC-005	*Supporting adult:* respond to a suggestion to go back a stage	Squigglr suggests an earlier stage after `DIFFICULTY_STREAK` failed attempts in a row.	The stage changes only if the adult accepts; a declined or ignored suggestion changes nothing.	G.3
UC-006	*Child:* create and edit artwork	The child opens the creative space.	The child's drawing is saved and stays editable.	G.4
UC-007	*Squigglr:* refine artwork and restore the original	The child finishes a drawing in the creative space.	The drawing is refined within `REFINE_TOLERANCE` and stays editable; refinement can be turned off; the original strokes can be restored; a failed refinement leaves the drawing unchanged.	G.4

ID	Actor: use case	Trigger	User goal	Goals
UC-008	*Squigglr:* record an attempt	The child ends an attempt at a lesson stage.	The record keeps the stage, every predefined difficulty observed, later corrections and the outcome.	G.5
UC-009	*Supporting adult:* review progress	The adult opens a child's progress summary.	The adult sees the child's progress over time, with attempts made without a guide separated from attempts made with one, and can open the attempts behind it.	G.6
UC-010	*Supporting adult:* review history and change settings	The adult opens a child's history or settings.	Only an authorised adult sees the history; the accessibility and audio settings, whether refinement is on, and the adult-set values are changed, and each adult-set value stays within its range.	G.3, G.6

8.3 Individual Product Use Cases (PUC's)

This subsection gives the full flow of each product use case in Table 16. Each use case lists its actor, preconditions, trigger, main flow, alternative flows, failure flows and postconditions, and the requirements and sources it depends on. Alternative flows are other normal paths; failure flows handle the invalid or missing inputs in Table 14. Both are numbered by the main flow step where they branch, so "3a" branches at step 3. The lesson behaviour in UC-003 and UC-005 is specified formally in Section 9.2.

8.3.1 UC-001: Create a child's profile

Actor: Supporting adult who holds the account.
Preconditions: The adult is signed in to their own account (BR-001).
Trigger: The adult adds a child to Squigglr.
Main flow:

1. The adult, or the child with the adult, chooses an avatar; Squigglr generates the profile's unique identifier (BR-001).
2. Squigglr asks for each adult-set value, shows its range and, for `STAGE_PASS`, shows example drawings of the minimum and maximum (BR-008).
3. The adult chooses each value.
4. Squigglr saves the profile and the values.

Alternative flows:

3a. The adult leaves a value unset. Squigglr sets it to the middle of its range (SAF-RAY-008).

Failure flows:

3b. A value is outside its range, `ATTEMPT_LIMIT` is not larger than `DIFFICULTY_STREAK`, or `ATTEMPT_LIMIT_MAX` is smaller than `ATTEMPT_LIMIT`. Squigglr does not save it and returns to step 3 (FR-LRN-017, SAF-RAY-008).

4a. The profile cannot be saved. Squigglr tells the adult, and no partial profile is created.

Postconditions: The profile exists, and every adult-set value is within its range or at its midpoint.

Requirements: FR-LRN-017; SAF-RAY-008; BR-001, BR-008.

Sources: G.3; SRC-008.

8.3.2 UC-002: Choose a lesson

Actor: Supporting adult.

Preconditions: The child has a profile (UC-001).

Trigger: The adult chooses a lesson for a child.

Main flow:

1. The adult selects the child, and Squigglr shows that child as the active learner (SAF-RAY-007).
2. The adult chooses a lesson from the catalogue (FR-LRN-001).
3. Squigglr checks the device and browser against `SUPPORTED_DEVICES` (SAF-RAY-002).
4. The adult completes the adult check and starts the lesson at its first stage (SAF-RAY-001, FR-LRN-016). The lesson continues in UC-003.

Alternative flows:

4a. The adult chooses a later starting stage (UC-004).

Failure flows:

3a. The device or browser is not supported. The adult sees a notice that results may be unreliable and decides whether to continue (SAF-RAY-002).

4b. A child tries to choose or start a lesson by tapping icons. The adult check is not completed, and no lesson starts (SAF-RAY-001).

Postconditions: The chosen lesson is running for the selected child.

Requirements: FR-LRN-001, FR-LRN-016, FR-LRN-018; SAF-RAY-001, SAF-RAY-002, SAF-RAY-007.

Sources: G.1, G.3; SRC-005.01.

8.3.3 UC-003: Practise a guided lesson

Actor: Child.

Preconditions: An adult has started the lesson for the child (UC-002 or UC-004).

Trigger: An adult starts a lesson for the child.

Main flow:

1. Squigglr shows the demonstration, which is not an attempt (FR-LRN-003), then moves to the starting stage.
2. Squigglr shows the stage's guidance: the full outline, the endpoints only, or no guide (FR-LRN-004–FR-LRN-006).
3. The child draws one or more strokes. Lifting the stylus does not end the attempt (FR-LRN-007, FR-LRN-008).
4. When the attempt is finalized, Squigglr classifies it as complete or incomplete and as assisted or independent, and passes it only if it is complete and meets `STAGE_PASS` (FR-LRN-011–FR-LRN-013).
5. Squigglr shows pre-approved feedback about the drawing, never about the child (FR-LRN-014, FR-LRN-022), and records the attempt (UC-008).
6. On a pass, Squigglr moves to the next stage and repeats from step 2. A pass at independent drawing completes the lesson (FR-LRN-023).

Alternative flows:

6a. The attempt does not pass and the stage's limit is not reached. The child retries from step 3 (FR-LRN-015).

6b. The child has `DIFFICULTY_STREAK` non-passing attempts in a row and is not improving. Squigglr offers the adult a suggestion to go back a stage (UC-005); practice continues meanwhile (FR-LRN-019, SAF-RAY-001).

6c. The unsuccessful attempts reach `ATTEMPT_LIMIT` while the child is improving. Squigglr allows up to `ATTEMPT_EXTENSION` more attempts, never more than `ATTEMPT_LIMIT_MAX` in total, and the child continues from step 3 (FR-LRN-020).

Failure flows:

3a. The attempt is interrupted or unfinished. It is not passed and does not count toward the limits (FR-LRN-009).

3b. The input is missing, invalid, too ambiguous to evaluate, or cannot be mapped reliably to the drawing surface. It is not passed and does not count (FR-LRN-010, SAF-AHM-002).

3c. The same stylus event is reported more than once. It counts once (SAF-AHM-001).

6d. The unsuccessful attempts reach `ATTEMPT_LIMIT` without an extension, an extension is used up, or the total reaches `ATTEMPT_LIMIT_MAX`. The lesson ends with no action from the child, and the progress so far is saved (FR-LRN-021, SAF-RAY-005, SAF-RAY-006).

*a. At any step, the connection is lost. Each completed attempt is kept on the device until the server confirms it is saved, and the adult is told which attempts could not be saved (SAF-RAY-003).

*b. At any step, there is no input for `LESSON_IDLE_PROMPT`. Squigglr shows a check-in the child can understand without reading; any touch dismisses it. If there is still no input at `LESSON_IDLE_END`, the lesson ends, the progress so far is saved, and the record shows that it ended for inactivity (SAF-RAY-009).

Postconditions: Every evaluable attempt is recorded, the furthest stage reached is saved, and no stage moved back without an adult's acceptance.

Requirements: FR-LRN-002–FR-LRN-015, FR-LRN-019–FR-LRN-023; SAF-RAY-003, SAF-RAY-005, SAF-RAY-006, SAF-RAY-007; SAF-AHM-001, SAF-AHM-002.

Sources: G.1, G.2, G.3; SRC-005.01.

8.3.4 UC-004: Start the child at a later stage

Actor: Supporting adult.

Preconditions: The child has a profile (UC-001).

Trigger: The adult judges that the child needs less help in a lesson, from knowing the child or from the records (UC-009).

Main flow:

1. The adult selects the child and the lesson, as in UC-002 steps 1 to 3.
2. The adult chooses a later starting stage.
3. The adult completes the adult check (SAF-RAY-001).
4. Squigglr starts the lesson at that stage, with the stage's counts at zero (FR-LRN-016; Section 9.2). The lesson continues in UC-003.

Alternative flows: None.
Failure flows:
3a. A child tries to change the starting stage. The adult check is not completed, and the stage does not change (SAF-RAY-001, FR-LRN-018).
Postconditions: The lesson starts at the stage the adult chose.
Requirements: FR-LRN-016, FR-LRN-018; SAF-RAY-001; BR-003.
Sources: G.3; SRC-005.01.

8.3.5 UC-005: Respond to a suggestion to go back a stage

Actor: Supporting adult.
Preconditions: A lesson is running (UC-003), and the child has `DIFFICULTY_STREAK` non-passing attempts in a row at an eligible stage without improving.
Trigger: Squigglr suggests going back a stage (FR-LRN-019).
Main flow:
1. Squigglr shows the suggestion and keeps it pending without blocking practice (SAF-RAY-001).
2. The adult completes the adult check and accepts.
3. Squigglr moves the lesson back to the earlier stage the adult chose, with the stage's counts at zero (Section 9.2).

Alternative flows:
2a. The adult declines. Nothing changes, and a new suggestion needs a new full streak (FR-LRN-019).
2b. The adult does not respond. The suggestion stays pending, nothing changes, and no second suggestion is offered while it is pending (SAF-RAY-001, BR-003). If the child passes the stage or the lesson ends first, the suggestion is cleared and recorded as ignored (FR-REC-010).
Failure flows:
2c. A child tries to accept the suggestion. The adult check is not completed, and nothing changes (SAF-RAY-001).
Postconditions: The stage has changed only if an adult accepted.
Requirements: FR-LRN-018–FR-LRN-020; SAF-RAY-001; BR-003.
Sources: G.3; SRC-005.01.

8.3.6 UC-006: Create and edit artwork

Actor: Child.
Preconditions: The child is the active learner (SAF-RAY-007).
Trigger: The child opens the creative space.
Main flow:

1. The child draws strokes on the canvas.
2. The child edits the drawing: moves, resizes, recolours, erases or undoes parts of it (G.4).
3. Squigglr saves the drawing, which stays editable. Refinement runs as in UC-007.

Alternative flows: None.
Failure flows:
2a. An edit cannot be applied. The drawing stays as it was.
3a. The drawing cannot be saved. The child keeps drawing, and the adult is told that it was not saved.
Postconditions: The drawing is saved and stays editable.
Requirements: FR-CRT-001–FR-CRT-008, FR-CRT-010, FR-CRT-013, FR-CRT-015, FR-CRT-020, FR-CRT-023.
Sources: G.4; SRC-009.01.

8.3.7 UC-007: Refine artwork and restore the original

Actor: Squigglr; the child or the supporting adult restores the original or turns refinement off.
Preconditions: A drawing exists in the creative space (UC-006), and refinement is on.
Trigger: The child finishes a drawing in the creative space.
Main flow:

1. Squigglr refines the child's own strokes, moving no point more than `REFINE_TOLERANCE` (BR-005, G.4).
2. The refined drawing stays editable (UC-006).

Alternative flows:
2a. The child or the adult restores the original. Squigglr returns the original strokes exactly (G.4, BR-005).
2b. The child or the adult turns refinement off. Later drawings are not refined (SRC-009.01).
Failure flows:

1a. Refinement fails. The drawing is left unchanged (G.4).
Postconditions: The drawing is refined or unchanged, stays editable, and its original can be restored. Refinement never reaches a guided lesson record (BR-004, BR-005).
Requirements: FR-CRT-010–FR-CRT-013, FR-CRT-017–FR-CRT-019, FR-CRT-020, FR-CRT-022, FR-CRT-027, FR-CRT-030; BR-004, BR-005.
Sources: G.4; SRC-009.01.

8.3.8 UC-008: Record an attempt

Actor: Squigglr.
Preconditions: A lesson is running for an identified learner (UC-003, SAF-RAY-007).
Trigger: The child ends an attempt at a lesson stage.
Main flow:

1. Squigglr records the child's actual strokes, never a corrected shape (BR-004), with the stage, the guidance and completeness classifications, every predefined difficulty observed, later corrections and the outcome (G.5; FR-LRN-011, FR-LRN-012).
2. Squigglr saves the record to the active learner only (SAF-RAY-007).
3. The server confirms the record is saved.

Alternative flows:
1a. The attempt ends the lesson at a limit, or an extension is granted. Squigglr also records that event (SAF-RAY-006).
Failure flows:
1b. The attempt is interrupted or its input is invalid. It is not recorded as passed and does not count (FR-LRN-009, FR-LRN-010).
1c. A stylus event is reported more than once. It is recorded once (SAF-AHM-001).
3a. The server does not confirm. The record stays on the device until it is saved, and the adult is told which attempts could not be saved (SAF-RAY-003).
Postconditions: The record matches what the child drew and holds no overall score, diagnosis or comparison (SAF-RAY-004).
Requirements: FR-REC-001, FR-REC-005–FR-REC-008; FR-LRN-009–FR-LRN-012; SAF-RAY-003, SAF-RAY-004, SAF-RAY-006, SAF-RAY-007; SAF-AHM-001.
Sources: G.5; SRC-001.

8.3.9 UC-009: Review progress

Actor: Supporting adult who holds the account.
Preconditions: The adult is signed in to their own account.
Trigger: The adult opens a child's progress summary.
Main flow:

1. Squigglr shows the child's progress over time, with attempts made without a guide separated from attempts made with one (G.6), the adult-set values in use (SAF-RAY-008), and a statement that the summary describes practice only and is not an assessment (SAF-RAY-004).
2. The adult opens an attempt behind the summary.
3. Squigglr shows that attempt's record (UC-008).

Alternative flows:
1a. The child has no practice yet. Squigglr says so.
1b. Lessons ended at a limit or were extended. Squigglr shows how many times for each lesson (SAF-RAY-006).
Failure flows:
1c. The request comes from anyone other than the account holder. It is refused, and nothing is shown (BR-001).
1d. Some attempts are not saved yet. The adult is told which (SAF-RAY-003).
Postconditions: The adult has seen the child's progress; nothing is changed.
Requirements: FR-PRG-008; FR-LRN-022; SAF-RAY-003, SAF-RAY-004, SAF-RAY-006, SAF-RAY-008; BR-001, BR-006.
Sources: G.6; SRC-005.02.

8.3.10 UC-010: Review history and change settings

Actor: Supporting adult who holds the account.
Preconditions: The adult is signed in to their own account, and the child has a profile (UC-001).
Trigger: The adult opens a child's history or settings.
Main flow:

1. Squigglr shows the child's history and current settings.
2. The adult changes the accessibility or audio settings, whether refinement is on, or the adult-set values.
3. The adult completes the adult check (SAF-RAY-008).
4. Squigglr saves the changes.

Alternative flows:
2a. The adult restores the original of a creative drawing (UC-007).
Failure flows:
1a. The request comes from anyone other than the account holder. It is refused, and nothing is shown (BR-001).
3a. A child tries to change a setting. The adult check is not completed, and nothing changes (SAF-RAY-008).
4a. An adult-set value is outside its range or inconsistent with the others. It is not saved (FR-LRN-017, SAF-RAY-008).
Postconditions: The changes are saved, and every adult-set value is within its range.
Requirements: FR-LRN-017; SAF-RAY-008; NFR-UX-010, NFR-UX-011; NFR-SEC-001; NFR-SEC-003; NFR-SEC-005; BR-001, BR-008.
Sources: G.3, G.6; SRC-008, SRC-009.01.

9 Functional Requirements

Functional requirements are grouped by externally observable product behaviour. Each requirement uses the identifier, fit criterion and traceability conventions defined in the Introduction.

9.1 Learning Activity Requirements

This subsection specifies the externally observable behaviour of guided lessons. It builds on G.1–G.3, CON-003, CON-007 and BR-002–BR-009, with source statements in Appendix 29. The core catalogue covers lines, shapes and simple drawings; letter tracing is deferred (ASM-010, LC-001). Delivery phase identifiers follow the conventions in the Introduction and the phase plan in Section 21.

Progression within lessons and adult control. A supporting adult selects each lesson and its starting stage and authorizes skips or backward changes (G.3, CON-007). Within the selected lesson, a passing drawing attempt automatically advances to the next guidance stage (G.2, BR-002). Passing the independent stage completes the current lesson; Squigglr does not automatically select another lesson.

Lesson content and guidance. FR-LRN-001 — Approved lesson catalogue

Requirement: Squigglr shall offer guided lessons for the approved catalogue of lines, basic shapes and simple drawings, excluding letter tracing activities from the core catalogue.
Rationale: These are the agreed categories for structured practice; letters are deferred (G.1, ASM-010). The lesson catalogue is maintained separately from the stage rules (LC-001).
Fit criterion: Inspect the lesson catalogue and verify that every approved catalogue entry is selectable, classified as a line, shape or simple drawing, and has the stages in FR-LRN-002; verify that no letter activity is required to complete the core catalogue.
Priority: Must
Phase: PH-1
Sources: G.1; SRC-001; SRC-009.04
Traceability: ASM-010; LC-001; UC-002

Table 17 lists the initial catalogue. The team chose it from the agreed categories; it can change (LC-001), and each lesson is stored as data (NFR-MAINT-001). The PoC (PH-1) offers the line, circle and square lessons, and the full catalogue is offered from PH-2. The required parts and endpoints define a complete attempt (Section 9.2).

Table 17: Initial lesson catalogue

Lesson	Category	Required parts and endpoints
Horizontal line	Line	One straight segment; its two endpoints.
Vertical line	Line	One straight segment; its two endpoints.
Diagonal line	Line	One straight segment; its two endpoints.
Circle	Basic shape	Four quarter arcs, which together close the circle.
Square	Basic shape	Four sides; four corners.
Triangle	Basic shape	Three sides; three corners.
House	Simple drawing	The four sides of the walls and the two sides of the roof; the two roof corners where the roof meets the walls, and its top.
Sun	Simple drawing	The four quarter arcs of the circle and eight short straight rays; the outer end of each ray.

FR-LRN-002 — Ordered guidance stages

Requirement: Squigglr shall make the following ordered guidance stages available for each guided lesson: demonstration, full outline tracing, endpoint-only guidance and independent drawing.
Rationale: The activity described to specialists removes guidance in this order (G.1–G.2).
Fit criterion: For every approved lesson, inspect all four stages in the specified order; confirm that the stage chosen by an adult is the stage presented when practice begins.
Priority: Must
Phase: PH-1
Sources: G.1–G.2; SRC-001; SRC-005
Traceability: BR-003; UC-003; UC-004

FR-LRN-003 — Demonstration

Requirement: Squigglr shall show an example of the intended drawing during a lesson's demonstration stage without treating that demonstration as a learner attempt.
Rationale: A demonstration provides an example before guided practice and must not inflate the learner's records.
Fit criterion: Start each approved lesson in its demonstration stage; verify that the intended drawing can be viewed and that watching it adds no learner attempt or passed stage record.
Priority: Must
Phase: PH-1
Sources: G.1; SRC-001; SRC-005
Traceability: BR-004; UC-003; UC-008

FR-LRN-004 — Full-outline guidance

Requirement: Squigglr shall display the lesson's complete reference outline during the full-outline tracing stage and accept stylus strokes drawn against it.
Rationale: Outline tracing is the first guided drawing stage in the approved progression.
Fit criterion: For each lesson, enter the full-outline stage, verify that the complete reference shape is visible, and draw an attempt using stylus input.
Priority: Must

Phase: PH-1
Sources: G.1; SRC-001; SRC-005
Traceability: CON-003; UC-003

FR-LRN-005 — Endpoint-only guidance

Requirement: Squigglr shall show the specified start and end points, without the full reference outline, during the endpoint-only stage and accept stylus strokes between those points.
Rationale: The middle drawing stage reduces visual assistance while retaining the destination cues.
Fit criterion: For each lesson, enter endpoint-only guidance; verify that the relevant endpoints are shown, the full outline is absent, and stylus drawing is accepted.
Priority: Must
Phase: PH-1
Sources: G.1–G.2; SRC-001; SRC-005
Traceability: CON-003; UC-003

FR-LRN-006 — Independent drawing

Requirement: Squigglr shall permit the drawing task to be attempted in the independent stage without displaying tracing outlines or endpoint guides.
Rationale: The final stage needs to be distinguishable from assisted practice (G.2, G.6).
Fit criterion: Enter the independent stage for each lesson; verify that neither an outline nor endpoint guides appear and that an attempt can still be made using the stylus.
Priority: Must
Phase: PH-1
Sources: G.2; SRC-001
Traceability: BR-004; UC-003; UC-008

Input, classification and feedback. FR-LRN-007 — Stylus stroke acquisition

Requirement: Squigglr shall accept stylus-drawn strokes as ordered sequences of input points for the active lesson attempt and preserve their order and stroke boundaries for subsequent classification.

Rationale: Stage checks and faithful attempt records require the original stroke structure (ASM-004, BR-004).
Fit criterion: Supply known one-stroke and multi-stroke stylus samples; verify that the captured point order, number of strokes and boundaries match the samples.
Priority: Must
Phase: PH-1
Sources: G.2, G.5; SRC-001; SRC-004
Traceability: CON-003; ASM-004; BR-004; UC-003; UC-008; H-AHM-001, H-AHM-002, H-AHM-003

FR-LRN-008 — Multi-stroke attempts

Requirement: Squigglr shall evaluate the collected strokes of a single attempt together rather than classify each stylus lift as a completed attempt.
Rationale: Lines and shapes can be composed of multiple strokes, and lifting a stylus alone does not establish that the task is finished.
Fit criterion: For a prepared two-stroke figure, lift the stylus after the first stroke and draw the second; confirm that one attempt is evaluated against the complete figure, not two independently completed attempts.
Priority: Must
Phase: PH-2
Sources: G.1–G.2; ASM-004; SRC-001
Traceability: FR-LRN-007; UC-003; UC-008; H-AHM-001, H-AHM-002

FR-LRN-009 — Interrupted attempt handling

Requirement: Squigglr shall distinguish an interrupted or unfinished drawing attempt from a completed drawing attempt, shall not record it as passed, and shall not consume an evaluable attempt from the stage limit.
Rationale: An input interruption or unfinished drawing is not evidence of a correctly completed task.
Fit criterion: Interrupt known partial attempts between and during strokes; verify that none is reported as passed, that the active stage does not advance, and that unfinished attempts are distinguishable from finalized attempts and that stage attempt counters are unchanged.
Priority: Must
Phase: PH-2
Sources: G.2, G.5; ASM-004; SRC-001

Traceability: FR-LRN-008; BR-004; UC-003; UC-008; H-AHM-001, H-AHM-006

FR-LRN-010 — Invalid and ambiguous input

Requirement: Squigglr shall withhold passing status from an attempt whose required stylus input is missing, invalid or too ambiguous to evaluate against the stage criteria, and shall not count such input toward the stage's evaluable attempt or unsuccessful attempt limits.
Rationale: Unreliable acquisition must not silently turn a malformed drawing into a successful result (ASM-004).
Fit criterion: Submit empty input, input with missing coordinate data and data for which the intended stroke order cannot be established; confirm that none passes, that a non-success status is available, and that the stage's attempt counters are unchanged.
Priority: Must
Phase: PH-1
Sources: G.2; ASM-004; BR-004; SRC-001
Traceability: FR-LRN-007; UC-003; H-AHM-001; H-AHM-003

FR-LRN-011 — Completeness classification

Requirement: Squigglr shall classify every evaluable drawing attempt as complete or incomplete relative to the selected lesson and guidance stage.
Rationale: Stage success must distinguish an unfinished line or missing side from a completed drawing (G.2).
Fit criterion: Evaluate adult-made or prepared examples with every required component present and with a deliberately missing side, segment or required endpoint; verify complete and incomplete classifications match the known examples.
Priority: Must
Phase: PH-1
Sources: G.2; BR-002; SRC-001
Traceability: FR-LRN-013; UC-003; UC-008; H-AHM-004, H-AHM-005

FR-LRN-012 — Guidance classification

Requirement: Squigglr shall classify each evaluated attempt as assisted when performed with outline or endpoint guidance and independent when performed without those guides.

Rationale: Adults need to distinguish guided practice from practice without a guide (G.6).
Fit criterion: Perform an evaluable attempt in each drawing stage; verify that outline and endpoint attempts are marked assisted and no-guide attempts are marked independent.
Priority: Must
Phase: PH-2
Sources: G.2, G.6; SRC-001
Traceability: FR-LRN-004–FR-LRN-006; UC-008; UC-009; H-MOU-005

FR-LRN-013 — Pass classification

Requirement: Squigglr shall report an evaluable attempt as passing its stage only when the drawing is classified as complete and its result meets the current learner's `STAGE_PASS` criterion.
Rationale: A pass evaluates the submitted drawing against defined criteria, not the child's ability (G.2, BR-006).
Fit criterion: For prepared complete and incomplete drawings whose results lie below, at and above the adult-selected threshold, verify that only complete drawings meeting the threshold pass.
Priority: Must
Phase: PH-1
Sources: G.2; BR-002; BR-006; SRC-008
Traceability: FR-LRN-010–FR-LRN-011; UC-003; H-AHM-004, H-AHM-005

FR-LRN-014 — Predefined difficulty feedback

Requirement: Squigglr shall present only pre-approved, stage-relevant feedback observations for evaluated attempts, including applicable missed tracing portions, incomplete lines or shapes and incorrect endpoints.
Rationale: Feedback must describe observable drawing difficulties and not generate personalized AI advice (BR-007).
Fit criterion: For each defined feedback category, evaluate a prepared matching drawing and verify that the corresponding approved observation appears; verify that no free-generated advice or diagnosis is shown.
Priority: Must
Phase: PH-2
Sources: G.1–G.2; BR-007; SRC-001; SRC-009.02
Traceability: FR-LRN-011; UC-003; H-AHM-005; H-MOU-004

FR-LRN-015 — Retry after non-passing attempts

Requirement: Squigglr shall allow the learner to retry a drawing stage following an attempt that does not pass, provided the stage has not reached its applicable attempt limit.
Rationale: G.1 requires opportunities to practise again without treating failure as a permanent stage result.
Fit criterion: Present a non-passing, evaluable sample before the limit and verify that another attempt can be started at the same drawing stage; repeat at the limit and verify that the lesson-end rule is followed instead.
Priority: Must
Phase: PH-1
Sources: G.1–G.3; BR-002; SRC-001
Traceability: FR-LRN-021; UC-003

Adult control, suggestions and stopping rules. FR-LRN-016 — Adult lesson and starting stage selection

Requirement: Squigglr shall use the lesson and starting guidance stage selected by the supporting adult for the learner's session, including an adult-selected later starting stage.
Rationale: The adult controls initial level selection rather than the product assigning readiness (G.3, CON-007).
Fit criterion: Select each available starting stage as an adult, launch the lesson, and verify that the selected stage is used; check that a child-only practice action, and an attempt to choose a lesson or starting stage without passing the adult check of SAF-RAY-001, does not change the configured lesson or starting stage.
Priority: Must
Phase: PH-1
Sources: G.3; CON-007; SRC-005.01; SRC-006
Traceability: BR-003; UC-002; UC-004; H-AHM-007

FR-LRN-017 — Adult-selected practice parameters

Requirement: Squigglr shall apply the learner's adult-selected values for `STAGE_PASS`, `DIFFICULTY_STREAK`, `ATTEMPT_LIMIT`, `IMPROVE_WINDOW`, `ATTEMPT_EXTENSION` and `ATTEMPT_LIMIT_MAX`, using the documented midpoint default for any unset value and rejecting out-of-range or inconsistent selections.

Rationale: BR-008 requires learner-specific settings and rejects contradictory limit relationships.
Fit criterion: Test every setting at the minimum, default and maximum values in Table 10, plus one value outside each bound; verify that selected values are used, defaults are applied correctly, and invalid combinations are rejected.
Priority: Must
Phase: PH-2
Sources: G.2–G.3; BR-008; SRC-008
Traceability: OI-05; UC-001; UC-010; H-AHM-007

FR-LRN-018 — Adult authorization for discretionary changes

Requirement: Squigglr shall require explicit authorization from the supporting adult before changing to another lesson, skipping a guidance stage or moving backward to an earlier stage; routine pass-based advancement to the next stage of the current lesson requires no additional adult action.
Rationale: Adults retain control of lesson selection and nonroutine guidance changes (G.3, CON-007), without interrupting the agreed forward progression on passing attempts (G.2, BR-002).
Fit criterion: Try selecting a new lesson, skipping ahead and applying a return suggestion without adult authorization, then repeat with authorization; verify that only authorized discretionary changes take effect. Verify independently that a passing attempt advances within the selected lesson without an additional adult action (FR-LRN-023).
Priority: Must
Phase: PH-2
Sources: G.2–G.3; CON-007; BR-002–003; SRC-005.01
Traceability: OI-04 (TA confirmation); FR-LRN-023; UC-004; UC-005; H-AHM-006

FR-LRN-019 — Difficulty suggestion

Requirement: Squigglr shall offer, without applying, an optional suggestion to return to an earlier drawing stage when the learner reaches `DIFFICULTY_STREAK` consecutive non-passing evaluable attempts at an eligible stage and the improvement rule does not suppress suggestions. When an adult declines the suggestion, another suggestion shall require a new full failure streak. A suggestion that no adult answers shall stay pending, without blocking practice, and no second suggestion shall be offered while it is pending.

Rationale: A supporting adult should decide whether earlier guidance is appropriate (G.3, BR-003).
Fit criterion: Replay known failing sequences immediately below and at the selected streak threshold at both eligible stages, including sequences with fewer than twice the configured `IMPROVE_WINDOW` results and sequences with qualifying improvement; verify that insufficient history does not delay the first suggestion, that qualifying improvement suppresses it, that a declined suggestion reappears only after another full failure streak, and that while a suggestion is unanswered, further failed attempts are counted but no second suggestion is offered.
Priority: Must
Phase: PH-2
Sources: G.3; BR-003; SRC-005.01; SRC-008
Traceability: FR-LRN-018; FR-LRN-020; UC-005; H-AHM-006, H-AHM-007

FR-LRN-020 — Improvement-based extension

Requirement: Squigglr shall suppress a difficulty suggestion while the learner qualifies as improving, and, when the unsuccessful-attempt count first reaches `ATTEMPT_LIMIT`, grant up to `ATTEMPT_EXTENSION` additional evaluable stage attempts if the average result of the most recent `IMPROVE_WINDOW` attempts exceeds the average of the preceding equally sized window by at least `IMPROVE_MARGIN`, subject to `ATTEMPT_LIMIT_MAX`. The extension shall be granted at most once per stage and expire after the allotted additional evaluable attempts, even if results continue to improve.
Rationale: The agreed improvement rule reduces premature interruption while retaining a hard cap (G.3, BR-002).
Fit criterion: Evaluate prepared improving and non-improving sequences exactly at the margin boundary and at the applicable limit; verify extension and suggestion suppression only for qualifying improvement, exactly the permitted number of extension attempts without regranting, and no attempt beyond the maximum.
Priority: Must
Phase: PH-2
Sources: G.3; BR-002–003; SRC-001; SRC-008
Traceability: FR-LRN-019; FR-LRN-021; OI-05; UC-003; H-AHM-006

FR-LRN-021 — Stage attempt limit and lesson end

Requirement: Squigglr shall end the active lesson when its unsuccessful-

attempt count reaches `ATTEMPT_LIMIT` without a qualifying extension, or a granted extension is exhausted, or the allowed total-attempt count reaches `ATTEMPT_LIMIT_MAX`, and make the furthest stage reached available for the lesson record.
Rationale: The limits prevent indefinite repetition and preserve usable progress (G.3, G.5, BR-002).
Fit criterion: Replay sequences at the base limit and at the maximum, with and without qualifying improvement; verify that a granted extension expires after the allotted attempts, does not renew, and is capped by the maximum. Verify that the highest stage reached is included in the saved lesson outcome.
Priority: Must
Phase: PH-2
Sources: G.3, G.5; BR-002; SRC-001
Traceability: FR-LRN-015; FR-LRN-020; UC-003; UC-008; H-AHM-007

FR-LRN-022 — No learner evaluation or diagnosis

Requirement: Squigglr shall express lesson feedback and completion results as properties of the drawing attempt rather than as assessments, diagnoses or comparisons of children.
Rationale: The product is a practice tool, not an assessment instrument (BR-006).
Fit criterion: Inspect all guided lesson feedback and completion outcomes, using prepared attempts; verify that each refers to the task, stage or observed strokes and that none assigns a diagnosis, ability label or inter-child ranking.
Priority: Must
Phase: PH-1
Sources: G.2; BR-006; SRC-001; SRC-008
Traceability: FR-LRN-013–FR-LRN-014; UC-003; UC-009; H-MOU-002

FR-LRN-023 — Automatic within-lesson advancement

Requirement: Squigglr shall automatically advance from full-outline tracing to endpoint-only guidance, and from endpoint-only guidance to independent drawing, upon a passing attempt at the active stage; a passing independent drawing attempt completes the current lesson without starting a different lesson.
Rationale: G.2 and BR-002 require predictable reduction of guidance after success. Adult consent governs lesson selection, skips and backward moves, not normal pass-based progression within the selected lesson (CON-007).

Fit criterion: For a selected lesson, finalize passing attempts at its outline, endpoint and independent stages; verify the exact transitions T to E, E to I and I to completed, respectively, without an adult interaction. Verify that a non-passing attempt stays at its current stage, a different lesson is not started, and no stage is skipped automatically.
Priority: Must
Phase: PH-1
Sources: G.2–G.3; BR-002–003; CON-007; SRC-001
Traceability: FR-LRN-013; FR-LRN-016; FR-LRN-018; UC-003; UC-004; OI-04 (TA confirmation); H-AHM-004, H-MOU-004

Source-to-requirement traceability. Table 18 provides source-side links for the most important learning-activity decisions; each requirement also links back to its evidence in its *Sources* field. The full system-wide traceability matrix is maintained by the owner of Section 28.

Table 18: Learning-activity evidence and derived requirements

Source or rule	Related learning requirements
G.1, SRC-001	FR-LRN-001, FR-LRN-002, FR-LRN-003, FR-LRN-004, FR-LRN-014, FR-LRN-015
G.2, BR-002	FR-LRN-005, FR-LRN-006, FR-LRN-011, FR-LRN-013, FR-LRN-020, FR-LRN-021, FR-LRN-023
G.3, BR-003, CON-007	FR-LRN-016, FR-LRN-018, FR-LRN-019, FR-LRN-023; OI-04
BR-008, SRC-008	FR-LRN-013, FR-LRN-017, FR-LRN-019, FR-LRN-020
ASM-004, BR-004	FR-LRN-007, FR-LRN-008, FR-LRN-009, FR-LRN-010
BR-006–007	FR-LRN-014, FR-LRN-022
SRC-005.01, SRC-005.05	FR-LRN-016, FR-LRN-018, FR-LRN-019
SRC-009.02, SRC-009.04	FR-LRN-001, FR-LRN-014

9.2 Formal Specification

The most safety-relevant learning behaviour is formalized here as a *labelled transition system*: a set of states and named events with rules describing when an event changes the state. The model specifies externally observable outcomes rather than a particular user interface or scoring algorithm. It covers guidance levels, attempt classification, optional suggestions, automatic pass-based advancement within a lesson, and the unsuccessful-attempt stopping rule. Every variable used below has a type or domain; arrows indicate transitions *caused by the event above the arrow*, not automatic passage of time. This notation applies only to guided learning activities.

Domains and state. For each selected lesson ℓ and learner profile p, let

$$\begin{aligned} L &= \{D, T, E, I\}, & A &= \{T, E, I\}, \\ R &= \{\mathsf{complete}, \mathsf{incomplete}, \mathsf{invalid}\}, \\ M &= \{\mathsf{assisted}, \mathsf{independent}\}, \\ Q &= \{\mathsf{active}, \mathsf{suggested}, \mathsf{ended}\}. \end{aligned}$$

The levels D, T, E, I mean *demonstration*, full-outline *tracing*, *endpoint-only* and *independent* drawing, respectively. Only levels in A admit scored drawing attempts; D is a non-scored demonstration. A state is

$$\sigma = (\ell, p, l, q, n_f, n_a, k, H, s, z, x, f),$$

where $l \in L$ is the current guidance level of the adult-selected lesson, $q \in Q$ is the session status, $n_f \in \mathbb{N}_0$ counts non-passing evaluable attempts at this stage, $n_a \in \mathbb{N}_0$ counts finalized, evaluable attempts at this stage, $k \in \mathbb{N}_0$ is the consecutive non-passing streak, and $H = (r_1, \ldots, r_m)$ is the ordered list of stage result values. The flag $s \in \{0, 1\}$ indicates a pending suggestion; $z \in \{0, 1\}$ indicates whether the single stage extension has been granted; $x \in \mathbb{N}_0$ counts evaluable attempts consumed *after* that grant, with $0 \le x \le e_p$ and $x = 0$ when $z = 0$. Finally, $f \in L$ is the furthest stage reached in the lesson. On a stage change (either normal pass-based progression or an authorized adult adjustment), the stage-specific counts, streak, history, pending suggestion and extension state reset to zero or empty; f retains the furthest stage reached so far. *No child ability estimate* is part of this state (BR-006).

Drawing input and attempt classification. An attempt is a finite ordered list of stylus strokes $X = (x_1, \ldots, x_j)$, where every x_i is an ordered list of reported points. Lifting the stylus closes one x_i, not the attempt. The terminal event Finalize(X) denotes the end of input collection for an attempt. This abstract event does not prescribe a particular user-interface control. A stylus lift alone does not finalize the attempt; interruption before finalization cannot produce a passed result (FR-LRN-008–010).

Each lesson has a *figure*: the reference drawing shown as the outline at stage T, made of the *required parts* and *required endpoints* listed for the lesson in Table 17. Distances are measured on the drawing area and compared with `TRACE_TOL`. At stage I, where no guide shows where to draw, the attempt is first moved and uniformly scaled so that its bounding box is centred on the figure's and its longer side matches the figure's; position and size on the screen therefore do not affect the result. At T and E the attempt is compared where it was drawn.

Let $\mathrm{Valid}_{\ell,l}(X)$ mean that all input required to interpret the strokes is available, and let $\mathrm{Complete}_{\ell,l}(X)$ mean that every required part has at least `PART_COVER` of its length within `TRACE_TOL` of a drawn stroke, and every required endpoint has a drawn point within `TRACE_TOL` of it. For example, an attempt at the square lesson with one side not drawn is incomplete, however accurate the other three sides are.

The stage result $v_{\ell,l}(X)$ of an evaluable attempt is

$$v_{\ell,l}(X) = 100 \cdot \min(\textit{drawn},\ \textit{on figure}),$$

where *drawn* is the share of the figure's length within `TRACE_TOL` of a drawn stroke, and *on figure* is the share of the drawn length within `TRACE_TOL` of the figure. It measures the drawing against the figure, not the learner. A result of 80 means that at least 80% of the figure was drawn and at least 80% of what was drawn lies on the figure; a square with one side missing scores at most 75. Stage result values use the normalized 0–100 scale specified in Table 10. Define

$$C_{\ell,l}(X) = \begin{cases} \mathsf{invalid} & \neg\,\mathrm{Valid}_{\ell,l}(X), \\ \mathsf{complete} & \mathrm{Valid}_{\ell,l}(X) \wedge \mathrm{Complete}_{\ell,l}(X), \\ \mathsf{incomplete} & \text{otherwise.} \end{cases}$$

The guidance classification uses the level of the actual attempt:

$$G(l) = \begin{cases} \mathsf{assisted} & l \in \{T, E\}, \\ \mathsf{independent} & l = I. \end{cases}$$

So $v_{\ell,l}(X) \in [0, 100]$ is the normalized stage result of an evaluable drawing. Its threshold is $\theta_p = \texttt{STAGE_PASS}(p)$, taken from the adult's approved profile value (or midpoint default under BR-008). The pass predicate is

$$\mathrm{Pass}(X, l, p) \iff C_{\ell,l}(X) = \mathsf{complete} \wedge v_{\ell,l}(X) \geq \theta_p.$$

A missing, malformed or interrupted attempt never satisfies this predicate. Non-passing but evaluable attempts contribute to the unsuccessful-attempt count. Invalid or interrupted input is never interpreted as success and does not contribute to a stage's evaluated-attempt or unsuccessful-attempt counters. The presented feedback is selected from the fixed set of observations in BR-007 rather than generated advice.

Improvement and suggestion predicates. Use the constants defined in Section 4.2:

$$\begin{aligned} d_p &= \texttt{DIFFICULTY_STREAK}(p), & b_p &= \texttt{ATTEMPT_LIMIT}(p), \\ w_p &= \texttt{IMPROVE_WINDOW}(p), & e_p &= \texttt{ATTEMPT_EXTENSION}(p), \\ h_p &= \texttt{ATTEMPT_LIMIT_MAX}(p), & \delta &= \texttt{IMPROVE_MARGIN}. \end{aligned}$$

For $H = (r_1, \ldots, r_m)$, define the total improvement predicate

$$\mathrm{Improving}(H, p) = \begin{cases} \mathsf{false}, & m < 2w_p, \\ \left(\dfrac{1}{w_p} \displaystyle\sum_{i=m-w_p+1}^{m} r_i - \dfrac{1}{w_p} \sum_{i=m-2w_p+1}^{m-w_p} r_i \geq \delta \right), & m \geq 2w_p. \end{cases}$$

Insufficient history therefore does not count as improvement and does not delay a difficulty suggestion. For the default $w_p = 2$, the first comparison uses four evaluable attempts; a shorter history still allows the suggestion at the selected failure-streak threshold.

For an eligible return to a preceding guidance level, the suggestion trigger is

$$\mathrm{SuggestBack}(\sigma) \iff q = \mathsf{active} \wedge s = 0 \wedge l \in \{E, I\} \wedge k \geq d_p \wedge \neg\, \mathrm{Improving}(H, p).$$

The rule applies to drawing levels E and I; T has no preceding drawing level. A suggestion changes q to suggested and s to 1, but *does not change* l, and the child keeps practising while it is pending (SAF-RAY-001). An adult declining a suggestion clears it and resets the streak counter k to zero, preventing an immediate re-offer; a new suggestion requires another d_p consecutive failures. A suggestion that no adult answers stays pending: it changes nothing and, because SuggestBack needs $s = 0$, prevents a second suggestion. If a pass, an adult selection or the end of the lesson clears it before any answer, it is recorded as ignored (FR-REC-010).

State-transition rules. Table 19 gives the level-changing and stage-stopping rules. Here σ' is the state after the event, "unchanged" means the active lesson and level remain exactly as before, and AdultAccept is a verified adult action rather than any child drawing input. Likewise, AdultSelect is the authorized action selecting a lesson and its starting level. A granted extension sets $z = 1$ once, with $x = 0$; each subsequently finalized evaluable attempt increments x. The extension cannot be granted again at this stage even if later scores qualify as improvement. After finalization, a passing result takes priority over stop and suggestion rules. Otherwise, reaching the base failure limit first triggers either the single extension grant (if improving and still below the hard cap) or lesson end. After a grant, exhausting $x = e_p$ or reaching $n_a = h_p$ ends the lesson. A suggestion is offered only when none of these higher-priority rules ends the lesson. For pass-based forward transitions define $\text{Next}(T) = E$ and $\text{Next}(E) = I$; I has no successor within its lesson. Passing the last stage ends *that* lesson.

Table 19: Guided lesson state transitions and authorization

Event	**Guard**	**Required state change or output**
Adult selects lesson / starting level (AdultSelect)	Supporting adult is authorized	Set (ℓ, l) to adult selection, q = active; reset stage counters, history, suggestion and extension state; keep any previously persisted lesson record.

Event	**Guard**	**Required state change or output**
Begin practice after demonstration	$q = \mathsf{active}$, $l = D$; child continues after viewing the example	Set $l = T$ within the same lesson; initialize drawing-stage counters. This does not start another lesson or require new adult authorization.
Finalize evaluable attempt	$l \in A$, $q \in \{\mathsf{active}, \mathsf{suggested}\}$	Classify $C_{\ell,l}(X)$ and $G(l)$; report pass/fail using the stage threshold; increment n_a and, for non-pass, n_f, k. If $z = 1$, increment x for this additional evaluable attempt. On pass, record the result and apply the prescribed forward-stage or lesson-end transition below.
Interrupted or invalid input	Missing finalization or $C_{\ell,l}(X) = \mathsf{invalid}$	Do not pass or change level. Neither n_a nor n_f is incremented, and no successful stage transition is permitted.
Offer suggestion	SuggestBack(σ)	Set $s = 1$, $q = \mathsf{suggested}$; leave ℓ, l unchanged.
Adult declines suggestion	$s = 1$ and an authorized adult declines	Clear s, reset $k = 0$, return q to active; leave ℓ, l unchanged and retain the attempt history H and cumulative stage counts.
Suggestion not answered	$s = 1$ and no adult response	Keep $s = 1$ and $q = \mathsf{suggested}$; leave ℓ, l unchanged. Attempts continue to be finalized and counted. If a pass, AdultSelect or the end of the lesson clears s, record the suggestion as ignored.

Event	**Guard**	**Required state change or output**
Adult accepts suggestion	$s = 1$ and an authorized adult chooses a preceding level	Activate the chosen earlier level, clear s, reset stage counts; retain the lesson's furthest stage reached.
End at limit	The finalized attempt did not pass and either $z = 0 \land n_f \geq b_p \land (\neg\,\text{Improving}(H, p) \lor n_a \geq h_p)$, or $z = 1 \land (x \geq e_p \lor n_a \geq h_p)$	Set $q = \mathsf{ended}$; report the furthest stage reached for saving; no more attempts in the ended session. A pass takes priority over this stop condition.
Extend at base limit	$z = 0$, $n_f \geq b_p$, $\text{Improving}(H, p)$, $n_a < h_p$, and the finalized attempt did not pass	Set $z = 1$, $x = 0$; allow at most e_p additional evaluable attempts, subject to $n_a < h_p$. Do not regrant this extension or offer a conflicting return suggestion.
Passing a nonfinal drawing stage	$\text{Pass}(X, l, p)$ and $l \in \{T, E\}$	Save the passed attempt at its original stage; automatically set $l' = \text{Next}(l)$ without adult action. Reset stage-specific counters, clear any pending suggestion (recorded as ignored) and update f; leave ℓ unchanged.
Passing the final drawing stage	$\text{Pass}(X, I, p)$	Save the passed independent attempt, set $q' = \mathsf{ended}$ and $f' = I$; complete the current lesson without choosing another lesson.

Invariants and verification obligations. For every reachable session state and every adult- or learner-generated event, the model requires:

I1: $\ell' \neq \ell \Longrightarrow$ AdultSelect;

I2: suggestion declined or not answered $\Longrightarrow (\ell', l') = (\ell, l)$;

I3: $C_{\ell,l}(X) \neq \mathsf{complete} \Longrightarrow \neg\,\mathrm{Pass}(X, l, p)$;

I4: pass/feedback never substitutes refined strokes for actual guided strokes;

I5: $n_a \leq h_p \ \wedge\ 0 \leq x \leq e_p \wedge (z = 0 \Rightarrow x = 0)$;

I6: $\mathrm{Pass}(X, T, p) \Longrightarrow (\ell', l') = (\ell, E)$;

I7: $\mathrm{Pass}(X, E, p) \Longrightarrow (\ell', l') = (\ell, I)$;

I8: $\mathrm{Pass}(X, I, p) \Longrightarrow q' = \mathsf{ended} \wedge \ell' = \ell$.

Outside ordinary pass-based progression and the demonstration-to-practice transition, a change to the active stage requires an authorized adult selection or acceptance (CON-007). Invariants I1 and I6–I8 ensure that a new lesson is never selected automatically and that passing an attempt advances only within the current lesson. I2–I5 cover suggestions, invalid input, faithful records, bounded stage attempts and the one-time extension. These rules are verified using prepared adult-made attempt sequences (CON-005–006), including threshold cases with insufficient improvement history, declined suggestions, a full extension and the hard cap in Table 10.

9.3 Creative Drawing Requirements

This subsection gives the requirements for the creative space, where a child draws freely beside the structured practice of the guided lessons. Creative drawing gives room for self-expression, and keeping the drawing editable keeps the child in control of the work (G.4; SRC-001). The requirements make G.4 and BR-005 precise: when the child finishes a drawing, Squigglr refines it automatically and minimally by adjusting the child's own strokes, the refined drawing stays editable, and the child, or the adult if the child cannot, can restore the original or turn refinement off. Nothing in this subsection applies to guided lessons, whose records keep the child's actual strokes (BR-004). The requirements serve two product use cases (UCs): UC-006, in which the child creates and edits artwork, and UC-007, in which Squigglr refines artwork and the original can be restored. The team checks every fit criterion on prepared, adult-made drawings, never with children (CON-005, ASM-014).

The following terms have a fixed meaning in this subsection. A *creative drawing* is one drawing in the creative space, made of strokes; each stroke has a path, a colour and a width. A *part* is one or more strokes that the child selects. A *change* is one stroke drawn, one editing action, one applied refinement or one restore. The child *finishes* a drawing by marking it as done; there is no separate control for asking for refinement. A *refinement* is one automatic adjustment of the drawing's strokes, and its result is the *refined drawing*. The *original* of a refinement is the drawing exactly as it was immediately before that refinement was applied: the same strokes in the same order, with the same points, colours and widths. The *supported set* is the agreed set of shapes and simple objects that Squigglr refines (G.4); its contents are open issue OI-03 (Table 20). `REFINE_TOLERANCE` is defined in Table 10.

The first group covers what the child can do to a creative drawing, whether or not it has been refined. These are the editing actions that the Problem Statement lists as inputs (SRC-001) and that G.4 measurement (b) requires for refined parts.

FR-CRT-001 — Draw strokes

Requirement: Squigglr shall add each stroke the child draws with the stylus in the creative space to the open creative drawing as a new stroke.
Rationale: Drawing strokes is the basic input of the creative space (SRC-001), and the stylus is the main input (CON-003).
Fit criterion: After an adult tester draws a prepared sequence of strokes, the drawing holds one new stroke for each stroke drawn, in the order drawn; each follows the points the browser reported for it (ASM-004) and can be selected (FR-CRT-002).
Priority: Must
Phase: PH-1
Sources: G.4; CON-003; SRC-001
Traceability: UC-006; FR-CRT-002

FR-CRT-002 — Select a part

Requirement: Squigglr shall let the child select one or more strokes of the open creative drawing as the part to edit.
Rationale: Moving, resizing, recolouring and erasing act on a chosen part of the drawing; the Problem Statement calls them editing actions on drawing

elements (SRC-001). No source names selection, but every editing action needs it.
Fit criterion: On a prepared drawing of at least two strokes, the tester selects one stroke and then a group of strokes; each time, the selected strokes are visibly marked, and the next editing action changes exactly the selected strokes.
Priority: Must
Phase: PH-1
Sources: G.4; SRC-001
Traceability: UC-006; FR-CRT-003 to FR-CRT-006

FR-CRT-003 — Move a part

Requirement: Squigglr shall let the child move a selected part to another place on the canvas.
Rationale: Moving is one of the editing actions of G.4 measurement (b) and the Problem Statement (SRC-001).
Fit criterion: After the tester moves a selected part of a prepared drawing, every point of the part has moved by the same offset, in the direction of the movement, and no other stroke has changed.
Priority: Must
Phase: PH-1
Sources: G.4; SRC-001
Traceability: UC-006; FR-CRT-002, FR-CRT-013

FR-CRT-004 — Resize a part

Requirement: Squigglr shall let the child make a selected part larger or smaller.
Rationale: Resizing is one of the editing actions of G.4 measurement (b) and the Problem Statement (SRC-001). Keeping the part's proportions keeps its shape as the child drew it.
Fit criterion: After the tester resizes a selected part of a prepared drawing, the part is a scaled copy of itself: the distance of each of its points from the centre of the part's bounding box is multiplied by the same factor, greater than one when enlarged and less than one when shrunk, and no other stroke has changed.
Priority: Must
Phase: PH-1
Sources: G.4; SRC-001

Traceability: UC-006; FR-CRT-002, FR-CRT-013

FR-CRT-005 — Recolour a part

Requirement: Squigglr shall let the child change the colour of a selected part to another colour that Squigglr offers.
Rationale: Recolouring is one of the editing actions of G.4 measurement (b) and the Problem Statement (SRC-001).
Fit criterion: After the tester recolours a selected part of a prepared drawing, every selected stroke has the chosen colour, its points are unchanged, and no other stroke has changed.
Priority: Must
Phase: PH-1
Sources: G.4; SRC-001
Traceability: UC-006; FR-CRT-002, FR-CRT-013

FR-CRT-006 — Erase a part

Requirement: Squigglr shall let the child erase a selected part.
Rationale: Erasing is one of the editing actions of G.4 measurement (b) and the Problem Statement (SRC-001).
Fit criterion: After the tester erases a selected part of a prepared drawing, the selected strokes are no longer in the drawing and no other stroke has changed; undoing the erase (FR-CRT-007) brings them back exactly.
Priority: Must
Phase: PH-1
Sources: G.4; SRC-001
Traceability: UC-006; FR-CRT-002, FR-CRT-007

FR-CRT-007 — Undo a change

Requirement: Squigglr shall let the child undo the changes to the open creative drawing one at a time, most recent first, back to the drawing as it was when the child opened it.
Rationale: G.4 measurement (b) requires that refined parts can be undone, and the Problem Statement lists undoing among the editing actions (SRC-001). A child who cannot read needs a way back from any mistake, including an unwanted refinement or restore, without calling an adult (SRC-005, P1, 34:13). Because an applied refinement and a restore each count as one change, one undo reverses either completely. Changes made before the drawing was

last opened cannot be undone, but the original stays restorable (FR-CRT-024).
Fit criterion: The tester makes a prepared sequence of changes to a prepared drawing, including strokes, each editing action, one refinement and one restore; after each undo the drawing is exactly as it was before the corresponding change, and after the last undo it is exactly as it was when opened.
Priority: Must
Phase: PH-1
Sources: G.4; BR-005; SRC-001; SRC-005, P1, 34:13
Traceability: UC-006, UC-007; FR-CRT-008, FR-CRT-017; H-MUH-001

FR-CRT-008 — Redo an undone change

Requirement: Squigglr shall let the child redo undone changes one at a time until the child makes a new change.
Rationale: Without redo, an accidental undo would lose part of the child's work. No source names redo; it is the usual partner of undo.
Fit criterion: After the tester undoes a prepared sequence of changes (FR-CRT-007), each redo returns the drawing exactly to the state before the matching undo; after any new change, no redo is available.
Priority: Could
Phase: PH-3
Sources: G.4; SRC-001
Traceability: UC-006; FR-CRT-007; H-MUH-001

FR-CRT-009 — Start a new drawing

Requirement: Squigglr shall let the child start a new, empty creative drawing.
Rationale: A child may want to draw something else; the learner persona PER-1 wants to finish a picture and show it to his family. The earlier drawing is kept (FR-CRT-023). No source names this action.
Fit criterion: With a saved drawing open, the tester starts a new drawing; the canvas is empty, and the earlier drawing can be reopened unchanged (FR-CRT-024).
Priority: Should
Phase: PH-2
Sources: G.4; SRC-001
Traceability: UC-006; FR-CRT-023, FR-CRT-024

The second group makes "automatic" and "minimal" in BR-005 testable, and lets refinement be turned off (UC-007, UC-010). Unless a fit criterion

says otherwise, it is checked on prepared, adult-made drawings of each item in the supported set.

FR-CRT-010 — Refine a finished drawing

Requirement: Squigglr shall refine the open creative drawing automatically each time the child finishes it while refinement is turned on, and at no other time.
Rationale: Refinement is automatic and needs no action from the child (BR-005), and UC-007 starts when the child finishes a drawing. Refining only then means that strokes never move while the child is still drawing, which could confuse a child or take the drawing out of their hands. Both questionnaire respondents suggested that refinement be optional or adjustable (SRC-009.01), so FR-CRT-030 lets it be turned off.
Fit criterion: With refinement turned on, finishing each prepared drawing starts exactly one refinement; drawing, editing, undoing, restoring and leaving the creative space without finishing start none.
Priority: Must
Phase: PH-1
Sources: G.4; BR-005; SRC-001; SRC-005, Moustafa, 6:03 (as presented by the team)
Traceability: UC-007; FR-CRT-011, FR-CRT-012, FR-CRT-014, FR-CRT-015, FR-CRT-030; H-MUH-002

FR-CRT-011 — Change only stroke paths

Requirement: Squigglr shall change only the path of each existing stroke when it refines a creative drawing.
Rationale: Refinement must be minimal and the work must remain the child's own (BR-005). The team presented refinement to the specialists as tidying the child's lines and curves, not swapping in a different picture (SRC-005, Moustafa, 6:03); the interview specialists did not comment on it (FACT-013), but the questionnaire respondents asked that refinement stay subtle and support the child's own drawing rather than replace or correct it (SRC-009.01). Adding, removing or replacing strokes, or changing their colours or widths, would make the drawing less the child's own.
Fit criterion: For each prepared drawing, the refined drawing has the same number of strokes as its original, in the same order, and each refined stroke has the colour and width of the corresponding original stroke.
Priority: Must

Phase: PH-1
Sources: G.4; BR-005; SRC-005, Moustafa, 6:03 (as presented by the team); SRC-009.01
Traceability: UC-007; FR-CRT-012, FR-CRT-020; H-MUH-002

FR-CRT-012 — Stay within the refinement tolerance

Requirement: Squigglr shall keep every refined stroke within `REFINE_TOLERANCE` of the corresponding original stroke.
Rationale: This makes G.4 measurement (a) precise. The check runs both ways, so a refinement that shortens a stroke or drops part of it also fails: points of the original would then lie far from the refined stroke. The team sets `REFINE_TOLERANCE` with adult-made drawings (BR-009).
Fit criterion: For each prepared drawing, every point of each refined stroke lies within `REFINE_TOLERANCE` of the corresponding original stroke, and every point of each original stroke lies within `REFINE_TOLERANCE` of the corresponding refined stroke.
Priority: Must
Phase: PH-1
Sources: G.4; BR-005; BR-009; ASM-014
Traceability: UC-007; FR-CRT-011, FR-CRT-020; H-MUH-002, H-MUH-006

FR-CRT-013 — Keep the refined drawing editable

Requirement: Squigglr shall let the child edit each stroke of a refined drawing with the same actions, and with the same results, as a stroke the child drew.
Rationale: G.4 measurement (b) requires that refined parts can be moved, resized, recoloured, erased and undone. A refined drawing returned only as a picture could not be edited, which the Development Plan already rules out (SRC-002).
Fit criterion: For each prepared refined drawing, each of FR-CRT-002 to FR-CRT-007 applied to refined strokes meets its own fit criterion.
Priority: Must
Phase: PH-1
Sources: G.4; BR-005; SRC-001; SRC-002
Traceability: UC-006, UC-007; FR-CRT-002 to FR-CRT-007; H-MUH-001

FR-CRT-014 — Do not refine a stroke twice

Requirement: Squigglr shall not refine again a stroke that an earlier refinement changed, unless the child has edited that stroke since.

Rationale: Each refinement stays within `REFINE_TOLERANCE` of its own input, so refining the same strokes every time the child finishes would let the drawing drift further and further from what the child drew. A child may finish the same drawing many times. No source states this case.
Fit criterion: Finishing a refined drawing again without any change leaves it exactly as it was; after the tester edits one refined stroke and draws one new stroke, finishing again changes no stroke other than those two.
Priority: Must
Phase: PH-2
Sources: G.4; BR-005
Traceability: UC-007; FR-CRT-010, FR-CRT-012; H-MUH-002

FR-CRT-015 — Leave unsupported drawings unchanged

Requirement: Squigglr shall leave a finished creative drawing unchanged if it has no strokes or is not a drawing of an item in the supported set.
Rationale: G.4 commits to refining drawings of items in the agreed supported set only, and the Development Plan keeps refinement of arbitrary drawings out of the proof of concept (SRC-002). The team presented a version that would still tidy the lines of a drawing it could not identify (SRC-005, Moustafa, 6:03); until the supported set is agreed (OI-03), leaving such drawings as the child made them is the choice that protects the child's work.
Fit criterion: Finishing an empty canvas, and finishing each drawing in a prepared set of adult-made drawings of items outside the supported set, leaves each drawing exactly as it was and editable.
Priority: Must
Phase: PH-2
Sources: G.4; BR-005; SRC-002
Traceability: UC-007; FR-CRT-010; H-MUH-002, H-MUH-006

FR-CRT-016 — Show whether a drawing is refined

Requirement: Squigglr shall show, while a creative drawing is open, whether it holds a refinement that has not been undone or restored.
Rationale: The child and the supporting adult need to tell a refined drawing from the child's original; otherwise an adult cannot see what Squigglr changed, and the child cannot see that restoring is possible. How the state is shown to a child who cannot read is a usability matter.

Fit criterion: For prepared drawings in each of these states (never refined; refined; refined then edited; refined then restored; refinement undone; refinement failed), what Squigglr shows matches the state in every case.
Priority: Should
Phase: PH-2
Sources: G.4; BR-005
Traceability: UC-007; FR-CRT-017; H-MUH-003, H-MUH-006

The third group covers restoring the original. BR-005 lets the child, or the adult if the child cannot, restore it. FR-CRT-017 defines what a restore returns, and FR-CRT-018 and FR-CRT-019 say who can restore.

FR-CRT-017 — Restore the exact original

Requirement: Squigglr shall return a creative drawing, on restore, exactly to the original of its most recent refinement that has not been undone or restored.
Rationale: This makes G.4 measurement (c) precise. After one refinement, a restore returns the child's drawing as it was before refinement. After several, each restore steps back one refinement, so no earlier version of the child's work is lost. A restore also removes edits made after that refinement; because a restore is one change, the child can undo it (FR-CRT-007) to get the refined and edited drawing back.
Fit criterion: For each prepared drawing: (a) after one refinement and a restore, the drawing equals its original stroke for stroke and point for point, in the same colours and widths; (b) after a refinement, an edit, a restore and an undo, the drawing equals the refined and edited drawing exactly; (c) after a refinement, new strokes, a second refinement and two restores, the drawing equals the original of the second refinement after the first restore and the original of the first refinement after the second; (d) no restore is offered when no refinement remains that has not been undone or restored.
Priority: Must
Phase: PH-1
Sources: G.4; BR-005; SRC-001
Traceability: UC-007; FR-CRT-007, FR-CRT-018, FR-CRT-019; H-MUH-001, H-MUH-003

FR-CRT-018 — Child restores the original

Requirement: Squigglr shall let the child restore the original of the open creative drawing.

Rationale: BR-005 lets the child restore the original, which keeps the child in control of the work (G.4). The team presented this to the specialists as part of the creative space (SRC-005, Moustafa, 6:03). A child who cannot read must be able to find the control, which is a usability matter.
Fit criterion: From the child's view of each prepared refined drawing, the tester restores the original with one action, and the result meets FR-CRT-017.
Priority: Must
Phase: PH-1
Sources: G.4; BR-005; SRC-001; SRC-005, Moustafa, 6:03 (as presented by the team)
Traceability: UC-007; FR-CRT-017; H-MUH-001, H-MUH-006

FR-CRT-019 — Adult restores the original

Requirement: Squigglr shall let the supporting adult, after passing the adult check of SAF-RAY-001, restore the original of any creative drawing saved under a learner profile in that adult's account.
Rationale: BR-005 lets the adult restore the original for a child who cannot, and BR-001 limits a learner's work to the adult who holds the account. The adult may need to do this when the child is not using the drawing, for example after a session. The adult check keeps a child from reaching this adult action by tapping icons (SAF-RAY-001).
Fit criterion: The tester, signed in as the adult of a prepared account and after passing the adult check of SAF-RAY-001, restores the original of a saved refined drawing; reopening the drawing under its learner profile shows a result that meets FR-CRT-017.
Priority: Must
Phase: PH-2
Sources: G.4; BR-001; BR-005
Traceability: UC-007; FR-CRT-017, FR-CRT-023, SAF-RAY-001; H-MUH-001

The fourth group covers failure and interruption. A refinement may fail, return an unusable result, or finish after the child has moved on, for example when the connection is slow or lost (ASM-005). How quickly refinement must respond, and how Squigglr copes when refinement is unavailable, belong to the performance requirements (Section 12).

FR-CRT-020 — Leave the drawing unchanged when refinement fails

Requirement: Squigglr shall make no change to a creative drawing when its refinement fails or the refinement result violates FR-CRT-011 or FR-CRT-012.
Rationale: This restates G.4 measurement (d) for automatic refinement, which the child never requests (BR-005). A partly applied or invalid result could change the drawing more than minimally, so it is applied whole or not at all. The Development Plan also requires that an invalid result leave the drawing unchanged (SRC-002). The drawing stays editable, and finishing it again starts a new refinement (FR-CRT-010).
Fit criterion: For each prepared drawing and each of these simulated cases (lost connection; an error during refinement; no result within the time limit of the speed requirements; a result that adds a stroke; a result with a point outside `REFINE_TOLERANCE`; a malformed result), the drawing afterwards equals the drawing as finished, can be edited (FR-CRT-002 to FR-CRT-007) and, from PH-2, when FR-CRT-016 applies, is shown as not refined.
Priority: Must
Phase: PH-1
Sources: G.4; BR-005; ASM-005; SRC-002
Traceability: UC-007; FR-CRT-011, FR-CRT-012; H-MUH-002

FR-CRT-021 — Discard an out-of-date result

Requirement: Squigglr shall discard a refinement result if the drawing has been changed or closed since that refinement started.
Rationale: A result computed from an older version of the drawing would remove the child's newer strokes, or change a drawing the child has left. Discarding it keeps the drawing as the child made it; the child can finish again. No source states this case.
Fit criterion: For each prepared drawing, with the result delayed by the test set-up: (a) if the tester draws a stroke before the result arrives, the drawing afterwards is the finished drawing plus that stroke, unrefined; (b) if the tester leaves the creative space before the result arrives, reopening the drawing shows it exactly as left, unrefined.
Priority: Must
Phase: PH-2
Sources: G.4; BR-005; ASM-005

Traceability: UC-006, UC-007; FR-CRT-022, FR-CRT-024; H-MUH-002

FR-CRT-022 — Keep drawing while refinement runs

Requirement: Squigglr shall let the child keep drawing and editing while a refinement is in progress.
Rationale: A child kept waiting may lose focus (PER-1), and children should be able to use Squigglr without constant adult help (SRC-005, P1, 34:13). No source states this case.
Fit criterion: With the result delayed by the test set-up, the tester draws a stroke and moves a part before the result arrives; both changes meet FR-CRT-001 and FR-CRT-003, and FR-CRT-021 then applies.
Priority: Should
Phase: PH-2
Sources: G.4; SRC-005, P1, 34:13
Traceability: UC-006, UC-007; FR-CRT-021

The fifth group covers saving and reopening. UC-006 ends with the drawing saved and still editable. One account can hold several learner profiles and a device may be shared by several children (BR-001; SRC-004), so each creative drawing belongs to one learner profile.

FR-CRT-023 — Save to the active learner

Requirement: Squigglr shall save each creative drawing, with the originals of its refinements, to the learner profile that was active when the child drew it, without any action from the child.
Rationale: A child who cannot read may leave without saving. Restoring after reopening needs the originals (FR-CRT-024). Saving to the wrong learner would show one child's work as another's (BR-001).
Fit criterion: For two learner profiles in one prepared account, the tester draws, refines and edits a drawing under each and leaves the creative space with its own controls; reopening each learner's creative space shows that learner's drawing exactly as left, and not the other learner's.
Priority: Must
Phase: PH-2
Sources: G.4; BR-001; BR-005
Traceability: UC-006; FR-CRT-024, FR-CRT-026; H-MUH-003, H-MUH-004, H-MUH-005

FR-CRT-024 — Reopen a drawing as it was left

Requirement: Squigglr shall reopen a saved creative drawing exactly as it was last saved, with every stroke editable and every refinement that has not been undone or restored still restorable.
Rationale: UC-006 requires the drawing to stay editable, and BR-005 requires the original to stay restorable; neither ends when the child leaves the creative space.
Fit criterion: For prepared drawings saved in each state listed in FR-CRT-016, reopening gives the same strokes and the same refined state, every stroke can be edited (FR-CRT-013), and a restore meets FR-CRT-017.
Priority: Must
Phase: PH-2
Sources: G.4; BR-005
Traceability: UC-006, UC-007; FR-CRT-017, FR-CRT-023; H-MUH-001, H-MUH-005

FR-CRT-025 — Keep changes through a lost connection

Requirement: Squigglr shall save, once the connection returns, every change the child made to an open creative drawing while the connection was lost.
Rationale: The connection may drop during normal use, and offline use is out of scope (ASM-005), so this covers a drawing that stays open until the connection returns. Reporting a save that fails is a robustness requirement.
Fit criterion: With a prepared drawing open, the test set-up cuts the connection; the tester draws strokes and edits a part, the connection is restored, and the tester leaves the creative space; reopening the drawing shows every change.
Priority: Should
Phase: PH-3
Sources: G.4; ASM-005
Traceability: UC-006; FR-CRT-023; H-MUH-005

FR-CRT-026 — Show only the active learner's drawings

Requirement: Squigglr shall show in the creative space only the creative drawings saved under the active learner profile.
Rationale: Several children may use one device and one account (SRC-004; BR-001); showing another child's drawing would expose their work and let it be changed by someone else. An adult selects the active learner (ASM-008).

Fit criterion: With drawings saved under two learner profiles of one prepared account, the creative space of each learner shows only that learner's drawings.
Priority: Must
Phase: PH-2
Sources: BR-001; ASM-008; SRC-004
Traceability: UC-006; FR-CRT-023; H-MUH-004

The last group keeps the creative space apart from the guided lesson records that adults rely on (BR-004; ULC-005 in Table 35).

FR-CRT-027 — No refinement in guided lessons

Requirement: Squigglr shall not refine any stroke drawn in a guided lesson.
Rationale: BR-005 limits refinement to the creative space, and a guided lesson record keeps the child's actual strokes (BR-004). A refined stroke in a lesson would show adults work the child did not do. How the record keeps those strokes is specified in Section 9.4.
Fit criterion: For prepared attempts at each drawing stage (full-outline tracing, endpoint-only guidance and independent drawing) of each catalogue lesson, no refinement starts, and the strokes shown during and after each attempt are point for point the strokes drawn.
Priority: Must
Phase: PH-2
Sources: G.4; G.5; BR-004; BR-005; SRC-005, Moustafa, 6:03 (as presented by the team)
Traceability: UC-003, UC-007; FR-CRT-010; H-MUH-002

FR-CRT-028 — Creative drawings are not attempts

Requirement: Squigglr shall not count a creative drawing as an attempt.
Rationale: Attempt records and progress summaries describe guided practice (G.5, G.6). A refined creative drawing counted as an attempt would make assisted work look like the child's own.
Fit criterion: For a prepared learner profile with saved attempts, drawing, refining and restoring prepared creative drawings leaves every attempt record, lesson record and progress summary of that learner exactly as before.
Priority: Must
Phase: PH-2
Sources: G.4; G.5; G.6; BR-004
Traceability: UC-006, UC-008, UC-009; FR-CRT-023; H-MUH-003

FR-CRT-029 — Show unsaved changes

Requirement: Squigglr shall show, while a creative drawing is open, whether every change to it has been saved.
Rationale: Offline use is out of scope (ASM-005), so changes made while the connection is lost are kept only while the page stays open (FR-CRT-025); an adult who can see that changes are not yet saved can keep the page open until they are. How the state is shown to a child who cannot read is a usability requirement.
Fit criterion: With a prepared drawing open, the tester edits it with the connection cut; the display shows unsaved changes; after the connection returns and the save completes, it shows that all changes are saved.
Priority: Should
Phase: PH-3
Sources: G.4; ASM-005; CA-003
Traceability: UC-006; FR-CRT-023, FR-CRT-025; H-MUH-005

FR-CRT-030 — Turn refinement off

Requirement: Squigglr shall let the child, and the supporting adult, turn automatic refinement off and on for a learner profile.
Rationale: UC-007, UC-010 and the data flows F2 and F9 let refinement be turned off, and both questionnaire respondents suggested that refinement be optional or adjustable (SRC-009.01). Whether children value automatic refinement is not verified (ASM-011); a child who does not accept refined drawings as their own needs a way to keep drawings unrefined (G.4).
Fit criterion: After the tester, once as the child in the creative space and once as the adult in the learner's settings, turns refinement off, finishing a prepared drawing of the supported set leaves its strokes exactly as drawn; after refinement is turned on again, finishing the drawing refines it.
Priority: Should
Phase: PH-2
Sources: G.4; BR-005; ASM-011; SRC-009.01
Traceability: UC-007; UC-010; FR-CRT-010; H-MUH-006

9.4 Attempt Record Requirements

This subsection specifies what Squigglr keeps about guided practice, so that a supporting adult can see how a child practised and not only what the

final drawing looks like (G.5). The requirements serve use cases UC-008 (record an attempt) and UC-003 (practise a guided lesson). An *attempt* is one try at one stage of a lesson. A stage (also called a guidance level in Table 9) is one of the stages of decreasing help listed in the Goals of the Project subsection; attempts are made at the drawing stages: tracing a full outline, drawing between endpoints only, and drawing with no guide. Squigglr keeps an *attempt record* for each attempt and a *lesson record* each time a learner starts a lesson. Creative drawings are not attempts (FR-CRT-028); the Creative Drawing Requirements subsection covers them.

Three safety requirements in the Safety-Critical Requirements subsection already govern records, and the requirements below do not repeat them. SAF-RAY-003 keeps each completed attempt on the device until the server confirms that it is saved, and tells the adult which attempts were not saved. SAF-RAY-006 records each lesson that ends at an attempt limit, either `ATTEMPT_LIMIT` or `ATTEMPT_LIMIT_MAX`, records each improvement extension, and keeps earlier attempts when the same lesson is started again. SAF-RAY-007 saves every attempt to the active learner only. The specialist interview did not cover how records are kept, shared or deleted (FACT-013), so the requirements below rest on the Problem Statement and the business rules. Who may view, share or delete records is decided in the Security Requirements section (Open Issue OI-02, Table 20). How a lesson moves between stages, and when an attempt ends, belong to the Learning Activity Requirements. Phases are the delivery phases PH-1, PH-2 and PH-3 defined, with their dates, in the Introduction.

FR-REC-001 — Attempt record

Requirement: Squigglr shall keep, for each attempt that ends with an outcome, an attempt record holding the lesson, the stage, the attempt's result, the outcome (whether the attempt passed its stage, FR-LRN-013) and the date and time the attempt ended.
Rationale: A finished drawing does not show how much guidance the child had (SRC-001). The stage tells the adult whether the attempt was made with a guide or without one, and the date and time place the attempt in the learner's history for progress summaries (G.6). The learner is fixed by SAF-RAY-007. The result is kept because the improvement rule compares the results of recent attempts (BR-002, FR-LRN-020), and the adult sees each attempt's result (ASM-006).
Fit criterion: For a set of prepared, adult-made attempts with known lessons,

stages, results and outcomes, including at least one passing and one failing attempt at each drawing stage, every saved attempt record shows exactly the known lesson, stage, result and outcome, and the records are in the order the attempts ended.
Priority: Must
Phase: PH-1
Sources: G.2, G.5; BR-002; SRC-001; SRC-002 (PoC plan, adaptive guidance experiment)
Traceability: UC-003, UC-008; SAF-RAY-007, FR-LRN-013, FR-LRN-020, FR-PRG-002, FR-PRG-003; H-MUH-003

FR-REC-002 — Actual strokes

Requirement: Squigglr shall keep in each attempt record the strokes the child drew in the attempt, in the order drawn, as they were captured.
Rationale: Adults judge progress from the records, and a record that shows a corrected or ideal shape would mislead them (BR-004). This principle was presented to the domain specialists in the interview (SRC-005, Moustafa, 6:03). Duplicate or unmappable input is handled by SAF-AHM-001 and SAF-AHM-002, so the strokes kept are those the child drew. Refinement never reaches guided lesson strokes (BR-005; FR-CRT-027).
Fit criterion: For prepared, adult-made attempts whose strokes are captured by the test harness as drawn, every saved attempt record holds the same strokes, point for point and in the same order, and no attempt record holds a stroke that was not drawn in that attempt.
Priority: Must
Phase: PH-2
Sources: G.5; BR-004; ASM-004; SRC-001; SRC-005, Moustafa, 6:03
Traceability: UC-003, UC-008; FR-LRN-007, SAF-AHM-001, SAF-AHM-002, FR-PRG-006; H-MUH-003

FR-REC-003 — Difficulties and corrections

Requirement: Squigglr shall keep in each attempt record every predefined difficulty observed during the attempt, each marked as corrected or not corrected within that attempt.
Rationale: Feedback lets the child go back and complete a missed part within the same attempt (BR-007). If the correction erased the difficulty, the record would hide what the child found hard; G.5 requires that no difficulty is lost

after a correction. The team presented records that keep both the miss and its later completion (SRC-005, Moustafa, 6:03).
Fit criterion: For prepared attempts with known predefined difficulties, some corrected within the attempt and some not, every saved attempt record lists every known difficulty once, with the correct mark, and lists no other difficulty.
Priority: Must
Phase: PH-1
Sources: G.5; BR-007; SRC-001; SRC-002 (PoC plan, adaptive guidance experiment); SRC-005, Moustafa, 6:03
Traceability: UC-003, UC-008; FR-PRG-004, FR-PRG-006; H-MUH-007

FR-REC-004 — Adult-set values in use

Requirement: Squigglr shall keep in each attempt record the value of each adult-set value that was in use for the learner when the attempt was made.
Rationale: The adult-set values are the eight values of BR-008 listed in Table 10, and the outcome of an attempt depends on the `STAGE_PASS` in use, and the adult may change the values at any time (BR-008). Keeping the values with each attempt means a later change does not alter what an earlier record means, and lets the adult see every value Squigglr records (ASM-006, LC-007). SAF-RAY-008 shows the values currently in use; this requirement keeps those in use for each attempt.
Fit criterion: For prepared attempts made before and after an adult changes each adult-set value, each attempt record holds the values in use when that attempt was made.
Priority: Must
Phase: PH-2
Sources: G.5; BR-008; ASM-006; LC-007; SRC-008
Traceability: UC-008, UC-010; SAF-RAY-008, FR-PRG-006; H-MUH-007

FR-REC-005 — Interrupted attempt

Requirement: Squigglr shall keep an attempt that an adult stops in Squigglr, by ending the session or switching learner, after at least one stroke but before its outcome is decided, as an attempt record marked as interrupted, holding the strokes and predefined difficulties up to that point and no outcome.
Rationale: An adult may stop a session (ASM-015) or switch learners on a shared device (ASM-008) in the middle of an attempt. The strokes drawn are still practice the adult may want to see, but the attempt has no outcome,

so it must not appear as passed or failed. An interrupted attempt does not consume an evaluable attempt from the stage limit (FR-LRN-009). An attempt cut off by closing the page or losing the connection may not reach Squigglr; completed attempts are protected by SAF-RAY-003.
Fit criterion: For prepared attempts stopped by an adult action after one or more strokes, every saved attempt record is marked as interrupted, holds the strokes drawn and has no outcome; an attempt stopped before any stroke leaves no attempt record.
Priority: Should
Phase: PH-2
Sources: G.5; ASM-008, ASM-015; CA-003, CA-004
Traceability: UC-003, UC-008; FR-LRN-009, FR-PRG-009; H-MUH-004

FR-REC-006 — Lesson record

Requirement: Squigglr shall keep, each time a learner starts a lesson, a lesson record holding the lesson, the starting stage, the furthest stage reached, the number of successful attempts at each stage and how the lesson ended.
Rationale: Caregivers and educators need to see how far the child got in each lesson and how many attempts succeeded at each stage (Table 5; BR-002). The starting stage shows whether a stage was reached by passing or chosen by an adult (BR-003). A new record each time the lesson starts keeps a repeated lesson from hiding earlier results (SAF-RAY-006).
Fit criterion: For prepared sequences of attempts in which a lesson is started at the first drawing stage and again at a later stage chosen by an adult, each lesson record shows the known starting stage, furthest stage, count of successful attempts at each stage and ending.
Priority: Must
Phase: PH-2
Sources: G.5; BR-002, BR-003; SRC-001
Traceability: UC-003, UC-004, UC-008; SAF-RAY-006, FR-PRG-002, FR-PRG-010; H-MUH-007

FR-REC-007 — Unfinished lesson

Requirement: Squigglr shall keep the lesson record of a lesson that stops before its last stage is passed and before it reaches an attempt limit, marked as unfinished, with the furthest stage reached and the successful attempts at each stage up to that point.

Rationale: BR-002 saves progress when a lesson ends at `ATTEMPT_LIMIT`, and SAF-RAY-006 records those endings. A lesson an adult stops early (ASM-015) would otherwise leave no record of the stages already passed.
Fit criterion: For prepared lessons stopped by an adult after known successful attempts, each lesson record is marked as unfinished and shows the known furthest stage and counts; for prepared lessons whose last stage is passed, no lesson record is marked as unfinished.
Priority: Should
Phase: PH-2
Sources: G.5; BR-002; ASM-015
Traceability: UC-003, UC-008; SAF-RAY-006, FR-PRG-002; H-MUH-007

FR-REC-008 — Each attempt saved once

Requirement: Squigglr shall save each attempt record exactly once, under the lesson record of the lesson in which the attempt was made.
Rationale: A save that is retried after the connection drops (CA-003, SAF-RAY-003) could store an attempt twice or under another lesson, and two sessions open for the same learner could mix their attempts. Either would change the counts an adult relies on.
Fit criterion: For prepared attempts saved with the connection cut and restored during each save, and for prepared attempts made for one learner in two sessions open at the same time, every attempt appears in exactly one attempt record, under the lesson record in which it was made, and the counts in each lesson record equal the known counts.
Priority: Must
Phase: PH-2
Sources: G.5; ASM-005; CA-003, CA-004
Traceability: UC-008; SAF-RAY-003, SAF-RAY-007; H-MUH-004, H-RAY-002, H-RAY-006

FR-REC-009 — Saved records unchanged

Requirement: Squigglr shall keep the contents of every saved attempt record, and of every lesson record whose lesson has ended, unchanged except by deletion.
Rationale: Later events, such as an adult changing an adult-set value, a starting stage or the lesson catalogue (LC-001, LC-002), must not rewrite what an earlier record says, or an attempt made with a guide could later

appear to have been made without one. Deletion is decided with the privacy requirements (OI-02).
Fit criterion: For a prepared history, after the adult changes every adult-set value, the starting stage of a lesson and the learner's other settings, and after a lesson is added to or removed from the catalogue, every saved attempt record and every lesson record of an ended lesson is identical to its contents before those changes.
Priority: Must
Phase: PH-2
Sources: G.5; BR-004, BR-008; LC-001, LC-002
Traceability: UC-008, UC-010; FR-REC-004; H-MUH-003, H-MUH-004, H-MUH-007

FR-REC-010 — Suggestion responses

Requirement: Squigglr shall keep in the lesson record each suggestion to go back a stage that it offers and the response of the adult to it.
Rationale: A go-back suggestion changes the stage only when an adult accepts it (BR-003). Keeping the suggestion and the response shows an adult why a later attempt was made at an earlier stage, and keeps a declined suggestion visible (SAF-RAY-001).
Fit criterion: For prepared sequences of attempts in which suggestions are accepted, declined and not yet answered, every lesson record lists each suggestion once, with the known response, and lists no other suggestion.
Priority: Should
Phase: PH-2
Sources: G.3; BR-003
Traceability: UC-005; UC-008; FR-LRN-019; SAF-RAY-001, SAF-RAY-007

9.5 Progress Summary Requirements

This subsection specifies the progress summary: the view in which a supporting adult reviews a learner's practice over time and opens the attempt records behind it (G.6). The requirements serve use cases UC-009 (review progress) and UC-010 (review history and change settings). A progress summary is built only from the attempt and lesson records of the Attempt Record Requirements (Section 9.4). It is meant for an adult such as PER-2, who reviews a child's history once a week to see which stages he reached and what keeps

going wrong, and PER-3, who works with several learners a day. Attempts *made with a guide* are those at the tracing and endpoint stages; attempts *made without a guide* are those at the stage with no guide. FR-LRN-012 calls these attempts assisted and independent.

The requirements below leave three matters to other requirements. SAF-RAY-004 forbids diagnostic labels, judgements of the child's abilities and comparisons with other children in any record or summary, and requires each summary to state that it describes practice only. SAF-RAY-006 shows how many times each lesson ended at a limit or was extended. Access to summaries is limited to the account-holding adult (BR-001, ULC-002) by the Security Requirements section, which also decides whether summaries can be shared or exported (OI-02).

FR-PRG-001 — One learner per summary

Requirement: Squigglr shall show in each progress summary only the records of the one learner profile that the summary names.
Rationale: One account can hold several learner profiles (BR-001), and an educator must never show one learner's records to another (PER-3). Naming the learner lets the adult check whose records are shown.
Fit criterion: For a prepared account with at least three learner profiles, each with a known history, every summary names one learner and contains every record of that learner and no record of another.
Priority: Must
Phase: PH-2
Sources: G.6; BR-001; ASM-008; SRC-004
Traceability: UC-009, UC-010; SAF-RAY-007; H-MUH-004, H-RAY-006

FR-PRG-002 — Progress over time by lesson

Requirement: Squigglr shall show, for each lesson in a progress summary, the learner's lesson records in the order they were started, each with its starting stage, furthest stage reached, successful attempts at each stage and how the lesson ended.
Rationale: Caregivers and educators need the furthest stage and the successful attempts at each stage (Table 5), and the order shows change over time (G.6). How the lesson ended (every stage passed, ended at a limit, or unfinished) keeps an unfinished lesson from looking like a lesson the child could not pass.

Fit criterion: For prepared histories showing improvement, a lesson ended at `ATTEMPT_LIMIT` and an unfinished lesson, every summary lists the lesson records in start order with the known values and endings.
Priority: Must
Phase: PH-2
Sources: G.6; BR-002; SRC-001
Traceability: UC-009; FR-REC-001, FR-REC-006, FR-REC-007, SAF-RAY-006; H-MUH-008

FR-PRG-003 — With and without a guide kept apart

Requirement: Squigglr shall show attempts made without a guide separately from attempts made with a guide in every part of a progress summary that counts or lists attempts.
Rationale: The adults need to see changes in independent drawing, which a combined figure would hide (SRC-001; G.6). Assisted work that appears independent could lead an adult to move a child on too soon.
Fit criterion: For prepared histories with known numbers of attempts made with and without a guide, every count or list in the summary shows the two kinds separately and correctly, and no figure adds them together; at least `ADULT_REVIEWERS` adult reviewers state which attempts were made without a guide, with at least `REVIEW_PASS_RATE` of their tasks succeeding.
Priority: Must
Phase: PH-2
Sources: G.2, G.6; SRC-001
Traceability: UC-009; FR-REC-001, FR-LRN-012; H-MUH-003

FR-PRG-004 — Recurring difficulties

Requirement: Squigglr shall list in each progress summary every predefined difficulty observed in at least `RECUR_MIN` of the learner's attempts at the same stage of the same lesson, with the number of those attempts and the number in which the child corrected it.
Rationale: Adults want to know what keeps going wrong (PER-2), and the Problem Statement promises summaries that show recurring difficulties (SRC-001). A fixed count makes "recurring" testable. The constant `RECUR_MIN` is defined in Table 10. Counting corrections shows difficulties that are being resolved as well as those that keep coming up (SRC-005, Moustafa, 6:03).

Fit criterion: For a prepared history in which one predefined difficulty occurs in exactly `RECUR_MIN` attempts at one stage and another in one attempt fewer, the summary lists the first, with the known counts, and not the second.
Priority: Must
Phase: PH-2
Sources: G.6; BR-007; SRC-001; SRC-005, Moustafa, 6:03
Traceability: UC-009; FR-REC-003; H-MUH-008

FR-PRG-005 — Open the attempts behind the summary

Requirement: Squigglr shall let the adult open, from each figure or item in a progress summary, the attempt records from which it was computed.
Rationale: Adults need to inspect the attempts behind a summary (G.6), and seeing the drawings lets them judge for themselves rather than rely on a number.
Fit criterion: For prepared histories, every figure or item opens exactly the attempt records it was computed from; at least `ADULT_REVIEWERS` adult reviewers locate the attempts behind given items, with at least `REVIEW_PASS_RATE` of their tasks succeeding.
Priority: Must
Phase: PH-2
Sources: G.6; SRC-001
Traceability: UC-009; FR-PRG-006; H-MUH-008

FR-PRG-006 — Attempt record view

Requirement: Squigglr shall show, for each attempt record the adult opens, every value kept in it under FR-REC-001 to FR-REC-005, with the strokes drawn as recorded.
Rationale: Squigglr shows the adult every value it records (ASM-006), and the strokes show what the child actually drew (BR-004).
Fit criterion: For each prepared attempt record, the view shows its lesson, stage, outcome or interrupted mark, date and time, predefined difficulties with their correction marks, adult-set values in use, and strokes identical to those recorded.
Priority: Must
Phase: PH-2
Sources: G.5, G.6; BR-004; ASM-006
Traceability: UC-009, UC-010; FR-REC-001 to FR-REC-005; H-MUH-008

FR-PRG-007 — Adult chooses the view

Requirement: Squigglr shall let the adult choose which lessons, dates and recorded values a progress summary shows.
Rationale: Which values each adult wants to see will differ and change (LC-007), and the specialists advised that useful progress measures be chosen with occupational therapists or educators (SRC-005.03), so the adult chooses rather than the team (ASM-006). Until the adult chooses, the summary covers all of the learner's records.
Fit criterion: For a prepared history, each choice of lessons, dates and values changes only what the summary shows: the records are unchanged, and attempts made with and without a guide stay separate.
Priority: Should
Phase: PH-3
Sources: G.6; ASM-006; LC-007; SRC-005.03
Traceability: UC-009; FR-PRG-003

FR-PRG-008 — Empty history

Requirement: Squigglr shall show, when the learner has no saved attempt record in the chosen lessons and dates, a progress summary that states that no practice is recorded and shows no stage, count or difficulty.
Rationale: A new learner has no history, and zeros or empty stages could be read as failure.
Fit criterion: For a new learner profile, and for a learner whose records all lie outside the chosen dates, the summary states that no practice is recorded and shows no stage, count or difficulty.
Priority: Must
Phase: PH-2
Sources: G.6; BR-006
Traceability: UC-009; FR-PRG-007; H-MUH-008, H-RAY-003

FR-PRG-009 — Interrupted attempts shown apart

Requirement: Squigglr shall show interrupted attempts in a progress summary separately from successful and unsuccessful attempts.
Rationale: An interrupted attempt has no outcome (FR-REC-005); counting it as failed would understate the child's practice, and counting it as passed would overstate it.
Fit criterion: For a prepared learner whose only attempts are interrupted, the summary shows them as interrupted and shows no successful or unsuccessful

attempt and no stage passed; adding prepared interrupted attempts to any history leaves its successful and unsuccessful counts unchanged.
Priority: Should
Phase: PH-2
Sources: G.6; ASM-015
Traceability: UC-009; FR-REC-005; H-MUH-008

FR-PRG-010 — Stages not attempted

Requirement: Squigglr shall mark each stage of a lesson that the learner has not attempted, including a stage before an adult-chosen starting stage, as not attempted, distinct from a stage attempted without success.
Rationale: An adult may start the child at a later stage (BR-003), and a learner with a short history has attempted only some stages. A stage left blank or shown as zero successes could be read as a stage the child could not draw.
Fit criterion: For a prepared history with a lesson started at a later stage and one stage attempted only without success, the summary marks the skipped stages as not attempted and shows the attempted stage with zero successful attempts.
Priority: Must
Phase: PH-2
Sources: G.6; BR-003; SRC-001
Traceability: UC-004, UC-009; FR-REC-006; H-MUH-003, H-MUH-008

FR-PRG-011 — Latest attempt included

Requirement: Squigglr shall state on each progress summary the date and time of the most recent attempt it includes.
Rationale: Attempts still held on the child's device (SAF-RAY-003) are not in a summary that an adult opens on another device. The date of the latest attempt lets the adult notice recent practice that is missing.
Fit criterion: For a prepared history with attempts saved and attempts held back by a cut connection, the summary shows the date and time of the latest saved attempt and includes none of the held-back attempts.
Priority: Should
Phase: PH-2
Sources: G.6; ASM-005; CA-003
Traceability: UC-009; SAF-RAY-003; H-MUH-008, H-RAY-002

FR-PRG-012 — No overall score

Requirement: Squigglr shall show no figure in a progress summary that combines the results of more than one stage into a single score, grade or rating.
Rationale: The needs of caregivers and educators exclude a single overall score (Table 5; PER-2), and the team cannot judge a child's abilities (BR-006; SRC-008). Results per stage and per lesson answer the adults' questions without a score.
Fit criterion: Inspection of the summaries for prepared histories finds no score, grade, rating or percentage computed from attempts at more than one stage.
Priority: Must
Phase: PH-2
Sources: G.6; BR-006; SRC-001, SRC-008; SRC-005, Moustafa, 6:03
Traceability: UC-009; SAF-RAY-004; H-RAY-003

10 Look and Feel Requirements

This section states how Squigglr looks (Section 10.1) and the mood its style creates (Section 10.2). A *child-facing screen* is any screen the child uses in a guided lesson or the creative space; an *adult view* is any screen for setup, settings, records or progress summaries. No requirement is gathered from children and no study with children takes place (CON-005, CON-006), so the fit criteria are checked by inspecting every screen or by adult reviewers. An adult review shows that adults find a design suitable, not that children can use it; checking that needs a study with ethics approval, which is out of scope (SRC-005.11).

The specialists advised starting from the Web Content Accessibility Guidelines (WCAG) 2.2 [45] (SRC-005.08), and the supervisor advised taking specific WCAG ideas as non-functional requirements rather than treating WCAG as a constraint (SRC-008). Each requirement below that is based on a WCAG success criterion names it in its rationale. Further WCAG-based accessibility criteria are in Section 11, and listing WCAG as a standard is open issue OI-08.

10.1 Appearance Requirements

NFR-UX-001 — Calm, uncluttered child screens

Requirement: Each child-facing screen shall offer the child no more than `CHILD_CONTROLS_MAX` controls and shall show no decorative animation or flashing while the child is drawing.
Rationale: Too many visual or auditory stimuli can overwhelm children, and specialists advised a calm, minimally stimulating interface (SRC-009.05). A child who cannot read navigates by images, so every extra control is something more to interpret (SRC-005.05). Banning flashing and moving content while the child draws follows WCAG 2.2 success criteria 2.3.1 Three Flashes or Below Threshold and 2.2.2 Pause, Stop, Hide [45].
Fit criterion: Inspect every child-facing screen in each lesson stage and in the creative space. Pass when each screen offers at most `CHILD_CONTROLS_MAX` controls and no element animates or flashes while a stroke is being drawn.
Priority: Must
Phase: PH-2
Sources: G.1; SRC-005.05; SRC-009.05
Traceability: UC-003; UC-006; NFR-UX-002; SAF-MOU-003; H-MOU-001

NFR-UX-002 — Images and icons instead of text

Requirement: Every control on a child-facing screen shall be identified by an image or icon, and no task on a child-facing screen shall require the child to read.
Rationale: Many learners cannot read yet; children navigate games through images and icons rather than text (SRC-005.05, FACT-007).
Fit criterion: Inspect every child-facing screen. Pass when every control shows an image or icon, and each child task in UC-003 and UC-006 can be completed with all on-screen text hidden.
Priority: Must
Phase: PH-2
Sources: G.1; FACT-007; SRC-005.05
Traceability: UC-003; UC-006; NFR-UX-001; H-MOU-003

NFR-UX-003 — Adult views distinct from child screens

Requirement: Adult views shall look clearly different from child-facing screens, so that an adult can tell at a glance which kind of screen is shown.

Rationale: Specialists described a separate adult interface for personalizing activities and viewing progress (SRC-005.02). A distinct look helps adults notice when a child has reached an adult-only screen, alongside the adult check (SAF-RAY-001).
Fit criterion: Show at least `UX_REVIEWERS` adult reviewers a shuffled set of screenshots of every child-facing screen and adult view, without titles. Pass when every reviewer sorts every screenshot correctly.
Priority: Should
Phase: PH-2
Sources: G.3, G.6; SRC-005.02
Traceability: UC-001; UC-002; UC-009; UC-010; SAF-RAY-001

10.2 Style Requirements

NFR-UX-004 — Feedback that never marks the drawing as wrong
Requirement: Feedback on an attempt, including a missed part or a non-passing attempt, shall use encouraging styling, shall not use error styling such as warning colours, crosses, alarm sounds or "wrong" symbols, and shall not show a missed part by colour alone.
Rationale: Children become frustrated when an activity keeps highlighting their mistakes or suggests that their drawing is wrong, so feedback should recognize effort instead (SRC-009.02). Feedback wording and colours should be positive (SRC-005.06). Feedback describes the drawing, not the child (BR-006, FR-LRN-022). Showing a missed part by more than colour follows WCAG 2.2 success criterion 1.4.1 Use of Color [45], so a child who does not see the colour still sees the missed part.
Fit criterion: For each predefined difficulty, and for a passing and a non-passing attempt, show a prepared drawing and inspect the feedback. Pass when every feedback includes an encouraging cue from the approved feedback list (FR-LRN-014), none uses an element on the error-styling list in this requirement, and every missed part is marked by an outline, pattern or icon as well as by colour.
Priority: Must
Phase: PH-1
Sources: G.1; SRC-005.06; SRC-009.02
Traceability: FR-LRN-014; FR-LRN-022; UC-003; H-AHM-005; H-MOU-002; H-MOU-003

NFR-UX-005 — Consistent look across child screens

Requirement: The same action shall use the same icon, colour and screen position on every child-facing screen.
Rationale: Children stay motivated when activities are predictable (SRC-009.02), and a child who navigates by images relies on each icon meaning the same thing everywhere (SRC-005.05). This follows WCAG 2.2 success criteria 3.2.3 Consistent Navigation and 3.2.4 Consistent Identification [45].
Fit criterion: List every action available on child-facing screens and inspect each screen where it appears. Pass when each action has one icon, one colour and one position across all screens.
Priority: Should
Phase: PH-2
Sources: SRC-005.05; SRC-009.02
Traceability: UC-003; UC-006; NFR-UX-002

NFR-UX-006 — Appealing but calm style

Requirement: Child-facing screens shall have a style that adult reviewers judge both appealing to children and calm.
Rationale: Specialists warned that the design may not be appealing enough for children, and that the right balance is a playful design in a calm, minimally stimulating environment (SRC-009.05).
Fit criterion: Show at least `UX_REVIEWERS` adult reviewers every child-facing screen and ask them to rate how appealing to children and how calm each is, on a five-point scale. Pass when at least `UX_PASS_RATE` of them answer 4 or 5 on each rating.
Priority: Should
Phase: PH-3
Sources: SRC-009.05
Traceability: NFR-UX-001; NFR-UX-007; H-MOU-001

NFR-UX-007 — Style reviewed with educators or therapists

Requirement: The colours, images and feedback style of child-facing screens shall be reviewed by at least `UX_REVIEWERS` adults, including at least one educator or occupational therapist, and each change they request shall be made or recorded with a reason.
Rationale: Specialists advised designing feedback wording and colours with educators and occupational therapists, built up bit by bit (SRC-005.06), and

involving them throughout design (SRC-009.06). None has been consulted yet (FACT-012).
Fit criterion: Pass when the review record lists the reviewers and their roles, includes at least one educator or occupational therapist, and states for every requested change either that it was made or the reason it was not.
Priority: Should
Phase: PH-3
Sources: SRC-005.06; SRC-009.06; FACT-012
Traceability: NFR-UX-004; NFR-UX-006; H-MOU-001; H-MOU-002

11 Usability and Humanity Requirements

This section states how easy Squigglr is to use, personalize and learn, how its wording treats children and adults, and which accessibility criteria it meets. As in Section 10, fit criteria are checked by the team or by adult reviewers, and an adult review does not show that children can use the product (SRC-005.11). The adult-set values for each child are specified in FR-LRN-017 and SAF-RAY-008 and are not repeated here.

11.1 Ease of Use Requirements

NFR-UX-008 — Practice without adult help

Requirement: Once an adult has started a lesson, Squigglr shall need no adult action through every stage and retry until the lesson ends, other than a response to a suggestion to go back a stage.
Rationale: Children should be able to practise without constant adult help, and parents quickly become annoyed if they must keep intervening (SRC-005.05). Within a lesson the child does not navigate; Squigglr moves between stages (CA-001, FR-LRN-023).
Fit criterion: Walk through every catalogue lesson with prepared passing and non-passing attempts and count the adult actions between the start and the end of the lesson. Pass when the only adult actions are responses to suggestions.
Priority: Must
Phase: PH-1
Sources: G.1, G.3; SRC-005.05

Traceability: UC-003; UC-005; FR-LRN-015; FR-LRN-023

NFR-UX-009 — Quick setup by a new adult

Requirement: An adult using Squigglr for the first time shall be able to create a child's profile and start a lesson for that child within `SETUP_TIME_MAX` minutes, with no help from outside Squigglr.
Rationale: Adults handle setup (SRC-005.05), and they are everyday web users rather than trained staff (Section 2). Setup that needs help delays practice.
Fit criterion: Give at least `UX_REVIEWERS` adults who have not used Squigglr the task "add a child and start a lesson for them". Pass when at least `UX_PASS_RATE` of them finish within `SETUP_TIME_MAX` minutes without help.
Priority: Should
Phase: PH-2
Sources: G.3; SRC-005.05
Traceability: UC-001; UC-002; NFR-UX-014

11.2 Personalization and Internationalization Requirements

NFR-UX-010 — Audio setting for each child

Requirement: The supporting adult shall be able to turn audio feedback on or off for each child, and the choice shall apply to every child-facing screen for that child.
Rationale: Audio feedback is useful, but users must be able to turn it off (SRC-005.07), and an adult can turn it off if sound upsets a child (ASM-015). The setting is per child because children's needs differ (FACT-003).
Fit criterion: Turn audio off for one prepared profile and on for another. Run a lesson and the creative space for each. Pass when no sound plays for the first profile and audio feedback plays for the second.
Priority: Must
Phase: PH-2
Sources: FACT-003; ASM-015; SRC-005.07
Traceability: UC-010; NFR-UX-021; NFR-UX-022; H-MOU-001

NFR-UX-011 — Refinement setting for each child

Requirement: The supporting adult shall be able to turn creative space refinement on or off for each child; while it is off, the child's strokes shall be kept exactly as drawn.
Rationale: Both questionnaire respondents suggested that refinement be optional or adjustable, and unclear lines can matter to a child's imagination (SRC-009.01). The child can also turn refinement off in the creative space (UC-007).
Fit criterion: Turn refinement off for a prepared profile and draw prepared drawings in the creative space. Pass when every saved stroke matches the input exactly; turn it on and verify the drawings are refined.
Priority: Should
Phase: PH-2
Sources: G.4; SRC-009.01
Traceability: UC-007; UC-010; H-MOU-006

NFR-UX-012 — Language-free child screens, one adult language

Requirement: Child-facing screens shall need no text to be used, and all text in adult views shall be in `INTERFACE_LANGUAGE`.
Rationale: Children do not need to read to use the child screens, while adult tasks such as reviewing records need text (ASM-007). Keeping language out of the child screens means another adult language would not change them (NFR-UX-002).
Fit criterion: Inspect every child-facing screen as in the fit criterion of NFR-UX-002, and every adult view. Pass when no child task needs text and all adult-view text is in `INTERFACE_LANGUAGE`.
Priority: Must
Phase: PH-2
Sources: ASM-007; SRC-005.05
Traceability: NFR-UX-002

11.3 Learning Requirements

This subsection covers learning to use Squigglr. Learning to draw is specified by the FR-LRN requirements.

NFR-UX-013 — Learning a lesson from its demonstration

Requirement: The demonstration of each lesson shall show, without words, where the drawing starts and which way it goes.
Rationale: Many learners cannot read and learn by watching (SRC-005.05), so the demonstration must show the task without an adult's explanation. Whether children learn from it can be checked only in a study with ethics approval, which is out of scope (SRC-005.11). The team's proposed demonstration shows where to start and which way to go (Moustafa, 6:03); the demonstration itself is FR-LRN-003.
Fit criterion: Show at least `UX_REVIEWERS` adults the demonstration of every catalogue lesson with sound and text hidden. Pass when at least `UX_PASS_RATE` of them mark the correct start point and direction for every lesson.
Priority: Must
Phase: PH-2
Sources: G.1; SRC-005.05
Traceability: FR-LRN-003; UC-003

NFR-UX-014 — Adult tasks learnable without training

Requirement: An adult shall be able to complete each adult task (creating a profile, choosing a lesson, responding to a suggestion, changing settings and reviewing progress) without training, using only help shown in Squigglr.
Rationale: Adults are not assumed to have training in visual-motor development or in Squigglr (Section 2), and the specialists expect adults to handle these tasks (SRC-005.05).
Fit criterion: Give at least `UX_REVIEWERS` first-time adults each task. Pass when at least `UX_PASS_RATE` of them complete every task without help from outside Squigglr.
Priority: Should
Phase: PH-3
Sources: G.3, G.6; SRC-005.05
Traceability: UC-001; UC-002; UC-005; UC-009; UC-010; NFR-UX-009

11.4 Understandability and Politeness Requirements

NFR-UX-015 — Short, encouraging messages to the child

Requirement: Every message to the child, including feedback on an attempt and the end of a lesson, shall be encouraging, shall describe the drawing

rather than the child, and shall be at most `FEEDBACK_WORDS_MAX` words when spoken.
Rationale: Feedback wording should be positive (SRC-005.06), and children become frustrated when their drawing is called wrong (SRC-009.02). Feedback describes the drawing, not the child (BR-006, FR-LRN-022). A lesson that ends at its limit must not feel like a failure (SAF-RAY-005). Messages are short because they are heard by children who may not read.
Fit criterion: List every approved feedback message (FR-LRN-014) and the lesson-end message. Pass when each has at most `FEEDBACK_WORDS_MAX` words, refers to the drawing and not the child, and uses none of the words "wrong", "bad", "fail" or "mistake".
Priority: Must
Phase: PH-1
Sources: G.1; BR-006; SRC-005.06; SRC-009.02
Traceability: FR-LRN-014; FR-LRN-022; NFR-UX-004; UC-003; H-AHM-005; H-MOU-002; H-MOU-003

NFR-UX-016 — Effort acknowledged

Requirement: The feedback on every evaluable attempt, including one that does not pass, shall acknowledge the child's effort before it shows what to change.
Rationale: Squigglr should recognize effort and progress, not only accuracy (SRC-009.02).
Fit criterion: For each predefined difficulty, show a prepared non-passing attempt. Pass when every feedback message begins with an acknowledgement from the approved feedback list.
Priority: Should
Phase: PH-2
Sources: SRC-009.02
Traceability: NFR-UX-015; FR-LRN-014; UC-003; H-MOU-002

NFR-UX-017 — Plain wording in adult views

Requirement: Adult views shall use plain language, shall explain each glossary term where it first appears in a view, and shall not use diagnostic, clinical or ranking words.
Rationale: The TA asked that terms such as "guidance level" be defined rather than left as jargon (SRC-006). Records describe practice only and are

not an assessment (BR-006, SAF-RAY-004), and adults are not assumed to have clinical training (Section 2).
Fit criterion: Inspect every adult view against the glossary (Section 4.1) and the words "diagnosis", "disorder", "delay", "score", "rank" and "below average". Pass when every glossary term used is explained in the view and none of the listed words appears.
Priority: Must
Phase: PH-2
Sources: BR-006; SRC-006
Traceability: SAF-RAY-004; UC-009; UC-010; H-MOU-005

11.5 Accessibility Requirements

These requirements take specific WCAG 2.2 success criteria [45] that the team can check without children, as the specialists and the supervisor advised (SRC-005.08, SRC-008); full WCAG conformance is not required. Together with NFR-UX-001, NFR-UX-004 and NFR-UX-005, they are the criteria Squigglr follows (OI-08). The stylus is the main input (CON-003); finger input is a stretch goal and is not required here.

NFR-UX-018 — Large touch targets

Requirement: Every control on a child-facing screen shall have a touch target of at least `TARGET_SIZE_MIN`.
Rationale: Children's grip and fine motor control vary (SRC-005.04), and the supervisor named large icons as a WCAG idea to adopt (SRC-008). The size follows WCAG 2.2 success criterion 2.5.5 Target Size (Enhanced) [45].
Fit criterion: Measure every control on every child-facing screen at the smallest supported screen size. Pass when every touch target is at least `TARGET_SIZE_MIN`.
Priority: Must
Phase: PH-2
Sources: SRC-005.04; SRC-008
Traceability: NFR-UX-002; UC-003; UC-006

NFR-UX-019 — Colour contrast

Requirement: On every screen, text shall have a contrast ratio of at least 4.5:1 with its background, and icons and graphics needed to use a control at least 3:1.

Rationale: Some children have sight challenges (SRC-005.09). The ratios follow WCAG 2.2 success criteria 1.4.3 Contrast (Minimum) and 1.4.11 Non-text Contrast [45].
Fit criterion: Check every screen with a contrast checker. Pass when every piece of text and every control graphic meets its ratio.
Priority: Must
Phase: PH-2
Sources: SRC-005.08; SRC-005.09
Traceability: NFR-UX-001; NFR-UX-004

NFR-UX-020 — Text alternatives for icons

Requirement: Every icon or image that acts as a control shall have a text alternative that assistive technology can read.
Rationale: An adult who uses a screen reader must still be able to set up and review practice, and designing for disability serves many groups (SRC-005.09). This follows WCAG 2.2 success criterion 1.1.1 Non-text Content [45]. The child does not need the text (NFR-UX-002).
Fit criterion: Run an automated accessibility checker on every screen and inspect each control. Pass when every control has a text alternative that names its action.
Priority: Should
Phase: PH-2
Sources: SRC-005.08; SRC-005.09
Traceability: NFR-UX-002; NFR-UX-012

NFR-UX-021 — Feedback not by sound alone

Requirement: Every message to the child shall be shown on screen and, unless audio is turned off, also played as sound; no message shall rely on sound alone.
Rationale: Offering more than one modality is encouraged in inclusive design, and the user can turn the audio off (SRC-005.07), so turning it off must not hide any feedback (NFR-UX-010). This follows WCAG 2.2 success criterion 1.3.3 Sensory Characteristics [45].
Fit criterion: With audio on and then off, show every approved feedback message and the lesson-end message. Pass when each appears on screen both times, and sound plays only when audio is on.
Priority: Must
Phase: PH-2

Sources: SRC-005.07
Traceability: NFR-UX-010; NFR-UX-012; NFR-UX-015; UC-003; H-MOU-003

NFR-UX-022 — Control of audio

Requirement: Any sound that plays automatically for more than 3 seconds shall have a control to pause or stop it.
Rationale: Too much auditory stimulation can overwhelm children (SRC-009.05). This follows WCAG 2.2 success criterion 1.4.2 Audio Control [45].
Fit criterion: List every sound Squigglr plays. Pass when each sound longer than 3 seconds can be paused or stopped from the screen it plays on.
Priority: Should
Phase: PH-2
Sources: SRC-005.08; SRC-009.05
Traceability: NFR-UX-010; H-MOU-001

NFR-UX-023 — Keyboard use of adult views

Requirement: Every function in adult views shall be usable with a keyboard alone.
Rationale: Adults may review records on a computer (Section 2), and some cannot use a pointer. This follows WCAG 2.2 success criterion 2.1.1 Keyboard [45]. Drawing on child-facing screens is excluded because it depends on the stylus path (CON-003).
Fit criterion: Complete every adult task in UC-001, UC-002, UC-005, UC-009 and UC-010 using only a keyboard. Pass when every task can be completed.
Priority: Should
Phase: PH-3
Sources: SRC-005.08; SRC-005.09
Traceability: NFR-UX-014

NFR-UX-024 — Resizable text in adult views

Requirement: Text in adult views shall be resizable to 200 percent without loss of content or function.
Rationale: Some adults have sight challenges (SRC-005.09). This follows WCAG 2.2 success criterion 1.4.4 Resize Text [45].
Fit criterion: Zoom the browser to 200 percent on every adult view. Pass when all text is visible and every function still works.

Priority: Could
Phase: PH-3
Sources: SRC-005.08; SRC-005.09
Traceability: NFR-UX-019

12 Performance Requirements

This section specifies performance and related quality requirements, including speed, safety, precision, robustness, capacity, scalability and longevity.

12.1 Speed and Latency Requirements

NFR-PERF-001 — Stroke feedback latency

Requirement: Squigglr shall display visual acknowledgement of an inputted stroke within `FEEDBACK_LATENCY_MAX` milliseconds of the stroke's final input point being received.
Rationale: Delay between stroke input and visual stroke generation creates a poor experience in general and confuses children with learning difficulties in particular.
Fit criterion: Pass when the time passed from the final input point to the first visible stroke render in each lesson stage is within `FEEDBACK_LATENCY_MAX`.
Priority: Must
Phase: PH-2
Sources: G.1; G.2
Traceability: FR-LRN-007; UC-003

NFR-PERF-002 — AI refinement response latency

Requirement: Squigglr shall display the refined result within `REFINE_LATENCY_MAX` seconds of the drawing being finished when AI refinement is included and the canvas remains open for interaction during that time.
Rationale: Long, unresponsive pauses break engagement and may be interpreted as an error (G.4).
Fit criterion: Pass when the time passed from the drawing being finished to the refined result display is within `REFINE_LATENCY_MAX` seconds and the canvas accepts new input without any blocking or error during that period.

Priority: Should
Phase: PH-2
Sources: G.4
Traceability: UC-007; NFR-UX-011

NFR-PERF-003 — Progress summary page load

Requirement: Squigglr shall display the summary page within `SUMMARY_LOAD_MAX` seconds of an adult request.
Rationale: Adults reviewing records expect performance similar to everyday web tools (G.6);
Fit criterion: Pass when time from request to fully visible summary is within `SUMMARY_LOAD_MAX` seconds for ten consecutive attempts.
Priority: Must
Phase: PH-2
Sources: G.5; G.6
Traceability: UC-009

12.2 Safety-Critical Requirements

This subsection holds safety requirements that come from the Hazard Analysis and do not fit another section. Each lists the hazards it mitigates.

SAF-RAY-001 Squigglr shall allow lesson choice, starting stage choice, learner switching and acceptance of a go-back suggestion only after an adult check that a child cannot complete by tapping icons, and shall keep a pending suggestion, without blocking practice, until an adult accepts or declines it, or a pass or the end of the lesson clears it.
Rationale: children should be able to practise without constant adult help (SRC-005, P1, 34:13), but progression belongs to adults (CON-007; SRC-005, P1, 28:37). The form of the check is `ADULT_CHECK`, TBD.
Fit criterion: in prepared walkthroughs, no adult-only action completes without passing the adult check, and a pending suggestion changes nothing until an adult acts.
Hazards: H-RAY-001, H-RAY-006.

SAF-RAY-002 Before a lesson starts, Squigglr shall check the device and browser against `SUPPORTED_DEVICES` (TBD) and, if they do not match,

show an adult-readable notice that results may be unreliable. When the first stroke of a lesson is drawn with a finger rather than a stylus, Squigglr shall show the adult the same kind of notice.
Rationale: the team does not control the device (CON-001–CON-003). A browser cannot tell whether a stylus is connected before it is used; it can only report the input type (pen, finger or mouse) of each stroke as it is drawn, so stylus use is checked at the first stroke (ASM-003).
Fit criterion: on each prepared unsupported combination the notice appears before the first attempt; on supported ones it does not. When the first stroke is drawn with a finger, the notice appears; when it is drawn with a stylus, it does not.
Hazards: H-RAY-002.

SAF-RAY-003 Squigglr shall keep each completed attempt on the device until the server confirms it is saved, and shall tell the adult which attempts could not be saved.
Rationale: offline access is not promised (SRC-005, Moustafa, 6:03), and G.5 needs complete records.
Fit criterion: with the connection cut during or after prepared attempts, no confirmed attempt is lost and every unsaved attempt is reported to the adult.
Hazards: H-RAY-002.

SAF-RAY-004 No record, summary or export shall show a diagnostic label, a judgement of the child's abilities or a comparison with other children, and every summary and export shall state that it describes practice only and is not an assessment or diagnosis.
Rationale: Squigglr makes no diagnosis (SRC-001; SRC-005, Moustafa, 6:03), and adults reading records may have no training.
Fit criterion: inspection of every summary and export template finds none of the excluded content and finds the statement present.
Hazards: H-RAY-003, H-MOU-005.

SAF-RAY-005 Squigglr shall end a lesson at `ATTEMPT_LIMIT`, or at `ATTEMPT_LIMIT_MAX` when extended, without any action from the child, and shall keep the progress saved up to that point.
Rationale: the child is not expected to exit a lesson, and no adult may be watching (CA-001), so repeated failure must end on its own (G.3).
Fit criterion: in prepared sequences of failed attempts, each lesson ends

at the limit in use with no child input, and the saved record matches the attempts made.
Hazards: H-RAY-004.

SAF-RAY-006 Squigglr shall record each lesson that ends at `ATTEMPT_LIMIT` or `ATTEMPT_LIMIT_MAX`, and each improvement extension, show the adult how many times these happened for each lesson, and keep earlier attempts when the same lesson is started again.
Rationale: each run of a lesson is a new lesson record (FR-REC-006); starting the lesson again could otherwise hide repeated failure from the adult.
Fit criterion: in prepared sequences that reach the limit and are then started again, the count and all earlier attempts appear in the adult view.
Hazards: H-RAY-005.

SAF-RAY-007 Squigglr shall identify the active learner on every practice screen in a form a child who cannot read can recognize, and shall save every attempt and suggestion to that learner only.
Rationale: school devices may be shared by several children (SRC-004).
Fit criterion: in prepared multi-learner sessions on one device, every attempt and suggestion is saved under the learner shown when it was made.
Hazards: H-RAY-006.

SAF-RAY-008 Squigglr shall save a child's adult-set values (Table 10) only within their team-set ranges, with `ATTEMPT_LIMIT` larger than `DIFFICULTY_STREAK` and `ATTEMPT_LIMIT_MAX` at least `ATTEMPT_LIMIT`; shall use the middle of each range when the adult does not choose; shall allow changes only after the adult check; and shall show the values in use alongside the child's records.
Rationale: the supervisor advised that the adult chooses these values (SRC-008); a wrong value can stop a child passing a stage they can draw, and a child must not change it.
Fit criterion: in prepared walkthroughs, every out-of-range value and every `ATTEMPT_LIMIT` not larger than `DIFFICULTY_STREAK` and every `ATTEMPT_LIMIT_MAX` below `ATTEMPT_LIMIT` is refused, an unset value equals the midpoint, and no change completes without the adult check.
Hazards: H-RAY-007.

SAF-RAY-009 During a guided lesson, when there is no input for `LESSON_IDLE_PROMPT`, Squigglr shall show a check-in that a child who cannot read can understand, without flashing (SAF-MOU-003), that any touch dismisses. When there is no input for `LESSON_IDLE_END`, Squigglr shall end the lesson, keep the progress saved up to that point, and record that the lesson ended for inactivity. Time without input shall not count as an attempt or toward `DIFFICULTY_STREAK` or `ATTEMPT_LIMIT`.
Rationale: a child may walk away during a lesson, and no adult may be watching (CA-001). On a shared device (CA-004), another child could then continue under the first child's profile. Ending an idle lesson before `IDLE_TIMEOUT` logs the adult out (SAF-SHA-006) gives the lesson a clear ending in the record. Some children pause often, so the adult sets both times for each child (BR-008).
Fit criterion: in prepared lessons with no input, the check-in appears at `LESSON_IDLE_PROMPT`, a touch dismisses it and the lesson continues, and with no further input the lesson ends at `LESSON_IDLE_END` with its progress saved and the ending recorded as inactivity; no attempt is counted for the idle time.
Hazards: H-RAY-006.

SAF-AHM-001 Squigglr shall prevent duplicate reports of the same stylus input event from being counted more than once during evaluation or recording of an attempt, while preserving legitimate separately drawn strokes, including overlapping strokes.
Rationale: duplicate input events can inflate drawing coverage, produce incorrect pass or failure decisions, and cause saved records to differ from what the learner actually drew (H-AHM-002; BR-004).
Fit criterion: replay a known one-stroke or multi-stroke input sequence with and without duplicated input events; verify that the recorded points, classification and pass result are identical. Replay two separate deliberately overlapping strokes and verify that both are preserved.
Related SRS requirements: FR-LRN-007, FR-LRN-008.
Hazards: H-AHM-002.

SAF-AHM-002 Squigglr shall preserve the relative positions and order of stylus points within the active drawing surface when the canvas size or supported device orientation changes, and shall reject input whose coordinates cannot be mapped reliably to that surface.

Rationale: an incorrect coordinate mapping can distort a learner's drawing, produce an inaccurate passing decision, and corrupt stored attempts (H-AHM-003; BR-004).
Fit criterion: replay the same prepared stroke sequences before and after resizing the drawing surface or changing supported device orientation; verify equivalent relative point positions, stroke order and pass classifications. Supply invalid or unmappable coordinate values and verify that none is classified as passing or counted as an evaluable attempt.
Related SRS requirements: FR-LRN-007, FR-LRN-010.
Hazards: H-AHM-003.

SAF-SHA-001 Squigglr shall authenticate data at every endpoint where it is accessed.
Rationale: Squigglr is a website with no installation (CON-001, CON-002) so access must be authenticated at server endpoints.
Fit criterion: with mock accounts, while logged in to one account, attempt to access a wide range of datatypes from another account using server commands; passes when attempts to access that accounts information are rejected.
Hazards: H-SHA-001.

SAF-SHA-002 Squigglr shall confirm that all essential third-party services do not transmit PII through their public interface.
Rationale: refinement services may receive raw private data; user data is protected by privacy commitments; third-party services are implemented as web dependencies (CON-001) which may transmit user data incongruously.
Fit criterion: confirm that network traffic for each third-party contains no PII data transmitted to a non-approved destination.
Hazards: H-SHA-002.

SAF-SHA-003 Squigglr shall report a save as successful only after the server confirms it, and shall inform the adult when a save fails.
Rationale: G.5 requires complete, meaningful activity records; Adults want to know which sessions contributed to their child's progress report in detail.
Fit criterion: for each of `FAILED_SAVES` (TBD) mock failed save requests, Squigglr does not report the attempt as saved and notifies the

adult; for each of `FAILED_SAVES` mock successful saves, Squigglr reports no failed save and does not notify the adult.
Hazards: H-SHA-003.

SAF-SHA-006 Squigglr shall log out the user after `IDLE_TIMEOUT` (TBD) of inactivity.
Rationale: Personal data, especially that of children's, is confidential. Unauthorized access to a user account or isolated user information violates privacy and undermines user trust; Logging out an account after a period of inactivity is a standard safety feature.
Fit criterion: passes when mock account is logged out after `IDLE_TIMEOUT`.
Hazards: H-SHA-001.

SAF-MOU-001 In guided lessons, feedback on a missed part shall show the child where the part is missing without drawing it for them.
Rationale: the aim of practice is drawing with gradually less help (G.2); if Squigglr completes the drawing, the child can come to rely on it (H-MOU-004). The child's actual strokes are what is recorded (BR-004).
Fit criterion: for each predefined difficulty, show a prepared attempt with a missed part and compare the strokes on screen and in the record before and after the feedback. Pass when the feedback adds no stroke to the attempt.
Related SRS requirements: FR-LRN-014, FR-LRN-023, BR-004.
Hazards: H-MOU-004.

SAF-MOU-002 Records and progress summaries shall keep attempts made with a guide separate from attempts made without one, shall not include creative space drawings, and shall not present the number of attempts as a measure of the child's skill.
Rationale: a summary that blurs assisted and independent attempts, counts refined drawings as practice or treats fewer attempts as better skill can lead an adult to move the child on too early or hold them back (H-MOU-005; SRC-005, Moustafa, 6:03). Refinement must never reach the records adults rely on (BR-005).
Fit criterion: for prepared histories with known assisted, independent and creative space work, check every summary and record. Pass when assisted and independent attempts are shown separately, no creative space drawing appears, and no view ranks or rates the child by the

number of attempts.
Related SRS requirements: G.6, FR-LRN-012, SAF-RAY-004, NFR-UX-017.
Hazards: H-MOU-005.

SAF-MOU-003 No content on any child-facing screen shall flash more than three times in any one-second period.
Rationale: flashing can trigger seizures in rare cases, the only physical harm in the hazard analysis (H-MOU-001). NFR-UX-001 bans flashing while the child is drawing, but feedback, the lesson-end message and screen transitions also appear on child-facing screens. This follows WCAG 2.2 success criterion 2.3.1 Three Flashes or Below Threshold [45].
Fit criterion: record every child-facing screen, including all feedback, the lesson-end message and every transition, and count the flashes in each one-second window. Pass when no window has more than three flashes.
Related SRS requirements: NFR-UX-001.
Hazards: H-MOU-001.

12.3 Precision or Accuracy Requirements

NFR-PERF-004 — Stage result scoring accuracy

Requirement: Squigglr shall compute a deterministic stage result score for all attempts: the same strokes and stage criteria shall produce the same evaluation.
Rationale: Adults use stage results to monitor a child's progress (G.5, G.6). Nondeterministic scoring may produce arbitrary or difficult-to- interpret results that are unsuitable for progress tracking. Deterministic scoring is also easier to implement and non-deterministic scoring has no clear benefits.
Fit criterion: Pass when identical programmed strokes on the same stage, replayed by an automated test script, produce identical scores for ten consecutive runs.
Priority: Must
Phase: PH-2
Sources: G.2; G.5; G.6
Traceability: FR-LRN-007; FR-LRN-008

NFR-PERF-010 — Stage result and completeness agree with their definition

Requirement: For every evaluable attempt, Squigglr's stage result shall differ by at most 1 point from the value defined in Section 9.2, and its complete or incomplete classification shall equal the one defined there.
Rationale: A threshold of 70 or 90 only means something if the result measures what the SRS says it does. Determinism (NFR-PERF-004) alone would allow a result that is repeatable but wrong (G.2).
Fit criterion: For prepared attempts at every stage of every catalogue lesson, including attempts with a missing part, a missed endpoint and results just above and below `STAGE_PASS`, compare Squigglr's stage result and classification with those computed by a separate reference script, written from Section 9.2 by a team member who did not write the evaluation code. Pass when every result is within 1 point and every classification matches.
Priority: Must
Phase: PH-2
Sources: G.2; BR-002
Traceability: FR-LRN-011; FR-LRN-013; NFR-PERF-004; H-AHM-004

12.4 Robustness or Fault-Tolerance Requirements

NFR-PERF-005 — Submit attempt on network loss and reconnection

Requirement: Squigglr shall submit each completed attempt not yet confirmed saved by the server but saved locally so it can be submitted after reconnection.
Rationale: A network interruption may outlast the application session; Squigglr can automatically submit the finished work for evaluation without asking the child to repeat it or submit it manually.
Fit criterion: Pass when an attempt completed offline is successfully submitted after connectivity is restored three consecutive times.
Priority: Must
Phase: PH-2
Sources: G.5
Traceability: FR-REC-009; SAF-RAY-003; UC-003

12.5 Capacity Requirements

NFR-PERF-008 — Concurrent session capacity

Requirement: Squigglr shall support a concurrent session target of `CONCURRENT_CAPACITY` (TBD; set by measurement in the VnV Plan) simultaneous active users without degrading response times below the thresholds stated in Section 12.1 (NFR-PERF-001, NFR-PERF-002, and NFR-PERF-003).
Rationale: The anticipated system deployment is to a small number of schools or households. The exact session target is deferred to the VnV Plan.
Fit criterion: Pass when the system sustains the latency thresholds outlined in Section 12.1 under the agreed concurrent load without error or failure.
Priority: Should
Phase: PH-3
Sources: G.1; G.5
Traceability: NFR-PERF-001; NFR-PERF-002; NFR-PERF-003

12.6 Scalability or Extensibility Requirements

Adding or removing lessons from the Squigglr lesson catalogue requires no change to the lesson execution logic or any other module's source code. Lesson content is stored as data and read at runtime (LC-001, Section 27). This is a design-level decision; no standalone NFR is required here because the constraint is already captured by NFR-MAINT-001.

12.7 Longevity Requirements

NFR-PERF-009 — Operational lifetime through capstone

Requirement: Squigglr shall remain fully functional on the supported browsers and devices (OI-09) through PH-3, the end of the capstone project period.
Rationale: After final evaluation and the exposition, post capstone maintenance is outside the project scope.
Fit criterion: Pass when Squigglr can be launched and all features exercised on the supported browsers and devices as listed in OI-09 at the PH-3 milestone; any browser or device that fails this check shall be removed from the supported list and OI-09 updated accordingly.
Priority: Must
Phase: PH-3

Sources: G.1; G.2; G.3
Traceability: CON-004; OI-09; NFR-OPS-003

13 Operational and Environmental Requirements

This section describes the environments, external systems, productization and release conditions under which Squigglr must operate.

13.1 Expected Physical Environment

Squigglr is meant for use on a touchscreen tablet or laptop (CON-003). A stylus is required for interaction with motor practice features. Finger input is a stretch goal (LC-006). Expected settings include the domicile, classroom, and therapy centres where children draw often with an adult nearby. The device is expected to be placed on a flat surface or held in the user's hands.

13.2 Wider Environment Requirements

NFR-OPS-001 — HTTPS-only communication

Requirement: Squigglr shall exchange all data between the browser client and server under encrypted HTTPS protocol.
Rationale: Account data travels over networks that the Squigglr team does not control (ASM-005). HTTPS prevents unauthorized access and tampering in transit. School and other facility networks may also block non-HTTPS traffic.
Fit criterion: Confirm that every request made by Squigglr uses the `https://` scheme. Verify that navigating to the `http://` URL either redirects to HTTPS or is blocked.
Priority: Must
Phase: PH-1
Sources: ASM-005
Traceability: NFR-SEC-001; NFR-SEC-004; UC-001

13.3 Requirements for Interfacing with Adjacent Systems

NFR-OPS-002 — Adjacent system interfaces

Requirement: Squigglr shall interact with the following external systems:

Authentication and session provider. Squigglr shall delegate account creation, login, and session management to a third-party authentication provider.

Refinement service. When refinement is on, Squigglr shall send the child's stroke data, without child PII (SAF-SHA-002), to the service that refines creative drawings and shall receive refined stroke data in return.

Hosting and data storage service. Squigglr shall be served, and its records kept, by a managed hosting and data storage service. For authentication and hosting, Squigglr shall display an error message in the event of provider unavailability until the availability returns.

Rationale: The adjacent systems are described by their role, not by vendor, so that a provider can be replaced without changing the requirements (LC-004). The candidate providers are compared in the Off-the-Shelf Solutions section.
Fit criterion: Confirm that the error message displays for a mock outage of the authentication and hosting services. (For AI refinement, provider unavailability protocol is described in FR-CRT-020)
Priority: Must
Phase: PH-2
Sources: ASM-005
Traceability: ASM-005; FR-CRT-020; UC-001

13.4 Productization Requirements

NFR-OPS-003 — No installation required

Requirement: Squigglr shall require no download, plugin, or browser extension and shall be usable directly from a supported browser.

Rationale: Installing applications on centrally administered device ecosystems is difficult or impossible without administrator approval (CON-001, CON-002) while a website URL is much easier to access.
Fit criterion: Pass when lesson and account features on Squigglr can be opened on each supported browser and device combination (OI-09) without installation
. *Priority:* Must
Phase: PH-1
Sources: CON-001; CON-002; SRC-003; SRC-004
Traceability: CON-001; CON-002; OI-09;

13.5 Release Requirements

Squigglr has no installation package. A release is a deployment to the hosting service, triggered by merging to the production branch.

14 Maintainability and Support Requirements

This section sets requirements for maintaining, supporting and adapting Squigglr and its lesson catalogue.

14.1 Maintenance Requirements

NFR-MAINT-001 — Lesson catalogue update without code change
Requirement: Squigglr shall require no change to the source code to add, remove, or edit a lesson from the lesson catalogue; all lessons shall be stored as data read at runtime.
Rationale: Lessons will need to be updated and the catalogue will need to be added to as the user base grows and the development team learns from previous experience. An extensible catalogue should be data-driven to make lessons easier to upload and eventually allow content creation to be offloaded to non-code teams.
Fit criterion: Confirm that lessons are stored as data objects and that the application renders them at runtime from these objects. Successfully upload, delete, and edit a lesson in a developer view.
Priority: Must

Phase: PH-2
Sources: LC-001
Traceability: FR-LRN-001; FR-LRN-002

NFR-MAINT-002 — Programmer locatability of stage behaviour
Requirement: A team member unfamiliar with the codebase shall be able to locate the module responsible for a given stage behaviour within `LOCATE_TIME_MAX` minutes.
Rationale: Future contributions must be easy to implement as the application will need many iterations with a growing user base.
Fit criterion: Passes if a team member unfamiliar with the codebase reads a description of a stage behaviour written by a team member familiar with the codebase and takes less than `LOCATE_TIME_MAX` minutes to find the corresponding module.
Priority: Should
Phase: PH-2
Sources: SRC-002
Traceability: FR-LRN-007; FR-LRN-008

14.2 Supportability Requirements

NFR-MAINT-003 — Module interface descriptions consistent with MG/MIS
Requirement: Each top-level Squigglr module shall have an interface description (exported functions, types, and pre/post-conditions).
Rationale: Defining the contract between modules via the service they provide and those using it improves comprehensibility as different teams work on multiple modules asynchronously.
Fit criterion: Confirm that every issue created for development of a module has a requirement for an interface description.
Priority: Should
Phase: PH-2
Sources: SRC-002
Traceability: NFR-MAINT-002

NFR-MAINT-004 — Consistent coding style

Requirement: Squigglr shall be developed along with a CI pipeline that rejects any commit that introduces a violation of pre-configured linter or formatter rules determined by the team.
Rationale: A consistent style is standard practice and reduces confusion among contributors.
Fit criterion: Confirm that the CI pipeline rejects commits that introduce a linter or formatter violation by testing two different violations.
Priority: Should
Phase: PH-1
Sources: SRC-002
Traceability: NFR-MAINT-002

14.3 Adaptability Requirements

The adaptability of Squigglr is addressed at the design level through the likely changes catalogue (LC-001 through LC-008) in Section 27. Some likely changes are already captured by requirements like LC-001 (stored lesson changes) in NFR-MAINT-001 and LC-004 in NFR-OPS-002 No standalone NFR is required here.

15 Security Requirements

This section specifies requirements for access control, data integrity, privacy, security auditing and protection against input-based attacks.

15.1 Access Requirements

NFR-SEC-001 — Adult account data access

Requirement: Squigglr shall only allow the adult owner of an account to read and write to their respective child's profile.
Rationale: Each adult account holds one or more child learner profiles (BR-001). Allowing unauthorized users to modify or read other childrens accounts violates privacy and undermines trust in the application.
Fit criterion: Using two test accounts, while authenticated as account A attempt to read or modify a child profile and attempt records belonging to

account B via direct server requests. Pass when every such request is rejected and account A's own data is unchanged.
Priority: Must
Phase: PH-2
Sources: G.5; G.6
Traceability: SAF-SHA-001; BR-001; UC-009; UC-010

NFR-SEC-002 — Session token invalidation on deliberate logout

Requirement: Squigglr shall invalidate the corresponding session token of an adult user when they deliberately log out so that a captured or replayed copy of that token cannot be used to access the account afterwards.
Rationale: A token captured during an active session (e.g. from a shared-device browser history or network log) could otherwise remain valid indefinitely (OI-01). Explicit logout is the adult's signal that the device may pass to another user.
Fit criterion: Obtain a valid session token while authenticated, log out deliberately, and attempt to make an authenticated API request by reusing the token. Pass when the request is rejected three times consecutively.
Priority: Must
Phase: PH-2
Sources: G.5; G.6
Traceability: NFR-SEC-001; OI-01

15.2 Integrity Requirements

NFR-SEC-003 — Audit for saved record modification

Requirement: Squigglr shall create a security audit event for each rejected request to modify saved records without storing the record's contents in the event.
Rationale: Saved records must remain unchanged; this requirement adds detection of attempted tampering while keeping child record contents out of security events.
Fit criterion: Pass when three modification attempts against a confirmed saved record are rejected, creating an audit, three consecutive times.
Priority: Should
Phase: PH-2
Sources: G.5; G.6

Traceability: FR-REC-009

15.3 Privacy Requirements

NFR-SEC-004 — Minimal personal data collection

Requirement: Squigglr shall not store any child-identifying information: a learner profile holds only a unique identifier generated by Squigglr and an avatar chosen by the adult or the child.
Rationale: Personal information about children is not necessary for Squigglr to function and minimizing collected data limits security risks. If identifying information was needed, it would make sense to collect the adult's information rather than the child; this supports Squigglr's accountability, consent, and access obligations under PIPEDA [13].
Fit criterion: Passes if there is no field during child registration other than choosing an avatar, and the stored child profile holds only the identifier and the avatar.
Priority: Must
Phase: PH-1
Sources: SRC-004
Traceability: BR-001; NFR-SEC-001

Note: Institutional self-hosting or per-institution data partitioning is out of scope for this project. Account-level access control is already enforced by NFR-SEC-001 and SAF-SHA-001, which is sufficient for a single-instance capstone project like Squigglr.

15.4 Audit Requirements

NFR-SEC-005 — Server-side logging of authentication events

Requirement: Squigglr shall record every failed login and session timeout event in a server-side audit log; the log entries shall not be accessible to adult users through any application interface.
Rationale: Audit logs are useful for security as they provide diagnostics after abnormal behavior like brute-force attacks. Logs are hidden from users because they reveal personal information and users have no legitimate use for them in most cases.

Fit criterion: Trigger ten failed login attempts and ten logouts by inactivity. Confirm that all events are recorded in the audit log with a time stamp and that a mock adult account is denied access to the audit log through request.
Priority: Should
Phase: PH-2
Sources: G.5
Traceability: SAF-SHA-001

15.5 Immunity Requirements

NFR-SEC-006 — Input sanitisation against injection

Requirement: Squigglr shall sanitise all text inputs supplied by a user to prevent injection attacks, including but not limited to SQL injection and cross-site scripting (XSS).
Rationale: Sanitizing user inputs protects against injection attacks.
Fit criterion: Passes if a pre-prepared set of SQL injections and XSS probe strings are sanitized through every text submission field in the application and API text submission endpoint.
Priority: Must
Phase: PH-2
Sources: G.5
Traceability: NFR-SEC-001; NFR-SEC-004

16 Cultural Requirements

This section requires review of symbols and imagery for cultural associations that could negatively affect users.

NFR-CULT-002 — Cultural Sensitivity of Symbols and Imagery

Requirement: Squigglr shall contain no symbols or imagery that carry a known negative cultural meaning in communities likely to use Squigglr.
Rationale: The style review required by NFR-UX-007 should assess such associations in aesthetic choices and clarity, because such associations may not be obvious to the development team; negative symbols can distress a child and create a negative experience for both the child and adult.

Fit criterion: Passes if style review finds no negative cultural associations or if concerns raised by the style review are addressed and approved by the reviewers.
Priority: Must
Phase: PH-3
Sources: G.1
Traceability: NFR-UX-007

17 Compliance Requirements

This section identifies legal and standards obligations that apply to Squigglr, including conditions relevant to adoption.

17.1 Legal Requirements

NFR-COMP-001 — PIPEDA

Requirement: Squigglr shall comply with applicable obligations under PIPEDA [13].
Rationale: PIPEDA applies because Squigglr is a Canadian project hosted for Canadian schools that collects personal information in the course of commercial-like activity.
Fit criterion: A review of every personal-information field Squigglr collects, and of every place it is stored or sent, against each of the ten PIPEDA fair information principles records each principle as met or not applicable, with none unmet.
Priority: Must
Phase: PH-3
Sources: [13]
Traceability: PIPEDA [13]

NFR-COMP-002 — Ontario MFIPPA Applicability

Requirement: Squigglr shall be compliant with the Ontario Freedom of Information and Protection of Privacy Act (MFIPPA) before adoption by any school boards.
Rationale: MFIPPA governs how Ontario local institutions, including school boards, collect, keep and disclose personal information. Adoption by a school

board is outside this project, so the requirement is checked before the end of the project, ready for any later adoption.
Fit criterion: A review of Squigglr's collection, retention and disclosure of personal information against MFIPPA Part II records no unmet obligation.
Priority: Must
Phase: PH-3
Sources: SRC-002
Traceability: NFR-SEC-001; OI-06

Note: GDPR does not apply because EU jurisdiction is out of scope for Squigglr. COPPA does not apply because US jurisdiction is out of scope for Squigglr and child-identifying data is not collected (BR-001).

17.2 Standards Compliance Requirements

NFR-COMP-005 — WCAG 2.2 Success Criteria

Requirement: Squigglr shall meet the selected WCAG 2.2 success criteria [45] encompassed by NFR-UX-001, NFR-UX-004, NFR-UX-005, and NFR-UX-018–NFR-UX-023.
Rationale: The selected criteria follow the specialist recommendation (SRC-005), supervisor advice (SRC-008), and OI-08. Full WCAG conformance is a stretch goal Section 25.
Fit criterion: Confirm that Squigglr is compliant with the listed WCAG 2.2 success criteria with a review team.
Priority: Must
Phase: PH-3
Sources: SRC-005
Traceability: WCAG 2.2 [45]

18 Open Issues

This section lists questions that are not yet settled and could change the requirements. Each issue has an identifier (OI-##), an owner, the decision or evidence needed to close it, the goals, constraints or requirements it affects, and a target date. Requirement identifiers are linked here as the requirements are written. Target dates come from the SRS schedule (review copy Oct. 17,

2026; submission Oct. 19, 2026) or from the course deadlines in Table 8 (CON-004) where an issue cannot be closed before submission.

Table 20: Open issues

ID	Issue and decision or evidence needed	Affects	Owner	Target
OI-01	*School and shared devices.* School devices may be shared by several children (SRC-004) and are often managed centrally (CON-002). One adult account holds several learner profiles, and an adult switches between them (BR-001). Still needed: whether an adult signs in on each shared device, and how a device is left when a session ends.	G.5, G.6, CON-001, CON-002; NFR-SEC-001, NFR-SEC-002, SAF-RAY-007, SAF-SHA-001, SAF-SHA-006	Shamil Canbolat	Oct. 17, 2026
OI-02	*Privacy of practice records.* The supervisor's advice to collect no information about children (SRC-004, #21) concerns personal information; storing practice attempts is acceptable. Still needed: the privacy requirements for attempt records, including who can see them, whether and how they can be shared, and how they are deleted.	G.5, G.6, CON-005; NFR-SEC-001, NFR-SEC-004, SAF-SHA-002, FR-REC-009	Shamil Canbolat	Oct. 17, 2026

ID	Issue and decision or evidence needed	Affects	Owner	Target
OI-03	*Scope of G.4.* A peer reviewer suggested moving creative drawing and refinement to the stretch goals (#70). The team may adopt the change or reply on the issue explaining why G.4 stays a core goal (SRC-006). Still needed: the reply on the issue and, if G.4 stays, a supported set of items small enough for the schedule.	G.4	Muhammad Huzaifah	Oct. 17, 2026
OI-04	*TA confirmation of revised G.3.* The team distinguishes adult selection of lessons, starting stages, skips and backward moves from automatic advancement after a passing attempt within the selected lesson (G.2, BR-002, CON-007). G.3 permits an optional suggestion to go back a stage after `DIFFICULTY_STREAK` failed attempts and a lesson end after `ATTEMPT_LIMIT`, held back while the child is improving (`IMPROVE_WINDOW`, `ATTEMPT_EXTENSION`). The TA agreed to the adult-control revision in principle and asked that the updated PSG be merged after marking the submitted version (SRC-006). Still needed: confirmation of the rule as clarified, not a new choice between automatic and adult-authorized forward progression.	G.3, CON-007	Shamil Canbolat	Oct. 17, 2026

ID	Issue and decision or evidence needed	Affects	Owner	Target
OI-05	*Verification of learning thresholds.* Section 4.2 defines the ranges and defaults for the eight adult-configurable values and the fixed `IMPROVE_MARGIN`. Verify the selected ranges with adult-made drawings, provide examples illustrating the endpoints of the `STAGE_PASS` range, and confirm that the drawing-evaluation method produces results on the specified 0–100 scale (SRC-004, SRC-008). No child testing is permitted.	G.2, G.3; FR-LRN-017, FR-LRN-020, FR-LRN-021	Ahmed Elzaria	Nov. 2, 2026 (V&V Plan)
OI-06	*Data isolation between institutions.* A peer reviewer asked how records from different schools or centres are kept apart (#75). Still needed: a decision on whether Squigglr supports separate institutions in this project and, if so, the access and privacy requirements that keep their records apart.	G.5, G.6; NFR-SEC-001; out-of-scope note after NFR-SEC-004	Shamil Canbolat	Oct. 17, 2026
OI-07	*Budget.* No budget has been set (Budget Constraints), and external services may have costs. Still needed: the TA's confirmation of whether any budget is available.	G.4 (refinement service)	Shamil Canbolat	Oct. 17, 2026

ID	Issue and decision or evidence needed	Affects	Owner	Target
OI-08	*Selected WCAG 2.2 success criteria.* The specialists recommended starting with WCAG (SRC-005, P2, 39:26), and the supervisor advised taking specific WCAG ideas as non-functional requirements rather than as a constraint (SRC-008). The criteria Squigglr follows are now listed in NFR-UX-001, NFR-UX-004, NFR-UX-005 and NFR-UX-018 to NFR-UX-024. Still needed: WCAG 2.2 listed in Standards Compliance Requirements.	G.1; NFR-UX-001, NFR-UX-004, NFR-UX-005, NFR-UX-018 to NFR-UX-024	Shamil Canbolat	Oct. 17, 2026
OI-09	*Supported devices and browsers.* CON-003 leaves the device and browser combinations TBD. Still needed: the list of combinations the team will support and test with a stylus.	CON-001, CON-003	Shamil Canbolat	Nov. 2, 2026 (V&V Plan)

19 Off-the-Shelf Solutions

This section surveys the products and components that already exist and records how closely each matches Squigglr's goals. The survey gives the comparison promised in the User Business subsection and answers two peer-review issues (SRC-007): what existing handwriting, tracing and motor-skill apps do and what gap remains (#69), and which existing models the team might use for refinement (#74). As the Volere template intends, the survey shows that alternatives were considered and how closely they match the needs; it is not a feasibility study [33]. Nothing here is a requirement or a design decision. Every product, library and model named is a candidate the team may evaluate, and the Development Plan leaves the choice of model open until the proof-of-concept work (SRC-002).

The findings are preliminary desk research, done on Oct. 15, 2026. The

team searched the web, Apple's App Store and Google Play for handwriting apps, letter and shape tracing apps (including web-based ones), children's drawing apps, and fine-motor or occupational therapy practice apps, then read each candidate's own store listing, website or documentation. Every statement about a product comes from those sources, cited below. No product was installed or tested, and no child took part (CON-005), so a missing feature means only that the sources read do not describe it. Prices are United States prices on that date, given only where the vendor states them. The specialist interview did not cover existing apps (FACT-013); the follow-up questionnaire (SRC-009) covered refinement in the creative space and letter activities.

Table 21 lists the dimensions used to compare products. Each comes from a goal, constraint or business rule, so the comparison measures fit to this project, not general quality. The Reusable Components subsection adds the Development Plan's criteria for choosing a refinement approach. The data a product collects about a child (BR-001) is not a dimension: vendors describe their data handling in different terms, and Squigglr's own privacy requirements are still open (OI-02). The privacy terms of candidate refinement services are compared in the Reusable Components subsection instead.

Table 21: Dimensions used to compare existing products with Squigglr

Dim.	What Squigglr needs	Source
D1	Graded guidance with a stylus: each lesson on a line, shape or simple drawing moves through stages that give less help, from a demonstration to drawing with no guide.	G.1, G.2, CON-003
D2	Adult control: the adult chooses the lesson and the starting stage, and the child goes back a stage only if an adult accepts.	G.3, CON-007, BR-003
D3	Records of the process: each attempt keeps the child's actual strokes, the stage (the guide used), every predefined difficulty, later corrections and the outcome.	G.5, BR-004
D4	Adult summaries without judgement: attempts made with a guide are shown apart from attempts made without one, with no score, diagnostic label or comparison with other children.	G.6, BR-006
D5	Runs in a web browser with no installation.	CON-001, CON-002
D6	Minimal, editable refinement: in a creative space the child's own strokes are adjusted within `REFINE_TOLERANCE`, stay editable, and the original can be restored.	G.4, BR-005

19.1 Ready-Made Products

The survey kept twelve products in five groups. Together they cover the kinds of product that the search found a supporting adult might use instead of Squigglr, plus the drawing-assistance features closest to refinement. Each product is described below from its vendor's sources, and Table 22 scores it against the dimensions of Table 21.

Handwriting and tracing apps. Kaligo, by Learn & Go, is a handwriting and literacy app for tablets that recommends a stylus, with a web portal for teachers [18, 20]. Teachers assign and create lessons, set the difficulty and visual guidance for each student, view each completed exercise as a replay or screenshot, and follow class progress. The vendor also offers a screening

use in which artificial intelligence (AI) assesses handwriting and fine motor skills to identify students who need support [19], and the App Store listing describes a learning experience that adapts to each child's pace [18]. The app is installed from the App Store, and an internet connection is needed to synchronise lessons and results [20]. The download is free; school prices are not stated on the pages read.

Wet-Dry-Try, from Learning Without Tears, is an iPad app based on the Handwriting Without Tears curriculum. Children practise capitals, numbers and lowercase letters in step-by-step sequences in a developmental order, a virtual coach gives spoken feedback on errors, up to three children can share one device, and the settings offer three sensitivity levels [14]. A classroom licence costs US$3.99 per student per year [21].

LetterSchool takes each letter, number or geometric shape through three games that give less help each time: tap the dots in order, trace the path, then write it from memory; earning a star unlocks the next game [25, 26]. A "Golden" level displays the child's exact writing so that adults can track progress, and progress is stored for up to three players on one device. On Android the first five letters, numbers and shapes are free, and the rest is a one-time purchase [26].

Writing Wizard, by L'Escapadou, traces letters, numbers, words and, for toddlers, shapes. Adults set the letter size, the difficulty and whether the model is shown; a five-star mode asks the child to trace an item five times, each harder than the last; and reports let adults replay and export what the child traced [1]. Installed versions exist for iPad, iPhone, Android, Kindle Fire and Windows [24], and the school edition costs US$13.99 on the App Store [1].

Fine-motor and occupational therapy practice apps. Dexteria, by BinaryLabs, is an iPhone and iPad app for practising fine motor skills and handwriting readiness. Dexteria creates performance and usage reports that can be emailed, its companion app compares performance with reference ranges and produces reports for school documentation, and the vendor states that it is not a medical device; the app costs US$6.99 [8]. Dynamilis, by School Rebound, is an iPad app for children aged five to twelve. Dynamilis analyses handwriting written with an Apple Pencil or Logitech Crayon, gives a handwriting profile and recommends a specialist when the score falls below a threshold, offers personalised practice activities, and has a replay mode for

its analysis; a subscription costs US$14.99 a month or US$89.99 a year [34].

Web-based tracing tools. Tracing Town's Tracing Playground runs in the browser with no account or download and accepts touch, stylus or mouse. A start dot, an arrow and numbered strokes guide each stroke of a shape or letter, the page says in words when the finger leaves the path, nothing is timed or scored, and nothing is recorded or sent from the device [41].

Children's drawing apps. The Crayola Create and Play app, which also has an online version, offers drawing, colouring and educational activities for children aged three to eight, by subscription after a free trial; the vendor states that it has no rules, points or time limits [9]. Tux Paint is a free, open-source drawing program released under version 2 of the GNU General Public License (GPL), installed on Windows, macOS, Linux, Android and other systems [40].

Drawing-assistance features. Google's AutoDraw is a free web tool that guesses what the user is drawing and suggests drawings made by artists [27]; choosing a suggestion replaces the user's sketch [16]. Smart Script, part of iPadOS 18, smooths and straightens text handwritten with an Apple Pencil while keeping the look of the writer's own handwriting [7]. QuickShape, in the Procreate drawing app, snaps a held stroke into a perfect line, arc, ellipse, triangle or quadrilateral that can then be edited through its nodes [31].

Table 22: Existing products scored against the comparison dimensions

Product	Delivery	D1	D2	D3	D4	D5	D6
Kaligo	Tablet app; teacher web portal	Part	Part	Part	No	No	–
Wet-Dry-Try	iPad app	Part	Part	–	–	No	–
LetterSchool	Tablet and phone app	Yes	–	Part	–	No	–
Writing Wizard	Installed app	Part	Part	Part	–	No	–
Dexteria	iPhone and iPad app	–	–	Part	No	No	–
Dynamilis	iPad app	–	–	Part	No	No	–
Tracing Playground	Web page	Part	–	No	No	Yes	–
Crayola Create and Play	App and online version	–	–	–	–	Part	–
Tux Paint	Installed program	–	–	–	–	No	–
AutoDraw	Web tool	–	–	–	–	Yes	No
Smart Script	iPadOS feature	–	–	–	–	No	Part
QuickShape	Drawing-app feature	–	–	–	–	–	Part

Yes: the sources describe the feature. Part: they describe a weaker or different form, explained in the text. No: they describe the opposite, such as an installed app, a score or no records at all. –: the sources read do not describe it, which does not show that the product lacks it.

The "Part" marks in Table 22 have these reasons. For D1, Kaligo lets teachers adjust visual guidance but describes no fixed sequence of stages, Wet-Dry-Try's step-by-step sequences cover only letters and numbers, Writing Wizard's five-star mode is mainly for letters and words, and Tracing Playground has a single guided level. For D2, Kaligo, Wet-Dry-Try and Writing Wizard let adults set difficulty, guidance or sensitivity, but none describes an adult choosing the starting point within a graded sequence, and Kaligo also adapts on its own. For D3, the products that keep anything keep a replay, a screenshot or the final writing (Kaligo, Writing Wizard, Dynamilis, LetterSchool) or usage reports (Dexteria). For D5, Crayola Create and Play has an online version beside its app. For D6, Smart Script refines handwritten text, not drawings, and QuickShape replaces a stroke with a perfect form, which is more than a minimal change.

Gap. No surveyed product combines the six dimensions, and three gaps stand out. First, records of the process (G.5): none of the sources read describes a record of which guide the child used, which predefined difficulties occurred and which were later corrected (D3). Second, adult sum-

maries without judgement (G.6): the products that analyse a child's work for adults screen skills (Kaligo), compare with reference ranges (Dexteria) or score handwriting against a threshold (Dynamilis), all of which BR-006 rules out, and none is described as separating attempts made with a guide from attempts made without one (D4). Third, refinement that keeps the drawing the child's own (G.4): the drawing-assistance features replace the sketch with an artist's drawing (AutoDraw) or a perfect shape (QuickShape), or refine handwritten text rather than drawings (Smart Script), and none of the children's products describes refinement at all (D6). Graded guidance (G.1, G.2) does exist, most closely in LetterSchool, but in installed apps centred on letters and numbers; the one browser-based tracing tool records nothing (D3, D5). Squigglr's differentiable value is therefore the combination: graded drawing practice in a browser, records that show the process behind each drawing, and adult summaries that report that process without judging the child. Each part exists somewhere; no product found offers them together. Whether supporting adults value this combination is not yet confirmed: the needs come from SRC-001 and SRC-005, and the value of refinement rests on ASM-011, which the specialists' questionnaire answers (SRC-009.01) support but do not establish.

Products that must not be used. No stakeholder has ruled out a product. Two observations follow from the business rules; they are not decisions. A product whose adult reports screen, score or compare children (Kaligo's screening, Dexteria's reference ranges, Dynamilis's profile) conflicts with BR-006, so it could not be offered to adults as Squigglr's records. A product that replaces the child's drawing (AutoDraw) conflicts with BR-005. Services excluded by their own terms are noted in the Reusable Components subsection.

19.2 Reusable Components

This subsection lists candidate building blocks: first approaches to refining a creative drawing, then supporting libraries and services. They are candidates, not mandates. The team chooses among them after the proof-of-concept experiments, and this specification requires none of them. The Development Plan names four criteria for choosing a refinement approach: drawing preservation, output validity, response time and cost (SRC-002). Three more come from this document. Refinement must stay minimal, with every refined point within `REFINE_TOLERANCE` of the original strokes (G.4,

BR-005). The provider may change, so it should sit behind one interface that takes stroke coordinates and returns adjusted coordinates at the same scale (LC-004). The child's drawings should reach as few outside parties as possible, in line with the supervisor's advice to collect no information about children (SRC-004) while the privacy requirements are still open (OI-02). Response times and request costs are not known yet; the proof-of-concept experiment on model-based enhancement will record them (SRC-002). Costs that are known appear in the Costs section. Table 23 summarises the candidates.

Does refinement need AI? The supervisor asked whether the corrections need AI at all (SRC-004, #21). The sources suggest that refinement as G.4 measures it does not. Geometric smoothing and simplification adjust the child's own path within a stated maximum error, need no training, and run in the browser, so the drawing need not leave the device [23, 2]. These methods do not know what a drawing depicts, so an adjustment that depends on the item drawn needs recognition, and published methods recognise strokes or infer geometric relations without a trained model [44, 15]. A pretrained model adds the ability to interpret a whole drawing, at the price of sending the drawing to a service or hosting a model, slower responses, fees, and validating every instruction before it is applied (SRC-002). The Development Plan already tries geometric refinement first, describes it as geometric processing rather than AI, and keeps it as the fallback if model work is deferred (SRC-002).

Geometric refinement. The Paper.js library, which the Development Plan expects to use, smooths paths by several methods, one of which (the centripetal Catmull–Rom spline) guarantees no self-intersections. Its `simplify` method fits as few curves as possible through a path's points within an allowed maximum error, and is documented for point data from a mouse or touch device [23]. Paper.js is released under the Massachusetts Institute of Technology (MIT) License [22]. Simplify.js reduces the points of a line with the Douglas–Peucker and radial-distance algorithms under a tolerance, and is released under a Berkeley Software Distribution (BSD) licence [2]. A tolerance bounds the fitting error, but whether every refined point stays within `REFINE_TOLERANCE` of the original strokes, as G.4 measures, still has to be checked on prepared drawings.

Published stroke methods. Interactive beautification [15] infers geometric relations such as perpendicularity, congruence and symmetry between a new stroke and existing ones, tidies one stroke at a time, and offers several candidate results when the intent is ambiguous. Candidates could let the child or the adult choose, but snapping to exact geometry may move points further than `REFINE_TOLERANCE` allows. The $1 recogniser [44] matches a single stroke against stored templates in about 100 lines of code, and in its authors' study exceeded 97% accuracy with one loaded template. A matcher of this kind could tell whether a drawing is one of the agreed supported items (G.4) without training or an outside service; drawings made of several strokes would need more than this method offers.

Pretrained models. The Development Plan plans an existing pretrained model that accepts a drawing preview and returns structured drawing instructions, which Squigglr validates before applying; no model has been selected and no custom training is planned (SRC-002). The candidates found fall into three kinds.

- *Hosted multimodal models*, reached through an application programming interface (API). The OpenAI API accepts images as input [28], and its Structured Outputs feature makes responses follow a JavaScript Object Notation (JSON) schema supplied by the developer [29], which suits instructions that must be validated. The Anthropic API also accepts images [6]. Their privacy terms differ. OpenAI does not train on API data by default and keeps abuse-monitoring logs for up to 30 days unless the customer is approved for zero data retention [28]; its guidance says developers should not process personal data of children under 13 without first setting up zero data retention [30]. Anthropic may not train models on customer content [3], deletes API inputs and outputs within 30 days except in listed cases such as enforcing its usage policy [4], and asks organisations whose products minors use to add safety measures, follow child-privacy law and disclose that users are interacting with AI [5]. Both charge by usage; fees are not estimated here.

- *Open-weight vision-language models* that the team could host itself, such as Qwen2.5-VL-7B-Instruct. Its model card lists the Apache License 2.0 and describes locating objects with bounding boxes or points

and giving stable JSON output for coordinates [32]. Self-hosting keeps drawings away from outside providers but needs a machine with a graphics processing unit (GPU), which the team would run and pay for; the model card's examples load the model onto one.

- *Stroke-based generative models*, such as Sketch-RNN, which generates vector drawings and was released with trained models and code in Google's Magenta project [10] under the Apache License 2.0 [39]. The model's training data is the Quick, Draw! dataset of 50 million drawings in 345 categories, stored as timestamped strokes and licensed under Creative Commons Attribution 4.0 (CC BY 4.0) [12]. Output in strokes stays editable, but the model generates drawings rather than adjusting the child's, so its fit with BR-005 is doubtful.

Supporting libraries and services. The Development Plan expects Paper.js with Pointer Events for drawing input, Supabase for records and adult authentication, and Vercel for hosting (SRC-002). Pointer Events is a World Wide Web Consortium (W3C) Recommendation for hardware-independent input from a mouse, pen or touchscreen [17]; which devices and browsers Squigglr supports is still open (OI-09). Supabase provides a Postgres database, authentication and file storage, is open source under the Apache License 2.0, and can be self-hosted [38]; the self-hosting option bears on data isolation between institutions (OI-06, #75). Vercel's Hobby plan is free but limited to non-commercial personal use, and when a usage limit is exceeded the feature usually stays unavailable until 30 days have passed [42].

Table 23: Candidate components, not mandates

Candidate	What it offers	Licence or cost	Main concern for Squigglr
Refinement: geometric, no AI			
Paper.js smoothing and simplification	Smoothing; curve fitting within a maximum error; editable paths	MIT; free	Error bound must be checked against `REFINE_TOLERANCE`; no recognition
Simplify.js	Point reduction under a tolerance	BSD; free	Reduces points only; no smoothing
Refinement: published stroke methods			

Candidate	What it offers	Licence or cost	Main concern for Squigglr
Interactive beautification	Inferred geometric relations; several candidates per stroke	Paper; no library	Snapping may exceed `REFINE_TOLERANCE`
$1 recogniser	Template matching of single strokes, no training	Paper with pseudocode	Single strokes only
Refinement: pretrained models			
OpenAI API	Image input; output that follows a supplied schema	Usage-based fees	Logs kept up to 30 days by default; children's data guidance
Anthropic API	Image input	Usage-based fees	Data deleted within 30 days, with exceptions; safeguards for minors
Qwen2.5-VL-7B-Instruct	Object location; JSON coordinates	Apache 2.0; hosting cost	GPU hosting run by the team
Sketch-RNN with Quick, Draw!	Generation of vector drawings	Apache 2.0; data CC BY 4.0	Generates rather than adjusts (BR-005)
Supporting libraries and services			
Pointer Events	Pen, touch and mouse input	Open web standard	Device support still open (OI-09)
Supabase	Database, authentication and storage; self-hosting	Apache 2.0; hosted plans	Access policies need configuration and testing (SRC-002)
Vercel	Hosting and preview deployments	Hobby plan free; paid plans	Hobby is non-commercial only and pauses at its limits

Services that should not be used. Two observations follow from the sources; they are not decisions. The Gemini API's terms forbid using it in a website or app that is directed towards, or likely to be accessed by, people under 18, so Squigglr could not use it as a refinement provider under those terms. The same terms let Google use content sent to the free tier to improve its products, with human review [11]; any free tier with such terms would expose children's drawings in a way the supervisor's advice (SRC-004) and the open privacy issue (OI-02) argue against. A service that returns only a bitmap image does not meet the need for editable strokes (SRC-002).

19.3 Products That Can Be Copied

Volere asks this question of every project and warns that copying relies on the base being of good quality [33]. The team has not evaluated the quality of any base listed here and found no published specification of any surveyed product that could be reused. The answer therefore has three parts: code and data that may legally be copied, interaction patterns worth imitating, and patterns to avoid.

Code and data whose licences allow copying. Paper.js and Simplify.js allow reuse as long as their copyright notices are kept [22, 2], and the Magenta code is under the Apache License 2.0 [39]. Tux Paint's source code may be copied under the GPL version 2 [40], whose conditions would then apply to the copied code; Tux Paint is built as an installed program, not a website. The Quick, Draw! drawings may be copied under the CC BY 4.0 licence, but they were contributed by players of a public game, the dataset records no player ages, and the drawings may still contain inappropriate content [12]. They therefore cannot serve as the adult-made drawings from which Squigglr's ranges are set (BR-009, ASM-014).

Interaction patterns worth imitating. Ideas, not code, are the most useful things to copy.

- LetterSchool's tap, trace and write sequence gives the same item with less help each time [25], as Squigglr's stages do (G.1, G.2).
- Tracing Playground marks the start with a dot, shows direction with an arrow, numbers the strokes, says in words when the finger leaves the path, has no timer or score, and follows the device's reduced-motion setting [41]; these suit children who may not read (FACT-007).
- Writing Wizard's and Kaligo's replays of what the child traced [1, 19] show adults the actual strokes behind a result (G.5, BR-004).
- QuickShape keeps a corrected shape editable [31], and Smart Script keeps the look of the writer's own hand [7]; both match the intent of BR-005.

Patterns to avoid. Skill screening, scores against a threshold and comparison with reference ranges conflict with BR-006, and replacing the child's drawing with an artist's conflicts with BR-005. No surveyed product can be copied whole, because none combines the dimensions in Table 21.

20 New Problems

This section records the problems that introducing Squigglr may create: in the places where it is used, for the systems it runs alongside, for its users, in the environment it is built for, and later in its life. Each problem is stated as a cause and its effect, followed by the precaution this specification already takes or the open item that will decide it. Links point to business rules (BR) and assumptions (ASM) in Section 5, open issues (OI) in Table 20 and likely changes (LC) in Table 34. Critical assumptions (CA) and hazard identifiers (H-RAY, H-AHM, H-SHA, H-MOU, H-MUH) from the Hazard Analysis, and the safety requirements (SAF-RAY, SAF-AHM, SAF-SHA, SAF-MOU) in Safety-Critical Requirements, are given as plain text. Nothing in this section is a requirement; each precaution names where the related requirement or decision is, or will be, recorded.

20.1 Effects on the Current Environment

Squigglr will be used at home, in classrooms and in places such as rehabilitation centres (Anticipated Workplace Environment). Table 24 lists how it changes the work of the adults and children in those places. The Volere template also asks what the product must not do there; the last row answers that.

Table 24: New problems: effects on the current environment

Problem	Cause and effect	Precaution and links
More work for supporting adults	Squigglr gives adults new tasks: creating a learner profile, choosing eight adult-set values, with example drawings to help with `STAGE_PASS` (BR-008), choosing lessons and starting stages, answering go-back suggestions (BR-003) and reviewing records. An educator with a short slot per learner (PER-3) or a parent who reviews once a week (PER-2) may leave defaults in place or let suggestions wait.	Each value defaults to the middle of its range (BR-008); a declined or ignored suggestion changes nothing (G.3); the user documentation in Section 24 explains each task. Links: ASM-016, CA-006; SAF-RAY-001, SAF-RAY-008; H-RAY-001, H-RAY-007.
Shared and centrally managed school devices	One school device may be used by several children (SRC-004), and educators cannot install software on managed devices (PER-3). A session left open, or a learner not switched, can show one child's records to another child or save attempts to the wrong learner profile.	An adult selects the right learner (ASM-008); how a shared device is signed in and left after a session is OI-01; SAF-RAY-007 identifies the active learner. Hazards: H-RAY-006, H-MUH-004.
Practice moves onto a screen	Practice with Squigglr is time on a device. Squigglr cannot see how long a child has used the device, or whether the child is distressed or uncomfortable (ASM-002), so long or repeated sessions can go unnoticed when no adult is watching.	The adult stays responsible for the child's time on the device (ASM-002) and can stop a session or mute audio (ASM-015); a lesson ends at `ATTEMPT_LIMIT`, or at `ATTEMPT_LIMIT_MAX` when extended (G.3). Hazards: H-RAY-004, H-RAY-005.

Problem	Cause and effect	Precaution and links
Relationship with professional support	Occupational therapists and educators may already support the child. Squigglr keeps practice records, but the specialists advised deciding with such professionals which measures help an adult (SRC-005, P2, 31:42). An adult could treat Squigglr as a substitute for that support.	Squigglr makes no claim that its practice improves any skill (BR-006; FACT-001, FACT-002); the user documentation states that Squigglr offers practice, not therapy or assessment.
What Squigglr must not do	Squigglr must not take over decisions that belong to the adults around the child. Squigglr never selects another lesson, skips a stage or moves a child back a stage without an adult's authorization (CON-007, BR-003), never gives a child personalized AI-generated feedback in a guided activity (BR-007), never replaces a child's strokes in a guided lesson record (BR-004), and never lets refinement act outside the creative space (BR-005). Breaking any of these would override an adult's judgement or mislead the adults who rely on the records.	These are business rules and unlikely changes (ULC-003 to ULC-005, Table 35). Hazards: H-MUH-002 (refinement reaching a guided lesson record), H-MUH-003 (assisted work shown as independent).

20.2 Effects on the Installed Systems

Squigglr installs nothing on the user's device (CON-002) and replaces no existing system (Implementation Environment of the Current System). The systems it must coexist with are the browsers, devices and networks that homes and schools already use, and the outside services Squigglr itself depends on. Table 25 lists the conflicts that can arise with each.

Table 25: New problems: effects on the installed systems

System	Cause and effect	Precaution and links
Browsers and devices already in use	Squigglr runs in whatever browser the adult's or school's device has (CON-001). Squigglr depends on that browser reporting stylus strokes in enough detail (ASM-004), on devices the team does not control (CA-002), and on browser updates not changing how input is reported. Strokes may then be captured wrongly, or not at all, on some devices.	The supported device and browser combinations are OI-09; SAF-RAY-002 shows a notice on an unsupported device; drawing input can be read through one input module (LC-003). Hazard: H-RAY-002.
Browser storage on a shared device	Keeping each completed attempt on the device until the server confirms that it is saved (SAF-RAY-003) means practice data may sit in the browser of a shared or school device. The next user of the device could see or overwrite another child's unsaved attempts, or the data could stay on the device after the session ends.	How a shared device is left after a session is OI-01; who may see records is part of the privacy requirements (OI-02). Hazards: H-RAY-006, H-MUH-004.
School networks and content filters	A managed school network may block the Squigglr website or the outside services it calls (ASM-013). Lessons then fail to load in class, or saves and refinement fail.	The documentation for school administrators lists what must be allowed (Table 31); a failed refinement leaves the drawing unchanged (G.4); unsaved attempts are kept and reported (SAF-RAY-003).

System	Cause and effect	Precaution and links
New outside services	Squigglr adds services that homes and schools did not depend on before: a hosting service, an authentication service for adult accounts and, possibly, an outside model for refinement (SRC-002). An outage, price change or change of terms at any of them affects every user at once, and a drawing sent to an outside model leaves the team's control.	The interfaces are specified in Requirements for Interfacing with Adjacent Systems and the data sent outside in Privacy Requirements; the refinement provider can sit behind one interface (LC-004); costs are in Section 23.

20.3 Potential User Problems

Squigglr has no existing users, so this subsection lists adverse reactions its intended users may have, drawn from the personas (PER-1 to PER-3) and the assumptions in Section 5.3. Table 26 gives each reaction and the precaution taken.

Table 26: New problems: potential user problems

Problem (user)	Cause and effect	Precaution and links
Frustration from repetition (child)	Failing the same stage again and again, or starting the same lesson again at once after it ends, can agitate a child who dislikes repeated tasks (PER-1). The child may become distressed or refuse the activity later.	A lesson ends at `ATTEMPT_LIMIT` (G.3, BR-002), and the adult sets the limits for each child (BR-008). Hazards: H-RAY-004, H-RAY-005.

Problem (user)	Cause and effect	Precaution and links
Reaction to sensory feedback (child)	Sounds, colours or animation used as feedback may upset some children, who react to sensory feedback in different ways (ASM-015).	Audio can be turned off (SRC-005, P2, 37:39); feedback is to be designed with educators and occupational therapists (LC-005) and is covered by the Usability and Humanity Requirements.
Refinement that does not feel like the child's own (child)	Refinement is automatic (BR-005), so the drawing changes without the child asking. Whether children value this, and whether they still see the result as theirs, is unverified (ASM-011, FACT-013). A child may be upset by the change or not know how to get the original back.	Refinement is minimal, within `REFINE_TOLERANCE`, and the result stays editable; the child, or an adult, can restore the original or turn refinement off (BR-005, G.4, UC-007, UC-010). The scope of G.4 is OI-03. Hazards: H-MUH-001, H-MUH-002, H-MUH-006.
Summaries read as a diagnosis (adult)	Adults reading records are not assumed to have training in visual-motor development (CA-005), and fewer attempts may be read as better skill. An adult may draw a wrong conclusion about the child and pass it on.	No record or summary gives a diagnosis, judgement or comparison (BR-006, SAF-RAY-004); the documentation explains how to read a summary (Table 31). Hazard: H-RAY-003.
Over-trust in default values (adult)	An adult may keep the default values, or pick values from the example drawings, without knowing whether they suit the child (CA-006, ASM-016). The child may be held at a stage too long or moved on too soon, and the records may then mislead the adult.	Example drawings show each end of the `STAGE_PASS` range (BR-008); the values in use are shown with the records (ASM-006); the ranges are OI-05. Hazard: H-RAY-007.

Problem (user)	Cause and effect	Precaution and links
Worry about the child's data (adult)	Parents want their child's data kept private (PER-2), and how records are shared or deleted is not yet decided (OI-02, FACT-013). Adults may refuse to use Squigglr, or share records more widely than they intend.	A learner profile holds only an identifier and an avatar (BR-001); privacy requirements follow from OI-02; a privacy notice is part of the user documentation (Table 31).

20.4 Limitations in the Anticipated Implementation Environment That May Inhibit the New Product

Table 27 lists the limits of the environment that Squigglr is built for and used in that may stop it from working as specified. Several are the implementation risks the Development Plan's Proof of Concept (POC) is designed to test (SRC-002).

Table 27: New problems: limitations in the implementation environment

Limitation	Cause and effect	Precaution and links
Stylus input in the browser	Whether supported browsers report stylus strokes in enough detail is untested (ASM-004). Stage checks and records could be wrong; if browser input proves unreliable, an installed app may be needed, which would change CON-001 (ULC-001).	A POC experiment draws on a target tablet and on a second device and browser combination and records any input failures (SRC-002); the supported combinations are OI-09. Hazards: H-AHM-001, H-AHM-003; safety requirement SAF-AHM-002.

Limitation	Cause and effect	Precaution and links
Devices without a stylus	Not every home or school may have a touchscreen device with a stylus (ASM-003). Children then draw with a finger, for which the stage checks and ranges are not tuned.	Finger input is a stretch goal (LC-006); SAF-RAY-002 tells the adult when a device is not supported.
Online use only	Squigglr needs a network connection to load lessons and to save and view records, and offline use is out of scope (ASM-005); a connection may be lost at any time (CA-003). Attempts may be lost, and an adult may decide from incomplete records.	SAF-RAY-003 keeps unsaved attempts and reports them; network loss is covered by the Robustness or Fault-Tolerance Requirements. Hazards: H-RAY-002, H-MUH-005, H-MUH-008.
Limits of free service plans	The planned hosting service (Vercel) and database and authentication service (Supabase) (SRC-002) have free plans with usage limits. The hosting free plan is for personal, non-commercial use and has one developer seat [43]. The database free plan includes a 500 MB database and pauses a project after one week of inactivity [37]. A limit reached, or a project paused during a break, makes Squigglr unavailable or stops saves.	The Development Plan requires service limits and costs to be monitored (SRC-002); the plan choice and its cost are in Section 23; the budget is OI-07.

Limitation	Cause and effect	Precaution and links
Refinement model not yet chosen	No model has been selected. A pretrained model may return invalid drawing instructions, misread the drawing or change it too much, and its response time and request cost are unknown (SRC-002).	A POC experiment records valid and invalid outputs, response times and estimated request costs, and checks that a rejected instruction leaves the drawing unchanged; geometric refinement can be used while model work is deferred (SRC-002). Links: LC-004, OI-03; hazard H-MUH-002.
No testing with children	No study with children may run (CON-006), so value ranges and `REFINE_TOLERANCE` are set from adult-made drawings (BR-009, ASM-014). They may not match how children draw.	The adult chooses each value for the child within a range (BR-008); ASM-014 is marked relevant to safety and is an input to the Hazard Analysis.
Course schedule and effort	The course fixes the deadlines (CON-004), and the team plans around about ten hours per member per week (SRC-002). Not every planned feature may be ready for each release.	The POC plan allows narrowing the supported drawing set, simplifying the guidance rules or using geometric refinement only (SRC-002); the phases are set in Planning of the Development Phases.

20.5 Follow-Up Problems

Follow-up problems are situations that Squigglr may create later and that the team may not be able to handle. Table 28 lists them with the precaution or the open item that covers each.

Table 28: New problems: follow-up problems

Problem	Cause and effect	Precaution and links
Demand for stretch features	Adults or schools may ask for finger input, camera finger tracing or letter activities, which are stretch goals (SRC-001; FACT-006; ASM-010). The team may face pressure to add features the schedule cannot support, or to add them without the checks the core features have.	Deferred features are kept in the Waiting Room section; new input types can be added through one input module (LC-003, LC-006).
A growing lesson catalogue	Lessons may be added or reordered, for example to add letters after shapes (LC-001, SRC-009.04). Each new lesson needs its stages, predefined difficulties and adult-made example drawings for `STAGE_PASS` (BR-008, BR-009), so content work grows with every lesson.	Lessons and their stages can be stored as data, so adding or reordering a lesson changes no code (LC-001).
Growing stored data	Every attempt keeps the child's actual strokes (BR-004) and every recorded value (LC-007), and each creative drawing keeps its original strokes (BR-005). Storage grows with use and could reach the database free plan's 500 MB limit [37], after which saves could fail.	Capacity is covered by the Capacity Requirements; storage use is monitored (SRC-002); failed saves are reported (SAF-RAY-003); how long records are kept follows from OI-02.

Problem	Cause and effect	Precaution and links
Use by institutions	Schools or centres may adopt Squigglr; a peer reviewer asked how their records would be kept apart (#75), and an educator may manage many learners (ASM-012). One shared store of records may not meet an institution's expectations.	OI-06 decides whether separate institutions are supported in this project and, if so, the access and privacy requirements that keep their records apart.
Maintenance after the course	Who maintains Squigglr after the project ends is not decided (Maintenance Users and Service Technicians). Hosting accounts, provider access and lesson content may lapse while adults' accounts and children's practice records stay stored with no one responsible for them.	Who maintains Squigglr after the project ends is an open item (Maintenance Users and Service Technicians), and how stored records are deleted is part of OI-02.
Changes by outside providers	Providers may change prices, limits or terms; one vendor states that its pricing may change [37]. A refinement provider may change or withdraw a model (LC-004). Costs may then rise above the course limit, or refinement may stop working.	The refinement provider can sit behind one interface (LC-004), with geometric refinement as a fallback (SRC-002); the budget is OI-07.
Pressure to evaluate children	If Squigglr is presented as an educational tool rather than as practice, adults or schools may expect it to evaluate children. The supervisor advised that any evaluation would need strong justification, because the team cannot judge a child's abilities (SRC-008).	Squigglr gives no diagnosis, judgement or comparison (BR-006, ULC-004).

Problem	Cause and effect	Precaution and links
Laws that may apply after wider use	Wider use, for example by schools or outside the course, may bring legal obligations about records of children that the project has not had to meet.	Which laws and standards apply is decided in the Legal Requirements section; this section assumes none.

21 Tasks

This section records the project-planning context and assigns the SRS requirements to delivery phases.

21.1 Project Planning

The Development Plan (SRC-002) defines the project's schedule, team responsibilities, development process, communication practices and deliverable management. Those arrangements are maintained in the Development Plan rather than duplicated in the SRS. This section assigns the SRS requirements to delivery phases; the course deadlines also appear under CON-004 (Table 8).

21.2 Planning of the Development Phases

The delivery plan distinguishes the following requirement targets:

PH-1 — Proof of Concept (PoC), November 23, 2026.
A demonstration of the core guided lesson flow, input capture, stage classification and pass-based progression using prepared examples.

PH-2 — Revision 0, February 1, 2027.
The learner-specific configuration, feedback, retry and stopping rules, and drawing-input and session safeguards required for a usable revision.

PH-3 — Final demonstration, March 15, 2027.
The remaining record-history and progress-presentation safeguards and final integrated functionality.

Future.
Enhancements outside the committed capstone delivery phases, recorded in Section 25.

Priority describes the requirement's delivery importance (*Must*, *Should* or *Could*); *phase* states the first milestone at which that behaviour is targeted. A PH-1 capability remains part of PH-2 and PH-3. The phase classification does not relax any safety requirement when its associated feature is in use: a phase demonstration shall not rely on a safeguard that has not been implemented for real learner use.

Table 29 lists every currently specified SRS requirement, including the safety requirements derived from the Hazard Analysis. The functional requirement priorities and phases agree with Section 9.1; the Hazard Analysis-derived requirements are classified here as *Must* and assigned to the phase containing the behaviour they protect. The 142 identified requirements in this SRS are scheduled in PH-1 through PH-3. Candidate enhancements are assigned to Future in Section 25.

Table 29: Requirement priorities and targeted delivery phases

Requirement ID	Required behaviour	Priority	Phase
Learning activity requirements (Section 9.1)			
FR-LRN-001	Approved lesson catalogue	Must	PH-1
FR-LRN-002	Ordered guidance stages	Must	PH-1
FR-LRN-003	Demonstration	Must	PH-1
FR-LRN-004	Full-outline guidance	Must	PH-1
FR-LRN-005	Endpoint-only guidance	Must	PH-1
FR-LRN-006	Independent drawing	Must	PH-1
FR-LRN-007	Stylus stroke acquisition	Must	PH-1
FR-LRN-008	Multi-stroke attempts	Must	PH-2
FR-LRN-009	Interrupted attempt handling	Must	PH-2
FR-LRN-010	Invalid and ambiguous input	Must	PH-1
FR-LRN-011	Completeness classification	Must	PH-1
FR-LRN-012	Guidance classification	Must	PH-2
FR-LRN-013	Pass classification	Must	PH-1

Requirement ID	Required behaviour	Priority	Phase
FR-LRN-014	Predefined difficulty feedback	Must	PH-2
FR-LRN-015	Retry after non-passing attempts	Must	PH-1
FR-LRN-016	Adult lesson and starting stage selection	Must	PH-1
FR-LRN-017	Adult-selected practice parameters	Must	PH-2
FR-LRN-018	Adult authorization for discretionary changes	Must	PH-2
FR-LRN-019	Difficulty suggestion	Must	PH-2
FR-LRN-020	Improvement-based extension	Must	PH-2
FR-LRN-021	Stage attempt limit and lesson end	Must	PH-2
FR-LRN-022	No learner evaluation or diagnosis	Must	PH-1
FR-LRN-023	Automatic within-lesson advancement	Must	PH-1
Creative Drawing Requirements (Section 9.3)			
FR-CRT-001	Draw strokes	Must	PH-1
FR-CRT-002	Select a part	Must	PH-1
FR-CRT-003	Move a part	Must	PH-1
FR-CRT-004	Resize a part	Must	PH-1
FR-CRT-005	Recolour a part	Must	PH-1
FR-CRT-006	Erase a part	Must	PH-1
FR-CRT-007	Undo a change	Must	PH-1
FR-CRT-008	Redo an undone change	Could	PH-3
FR-CRT-009	Start a new drawing	Should	PH-2
FR-CRT-010	Refine a finished drawing	Must	PH-1
FR-CRT-011	Change only stroke paths	Must	PH-1
FR-CRT-012	Stay within the refinement tolerance	Must	PH-1
FR-CRT-013	Keep the refined drawing editable	Must	PH-1
FR-CRT-014	Do not refine a stroke twice	Must	PH-2
FR-CRT-015	Leave unsupported drawing unchanged	Must	PH-2

Requirement ID	Required behaviour	Priority	Phase
FR-CRT-016	Show whether a drawing is refined	Should	PH-2
FR-CRT-017	Restore the exact original	Must	PH-1
FR-CRT-018	Child restores the original	Must	PH-1
FR-CRT-019	Adult restores the original	Must	PH-2
FR-CRT-020	Leave the drawing unchanged when refinement fails	Must	PH-1
FR-CRT-021	Discard an out-of-date result	Must	PH-2
FR-CRT-022	Keep drawing while refinement runs	Should	PH-2
FR-CRT-023	Save to the active learner	Must	PH-2
FR-CRT-024	Reopen a drawing as it was left	Must	PH-2
FR-CRT-025	Keep changes through a lost connection	Should	PH-3
FR-CRT-026	Show only the active learner's drawings	Must	PH-2
FR-CRT-027	No refinement in guided lessons	Must	PH-2
FR-CRT-028	Creative drawings are not attempts	Must	PH-2
FR-CRT-029	Show unsaved changes	Should	PH-3
FR-CRT-030	Turn refinement off	Should	PH-2
Attempt Record Requirements (Section 9.4)			
FR-REC-001	Attempt record	Must	PH-1
FR-REC-002	Actual strokes	Must	PH-2
FR-REC-003	Difficulties and corrections	Must	PH-1
FR-REC-004	Adult-set values in use	Must	PH-2
FR-REC-005	Interrupted attempt	Should	PH-2
FR-REC-006	Lesson record	Must	PH-2
FR-REC-007	Unfinished lesson	Should	PH-2
FR-REC-008	Each attempt saved once	Must	PH-2
FR-REC-009	Saved records unchanged	Must	PH-2
FR-REC-010	Suggestion responses	Should	PH-2

Requirement ID	Required behaviour	Priority	Phase
Progress Summary Requirements (Section 9.5)			
FR-PRG-001	One learner per summary	Must	PH-2
FR-PRG-002	Progress over time by lesson	Must	PH-2
FR-PRG-003	With and without a guide kept apart	Must	PH-2
FR-PRG-004	Recurring difficulties	Must	PH-2
FR-PRG-005	Open attempts behind the summary	Must	PH-2
FR-PRG-006	Attempt record view	Must	PH-2
FR-PRG-007	Adult chooses the view	Should	PH-3
FR-PRG-008	Empty history	Must	PH-2
FR-PRG-009	Interrupted attempts shown apart	Should	PH-2
FR-PRG-010	Stages not attempted	Must	PH-2
FR-PRG-011	Latest attempt included	Should	PH-2
FR-PRG-012	No overall score	Must	PH-2
Appearance Requirements (Section 10.1)			
NFR-UX-001	Calm, uncluttered child screens	Must	PH-2
NFR-UX-002	Images and icons instead of text	Must	PH-2
NFR-UX-003	Adult views distinct from child screens	Should	PH-2
Style Requirements (Section 10.2)			
NFR-UX-004	Feedback that never marks the drawing as wrong	Must	PH-1
NFR-UX-005	Consistent look across child screens	Should	PH-2
NFR-UX-006	Appealing but calm style	Should	PH-3
NFR-UX-007	Style reviewed with educators or therapists	Should	PH-3
Ease of Use Requirements (Section 11.1)			
NFR-UX-008	Practice without adult help	Must	PH-1
NFR-UX-009	Quick setup by a new adult	Should	PH-2
Personalization and Internationalization Requirements (Section 11.2)			

Requirement ID	Required behaviour	Priority	Phase
NFR-UX-010	Audio setting for each child	Must	PH-2
NFR-UX-011	Refinement setting for each child	Should	PH-2
NFR-UX-012	Language-free child screens, one adult language	Must	PH-2
Learning Requirements (Section 11.3)			
NFR-UX-013	Learning a lesson from its demonstration	Must	PH-2
NFR-UX-014	Adult tasks learnable without training	Should	PH-3
Understandability and Politeness Requirements (Section 11.4)			
NFR-UX-015	Short, encouraging messages to the child	Must	PH-1
NFR-UX-016	Effort acknowledged	Should	PH-2
NFR-UX-017	Plain wording in adult views	Must	PH-2
Accessibility Requirements (Section 11.5)			
NFR-UX-018	Large touch targets	Must	PH-2
NFR-UX-019	Colour contrast	Must	PH-2
NFR-UX-020	Text alternatives for icons	Should	PH-2
NFR-UX-021	Feedback not by sound alone	Must	PH-2
NFR-UX-022	Control of audio	Should	PH-2
NFR-UX-023	Keyboard use of adult views	Should	PH-3
NFR-UX-024	Resizable text in adult views	Could	PH-3
Speed and Latency Requirements (Section 12.1)			
NFR-PERF-001	Stroke feedback latency	Must	PH-2
NFR-PERF-002	AI refinement response latency	Should	PH-2
NFR-PERF-003	Progress summary page load	Must	PH-2
Safety-Critical Requirements (Section 12.2)			
SAF-RAY-001	Adult-only action verification	Must	PH-1
SAF-RAY-002	Unsupported-device notice	Must	PH-2
SAF-RAY-003	Preservation of unsaved attempts	Must	PH-2

Requirement ID	Required behaviour	Priority	Phase
SAF-RAY-004	Non-diagnostic summaries and exports	Must	PH-2
SAF-RAY-005	Automatic lesson termination	Must	PH-2
SAF-RAY-006	Limit-ending and extension history	Must	PH-2
SAF-RAY-007	Active-learner identification and isolation	Must	PH-2
SAF-RAY-008	Adult-setting bounds and authorization	Must	PH-2
SAF-RAY-009	Lesson inactivity check-in and ending	Should	PH-2
SAF-AHM-001	Duplicate input-event integrity	Must	PH-1
SAF-AHM-002	Drawing-coordinate integrity	Must	PH-1
SAF-SHA-001	Authentication	Must	PH-2
SAF-SHA-002	Prevent PII transmission to third parties	Must	PH-2
SAF-SHA-003	Verify saves and report save failures	Must	PH-2
SAF-SHA-006	Session timeout	Must	PH-2
SAF-MOU-001	Show missed parts without drawing them	Must	PH-2
SAF-MOU-002	Keep assisted and independent records separate	Must	PH-2
SAF-MOU-003	Limit flashing on child-facing screens	Must	PH-2
Precision or Accuracy Requirements (Section 12.3)			
NFR-PERF-004	Stage result scoring accuracy	Must	PH-2
NFR-PERF-010	Stage result and completeness agree with their definition	Must	PH-2
Robustness or Fault-Tolerance Requirements (Section 12.4)			
NFR-PERF-005	Submit attempt on network loss and reconnection	Must	PH-2
Capacity Requirements (Section 12.5)			
NFR-PERF-008	Concurrent session capacity	Should	PH-3
Scalability or Extensibility Requirements (Section 12.6)			

Requirement ID	Required behaviour	Priority	Phase
Longevity Requirements (Section 12.7)			
NFR-PERF-009	Operational lifetime through capstone	Must	PH-3
Wider Environment Requirements (Section 13.2)			
NFR-OPS-001	HTTPS-only communication	Must	PH-1
Requirements for Interfacing with Adjacent Systems (Section 13.3)			
NFR-OPS-002	Adjacent system interfaces	Must	PH-2
Productization Requirements (Section 13.4)			
NFR-OPS-003	No installation required	Must	PH-1
Release Requirements (Section 13.5)			
Maintenance Requirements (Section 14.1)			
NFR-MAINT-001	Lesson catalogue update without code change	Must	PH-2
Supportability Requirements (Section 14.2)			
NFR-MAINT-002	Programmer locatability of stage behaviour	Should	PH-2
NFR-MAINT-003	Module interface descriptions	Should	PH-2
NFR-MAINT-004	Consistent coding style	Should	PH-1
Adaptability Requirements (Section 14.3)			
Access Requirements (Section 15.1)			
NFR-SEC-001	Adult account data access	Must	PH-2
NFR-SEC-002	Session token invalidation on logout	Must	PH-2
Integrity Requirements (Section 15.2)			
NFR-SEC-003	Audit for saved record modification	Should	PH-2
Privacy Requirements (Section 15.3)			
NFR-SEC-004	Minimal personal data collection	Must	PH-1
Audit Requirements (Section 15.4)			
NFR-SEC-005	Server-side logging of authentication events	Should	PH-2
Immunity Requirements (Section 15.5)			

Requirement ID	Required behaviour	Priority	Phase
NFR-SEC-006	Input sanitisation against injection	Must	PH-2
Cultural Requirements (Section 16)			
NFR-CULT-002	Cultural sensitivity of symbols and imagery	Must	PH-3
Legal Requirements (Section 17.1)			
NFR-COMP-001	PIPEDA	Must	PH-3
NFR-COMP-002	Ontario MFIPPA applicability	Must	PH-3
Standards Compliance Requirements (Section 17.2)			
NFR-COMP-005	WCAG 2.2 selected success criteria	Must	PH-3

The phase allocation is a planning baseline rather than an alternative source of requirement definitions. If a requirement's scope or phase changes, its entry in this table and, for requirements with a `Phase` field, its individual specification are updated together.

22 Migration to the New Product

This section asks what must be converted or carried over before Squigglr can be used. Squigglr is a new product (Implementation Environment of the Current System), so both subsections find nothing to migrate. Each gives its reason and the condition that would change the answer.

22.1 Requirements for Migration to the New Product

Not applicable. There is no existing product to convert, decommission or run in parallel, because Squigglr is new and replaces no system. Getting started is part of the product, not a migration task: an adult creates a learner profile and chooses the adult-set values (UC-001, BR-001, BR-008), and these steps are covered by the use cases and by the user documentation in Section 24. Between the releases set in Planning of the Development Phases, Squigglr holds only prepared or adult-made data (CON-005, BR-009, ASM-014). Because every release is tested again in full (SRC-008), that data can be created again for each release instead of being converted. If adults keep real records in one release and a later release changes how records are stored, for

example to show new values (LC-007) or to measure improvement differently (LC-008), migrating those records would become a requirement here.

22.2 Data That Has to be Modified or Translated for the New System

Not applicable. No data held elsewhere is imported into Squigglr: no use case from UC-001 to UC-010 enters children's earlier drawings or adults' own notes. Every attempt record, lesson record and creative drawing is created inside Squigglr as the child practises (G.4, G.5), so nothing has to be translated into a new format.

23 Costs

This section lists the costs of building and running Squigglr that are known from a source, and names the costs that are not yet known, with who will settle each and when. No figure here is an estimate by the team. Table 30 gives each item. Amounts are copied as the sources state them; vendor amounts are list prices before tax on the date the page was accessed and may change.

Table 30: Known and unknown costs

Item	Cost or limit	Basis and source	Status
Course spending limit	At most $500 per team	Spending on equipment, supplies or digital services by a team should not exceed this amount, and receipts are kept [36].	Limit set by the course.
Expense reimbursement	Up to $150 per group	The department reimburses approved expenses; each needs the written approval of the administrative TA [36]. Tokens for AI services are not eligible [35].	Limit set by the course.

Item	Cost or limit	Basis and source	Status
Expo poster	Printing up to $60 per group covered	Covered by the department [36].	Known; no cost to the team within the limit.
Hosting (Vercel)	Free plan: $0 per month. Paid plan: $20 per month per developer seat	Planned host (SRC-002). The free plan is for personal, non-commercial use, has one developer seat and monthly limits, such as 100 GB of data transfer; the paid plan includes $20 of usage credit [43].	Plan TBD: Development and Operations (DevOps) Lead (Shamil Canbolat), after OI-07, before the POC demonstration window.
Database and adult accounts (Supabase)	Free plan: $0 per month. Paid plan: from $25 per month	Planned database and authentication service (SRC-002). The free plan includes a 500 MB database and 50,000 monthly active users, pauses a project after one week of inactivity and allows two active projects; the paid plan includes 8 GB of disk per project and has a spend cap on by default [37].	Plan TBD: as for hosting.

Item	Cost or limit	Basis and source	Status
Refinement model requests	Unknown	No model has been selected; a POC experiment records estimated request costs (SRC-002). Tokens are not reimbursable [35]. Geometric refinement uses no outside model (SRC-002).	TBD from POC evidence: Testing Lead (Muhammad Huzaifah) coordinates the evaluation (SRC-002), by the POC demonstration window.
Devices for development and testing	Unknown	The POC needs a target tablet with a compatible stylus and a second device and browser combination (SRC-002). Whether team members already own suitable devices is not recorded.	TBD with the supported devices (OI-09): Shamil Canbolat, Nov. 2, 2026.
Devices and network for users	Not a project cost	Homes and schools use their own touchscreen device with a stylus and their own network (ASM-003, ASM-005); no sale or purchase of Squigglr is planned (Customer).	Known.
Team effort	About ten hours per member per week	Planning average for each of the five members, including scheduled course activities (SRC-002). This is a planning assumption, not measured effort.	Known as a plan; effort per phase belongs in the Tasks section.

The total monetary cost is TBD, which matches Budget Constraints.

With both free plans and geometric refinement only, the known service costs are $0, subject to the free plans' limits and terms; device costs are still unknown. Any paid plan or model requests must fit within the $500 course limit, of which at most $150 can be reimbursed. Whether any budget is available is OI-07 (Shamil Canbolat, Oct. 17, 2026). The Volere template suggests estimating cost from counts of use cases and requirements [33]; those counts are not final while the requirements are being written, so no such estimate is given here.

24 User Documentation and Training

This section describes the user documentation and training Squigglr needs, not the documents themselves. Following the Volere template [33], each item is derived from the use cases (UC-001 to UC-010) and the business rules and assumptions the adult must understand to carry them out.

24.1 User Documentation Requirements

Table 31 lists the documentation supporting adults need. Every use case from UC-001 to UC-010 is covered by at least one item.

Table 31: User documentation items

Item	Purpose and contents	Use cases and sources	Audience
Getting started	Create a learner profile by choosing an avatar; Squigglr stores no name. Choose each adult-set value: what it means, its range, the example drawings for `STAGE_PASS`, and that a value not chosen is set to the middle of its range, which may not suit the child. Change the values later.	UC-001, UC-010; BR-001, BR-008; ASM-016	Supporting adults

Item	Purpose and contents	Use cases and sources	Audience
Lessons and stages	The lesson catalogue; the stages from demonstration to drawing with no guide; choosing a lesson and a starting stage; what happens when a lesson ends at `ATTEMPT_LIMIT` or is extended while the child improves.	UC-002, UC-003, UC-004; G.1 to G.3; BR-002, BR-003	Supporting adults
Go-back suggestions	What a suggestion means, that nothing changes unless an adult accepts it, and how to see the attempts behind it before deciding.	UC-005; BR-003; CON-007	Supporting adults
Reading records and summaries	What each attempt and lesson record holds; how attempts made with a guide are told apart from those made without one; how to open the attempts behind a summary; the values in use for each attempt; and that records describe practice only and are not a diagnosis or a measure of ability, including when shared with others.	UC-008, UC-009, UC-010; G.5, G.6; BR-006; ASM-006; CA-005; SAF-RAY-004	Supporting adults, and those they share records with
Creative space and refinement	That refinement happens automatically and changes the drawing only a little; which items are supported; how the child or an adult restores the original or turns refinement off or on; that refinement never reaches guided lesson records; and what, if anything, is sent to an outside service.	UC-006, UC-007, UC-010; G.4; BR-004, BR-005; OI-03	Supporting adults

Item	Purpose and contents	Use cases and sources	Audience
Safe use and shared devices	Selecting the right learner on a shared device and leaving the device after a session; muting audio and stopping a session; the adult's responsibility for the child's safety and time on the device; supported devices and browsers; the need for a network connection, and what happens to attempts that could not be saved.	ASM-002, ASM-005, ASM-008, ASM-015; OI-01, OI-09; SAF-RAY-003	Supporting adults
Privacy notice	What Squigglr stores about the adult and each child, who can see it, and how records are shared and deleted.	BR-001; ASM-009; OI-02	Supporting adults
Notes for school administrators	The web address and the outside services a managed network must allow, and confirmation that nothing is installed.	CON-002; ASM-013	School administrators

Each item appears as help inside the adult views, where the task is done, and is collected in a written User Guide, which the course template provides as a separate document. As the Volere template advises, the guide uses the terms defined in this specification and turns the use case scenarios into its step-by-step instructions. The guide is written in the interface language (ASM-007). Developer documents, such as the design documents, are not user documentation and are not listed.

Children receive no written documentation, because many cannot read (ASM-007). Instead, each lesson starts with a demonstration (G.1), and children move through Squigglr using images and icons rather than text (FACT-007; SRC-005, P1, 34:13). How quickly children and adults must be able to learn Squigglr is specified in Learning Requirements and is not repeated here.

The user interface and user experience (UI/UX) Lead coordinates the documentation, as the lead for user flows and usability (SRC-002). Each

item is first written for the release that delivers the use cases it covers, and is updated whenever those use cases, the lesson catalogue (LC-001) or the adult-set values (LC-002) change. Who keeps it current after the project ends follows the maintenance decision in Table 28. No documentation requirement is stated here: the behaviour the documentation explains is already specified by the use cases and business rules, the practice-only statement is SAF-RAY-004, and learnability is in Learning Requirements, so a separate requirement would repeat them.

24.2 Training Requirements

No training sessions are planned for any user. Supporting adults are assumed to be everyday web users (Hands-On Users of the Project), and the documentation in Table 31, shown where each task is done, is meant to let them set up and use Squigglr without help. Whether it is enough is checked, not assumed: the adult-reviewer tasks in the G.6 measurement and the planned usability sessions with adult reviewers (SRC-001) test whether adults can complete their tasks with only this help. Children receive no training beyond the demonstration that starts each lesson (G.1). The adult introduces Squigglr to the child, and the getting-started item suggests how to begin a first session, for example by choosing an early stage and checking the audio setting. If schools adopt Squigglr and an educator manages many learners (ASM-012), a short walkthrough for educators may be needed; it is not planned in this project.

25 Waiting Room

The waiting room records candidate capabilities excluded from the PH-1, PH-2 and PH-3 delivery commitments. They may be considered after the core project goals and their safety requirements have been satisfied. Their classification as *Future* is not a requirement to implement them during the capstone project.

Table 32: Capabilities deferred to Future

Capability	**Reason for deferral and source**	**Phase**
Camera-based finger tracing in the air	The specialists distinguished this gross-motor activity from the fine-motor practice supported by stylus-on-screen drawing and advised prioritizing touchscreen work (FACT-006; SRC-005). It is not part of the core guided lesson catalogue (G.1, CON-003).	Future
Letter and handwriting tracing	The agreed core catalogue covers lines, basic shapes and simple drawings (FR-LRN-001). The follow-up questionnaire places letters after shape practice (ASM-010, SRC-009.04; LC-001).	Future
Finger drawing and input-size progression	Stylus input is the core interaction (CON-003). Finger input, or progression between finger, large-stylus and smaller-stylus interaction, requires additional input-specific verification and calibration (FACT-005; LC-006).	Future
Full WCAG conformance	The capstone scope includes selected, verifiable WCAG 2.2 success criteria. Full conformance across all applicable criteria is a stretch goal, rather than a claim for the capstone release (OI-08; SRC-005).	Future

Offline lesson use and synchronization	The initial scope assumes network access for lessons and records (ASM-005). Starting or running lessons with no connection, and synchronizing them later, would extend this scope. Keeping work through a temporary loss of connection (SAF-RAY-003, NFR-PERF-005, FR-CRT-025) is part of the committed product.	Future

Creative space refinement remains within goal G.4 unless the team formally changes that goal (OI-03); it is not moved into the waiting room here. Automatic selection of new lessons, diagnostic assessment, and AI-generated personalized advice are excluded by CON-007 and BR-003, BR-006 and BR-007, respectively; they are not implicitly promised as future features.

26 Ideas for Solution

This section is non-normative: it records ideas for how the team might build Squigglr, so that they are not lost and are not mistaken for requirements; none of them binds the design. Ready-made products and components that could be reused are in the Off-the-Shelf Solutions section; this section is about how the team might put the product together. Table 33 gives each idea with its trade-off and source.

Table 33: Ideas for solution (non-normative)

Idea	**Trade-off**	**Source**
Geometric refinement first. Smooth and simplify the child's own paths, and try a pretrained model only for changes that geometry cannot make.	No outside service, request cost or data leaving Squigglr, and results are predictable. Geometry can only tidy strokes, not interpret a drawing, and the supervisor's question of whether AI is needed (SRC-004) stays open until POC evidence exists.	SRC-002 (Drawing Assessment and Enhancement; POC experiments); SRC-004

Idea	Trade-off	Source
Check model output before applying it. Treat each model response as untrusted: apply it only if every operation is from a known set, refers to the current drawing and keeps every point within `REFINE_TOLERANCE`; otherwise leave the drawing unchanged.	Adds checking logic and rejects some useful responses, but makes the G.4 check that a failed refinement leaves the drawing unchanged simple to test.	SRC-002; G.4; H-MUH-002
Keep the original strokes apart. Store the child's original strokes separately from the refined drawing, so that restoring switches back to them rather than undoing each change.	More stored data per creative drawing (Growing stored data, Table 28); restoring is exact however many edits followed.	SRC-002; BR-005; H-MUH-001, H-MUH-003
One refinement interface. Put every refinement method behind one interface that takes stroke coordinates and returns adjusted coordinates at the same scale.	A provider or a geometric method can be swapped in one module. An interface that returns only adjusted coordinates cannot express adding a shape, which the Development Plan lists as a candidate model instruction; adding content would also go beyond minimal refinement (BR-005).	LC-004; SRC-002

Idea	Trade-off	Source
Lessons and feedback as content. Store lessons, stages, predefined difficulties, example drawings and feedback wording as data that the lesson logic reads.	Adding a lesson or rewording feedback changes no code, but the data needs its own checks so that no lesson is released without its difficulties or example drawings.	LC-001, LC-005; BR-008, BR-009
One input module. Turn browser pointer events into strokes in one place, so that the rest of Squigglr sees only strokes.	A new device or finger input affects one module, but that module must be tested on each supported device, because emulation cannot replace stylus testing (SRC-002).	LC-003, LC-006; SRC-002
Local queue for unsaved attempts. Hold each completed attempt in the browser until the server confirms the save, then delete the local copy.	Keeps attempts through a lost connection (CA-003), but copies left on a shared device can be seen by the next user (H-RAY-006), so when they are cleared must be settled with OI-01.	SAF-RAY-003; CA-003; OI-01
Spending controls. Use the vendors' own controls: the database service turns a spend cap on by default on its paid plan [37], and the hosting service gives each new team a usage budget with notifications and can pause all projects when it is reached [43].	Keeps spending within the course limit (Section 23), but a reached cap can pause the service, which is the availability problem in Table 27.	Vendor pages; OI-07

Idea	Trade-off	Source
Model on university computing. Run a pretrained model on the department's graphics processing unit (GPU) cluster, which the course outline says is available for large language model computing, instead of a paid outside service.	No per-request fee, and drawings stay with the university. The team would run the model itself, and whether a public website may call the cluster, and for how long after the course, is unknown.	[36]; SRC-002 (no custom model training planned)

27 Likely and Unlikely Changes

This section is added to the Volere template because the course asks for it. It records which parts of the requirements are expected to change, so that the design can keep them easy to change, and which are not expected to change, so that the design can rely on them. Each change has an identifier: LC-### for a likely change and ULC-### for an unlikely one.

27.1 Likely Changes

Table 34: Likely changes

ID	Change and why it is likely	Affects	How the design can hide it
LC-001	The lesson catalogue (which lines, shapes and drawings, in what order) may change, for example to add letters after shapes (SRC-009.04).	G.1, G.2	Lessons and their stages can be stored as data that the lesson logic reads, so adding or reordering a lesson changes no code.

ID	Change and why it is likely	Affects	How the design can hide it
LC-002	The adult-configurable ranges in Table 10 may be recalibrated after adult-made verification (OI-05), and adults may change their chosen values at any time.	G.2, G.3, CON-007	Each range is defined in one place, each child's value is stored in their profile, and the activity logic reads only the profile value.
LC-003	The supported device and browser combinations are TBD and will change as browsers and tablets change (OI-09).	CON-001, CON-003	Drawing input can be read through one input module that turns browser pointer events into strokes; the rest of the product sees only strokes.
LC-004	The service that refines creative drawings may change because of cost (OI-07), quality or availability. The refinement is planned to use a lightweight AI model (SRC-005, Moustafa, 6:03).	G.4	Refinement can sit behind one interface that takes stroke coordinates and returns adjusted coordinates at the same scale, so a different provider replaces one module.
LC-005	Feedback wording, colours and audio will change: the specialists advised designing them with educators and occupational therapists, bit by bit (SRC-005, P1, 36:26; P2, 37:39).	G.1	Feedback text, colours and sounds can be kept as content separate from the rules that decide which feedback to give.

ID	Change and why it is likely	Affects	How the design can hide it
LC-006	Finger input may be added later; it is a stretch goal, and the specialists suggested finger, large stylus and small stylus as possible difficulty levels (SRC-005, P1, 26:46).	CON-003, G.1	The input module of LC-003 can accept a new input type without changing the lesson logic or the records.
LC-007	Which recorded values each adult wants to see will change from adult to adult and over time.	G.6	Squigglr records every value and lets the adult choose what their view shows, so a new view needs no change to what is recorded.
LC-008	The way improvement is measured (average of recent attempts compared with earlier ones) may be replaced once it is tried.	G.3, BR-002	The measure is one function that reads the stored attempt results; the lesson logic only asks whether the child is improving.

27.2 Unlikely Changes

Table 35: Unlikely changes

ID	Fact not expected to change	Affects	Justification and source
ULC-001	Squigglr is delivered as a website, with no installation.	CON-001, CON-002	Imposed by the supervisor and TA because installing apps on school-managed devices is difficult (SRC-003, SRC-004). This may change if browser input proves unreliable (ASM-004).

ID	Fact not expected to change	Affects	Justification and source
ULC-002	Only supporting adults view and manage practice records; children do not.	G.5, G.6	The records exist to help adults understand practice (SRC-001), adults handle setup (SRC-005, P1, 34:13), and the records concern children, whose privacy the product must protect (OI-02).
ULC-003	An adult chooses the lesson, starting stage, and any skip or backward move. Passing an attempt advances to the next stage within the chosen lesson, never to a new lesson.	G.2, G.3, CON-007	Deciding which lesson or exceptional level a child needs automatically would require assessment outside the project scope (SRC-005, P1, 28:37; P2, 29:18); the TA agreed to the change (SRC-006).
ULC-004	Squigglr gives no diagnosis, no judgement of a child's abilities, no comparison with other children and no effectiveness claims.	G.2, G.6	Excluded from the start (SRC-001) and presented to the specialists as deliberately left out (SRC-005, Moustafa, 6:03); the team cannot judge a child's abilities (SRC-008), and effectiveness claims would need studies the project cannot run (CON-006).
ULC-005	In guided lessons, the record keeps the child's actual strokes; refinement happens only in the creative space, the original can be restored, and refinement can be turned off.	G.4, G.5	A record of practice is meaningful only if it shows what the child drew (SRC-001; SRC-005, Moustafa, 6:03).

ID	Fact not expected to change	Affects	Justification and source
ULC-006	The project gathers no information from or about children and runs no study with children.	CON-005, CON-006	Directed by the supervisor (SRC-004); a study with children would need ethics approval that takes months (SRC-005, P2, 22:56).

28 Traceability

This section provides three traceability views of the requirements links: sources through product use cases to requirements (Section 28.1), functional-to-nonfunctional links (Section 28.2), and hazards to their mitigating requirements (Section 28.3). Entries are based on the requirements' own *Sources*, *Traceability* and *Hazards* fields and on the corresponding Product Use Case and FMEA tables. Discussion of similar behaviour does not imply a requirements link.

28.1 Goal/source to use case to requirement

Table 36 follows each product use case's stated project goals and sources to requirements whose own *Sources* field names at least one of those goals and whose *Traceability* field names that use case.

Table 36: Goal/source to product use case to requirement

Goal and source in use case	Use case	Requirements with matching Sources and Traceability fields
G.3; SRC-008	UC-001	FR-LRN-017; NFR-UX-003; NFR-UX-009; NFR-UX-014
G.1, G.3; SRC-005.01	UC-002	FR-LRN-001; FR-LRN-016; NFR-UX-003; NFR-UX-009; NFR-UX-014

Goal and source in use case	Use case	Requirements with matching Sources and Traceability fields
G.1–G.3; SRC-005.01	UC-003	FR-LRN-002–FR-LRN-015; FR-LRN-020–FR-LRN-023; NFR-UX-001; NFR-UX-002; NFR-UX-004; NFR-UX-008; NFR-UX-013; NFR-UX-015; NFR-PERF-001
G.3; SRC-005.01	UC-004	FR-LRN-016; FR-LRN-018; FR-LRN-023
G.3; SRC-005.01	UC-005	FR-LRN-018; FR-LRN-019; NFR-UX-008; NFR-UX-014
G.4; SRC-009.01	UC-006	FR-CRT-001–FR-CRT-030
G.4; SRC-009.01	UC-007	NFR-UX-011; NFR-PERF-002; FR-CRT-010; FR-CRT-030
G.5; SRC-001	UC-008	FR-REC-001–FR-REC-010; FR-PRG-006; FR-LRN-007; FR-LRN-009; FR-LRN-021
G.6; SRC-005.02	UC-009	FR-PRG-001–FR-PRG-012; NFR-UX-003; NFR-UX-014; NFR-PERF-003; NFR-SEC-001
G.3, G.6; SRC-008, SRC-009.01	UC-010	FR-LRN-017; NFR-UX-003; NFR-UX-014; NFR-PERF-003; NFR-SEC-001

28.2 Functional to nonfunctional requirements

Table 37 records cross-type references in a requirement's *Traceability* field. One-way references should not be mistaken for a reciprocal links.

Table 37: Direct functional-to-nonfunctional requirement links

Functional requirement	Nonfunctional requirement	Declared by
FR-LRN-003	NFR-UX-013	NFR Traceability

Functional requirement	Nonfunctional requirement	Declared by
FR-LRN-007	NFR-PERF-001; NFR-PERF-004; NFR-MAINT-002	NFR Traceability
FR-LRN-008	NFR-PERF-004; NFR-MAINT-002	NFR Traceability
FR-LRN-011	NFR-PERF-010	NFR Traceability
FR-LRN-013	NFR-PERF-010	NFR Traceability
FR-LRN-014	NFR-UX-004; NFR-UX-015; NFR-UX-016	NFR Traceability
FR-LRN-022	NFR-UX-004; NFR-UX-015	NFR Traceability
FR-LRN-001; FR-LRN-002	NFR-MAINT-001	NFR Traceability
FR-CRT-010	NFR-PERF-002; NFR-UX-011	NFR Traceability
FR-REC-006; FR-PRG-002	NFR-UX-017; NFR-PERF-003	NFR Traceability
FR-PRG-004; FR-PRG-005; FR-PRG-006	NFR-SEC-001; NFR-UX-003	NFR Traceability

28.3 Hazards to mitigating requirements

Table 38 transcribes requirement IDs from the Hazard Analysis FMEA's *Mitigating requirement IDs* column. The final column lists only requirements whose own *Hazards* or *Traceability* field names that hazard; a dash indicates a mitigation link that the SRS does not currently declare back to the hazard.

Table 38: Hazards and requirement links

Hazard	FMEA mitigating requirements	Requirements that declare this hazard in the SRS
H-RAY-001	SAF-RAY-001	SAF-RAY-001
H-RAY-002	SAF-RAY-002; SAF-RAY-003; FR-REC-008; FR-PRG-011	SAF-RAY-002; SAF-RAY-003; FR-REC-008; FR-PRG-011
H-RAY-003	SAF-RAY-004; FR-PRG-008; FR-PRG-012	SAF-RAY-004; FR-PRG-008; FR-PRG-012
H-RAY-004	SAF-RAY-005	SAF-RAY-005
H-RAY-005	SAF-RAY-006	SAF-RAY-006
H-RAY-006	SAF-RAY-001; SAF-RAY-007; FR-REC-008; FR-PRG-001	SAF-RAY-001; SAF-RAY-007; FR-REC-008; FR-PRG-001
H-RAY-007	SAF-RAY-008	SAF-RAY-008
H-AHM-001	FR-LRN-007–FR-LRN-010	FR-LRN-007–FR-LRN-010
H-AHM-002	FR-LRN-007; FR-LRN-008; SAF-AHM-001	FR-LRN-007; FR-LRN-008; SAF-AHM-001
H-AHM-003	FR-LRN-007; FR-LRN-010; SAF-AHM-002	FR-LRN-007; FR-LRN-010; SAF-AHM-002
H-AHM-004	FR-LRN-011; FR-LRN-013; FR-LRN-023	FR-LRN-011; FR-LRN-013; FR-LRN-023

Hazard	FMEA mitigating requirements	Requirements that declare this hazard in the SRS
H-AHM-005	FR-LRN-011; FR-LRN-013; FR-LRN-014; NFR-UX-004; NFR-UX-015	FR-LRN-011; FR-LRN-013; FR-LRN-014; NFR-UX-004; NFR-UX-015
H-AHM-006	FR-LRN-009; FR-LRN-018–FR-LRN-020	FR-LRN-009; FR-LRN-018–FR-LRN-020
H-AHM-007	FR-LRN-016; FR-LRN-017; FR-LRN-019; FR-LRN-021	FR-LRN-016; FR-LRN-017; FR-LRN-019; FR-LRN-021
H-SHA-001	SAF-SHA-001; SAF-SHA-006	SAF-SHA-001; SAF-SHA-006
H-SHA-002	SAF-SHA-002	SAF-SHA-002
H-SHA-003	SAF-SHA-003	SAF-SHA-003
H-MOU-001	SAF-MOU-003; NFR-UX-001; NFR-UX-006; NFR-UX-007; NFR-UX-010; NFR-UX-022	SAF-MOU-003; NFR-UX-001; NFR-UX-006; NFR-UX-007; NFR-UX-010; NFR-UX-022
H-MOU-002	NFR-UX-004; NFR-UX-007; NFR-UX-015; NFR-UX-016; FR-LRN-022	NFR-UX-004; NFR-UX-007; NFR-UX-015; NFR-UX-016; FR-LRN-022
H-MOU-003	NFR-UX-002; NFR-UX-004; NFR-UX-015; NFR-UX-021	NFR-UX-002; NFR-UX-004; NFR-UX-015; NFR-UX-021

Hazard	FMEA mitigating requirements	Requirements that declare this hazard in the SRS
H-MOU-004	SAF-MOU-001; FR-LRN-014; FR-LRN-023	SAF-MOU-001; FR-LRN-014; FR-LRN-023
H-MOU-005	SAF-MOU-002; FR-LRN-012; NFR-UX-017; SAF-RAY-004	SAF-MOU-002; NFR-UX-017; FR-LRN-012; SAF-RAY-004
H-MOU-006	NFR-UX-011	NFR-UX-011
H-MUH-001	FR-CRT-007; FR-CRT-008; FR-CRT-013; FR-CRT-017; FR-CRT-018; FR-CRT-019; FR-CRT-024	FR-CRT-007; FR-CRT-008; FR-CRT-013; FR-CRT-017; FR-CRT-018; FR-CRT-019; FR-CRT-024
H-MUH-002	FR-CRT-010; FR-CRT-011; FR-CRT-012; FR-CRT-014; FR-CRT-015; FR-CRT-020; FR-CRT-021; FR-CRT-027	FR-CRT-010; FR-CRT-011; FR-CRT-012; FR-CRT-014; FR-CRT-015; FR-CRT-020; FR-CRT-021; FR-CRT-027
H-MUH-003	FR-REC-001; FR-REC-002; FR-CRT-016; FR-CRT-017; FR-CRT-023; FR-CRT-028; FR-REC-009; FR-PRG-003; FR-PRG-010	FR-REC-001; FR-REC-002; FR-CRT-016; FR-CRT-017; FR-CRT-023; FR-CRT-028; FR-REC-009; FR-PRG-003; FR-PRG-010

Hazard	FMEA mitigating requirements	Requirements that declare this hazard in the SRS
H-MUH-004	FR-REC-005; FR-REC-008; FR-REC-009; FR-CRT-023; FR-CRT-026; FR-PRG-001	FR-REC-005; FR-REC-008; FR-REC-009; FR-CRT-023; FR-CRT-026; FR-PRG-001
H-MUH-005	FR-CRT-023; FR-CRT-024; FR-CRT-025; FR-CRT-029	FR-CRT-023; FR-CRT-024; FR-CRT-025; FR-CRT-029
H-MUH-006	FR-CRT-012; FR-CRT-015; FR-CRT-016; FR-CRT-018; FR-CRT-030	FR-CRT-012; FR-CRT-015; FR-CRT-016; FR-CRT-018; FR-CRT-030
H-MUH-007	FR-REC-003; FR-REC-004; FR-REC-006; FR-REC-007; FR-REC-009	FR-REC-003; FR-REC-004; FR-REC-006; FR-REC-007; FR-REC-009
H-MUH-008	FR-PRG-002; FR-PRG-004; FR-PRG-005; FR-PRG-006; FR-PRG-008; FR-PRG-009; FR-PRG-010; FR-PRG-011	FR-PRG-002; FR-PRG-004; FR-PRG-005; FR-PRG-006; FR-PRG-008; FR-PRG-009; FR-PRG-010; FR-PRG-011

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29 Appendix — Requirement Sources

This appendix is the requirement sources register. Each source has a fixed identifier (SRC-001 to SRC-009) that is never renumbered. A requirement names the sources it comes from in its rationale or source field. A statement from the specialist interview is cited by its source identifier, participant code and transcript timestamp, for example "(SRC-005, P1, 28:37)"; the statement identifiers in Section 29.1 (for example SRC-005.01) can be cited instead. The interview is expert-reported evidence only. The three participants are de-identified domain specialists (P1, P2 and P3). No child took part in the interview or in any other source listed here.

Table 39 lists the project documents and meeting sources, and Table 40 lists the interview, questionnaire and peer-review sources. Requirements identify their supporting sources through stable source identifiers. Each source and statement has a LaTeX label (for example `src:005.01`), so a requirement can link to it and the link works in the PDF.

The sources represent users in different ways. Children are the main beneficiaries, but by the supervisor's direction (SRC-004) no requirement is gathered from children and no information about them is collected. Their needs enter the requirements only through adults who know them. The domain specialists' interview and questionnaire (SRC-005, SRC-009) are the outside evidence for these requirements. Their advice established adult control of lesson selection and optional stage changes, stylus as the main input, finger input as a stretch goal, and camera tracing as a stretch goal. Parents, educators and occupational therapists have not been consulted yet. The specialists advised doing so (SRC-005.03, SRC-005.06), and until then the requirements that depend on their views rest on the expert statements below and are open to change.

Table 39: Requirement sources: project documents and meetings

ID	Type	Date	Reference	Summary	Limitations
SRC-001	Project document	–	`docs/Problem StatementAnd Goals/ Problem Statement.tex`	Problem, stakeholders, goals and stretch goals.	Written by the team; some assumptions changed after SRC-005.

ID	Type	Date	Reference	Summary	Limitations
SRC-002	Project document	–	`docs/Development Plan/Development Plan.tex`	Team process, tools and schedule.	Written by the team; plans, not user needs.
SRC-003	TA meeting	Sep 23, 2026	Issue #10	Website instead of an installed app; teachers and students as primary stakeholders; specific, measurable metrics.	Short minutes; not a user source.
SRC-004	Supervisor meeting	Sep 23, 2026	Issue #21	Collect no information about children; do not gather requirements by asking children; ethics approval for studies; start with a web app; traceability.	Advice from the client, not observed user data.

ID	Type	Date	Reference	Summary	Limitations
SRC-008	Supervisor meeting	Oct 14, 2026	Issue #107	Let the supporting adult choose child-dependent thresholds within a team-set minimum and maximum, with example drawings of each end; default to the middle of the range. Take specific WCAG ideas, such as large icons, as non-functional requirements rather than treating WCAG as a constraint.	Advice from the client, not observed user data.

Table 40: Requirement sources: interview and reviews

ID	Type	Date	Reference	Summary	Limitations
SRC-005	Specialist interview	Oct 1, 2026	Appendix 30	Three de-identified specialists on progression, input method, child-friendly design, feedback, accessibility and ethics.	Three experts, one session; expert opinion, not child observation; the experts advised confirming details with occupational therapists and educators.
SRC-006	TA meeting	Oct 7, 2026	Issue #78	State machine for the formal specification; define jargon; named TBD constants; specialists as de-identified secondary stakeholders.	Course guidance, not user needs.
SRC-007	Peer review	–	Issues #69–#75	Team 13 review of SRC-001 and SRC-002.	Reviewers' opinions; not all suggestions are adopted.
SRC-009	Specialist questionnaire	Oct 2–4, 2026	Appendix 31	Written answers on the creative space and refinement, motivation and frustration, letters, and concerns.	Two of three specialists; anonymous, so answers cannot be matched to P1–P3; expert opinion, not child observation.

29.1 Key Interview Statements

This subsection lists the interview statements most relevant to the requirements. Each is a short paraphrase, not a quote; the exact words are in

Appendix 30 at the given timestamp.

SRC-005.01 (P1, 28:37; P2, 29:18; 30:20–31:02)
Deciding automatically when a child is ready for the next level would need assessment and is too complex for the time frame. An adult who knows the child (a parent or educator) should choose the level. The team agreed to leave progression to the supervising adult.

SRC-005.02 (P2, 29:18)
In a past project, a teacher personalized each child's activities through a separate adult interface that also showed progress.

SRC-005.03 (P2, 31:42)
Which progress measures help an adult should be decided with occupational therapists or educational specialists, using milestones from the literature if they exist.

SRC-005.04 (P3, 24:34; P3, 25:36; P1, 26:46)
The specialists disagreed on input. P3 said that how a child holds a pen affects how they draw and the usability of the app, and suggested finger input unless fine motor practice is the goal; R2 later agreed that finger drawing may be simpler (SRC-009.03). P1 agreed that a finger is the easiest start, but said fine motor practice needs a held tool, moving from a larger stylus to a smaller one as later difficulty levels.

SRC-005.05 (P1, 34:13)
The app should be child-friendly: a child who cannot read yet should be able to enter a lesson, practise, restart and move back and forth using images and icons rather than text, without constant adult help. Adults handle setup.

SRC-005.06 (P1, 36:26)
Positive feedback wording and colours should be designed with feedback from educators and occupational therapists, built up bit by bit.

SRC-005.07 (P2, 37:39)
Audio feedback is a good idea; offering more than one modality is encouraged in inclusive design, with the user able to turn the audio off. Design it carefully with therapists' feedback, and consider adults with ADHD or autism as reviewers when children cannot take part.

SRC-005.08 (P2, 39:26)
Start with the Web Content Accessibility Guidelines (WCAG), then apply more specialized design heuristics for autism from the literature.

SRC-005.09 (P2, 19:05; P1, 20:27; P2, 20:47)
From a universal design view, the skills apply to all children, including those with motor or sight challenges; designing for disability serves many groups.

SRC-005.10 (P1, 13:42; P1, 15:12; P1, 16:13; P2, 16:45)
Camera finger tracing in the air practises gross rather than fine motor skills. It is useful but should come after the core touchscreen tracing when time is limited.

SRC-005.11 (P2, 22:56; P1, 24:03; P1, 33:00)
Testing with children needs ethics approval (about three to four months where P2 works). Meeting experts for feedback without involving children does not.

SRC-005.12 (41:51–42:48)
All three participants agreed to the use of the transcript for eliciting requirements.

SRC-005.13 (P2, 35:36)
How the child and the caregiver share the app, for example with the child as the user and the caregiver giving access, is the team's design decision; there is no single right workflow. Many drawing apps for children already exist.

29.2 Key Questionnaire Statements

This subsection lists the questionnaire statements most relevant to the requirements. Each is a short paraphrase; the full answers are in Appendix 31, labelled by respondent (R1, R2) and question (Q1–Q4).

SRC-009.01 (R1, Q1; R2, Q1)
The free drawing space is valuable. Refinement should stay subtle and support the child's own drawing rather than replace or correct it, so the child keeps ownership. Both respondents suggested that refinement be optional or adjustable.

SRC-009.02 (R1, Q2)
Activities motivate when they are achievable, predictable and visually engaging, with immediate positive feedback and some choice. They frustrate when they are too demanding, repeatedly highlight mistakes, need very precise movements or change the child's work unexpectedly. Squigglr should recognize effort and progress, not only accuracy.

SRC-009.03 (R2, Q2)
Requiring a pen or stylus rather than a finger may make drawing harder; finger drawing may be simpler.

SRC-009.04 (R1, Q3; R2, Q3)
Letters are a useful later stage, introduced gradually after shapes, with some flexibility in how precisely a letter must be traced. R1 suggested Arabic as well as English letters; R2 suggested letters and numbers, then tracing letters to form a word.

SRC-009.05 (R2, Q4)
The interface should be appealing and child-friendly while avoiding excessive visual or auditory stimuli.

SRC-009.06 (R1, Q4)
Involve children, parents, educators and occupational therapists in design and evaluation, and look beyond accuracy at engagement, frustration, independence, confidence and transfer to drawing without the app.

SRC-009.07 (R1, Q4)
Squigglr could serve a broader population, such as other neurodevelopmental conditions and rehabilitation; avoid restricting it to one condition too early.

30 Appendix — Specialist Interview Transcript

This appendix contains the transcript of a 43-minute remote group interview held on October 1, 2026, with three domain specialists. The specialists develop software for children with autism, ADHD and other learning needs, and hold doctorates in human-computer interaction and related engineering fields. The interview was conducted by Rayan Nasrallah and Moustafa Moustafa: a 10-minute presentation of the proposed application, followed by a semi-structured discussion. All three participants consented to the use of the transcript for eliciting requirements (see 41:51–42:48 below).

The transcript is de-identified at the participants' request. They are referred to as P1, P2 and P3. Their names, institutions, locations and other identifying details are replaced with bracketed placeholders. A personal health detail is removed. The automatic transcription was corrected where it misheard words, and filler words, false starts and short acknowledgements ("mm", "yeah") were removed. Each turn keeps the timestamp from the original recording (minutes:seconds), which is how requirement sources in this document cite it (for example, P1, 28:37).

Rayan (0:03)

First, I want to introduce myself and also Moustafa. I'm a student at McMaster University. I'm in my final year and I'm studying software engineering with Co-op. So it's an engineering program where we also take a year or so to do an internship program at a company here in Canada. So I'm on my final academic year, and we reached out to you guys because we're doing a project for our final year. It's called the Capstone. And we'll just introduce the project very briefly, and then we want to hear from you guys mostly and get your expertise on the topics we'll be discussing. I'll let Moustafa introduce himself as well.

Moustafa (0:54)

Yes, very similar to Rayan. Also, my name is Moustafa, 4th year software engineering as well. We're in the exact same program. Both kind of had the exact same story. We lived in [location redacted] our whole lives, or for Rayan, it's half of his life, the second half. And then we came to Canada for university. We've been here for five years now. Yeah, Rayan can start.

Rayan (1:16)
Yeah, I'll start sharing my screen. Can you guys see it?

P1 (1:26)
Yes. So both of you are also like final year students in university. Okay.

Moustafa (1:32)
Yeah, we're both on our 4th academic year, but 5th year with the 16 months we took in the middle for the co-op.

P1 (1:39)
Okay, I see.

Rayan (1:41)
Yeah. So last year, we both went for our internship for 16 months, like between our third year and our 4th academic year. So right now we're on our fifth year of university, but final, final year hopefully. Inshallah.

Rayan (2:04)
So I'll just get started. Today, we're going to introduce Squigglr, the application that we're building for our final year engineering project. And over the next 10 minutes or so, we're going to walk you through what we're planning and what we envision for this project moving forward. Let's start with who this app is for. So Squigglr is meant for a pretty broad group of children. That includes kindergarten and primary school kids who are still building their foundational drawing skills and children with autism, ADHD, or other support needs. And one thing we want to be careful about is that we don't want to assume two children with the same diagnosis need the same kind of support. Every child has their own abilities and their own preferences. Furthermore, around each child, you've got the adults who support them, like parents, guardians, and caregivers, educators and learning support staff, and school administrators. We want Squigglr to help them too. Before we jump into the concept of our application, I want to go through some of the research that shaped our thinking. First, when we say visual motor integration, what we mean is coordinating what you see with how you move. For example, seeing a square and then guiding your hand to draw it. The first paper is Daly, Kelley, and Krauss. They looked at kindergarteners and found that kids with stronger visual motor integration tended to be much better at copying letters legibly. So basically, these skills are closely tied to things like

handwriting. It's a correlation though, so it doesn't prove that practicing one improves the other. The second is Taverna and colleagues. They ran a 10-week program with kindergarten and 1st grade children built around short games like completing paths and tracks. And every activity had easy, medium, and hard levels. After the program, the kids' visual motor integration scores went up. There wasn't a control group, so we can't say for sure that the program caused it, but it's a good example of structured practice with graded difficulty, which is pretty much what we're trying to go for. So putting these together, they support the idea behind structured, graded drawing practice, but they don't show that Squigglr itself will get those results. That still has to be evaluated. The next two papers are about why we're paying attention to motor-related needs in children we're designing for. The 3rd paper compared handwriting between autistic children and children without autism. The autistic group had more trouble specifically with forming letters. So basically getting the shapes of the letters right. And within that group, their motor skills predicted how well they wrote. Then the final paper did a systematic review on children with ADHD. They found that fine and gross motor difficulties come up a lot, but the severity ranges from mild to severe. So for us, the big takeaway is really the variation. Not every autistic child or child with ADHD has motor difficulties. And the ones who do are all different. So instead of building one path per diagnosis, we want guidance and difficulty that can be adjusted. And we want kids to be able to practice as much as they need at their own pace. All right, so let's take the first look at the product. Squigglr has three parts connected to each other. First, there are guided activities. That's where kids practice lines, basic shapes, and simple drawings, and they gradually move from tracing to drawing without any guides. Then there's the creative space, where they can just draw freely and use those skills they've built outside of the structured exercises. And lastly, there's practice history, which keeps a record of attempts so parents and educators can actually see what's being practiced. And across all three, you've got adjustable support, history that's saved between sessions, and everything runs in the browser. Now, Moustafa will go through each of these, starting with how an activity progresses.

Moustafa (6:03)

Yes, thank you, Rayan. So we're going to be using the activity of learning

how to draw a square as our example here. First, the child would see the demonstration, so where to start and which way to actually go to draw it. Then they trace over a visible outline. After that, we can start taking guidance away step by step. So maybe they only get the corners or part of the outline. And finally, they would draw the square completely on their own with no guides at all. Kids can repeat any of these stages as many times as they want, and the support can be turned up or down. Now, one thing we haven't done yet is decide exactly when a child moves from one stage to the next. So we're not setting any accuracy thresholds or a required number of attempts. We think that's something that should be informed by specialists, not just [inaudible]. So on the next slide, this is a concept of a tracing activity in progress. So in this example, you can see the child has traced three sides of the square, but they missed the right side, for example. Instead of failing the attempt or just finishing it for them, the app highlights the part that's unfinished and gives a short prompt so the child can go back and complete it in the same attempt. There are two main principles here. First, we keep the child's actual drawing. We never swap in a perfect square and count that as them doing it on their own. And the support can be adjusted too. For example, how much of the outline is shown, or if it's just the corners, as mentioned on the slide before. Second, the feedback is based on predefined observations that we are manually deciding. We're not proposing an AI model that gives personalized advice on each drawing. It would just be basic math and algorithms to figure out what's wrong with the shape they drew and comment on it if needed. However, the exact implementation details are still a work in progress. Now, so alongside the lessons, kids also get a free creative space where they can draw whatever they want. In this example on the left, a child has drawn something that looks like a bird. The idea is that the app would take that and tidy it up just a little bit. So basically the lines would get a bit straighter, the curves get a bit rounder, but it's still the child's drawing. We're not swapping it out for a perfect clip art bird. And if the model can tell what it is, like here it's clearly a bird, it can use that to guide the cleanup. But even if it can't, it will still just clean up the lines and curves a little bit. The way we're planning to do it is we send the drawing coordinates to a lightweight AI model, and then it sends it back slightly adjusted, like it sends back the same thing with slightly adjusted coordinates at the same scale, and we rebuild the drawing inside the application itself. That way, everything stays editable

so the child can still move things, resize them, and change the colour or erase them, and they can always restore their original drawing, so they stay in control of their own work. This only happens in the creative space, though. In the structured activities that we talked about in the previous slides, we always keep the child's real drawing. The main goal for this creative drawing area is for it to be fun and to give kids a chance to use their skills that they learned outside of the exercises. Now here is the concept of what a parent or educator might see on our app. For every structured attempt or exercise, the record includes the activity and its guidance level, the child's drawing, any difficulties they faced, and the final outcome. So for example, in attempt one, the child missed 2 sections at first. But when they went back and completed them, the record keeps both of those. Attempt 2 was the same outline guidance, nothing was missed, and then a later attempt, so attempt 3 near the bottom, it said that was completed with fewer guides and it was fine. Adults can look at which activities were practiced, attempts over time, what guidance was used, and the difficulties that kept coming up or got resolved, and the actual drawing of the child. And there are some things we're deliberately leaving out. So no scores, no comparisons with the general population, no diagnostic labels, and no recommendations for special classes or treatment. We also don't want to suggest that fewer attempts automatically means better visual motor integration. The whole point is to just make practice easier to understand and to support better conversations about what a child needs. So now moving on to how we're thinking of building it, Squigglr is planned to be a web application. Basically, a child practices and draws on a compatible device that supports drawing, like an iPad or a tablet. Their attempts get saved to an account, and then adults can review that history later from any device. Part of the reason we went with the browser is that installing separate apps on school-managed devices can be pretty difficult, according to our supervisor who has some experience working with schools in Canada. We'll be targeting supported browsers and compatible devices, but we're not promising offline access or that it will work on every device just yet. Okay, so just to sum up the core idea, we have structured practice with lines, shapes, and simple drawings, progression from tracing to drawing without guides, guidance and difficulty suited to each child, free drawing with automatic refinement, accurate records of attempts and difficulties, and a clear practice history for parents and educators. There are also two extra features that

are not part of the main requirements, but we wanted to get your feedback on them as well. The first is camera-based finger tracing, where the shapes and guides show up over a live camera view, and the child traces them with their finger in the air. So instead of having to draw on the screen, they would just do it like this in the air, and it would work the exact same way. The second is taking the same approach of the guided practice we talked about for the shapes and the drawing, but applying it to letters as well. So our main vision is a welcoming place for children to practice drawing at their own pace, with a clear picture of that practice for adults who support them. So that's the whole app idea in summary. Thank you so much for your time. And before we get into our questions for you, please let us know if you have any questions or if anything we chose so far wasn't clear.

P2 (12:07)
Thank you very much.

P1 (12:07)
So thank you very much. Actually, I have a couple of questions. It's a very nice application. One of my questions, actually I have three questions. The first one would be, did you guys co-design this app with educators or therapists or people, like basically the end users who would ultimately utilize this application?

Rayan (12:27)
Moustafa, should I answer?

Moustafa (12:29)
Yeah, go, yeah, for sure.

Rayan (12:31)
So basically, like, we just started this project. We haven't built anything. We haven't even come up with the requirements. So our first kind of thing that we wanted to do is reach out to specialists like you guys that have expertise building apps like this. So the whole point of the engineering project is like we actually build a detailed requirements document over the next month or so. And then we start with the implementation towards December and hopefully finish it towards February and then do a usability test before the end of the second semester, before the end of April. So

we're still building the application, like we're still getting the requirements and everything. And the problem with working with kids, I'm sure you guys know, is there's like ethics involved in everything. So our strategy for eliciting requirements was more to go and ask people like yourselves and also like parents and caregivers and so on. So we haven't worked with kids or anything or educators yet, but like this is our first kind of step, if that makes sense.

P1 (12:42)
Okay.

Moustafa (13:41)
Yeah, and also just to add a small note, we also have our supervising professor for the Capstone Project, who also has some experience in this field. So he's partnered with a couple of organizations in Canada, and he has extensive experience with working with kids with autism and stuff like that. So he's going to be able to help us out as well.

P1 (13:42)
Okay. Okay, sounds great. I don't know why I thought that maybe this app was already being developed. The second question I have is a question that you actually proposed in the last slide about the camera, whether they should trace it using the camera. I mean, the camera-based finger tracing, exactly. This idea is nice, but I would say from my viewpoint, I think it might target a very different task. So whenever they're doing it in the air, that might be more of like a gross motor skill. And then whenever they're drawing on an iPad, that's more on the fine motor skills. So this is something that you could probably check with, you know, either therapists or educators and things. But this is my feedback on that. And you could integrate both. I mean, the camera-based finger tracing is more kind of like an augmented reality component that you would be including. And it's very interesting, especially with children who are more neurodiverse. But you just want to focus; they might be targeting different aspects.

Moustafa (14:55)
Okay. Okay, makes sense. But do you think it might be a good idea in general though to have it? Like, do you think there's actually something helpful about this feature if we do end up implementing it?

P1 (15:12)

I do, because a lot of times, even just looking at child development, before they start the very fine tracing, they usually have big markers, big things, and they're kind of tracing it very largely. So this would replicate that. And that kind of works more on the visual motor, even if it's not fine motor, it's more on the gross motor skills. And it helps them also in the visualization. So if they can visualize tracing a square, it might make it easier for them to, you know, trace it when it comes on an iPad and utilize their fine motor skills. So I think it's definitely an important feature to incorporate.

Rayan (15:48)

For now, though, because we might have a limited scope, like we have a certain amount of time to implement this app, do you think our focus with the shapes and the tracing and teaching them to draw is a good starting point and that we should focus on that before implementing this camera feature? Or do you think it should be part of this application from the get-go?

P1 (16:13)

I mean, if you're limited in time, I would say just focus on the first part, and then you could progress later on to extend it to the camera-based finger tracing.

Rayan (16:23)

Sounds good. Thank you for the feedback.

P1 (16:28)

I have a couple more, like some feedback, but I'll give maybe [P2] an opportunity if she has any questions and then I can continue.

P2 (16:40)

Thank you. It's really very nice.

P1 (16:43)

Your voice is very far, [P2].

P2 (16:45)

Can you hear me now? Can you hear me well? So it's really very nice work, both. It's just very inspiring what you're doing. So I agree with [P1]

in terms of focusing and leaving the camera and the tracking for maybe something to extend beyond this project. That's one thing. A second aspect, actually, while you were introducing it, I was thinking about my own self when I was very young in elementary school. I've always had difficulty in drawing a line. So when I saw this project, it's very nice. I think it's a very relevant problem. That's first. And my question to you: are you targeting specific groups of children with learning needs, or is it kind of broader? I would like to hear from you, who is your sample population?

Rayan (17:45)

So the idea started for kids with special needs. So kids with autism and ADHD, because the whole thing is each case differs. But from what I've seen personally, even as kids tend to get older, they still struggle with these basic skills that really young kids like toddlers and kids in kindergarten learn to develop early on in their lives, right? So our idea is we're mainly focusing on kids with autism and ADHD, but we also want to extend it to very young children, right? Like this is a skill that, like you said, you remembered when you were young. A lot of kids don't practice this when they're young. It's not integrated into learning. And I feel like it is a very necessary skill. So not just kids with special needs that are a little bit older, but also very young kids as well. We think they can use this application. And one big component of our design, which we wanted to ask about later, was how we can let the guidance and the support be controlled by the caregiver or the parent that's there supporting the activity. Or even if we should have a parent or a caregiver supporting the activity in the first place. So that's where it goes on to.

P2 (19:05)

Thinking about it from an inclusive, universal design perspective, this could be relevant to, yes, I agree, autism and ADHD, but then also children with gross motor and fine motor challenges, and also kids with sight challenges. [Personal account of a childhood vision condition removed.] So I believe this could be a training tool for any child in need of it, regardless of their situation, because there are many conditions that could affect our visual perception, and also affect even our handwriting. So just like when I saw this, I felt like...

Rayan (20:13)
Yeah. That's the plan, hopefully.

Moustafa (20:14)
So similar to what Rayan said, you don't think we should only limit it to children with special needs or autism or ADHD or anything like that? It should be general.

P2 (20:14)
Yeah, because...

P1 (20:27)
Yeah, I agree. I don't think you should limit it, because the skills that you're trying to teach now are applicable to all children, right? But I guess depending on their disability, they might achieve it at different stages in life. But if you go to nurseries that are looking at children at two and three years old, this is what they're focused on.

P2 (20:47)
Exactly. That's true. Actually, there is a saying in the world of accessibility: if you design for people with disability, you are actually designing for all. So whatever serves a certain population could be of use for many other populations. Then, thinking even of elderly people, people who get a stroke at some point in life, this is also very effective. I would see it in a rehabilitation centre, for example.

Moustafa (21:18)
Yeah, I didn't think of it, yeah, that's true.

P2 (21:25)
So the use cases could be endless, really. And I think that's also, if it's a startup, a selling point as well, because it could be put in a rehabilitation centre, it could be put in a kindergarten, it could be put in a school or at home.

Rayan (21:45)
Inshallah.

Moustafa (21:47)
Something happened to your mic. Yeah, yeah. I think it disconnected.

P2 (21:57)
We can see you, but we can't hear you.

Rayan (22:00)
Hello!

Moustafa (22:03)
Talk again.

Rayan (22:04)
Okay. Can you guys hear me?

Moustafa (22:06)
Yeah, that was it.

Rayan (22:07)
Yeah, yeah, I forgot what I was saying. Yeah, we wanted to ask actually if at some point we are able to build this app, would you guys, because you guys actually do research, would you guys see yourselves kind of looking into this area, or using this app, or doing further research? Like, if we just start with the idea, like a proof of concept, this app could serve as a proof of concept. Would you guys be able to take it beyond that in the future? And hopefully maybe in February, be able to actually get children to use it and do some observation, not necessarily a trial or research, but just observation activities, because it's difficult for us to do it here in Canada. That's why I wanted to ask.

P2 (22:56)
Yeah, but actually, it's equally difficult [here]. You need to get the ethical approval and the process is quite... Yeah, I think the laws have been tightening over the last 10 years, I would say, especially in this part of the world and in research. But it's easy for us to obtain approvals because we have experience, and an approval would take about three to four months really to obtain. But then actually, as you also speak, I am aware of a group at [another local university] that is looking at handwriting and dyslexia and other learning [needs]. So there is also a very active group there that looks at learning needs and technology, the use of image processing in particular. I can share their contact with you if you're also interested to meet more expertise in the field. And I also

wanted to say something else. Well, I forgot, but these are the two points from my side, but this is really nice, very nice work.

Moustafa (22:59)
Yeah, I think it's difficult everywhere.

Rayan (24:03)
Thank you.

Moustafa (24:03)
Thank you so much.

P1 (24:03)
Yeah, absolutely. I think it's great work. I think it's much needed. And I mean, we would be happy to kind of help guide you. I think there is some merit in trying to, you know, pilot or test this application. Of course, you would need an IRB, but maybe you could do this where you're at, through your university, right? And we could support you through the process and kind of help guide you. So I think it's a great application.

Rayan (24:26)
Yeah, that would be perfect. Yeah. Thank you guys. Appreciate it.

P2 (24:31)
Thank you.

Rayan (24:32)
[P3], you.

P3 (24:34)
Yes, I just want to ask one thing. Honestly, I'm curious, I'm not sure if I missed it or not while you were presenting. Will you ask the children to hold the pen in order to do the line, like the drawing, or will it be their finger? Like, will it be touch, or are you thinking of a pen?

Rayan (24:56)
I think it can be either. We haven't really got into that decision. Our initial thought was a pen. Like, I have this iPad here. So like this, for example. And then maybe as well, there's attachments you can buy for these that make the grip wider. Because I know some kids have trouble holding it like this. So I know there's grips that you can buy that make

it large. So it's very flexible. I feel like it should support tracing with the finger and pen. But what do you think?

Moustafa (25:03)
Thinking about it now, a pen is probably not a good idea, right?

P3 (25:11)
Exactly, yeah.

P3 (25:36)
And that's why I ask, because as [P1] was saying and [P2] also was saying, regarding the fine motor skills of the children, like how they will hold the pen? Will they hold it from the top, or some will hold it like this? You know, I don't have a pen near me, but they will hold it like this. So this might affect how they will draw, which might affect maybe the usability of the app itself. So if your goal is not about the fine motor or something, we could remove the pen. If your goal is more about the drawing itself, only the drawing idea, I guess you can use the finger thing. Not sure if [P2] and [P1] agree with me on this idea, but just to make your app more reliable in real life for the children.

P1 (26:35)
So you mean to draw with their finger on the application?

P3 (26:40)
Yeah, it will be like a touch. They will draw as Moustafa was saying in space using the camera. So it will be the same but on the tablet or iPad itself, rather than holding a pen and having more challenges.

P1 (26:46)
In the air, yeah. That may be a start. I see that. That's a good option for a start. But again, if the idea is to kind of improve the fine motor skills, they have to practice with being able to hold something. So usually they start with something large, because it's easier for their hands, and then they have to go to the smaller ones, like pencils. That's more difficult, so that actually takes some training for them to do. So you could embed difficulty levels, you know, where they start with their finger, which is the easiest, and then with a larger kind of marker or pencil, and then something smaller that's more reflective of what they would use in schools.

Rayan (27:31)

So on that point, we had a question about how we should adjust, because in our thought process, we don't want the app to judge when a child is ready to move from one stage to the next. So the example we gave throughout the slides was the square. But of course, a square is a very simple shape. And then they could move to triangles and then stars and more complex objects that they can trace and then draw by themselves, right? And we'd have an accuracy threshold. But our concern was, should we dictate when the child is ready to move from a square to a triangle or a square to a star? Or is that something we should leave fully to the supervising adult during the activity? And what determines when a child is ready to move on? Because now you introduce an extra degree of complexity, which is holding an object in and of itself; decreasing the size of that is also progression, but that's not something we can necessarily track.

P1 (28:37)

So if you're going to look at that, then you're going to open up a door of assessments, right? Because there needs to be assessment. And I think that, given your time frame, it might be too complicated at this stage. I would keep it user interactive, so the user can decide whether they're going to go from, I don't know, level one, level 2, level 3, right? And then ultimately who decides what they're going to do would be either their parent or the educator or somebody who knows where their level is at. And then they could tell them, you know, you need to practice level one until you're proficient in it, and then they can ask them to move. But if you want the app to do it automatically, that's going to be more advanced.

P2 (29:18)

Actually, just something we had in one of our projects: we had an app for children to learn vocabulary, and we co-designed it with a centre [location redacted] with speech therapists in the centre, and we allowed the teacher to kind of personalize the journey for each child separately, and that's not really automated. The teacher would have a back-end kind of interface to do that. But as [P1] said, that actually will bring you to the other side of the story, which is assessment. But then you might want to think about just allowing the teacher or the caregiver to kind of change the activity

and giving them a back-end view of the progress. But then, thinking also, if you do that, then one would ask, oh, what kind of assessment are you applying here?

Rayan (30:20)

Yeah. So I think maybe the best thing is to do what you guys recommend, which is to kind of leave it to the caregiver or the supervising adult, and we shouldn't worry about assessment right now. It shouldn't be in the scope given the time frame. Maybe something in the future, Moustafa, I don't know if you agree.

Moustafa (30:38)

Yeah, I feel like it's very good to have, but maybe not in these short months. It would be nice, but I don't think we have time to do it right now. So right now, as you guys recommended, it would probably be the adults who would say, okay, this is fine for now, move on to level 2 or level 3, instead of us having to figure out how to map when this child is ready for level 2 or level 3. We can figure that out later.

Rayan (30:52)

Hundred percent, yeah.

Rayan (31:10)

Speaking of that, in the history that we showed, what statistics or what values or what information would actually be useful to the parent or caregiver, so they can say, okay, this progress history is actually benefiting my judgment of whether the child is ready to progress? For example, should we have a percentage, like, okay, the accuracy threshold is above 80%, we passed it, but we're still showing you that this is the accuracy of the kid's drawing?

P2 (31:42)

That's a very good question. Really, I think this is a question to ask the therapists or the teachers, if there are certain defined milestones in the literature that you could kind of have as your benchmark or reference there. And that's something to discuss really with, I think, either an OT, an occupational therapist, or an educational specialist.

Rayan (32:12)

Yeah. Do you guys know any occupational therapists or educational

specialists that we can speak to about something like that, that you could connect us with?

P2 (32:21)

I think the easier way to do that is reaching out to private centres [location redacted]. I think [P3] can share them with you step by step. They are very good, they are very approachable. You might want to introduce this, and they are very interested in technologies. So, maybe that.

P1 (32:42)

And [a local hospital] as well. They have a therapy program where they have occupational therapists as well.

Rayan (32:50)

We're scared that if we email them, we might scare them or something. They might start talking about ethics and stuff, but it's just a small project. It doesn't seem like it, I guess, when we reach out to them, but it's hard to kind of...

P1 (33:00)

Well, the ethics only comes in whenever you say that you want to test your product, you want to meet with the kids and stuff. But if you're asking just to meet with them and get their expertise and get their feedback, it should be okay as long as you're not really interacting with the children at this point. Just to get feedback through a meeting, like what we're doing now, it should be fine.

Rayan (33:26)

Okay. Sounds good. Moustafa, do you have any more questions you wanted to ask?

Moustafa (33:34)

Yeah, this was on a point that we were mentioning before. So with deciding when the child goes on to the next level and stuff like that, a similar question: how should this app be made? Should this app be built in a way that the child by himself can go in and kind of know how to play around and move from level one to level 2, or even go to the creative mode and draw? Or are we expecting that there's always a parent present? Because we've never worked with kids with disabilities, or not even kids with disabilities, just children in general, very young

people. We don't know how to give feedback, like, oh, good job, now try this. Or how do we tell them to draw a square? Or how do we tell them to do this? Or are we assuming that there's a, yeah?

P1 (34:13)

That's a very good question. Yeah, so to answer the first part of your question, I think you should design the application so that it's child-friendly. Because if you're going to design it in a way that the parent has to keep, you know, interfering, parents are going to quickly get annoyed, and, you know, if they give them the iPad to practice, they want them to practice without their constant interference. So then you have to think about how you design the application so that it's child-friendly, meaning that, you know, both the child and the adult can understand the interaction. So a child who's three years old maybe doesn't know how to read yet, but, you know, kids these days are very tech-savvy, so they know how to navigate applications and things. But if you look at the games that they play, it's usually not very text heavy or oriented, so they have images and icons that they navigate in order to go and play. So you would have to design it from the get-go in that kind of manner. Adults obviously will know how to navigate through and install and do the initial parts, but you know, you should also design it in a way so that children can navigate and be able to enter the lesson that they're on, be able to practice, to restart, to go back and forth. Because otherwise, if you make it too high level, where they require constant adult supervision, I think you're going to limit the usability of it.

Moustafa (35:31)

Okay, so it should be somewhat expected that the child can use it alone, play around with it.

P1 (35:36)

Yes, and then, yes.

P2 (35:36)

I think also, sorry, [P1]. It's just, I think also it's your decision on what you want the app to look like. Like, you know, there are many drawing apps in the App Store, right? For children, for mainstream, or maybe not even in mainstream school, it could be used in a centre. But it's your decision to say whether you want this app to kind of follow a process and

have, yes, a child as a user, but then the caregiver can give them the access. So I think these different workflows, you can put whatever you like, because there is no right and wrong, sort of. There's only the way you present it and the angle in which you're presenting it.

Moustafa (36:22)
I see, I see.

P1 (36:26)
No, no, there's no problem at all. I think the next step would be for you guys to meet the end users and the stakeholders, which would be the educators and occupational therapists. They can also give you more feedback. And they're the ones who will be able to guide you, because we're developing similar, not similar, but I mean, similar types of applications, one for occupational therapy. And even the way that we give them positive reinforcement, like if they get something correct, what we say and how we say it and the colours that we use. I mean, we go through the literature, of course, but we also get the feedback from, you know, the experts, because we're doing the design and the technology aspect, but they're the ones who deal with the patient side or the children's side on a day-to-day basis. So these kinds of design aspects, you can build bit by bit.

Moustafa (37:15)
I see. Okay, thank you so much. Rayan, you had something to add.

Rayan (37:18)
I just had a follow-up, yeah, so... The first thing is, what do you guys think about voice, like dictation, like voice support for the kids? Like saying, oh, you did a good job, with like some sound? Or do you guys generally see that that's not a good idea?

P2 (37:39)
No, actually, it is a very good idea. We had an app, the vocabulary learning one, that had a voice input as well. And multi-modality in general, like having two modalities to communicate, and then the user could choose; he could turn off the audio. So multi-modality is very much encouraged in the field of design, right? If you want to design an app... now if you look at the Apple products and the phones in particular, they

also have the haptic modality, so the phone can shake sometimes if there is a notification. So multi-modality is always encouraged, even in the realm of inclusive design and universal design. It's always encouraged, but it's just, as [P1] said, designing it carefully, getting the feedback from the therapists, and even, if you have the chance, getting the feedback from the users themselves. Even if you cannot reach the children as a sample or a group, you could reach an adult group with ADHD and autism. And if you want to expand it to different adult groups, then that's also useful, because then you based your decision on real experiences.

Moustafa (38:58)

It's a very good idea, okay.

Rayan (38:58)

Okay. And then I have a follow-up to that as well. In general, what design choices in apps tend to cause problems for kids with autism and ADHD? And also, what are best practices for these types of apps? Because you all have worked on HCI and these types of apps, making sure the interface is specific to their needs. So what general patterns do you guys recommend we should follow?

P2 (39:26)

So yeah, this is a very good question. I teach an elective course for the undergrads in the spring on ICT accessibility, so how to design accessible interfaces and apps, and the main set of guidelines is called the Web Content Accessibility Guidelines. We call it the WCAG. And it's very famous. It actually is adopted as law in many countries, including the US, and [location redacted] adopted it last year. So I think start with the WCAG and then move forward to more specialized guidelines too. Start with the WCAG and then look at design guidelines for autism in particular; we call them heuristics. So heuristics for autism, app design for autism. There is a lot in the literature; just write in Google Scholar "heuristics for app design for autism" and you'll find a lot. But I would suggest starting with the WCAG and then moving to the more detailed ones.

Rayan (40:39)

Okay, perfect. Thank you. Thank you.

Moustafa (40:39)
Your course would be very helpful for us right now, the accessibility course. I wish, I wish.

P2 (40:44)
Yeah, you're welcome to attend, welcome to come [location redacted].

Rayan (40:44)
Yeah. Yeah. Hopefully, that's the dream. Inshallah. Do you guys have any more comments about that? Like, [P1], [P3]? Do you guys have any advice or any patterns that you guys think of?

P1 (41:08)
I mean, from my end, no. I think what you're doing is really great and it's very timely, and we would be happy, maybe after a couple of weeks or a month, to meet again. And if you have questions, we can again give you our view and feedback, just to kind of help guide you through the process.

Rayan (41:26)
Yeah, that would be perfect. I mean, we have a lot more questions, but because we're running short on time, would it be okay if we sent you guys a form or something that you can answer on your own time, whenever you have time?

Moustafa (41:26)
That would be amazing, yeah. It would probably be just very, very short questions.

Rayan (41:44)
Like, they're very specific.

P2 (41:49)
Yes, of course.

P1 (41:51)
Yeah, that's fine.

Rayan (41:51)
Okay, perfect. Well, we'll create a form then, and we'll send that out to you guys, and it has some more questions that we didn't really get to ask.

And also, the whole purpose of this interview, like we said, was to kind of get your feedback and then elicit requirements from that. I guess now I'm realizing we have a lot more work to do. We have to talk to some more people before we can actually elicit the requirements properly. But I was going to ask you guys if you could sign a form, a consent form, that we can use this transcript to basically do that work. We'll send it to you guys as well, with the questions, at the same time, by email, if that's fine.

P2 (42:37)
Yeah, of course.

P1 (42:38)
No, it's fine.

Moustafa (42:39)
Yeah, it's basically just saying that we can use the transcript of what we talked about here to elicit the requirements, and that we had this meeting, basically. That's all.

P3 (42:40)
Sure, no problem.

P1 (42:47)
Sure.

Rayan (42:48)
Okay, perfect. Thank you guys so much.

Moustafa (42:49)
Thank you so much. Also, does anyone have any other questions before we end?

P1 (42:52)
No problem.

P3 (42:56)
Thank you so much. That's amazing work.

P1 (42:58)
No problem. Best of luck [inaudible].

Rayan (43:00)
Well, we'll be reaching out to you guys again, hopefully in the near future. Inshallah.

Moustafa (43:00)
Thank you guys so much. This was very helpful. Inshallah.

P3 (43:06)
Inshallah, Inshallah.

Moustafa (43:08)
Thank you so much. Thank you so much.

P1 (43:09)
Thank you.

Rayan (43:10)
Thank you guys for your time. Bye-bye.

P3 (43:11)
Thank you, bye.

P1 (43:14)
Bye.

31 Appendix — Specialist Questionnaire Responses

This appendix contains the written responses to a follow-up questionnaire sent to the three domain specialists who took part in the interview (Appendix 30). Two of the three responded. The form was anonymous, so the responses are labelled R1 and R2 and cannot be matched to P1–P3. The responses are reproduced as submitted. The form export cuts off the end of the first and fourth questions, shown here as "…".

Q1

Regarding the free drawing space we showed, where a child can draw anything and AI gently tidies it up; do you think this is helpful for children, or unnecessary? Its meant to be a fun area where…

R1 I think the free drawing space could be very valuable, particularly because it gives children an opportunity to explore and practice without the feeling that there is one "correct" answer. However, I would be cautious about how much the AI modifies the child's drawing. The aim should be to support the child's own effort rather than replace or correct it. I would prefer the AI to provide subtle assistance, suggestions, or optional touch-ups that the child can accept or reject. Preserving the child's sense of ownership and agency is very important. It may also be useful to allow the level of AI assistance to be adjusted depending on the child.

R2 I feel it is a very nice idea so that the children could draw anything and the AI could let's say make it better. However, sometimes the unclear lines are important for the imagination of the child. For example, a family picture with unorganized lines will look better (in the child's perspective) than a very clean drawing. Hence, I feel this should be also optional feature and not mendatory.

Q2

From your experience, what makes practice activities like this motivating for these children, and what makes them frustrating?

R1 Motivation often comes from activities that are achievable, predictable, visually engaging, and provide immediate positive feedback. Choice is also important—for example, allowing the child to choose what to draw, the theme, or the level of difficulty. Small signs of progress and encouragement can be motivating without making the activity overly competitive.

Frustration can arise when the system is too demanding, repeatedly highlights mistakes, requires very precise movements, or changes the child's work in ways they do not expect. It is particularly important that the AI does not unintentionally communicate that the child's drawing is "wrong." The system should accommodate different motor abilities and drawing styles and should recognize effort and progress, not simply accuracy.

R2 The drawing activity itself is likely to be a key source of motivation. However, the experience may become frustrating if the activity includes too many demanding or overwhelming elements. For example, requiring children to draw using a pen or stylus rather than their fingers could make the interaction more difficult, whereas finger-based drawing may provide a simpler and more intuitive experience.

Q3

Would extending Squigglr from shapes, to tracing and forming letters be useful?

R1 Yes, definitely. I think this could be a very useful extension, particularly if it is introduced gradually from simple shapes and strokes to more complex letter forms. There is a natural progression from basic pre-writing shapes to letter formation. However, I would recommend maintaining flexibility rather than expecting one very precise tracing pattern. It could also be interesting to explore Arabic as well as English letters, particularly in our context, as the motor patterns and directionality are different.

R2 Yes, I believe this could represent a natural second stage in the development of the application. The initial phase could focus on establishing the effectiveness and usability of free drawing and basic shape-tracing

activities in supporting children's engagement and learning. Once this concept has been validated, the application could be extended to include letters and numbers.

A more advanced mode could then introduce sequential tracing activities in which children follow individual letters to form a complete word. For example, the child could trace the letters "B-O-Y," after which the application displays an illustration of a boy. To further enhance engagement and reinforce the connection between the written word and its meaning, the illustration could also include a simple animation, such as the boy waving.

Q4

Now that we've talked through everything, what stands out to you most about Squigglr: what worries you, and what seems most promising? Anything we didn't ask about that you think we should consider...

R1 I would also strongly recommend involving children, parents, educators, occupational therapists, and other relevant professionals throughout the iterative design and evaluation process. Beyond measuring whether drawings become more accurate, it would be valuable to examine engagement, frustration, independence, confidence, and whether the skills developed through Squigglr transfer to drawing and writing without the system.

As I also mentioned during the meeting, I see potential for Squigglr to serve a much broader population. Its underlying concept could be relevant across a range of neurodevelopmental and neurodegenerative conditions, as well as rehabilitation contexts such as stroke and acquired brain injury, and for individuals with learning difficulties or motor challenges.

For this reason, I would suggest not restricting the development of the app too early to one specific condition or case study. Instead, I would focus on clearly defining the core capabilities and value that Squigglr can offer—such as supporting fine motor skills, visuomotor coordination, tracing, drawing, and handwriting—and then explore how these capabilities can be personalized and evaluated across different

user groups. This could significantly broaden both the impact and future potential of the application.

R2 The application itself is very promising. However, one concern I have is that the current design may not be sufficiently appealing or engaging for children. I would therefore recommend dedicating substantial attention to the user interface and overall visual design of the application.

The aim should be to create an interface that is engaging, enjoyable, and child-friendly, while at the same time maintaining a simple and comfortable interaction experience. In particular, it would be important to avoid excessive visual or auditory stimuli that could make the application overwhelming. Finding the right balance between an attractive and playful design and a calm, minimally stimulating environment could significantly improve the children's overall experience and willingness to engage with the application.

32 Appendix — Generative AI Use Disclosure

This appendix describes how generative AI tools assisted the team in preparing and reviewing this document. The ideas and project decisions in this SRS came from the team's own discussions, the specialist interview and follow-up questionnaire, and our meetings with our supervisor and teaching assistant. AI tools helped us carry out and express that work; they did not make decisions for us. Each member's own use is described in the Individual Disclosures at the end of this appendix. The team remains responsible for the content of the final document.

32.1 AI Tools Used

The team used the following tools while preparing this SRS:

- **Anthropic's Claude Opus 5.5**, accessed through Claude Code at medium and high effort (Rayan, Moustafa) and through the omp coding agent at high effort (Huzaifah);
- **Anthropic's Claude Sonnet 5.5** at high effort, run as helper agents that rebased and reviewed branches (Huzaifah);
- **OpenAI's ChatGPT (GPT-6)** (Ahmed);
- **the Kiro IDE**, which automatically selects among several versions of Anthropic's Claude, and **VS Code Copilot** using ChatGPT 6 Luna (Shamil);
- **GitHub Copilot code review**, which the team uses to provide automated reviews on pull requests to the project repository.

32.2 How and Where Used

The uses below materially contributed to this document. The sections each member wrote are listed in the revision history and in the Individual Disclosures.

- **Wording and formatting.** Members worked out their ideas and decisions first, and then used AI to help express them clearly and consistently: improving grammar, sentence structure, clarity and flow, and helping with LaTeX formatting. Some members also used AI-generated starting text for brainstorming the layout of a section, then wrote the content in their own words or reworded and rewrote the generated text. This use affected the sections that each member wrote throughout the document.

- **Organizing requirement sources.** AI helped sort through the specialist interview transcript, questionnaire responses and meeting notes, and locate the specific statements, with their timestamps, that support the points the team wanted to make. This mainly affected the stakeholder citations in Sections 1, 2 and 5 and Appendix — Requirement Sources (Section 29).

- **References and tables.** AI helped format citations for the compliance documents in the references and generate the table in the Traceability section (Section 28).

- **Planning the work.** AI helped divide the SRS workload fairly and evenly among members based on the scope and complexity of each task. Some members also used it to split their assignments into smaller GitHub sub-issues and to prepare notes and guides for each one from the course lectures, rubrics, checklists and example documents.

- **Reviewing for consistency.** As sections were merged, members used AI to re-read the document and check that their additions agreed with the rest of the SRS, the course rules and the work teammates had already merged. It flagged missing cases and contradictions. One member followed the three-pass review method from our September 23 guest lecture: one pass as a critic, one looking for missing cases and one checking the course rules.

- **Pull request reviews.** The team uses GitHub Copilot for automated reviews on pull requests, and Copilot left review comments on several of the pull requests for this SRS. Members also used Claude through Claude Code to help review teammates' pull requests by checking the changes against our decisions and the rest of the document. In every

case, the reviewing member decided which comments were valid, which to post, and how to resolve them.

- **Repository management.** Through Claude Code and the omp agent, AI helped manage branches, commits and merges, resolve merge conflicts, and write pull request descriptions.

This team-level part of the appendix was drafted with Claude Code from the Individual Disclosures below, and the team reviewed it.

32.3 Verification of AI-Generated Content

Every AI-assisted passage was reviewed by its author against their intended meaning and the decisions the team had agreed on before it was included. Any suggested wording that introduced commitments or assumptions the team had not agreed on was removed or rewritten, so the document reflects only what the team decided.

Statements attributed to stakeholders were checked against the original interview transcript, questionnaire responses or meeting notes to confirm that each cited statement and timestamp says what the SRS claims. Citations and references that AI formatted or suggested were checked against the source documents.

Review comments from GitHub Copilot, and findings from AI-assisted reviews, were treated as suggestions. The reviewing member judged whether each one was valid before acting on it. Teammates also reviewed each other's pull requests before merging, and members checked their sections against the parts already merged so that the document stayed consistent.

The team is responsible for the accuracy, completeness, consistency, and appropriateness of this document, regardless of whether material was written by a team member or with the help of an AI system.

Individual Disclosures

Rayan: I used Anthropic's Claude Opus 5.5 at medium effort, accessed through Claude Code, while preparing my sections of this document. I developed the ideas and made the project decisions in my sections myself, and Moustafa and I conducted the specialist interview and follow-up questionnaire. A major use of AI was organizing our sources: I used the model to

sort through the interview transcript, questionnaire answers, and meeting notes, and to extract the specific statements, with their timestamps, that supported the ideas I wanted to include. This would have been very difficult to do by hand given the length of the material. I also used it to turn my notes and decisions into draft wording, to improve sentence structure, clarity, and flow, and to help with LaTeX formatting. Through Claude Code, it also helped manage the repository (branches, commits, and merges) and helped me review my teammates' pull requests by checking their changes against our decisions and the rest of the document; I decided which review comments to post.

All ideas and decisions in my sections are my own. I reviewed every AI-assisted passage against my intended meaning and checked each cited statement against the original transcript, questionnaire, or meeting notes to confirm that it says what we claim it says.

Ahmed: I used OpenAI's ChatGPT (GPT-6) while preparing my sections of the SRS, particularly the document structure, learning activity requirements, formal lesson specification and development-phase planning. All the ideas behind my contributions are my own, and project decisions were made through discussion with my teammates. I used AI to help express those ideas in a clear and grammatically correct way, including turning my notes into requirement statements, improving sentence structure and making technical explanations easier to follow. It also helped with LaTeX formatting and with checking the clarity of section references and wording across related requirements. Our team also used AI to help divide the workload fairly and evenly among members, taking into account the scope and complexity of each task. I personally used it to help write clear and descriptive pull request descriptions that accurately summarized the changes I made, making them easier for my teammates to review.

I reviewed the AI-assisted text to make sure it expressed the decisions we had made and stayed consistent with the SRS, our meeting discussions and the parts of the document written by other team members. AI was a writing and formatting aid; the final content and decisions remained my responsibility and the team's.

Shamil: For the SRS I used the Kiro IDE which automatically selects an optimal LLM for consistency among multiple versions of Claude, along with VS Code Copilot which uses ChatGPT 6 Luna. I used the chatbot to generate a skeletons of my sections with starting text based on analysis of the document. I used this starting text for idea generation and brainstorming,

using my own words for the written content, but sometimes repurposing generated text with rewording, paraphrasing, or by completely rewriting. For short standalone sentences I may have kept them as they are. When rewriting, I removed sections based on false assumptions and made sure it was consistent with the rest of the document. None of the text for the reflection questions is AI-generated. I also used AI to make citations for compliance documents in the references document and generate the table for the traceability section. All decisions in my sections are my own and I reviewed all AI-assisted passages.

Moustafa: I used Anthropic's Claude Opus 5.5, accessed through Claude Code, alternating between medium and high effort depending on the task. Because the document was so large, I relied on it heavily to save time on project management. Connected to the GitHub CLI, it split my assignment into ten sub-issues so I could track my work in smaller pieces, checked which tasks I still had left, and made my commits and pull requests. For writing, I decided what each part should say and gave it my ideas, sometimes after brainstorming with it. It then wrote a first draft and the LaTeX for it, which I refined until it said exactly what I intended. In this document, that covered Sections 6, 8, 10 and 11, the SAF-MOU safety requirements in Section 12.2, and my changes to Appendix A. I also used it for review: as teammates merged their sections, I had it read the document again to check that my additions stayed consistent with the rest and contradicted nothing, and it helped me review my teammates' pull requests. GitHub Copilot also gave a light automated review on a few of my pull requests.

All ideas in my sections are my own. I reviewed every AI-drafted passage against my intended meaning before including it, and I decided which flagged inconsistencies and review comments were valid and how to resolve them.

Huzaifah: I used Anthropic's Claude Opus 5.5 at high effort through the omp coding agent, along with Claude Sonnet 5.5 (high) helper agents that rebased and reviewed my branches. GitHub Copilot also left an automated review on one of my pull requests. I used AI in three ways. For planning, it helped me split my assignment into eight smaller sub-issues and build a set of notes and a guide for each one from the course lectures, rubrics, checklists and example documents. For review, I followed the method from our September 23 guest lecture: it checked my sections in three passes, one as a critic, one looking for missing cases and one checking the course rules. It also compared my sections with the work my teammates had already merged. For Git, it managed my branches and commits, merged main into my branches, fixed

merge conflicts and wrote my pull request descriptions. In this document, this covered Sections 7, 9.3 to 9.5, 19, 20, 22 to 24 and 26. All decisions in my sections are my own.

33 Appendix — Reflection

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?

   Rayan: What went well was that most of my sections could be built on what our stakeholders had actually told us, rather than on our own guesses. Moustafa and I interviewed three domain specialists on October 1, 2026, and the de-identified transcript is now an appendix to the SRS. Because of this, Sections 1, 2 and 5 and the requirement sources in Appendix A cite the specialists (P1 to P3) by timestamp, so a reader can check exactly where each statement came from. Our meetings with our supervisor, Dr. Anand, on September 23 and October 14, 2026, and with our TA on September 23 and October 7, 2026, gave us the same kind of traceable source for the constraints and business rules. It also helped that I split my part of the SRS into one issue and one pull request per section, which kept each review small and let each section be checked and merged on its own. The main thing I took from this is that recording sources as we go makes the requirements much easier to justify and to change later.

   Ahmed: What went well for me was being able to build the learning activity requirements around a flow we had already discussed as a team. We knew that an adult would choose the lesson and starting stage, while the child would move through progressively less guided drawing stages based on their attempts. Having those decisions in place made it easier

to write requirements that describe exactly what the application should do instead of leaving the behaviour open to interpretation.

I also think splitting my work into smaller, focused contributions helped a lot. I worked on the SRS structure first, then the learning requirements and formal specification, and later the development phases. That gave the team something specific to review each time. The formal specification was especially useful because it forced me to think through cases like failed attempts, improvement extensions and adult-approved changes. I could then compare those rules with the written requirements and the hazards in the other document. It was satisfying to see the different sections start fitting together rather than feeling like separate pieces of writing.

Shamil: Producing requirements for each of my sections went well for me. I had worked on the Proof of Concept (POC) section for the previous deliverable and one point of feedback we received was that this section was too ambitious and that the POC should not be a smaller version of the final project. Having gone into more detail than necessary in that deliverable, I had a decent idea of what this project required and could express that in words. When my partners were preparing the GitHub issues for us, they included suggestions, examples, and additional details for some of the sections which I used to generate the requirements. Some sections were obviously not required in our project so those were easy as well.

Moustafa: What went well was that we did most of the thinking before we started writing. Rayan and I interviewed three domain specialists on October 1, and they later filled out a follow-up questionnaire for us. Along with our meetings with our supervisor, Dr. Anand, and our TA, that shaped a lot of the app before we assumed anything ourselves. For example, the specialists are why the adult decides when a child moves on, and Dr. Anand is why adults set the thresholds within a range we define. So when it came time to write my sections, it felt more like organizing than coming up with things from scratch. For most requirements I could just ask myself what the specialists, Dr. Anand or the TA had said, and put it there with the source.

Huzaifah: What went well for me was Section 19, off-the-shelf Solutions, because it settled questions that were still open. In peer review,

issue #69 asked what gap Squigglr fills that existing apps do not, and issue #74 asked which existing models we could use for refinement. Our supervisor had also asked whether refinement needs AI at all. Comparing existing products and components let me answer all three in one place: no product we found covers everything Squigglr aims to do, and simple geometric smoothing can handle refinement first, with AI kept as an option.

2. What pain points did you experience during this deliverable, and how did you resolve them?

   Rayan: The main pain point was that decisions kept changing as we learned more, and each change affected several sections at once. In our interview on October 1, 2026, the specialists said that the supporting adult, not the app, should decide when a child moves to an easier or harder stage, so we changed our progression goal from automatic to adult-controlled. In our meeting on October 14, 2026, Dr. Anand advised that thresholds such as the pass criteria and the number of failed attempts should be set by the adult for each child within a range the team defines, which changed several constraints, business rules and goals. The answers to the follow-up questionnaire that Moustafa and I sent the specialists, returned between October 2 and 4, 2026, then confirmed that letters belong in a later stage after shapes, so we kept them as a stretch goal and recorded the order of stages as a likely change. I resolved this by updating every affected section in the same pull request as each change, and by recording anything still unsettled in Open Issues (Section 18), so that nothing was left half-changed. A smaller problem was that several of my branches were open at once and each added a row to the revision history, so later pull requests conflicted on that table; I merged main into each branch and kept every row. I also found that some early drafts included details nobody had actually discussed, so I went back through each section and removed any claim I could not trace to a source or a team decision. Looking back, I learned that requirements are never really finished early in a project, and that keeping each change in one place and tracing it to its source is what makes them manageable.

   Ahmed: My biggest difficulty was keeping everything consistent as the requirements changed. A rule that sounded simple in a meeting often

had several consequences once I tried to specify it. For example, we had to distinguish between automatically moving to the next stage after a passing drawing and changing lessons or moving backward, which should require an adult's approval. We also had to make sure that interrupted attempts did not count toward the attempt limit and that a child could not keep receiving extensions indefinitely. I ended up revisiting the same requirements and state transitions multiple times to make those distinctions clear.

The LaTeX and Git workflow was another challenge. Some section references initially appeared as question marks until I checked the labels and rebuilt the document, and integrating changes from multiple branches meant I had to be careful not to replace newer work from my teammates. One review caught an issue where a previous version would have removed safety requirements that were already on main. I corrected that and became much more deliberate about checking diffs before submitting a pull request. It was a good reminder that writing a correct section is only part of the job; it also has to integrate safely into the team's document.

Shamil: The main pain point was traceability, which was caused by our team's asynchronous progress and work split. We split the work evenly using AI to analyze the sections, but we did not properly take into account which sections would be based off of other sections. Due to this, for example, Ahmed created constants in Section 4.2 attached to my sections before I had even made those sections. Huzaifah realized while doing starting his assigned section that my section aligned better with the work he had already done, so we switched cases mid-project. These problems were compounded by the fact that some started the project and were communicating much earlier than others like Huzaifah and I, so communication was strained and we had to make assumptions in our work. In an attempt to prioritize an even work split, we overlooked that assigning sections based on their link to our work in the Project Statement and Goals would have made everyone's job easier.

Moustafa: The biggest pain point was how long and repetitive the document was. With five of us writing different sections, there was a good chance two people were writing about the same thing in different places, and even a slightly different idea of how something should work meant we would contradict each other. As Ahmed and Rayan merged

their sections while I wrote mine, I kept noticing things that did not line up. For example, my use cases had the child opening a lesson, while Rayan's assumptions said only the adult starts one. Each time, one of us had to change our part, so we would message or call to decide which version made more sense. Checking a document this size by hand was really tedious, so we used AI to review it and flag inconsistencies, and then we decided together which version was right.

Huzaifah: My main pain point was merge conflicts. I worked on eight sub-issues at the same time, each on its own branch, and every one of them added a row to the revision history, while some also added entries to the shared references file. Each time one pull request was merged, the others conflicted on those same lines. I resolved this by merging main into each remaining branch and keeping every row from both sides, so that no one's changes were lost.

3. How many of your requirements were inspired by speaking to your client(s) or their proxies (e.g. your peers, stakeholders, potential users)?

   Rayan: In my sections, I wrote 24 requirements: 7 mandated constraints, 9 business rules and 8 safety requirements. Of these, 18 cite at least one conversation with our stakeholders or their proxies. Ten come from our interview with the specialists on October 1, 2026, including that adults choose the lesson and stage, that guided lesson records keep the child's actual strokes, and that Squigglr gives no diagnosis. Eleven come from our meetings with Dr. Anand on September 23 and October 14, 2026, including that Squigglr is a website, that no information is gathered from or about children, and that the adult sets each child's thresholds within a team-set range. Some cite both. The other six come from our own project documents and the course requirements, such as the deliverable deadlines. Since none of these stakeholders are the children themselves, nearly every requirement that affects how a child uses Squigglr rests on what adults and specialists told us, which is why we plan to check them with the domain specialists again later.

   Ahmed: I contributed 25 individually identified requirements: 23 learning activity requirements (FR-LRN-001 through FR-LRN-023) and two safety-critical requirements (SAF-AHM-001 and SAF-AHM-002). At least 13 of the learning requirements explicitly cite the domain-specialist interview, follow-up questionnaire, or meetings with our su-

pervisor and TA as sources. For example, the specialists' advice shaped the distinction between adult-controlled lesson decisions and routine progression within a lesson. Our supervisor's feedback about letting adults choose thresholds within team-defined ranges also affected the pass, settings and improvement-related requirements.

The remaining requirements mainly came from working through what was needed to make those agreed behaviours reliable and testable. When we specified stylus capture, for example, I had to account for multiple strokes, interrupted input and incorrect classifications even though nobody had handed us a detailed rule for each case. The two additional safety requirements came from my failure-mode analysis of duplicate stylus events and coordinate-mapping errors. I learned that talking to stakeholders establishes the direction, but turning their needs into a complete specification also requires careful engineering judgement about failure cases and edge conditions.

Shamil: I contributed nine performance, four safety-critical, three operational and environmental, four maintainability and support, six security, two cultural, and two compliance requirements for a total of 30 requirements. I did not personally consult any clients or proxies to generate my requirements, but some of my requirements are traceable from other sections that were solicited from stakeholders. For example, NFR-OPS-003 (no installation required) is based on constraints chosen after speaking with clients/proxies. Including the one just mentioned, five of my requirements are based on such sections. NFR-SEC-001 and NFR-SEC-004 cite SRC-004, which is a supervisor meeting (our supervisor is also a client). NFR-CULT-002 draws on NFR-UX-007 which was a solicited UX requirement.

Moustafa: I wrote 27 requirements: 24 look-and-feel and usability requirements (NFR-UX-001 to NFR-UX-024) and three safety requirements (SAF-MOU-001 to SAF-MOU-003). Of these, 25 cite something our stakeholders or their proxies told us. Most come from the specialists. The interview gave us things like icon-based navigation for kids who can't read yet, positive feedback wording, and audio that can be turned off. The questionnaire added that the app should stay calm and not overwhelming, that refinement should be optional, and that feedback should recognize effort. Our supervisor told us to borrow specific WCAG ideas, like large icons, instead of treating WCAG as a

constraint, which shaped the whole accessibility section, and our TA's comment about defining jargon became the plain-wording requirement for adult views. The two that don't cite anyone, SAF-MOU-001 and SAF-MOU-003, came out of our hazard analysis.

Huzaifah: I wrote 52 requirements: 30 creative drawing requirements (FR-CRT-001 to FR-CRT-030), 10 attempt record requirements (FR-REC-001 to FR-REC-010) and 12 progress summary requirements (FR-PRG-001 to FR-PRG-012). Of these, 16 directly cite something our stakeholders or their proxies told us: 12 cite the specialist interview, 3 cite the follow-up questionnaire and 4 cite our meetings with our supervisor, with some citing more than one. The other 36 do not cite a conversation directly, but all 52 build on a goal, constraint or business rule in the SRS, many of which came from those same conversations.

4. Which of the courses you have taken, or are currently taking, will help your team to be successful with your capstone project.

   Rayan: Two courses help directly. The first is SFWRENG 4HC3, Human-Computer Interaction, which I am taking now and where our group's term project is Squigglr itself. The user research we do there, such as planning interviews with caregivers and educators and building personas, feeds straight into the stakeholders and requirements in this document. The second is SFWRENG 3RA3, Software Requirements, which I took in my third year. The requirements document we wrote for that course followed Meyer's template rather than Volere, but the concepts carry over directly. In 3RA3 we learned to separate hard goals, which need a clear, measurable and verifiable satisfaction criterion, from soft goals, which have to be broken down until each part can be measured. That is the same idea as the fit criterion Volere attaches to each requirement. We also learned to check every requirement against three qualities, that it is necessary, verifiable and attainable, and to write requirements with the EARS sentence patterns, such as "When an event occurs, the system shall. . . ", so that the wording stays precise. Being able to transfer those concepts helped me a lot with this deliverable in particular, because I already knew how to write goals and requirements that are precise and checkable and only had to adapt them to a new template.

   Ahmed: SFWRENG 4HC3, Human-Computer Interaction, has been

very relevant because it keeps bringing the discussion back to what users actually need and how they will experience the product. The work on personas, scenarios and requirements elicitation helped me think about the difference between a child drawing in a lesson and an adult setting it up or reviewing the results. That distinction became important when writing our lesson-control requirements.

Our Machine Learning course has also been useful, even though Squigglr is not meant to diagnose or assess children. Learning how a model or classifier produces an output made me think more carefully about what it means for a drawing to pass a fixed criterion and what can go wrong if the classification is inaccurate. The formal models and state-based reasoning we encounter in software engineering and real-time systems are relevant too. I used that way of thinking when writing the guided lesson transitions, such as exactly when the system should advance, retry, suggest an earlier stage or end the lesson. These courses helped me move from describing a feature to specifying behaviour that somebody can actually verify.

Shamil: My software requirements course SFWRENG 3RA3 and my software testing course 3S03 will help most in being successful for the capstone project. For the requirements course, we also had a SRS project that we had to complete by the end of the semester. I gained experience in using PDF version control, Github issues and collaboration, using a template, and writing many identical or similar requirements sections. The testing course taught me the different classifications of testing techniques that I had to consider while choosing mitigation strategies and fit criteria.

Moustafa: Two courses help me the most. The first is SFWRENG 4HC3, Human-Computer Interaction. A lot of what it teaches feels obvious at first, but once you actually go through the concepts you realize how much thought a good interface needs, and I had never thought about UI that deeply before. That matters a lot for Squigglr, because it is built for young kids and kids with disabilities, so being able to navigate the app easily is one of the most important parts of the whole project. Most of my accessibility and usability sections came from that way of thinking. The second is SFWRENG 3A04, Software Design III: Large System Design. In that course we worked with business use cases and wrote use cases with a main flow and

alternative flows, which is exactly what I did for the business use cases and product use cases here. We did the same thing in 3RA3. 3A04 will also help when we plan the architecture for the design document.

Huzaifah: Three courses help the most. SFWRENG 3RA3, Software Requirements, taught me to write requirements with clear fit criteria and to check that each one can be tested, which I did for every requirement in my sections. SFWRENG 4HC3, Human-Computer Interaction, keeps me focused on who the users are, the children and the adults who support them, and how they will actually use the interface. SFWRENG 3A04, Software Design III: Large System Design, covered use cases and system architecture, which will help when we move from requirements to the design document.

5. What knowledge and skills will the team collectively need to acquire to successfully complete this capstone project? Examples of possible knowledge to acquire include domain specific knowledge from the domain of your application, or software engineering knowledge, mechatronics knowledge or computer science knowledge. Skills may be related to technology, or writing, or presentation, or team management, etc. You should look to identify at least one item for each team member.

   Team: The team needs a mix of technical skills and an understanding of the people we are building for. For Ahmed's part, he needs to keep improving at turning a user flow into a clear interface and a set of consistent, testable requirements. Rayan's backend work will require reliable data modelling and access control, especially since one adult account may have several learner profiles and practice histories. Moustafa will need strong experience with browser drawing input and responsive canvas behaviour, because the accuracy of stylus events affects both evaluation and saved records, and will need to learn to build and check the accessibility criteria of Sections 10 and 11, such as contrast, touch target sizes and text alternatives.

   Huzaifah will need a good verification and testing approach, including ways to reproduce edge cases such as interrupted strokes, threshold boundaries and failed saves. Shamil's role will benefit from strong stakeholder communication and documentation skills so that decisions, questions and feedback do not get lost between meetings. As a team, we also need more domain knowledge about early visual-motor practice

and accessible interfaces, and we need to get good at testing without children: no study with children is possible (CON-006), so our reviews with adults and prepared drawings must still tell us something useful. None of us should assume that something which seems intuitive to us will necessarily work well for the children or adults who would use Squigglr.

Thus, the most important knowledge the team needs is domain knowledge about the children Squigglr is for: how autistic children and children with ADHD develop visual-motor skills, how they use apps, and what helps or frustrates them. None of us has a background in this area, and it is not something we could safely guess, because a wrong assumption could make the app harder or more upsetting for the children it is meant to help. That is why it was necessary to get input from people who have designed software for autistic children and children with ADHD, rather than relying only on our own ideas. Their experience is what turned our general idea into requirements we can actually justify.

6. For each of the knowledge areas and skills identified in the previous question, what are at least two approaches to acquiring the knowledge or mastering the skill? Of the identified approaches, which will each team member pursue, and why did they make this choice?

   Team: We see two useful approaches for each of these areas: learning from established sources and then testing what we learn in something concrete. For Ahmed's UI and requirements work, that means reviewing HCI guidance and examples from similar applications, then building prototypes and asking adults to walk through the flows. For Rayan and Moustafa, this was in the form of interviews and questionnaires with domain specialists. They interviewed three specialists on October 1, 2026, because a live conversation let them ask follow-up questions and explore unexpected topics, such as who should control a child's progression. The written questionnaire, answered between October 2 and 4, 2026, covered the topics the interview did not reach, such as refinement in the creative space and letter activities. Using both worked well together: the interview showed the team what it did not know, and the questionnaire filled in the gaps it left.

   The second approach matters because something that looks clear on pa-

per may still be confusing when somebody tries to use it. For Rayan's data and access-control work, official documentation and small integration prototypes can complement each other, and the team will test the multi-learner and saving flows early. For Moustafa's stylus and canvas work, browser input documentation is a starting point, but trying the same prepared strokes on different touchscreen sizes will reveal problems that documentation cannot. For accessibility, Moustafa can learn the tools that check the WCAG 2.2 success criteria we selected, or study how existing children's apps handle icons and touch targets; he will start with the criteria and their checking tools, because our requirements are written against them. For testing without children, the team can learn from how other projects ran reviews with adults, or ask the specialists how they evaluate apps when children cannot take part; we will ask the specialists, who have done this before and offered to help guide a pilot (SRC-005). For Huzaifah, studying testing methods and building repeatable synthetic test cases provide both the theory and evidence that our requirements are met. For Shamil, reading relevant research and following up with adult stakeholders through interviews or questionnaires would help check our assumptions. These approaches let us learn efficiently while still checking that what we build works as intended. All of these approaches fit the work that each person was assigned: Ahmed can identify otherwise hidden user flows by walkthroughs with adults; Rayan can verify that data access works properly with prototypes; Moustafa can compare stylus functions for different devices; Huzaifah can prove that requirements are met reliably; and Shamil can assess the team's current assumptions against research.